Advanced Quantum Materials

Lecture Notes

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Compiled from the course's recorded lectures and slides.

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Session 1

Second Quantization, Mean-Field, and Spontaneous Symmetry Breaking

This is the first session of the course, so there is no previous cliffhanger to pick up. Instead let us open with the question the whole course is organized around: how do we get from a single atom, an object we understand rather well, to a *quantum material*, where electrons moving together produce phenomena no individual atom could ever show on its own? Answering that question is going to require a new piece of mathematical language, second quantization, and a first tool for actually using it, mean-field theory. By the end of today we will have used that tool three times, to build a charge density wave, a magnet, and a superconductor out of the very same starting Hamiltonian, just by changing the sign and the geometry of one interaction term. That three-way split (broken translational symmetry, broken time-reversal symmetry, broken gauge symmetry) is the skeleton the rest of the course hangs on, so it is worth having it clearly in view from today. It is also worth flagging early that today is the *only* place in the course where the general logic of spontaneous symmetry breaking gets developed at length: next session moves straight into symmetry and band structure without revisiting it, and magnetism and superconductivity only get their full, dedicated treatments much later, in sessions 6 and 5 respectively. What we build here is meant to carry that conceptual load on its own.

1.1 Why quantum materials, and why now

Let us start with a bit of perspective on why this subject matters before we get into the mathematics. Every age of human technology has been named after the material that defined it: the stone age, the copper age, and, very recently, the silicon age, which gave us the entire information revolution we are currently living through. If we look ahead, the next age is plausibly a *quantum* age, one in which our technology relies directly on harnessing quantum mechanical behavior at microscopic scales rather than working around it. Quantum materials are the physical platform for that shift: they are what actually realizes quantum matter in the lab.

By the end of today, concretely, we should be able to describe what a quantum material is, explain why quantum mechanics (rather than classical physics) is needed to understand it, work with the basics of second quantization, and rationalize why mean-field theory gives rise to spontaneous symmetry breaking.

This is not just a rhetorical framing. Quantum materials already sit behind technologies we use today: superconductors make MRI machines possible by generating the very large magnetic fields nuclear magnetic resonance needs, and semiconductors are the basis of the solar cells much of our renewable energy strategy depends on. Looking forward, essentially every

serious quantum computing platform, whether built from superconductors, semiconductors, or combinations of the two, is a quantum materials problem at its core. It is also worth noting, briefly, that this connects to several of the UN Sustainable Development Goals: quantum computing to goals 8 and 9 (economic growth, industry and innovation), solar energy and low-consumption superconducting or topological electronics to goals 7 and 13 (clean energy, climate action), and the inherently international, cross-disciplinary nature of the field to goal 17 (partnerships). None of this is mathematics, but it is worth remembering why we are doing the mathematics.

Within the course, we will meet two kinds of examples that make a useful contrast. Semiconductors are, in a sense, the easy case: their properties follow from single-particle physics, which you have likely already seen in an earlier materials course. Superconductors are the harder case: understanding them requires accounting for how electrons interact with each other collectively, which is exactly the many-body machinery we start building today.

1.2 A tour of the quantum materials we will meet

It is worth previewing, briefly, the range of materials this formalism is eventually going to let us understand, so that today's abstract tools have concrete targets to attach to later.

- **Insulators**, conventional ones like magnesium oxide, sodium chloride, or boron nitride, where a finite band gap simply prevents electrons from propagating. *Topological* insulators look identical in the bulk but behave completely differently at the edge, a distinction we will need a dedicated session to explain.

- **Semiconductors** (gallium nitride, silicon, gallium arsenide), which share the single-particle description of insulators but are lightly doped, so they carry free carriers that do the actual work in electronics and solar cells.

- **Magnets**: ferromagnets, antiferromagnets, and spin spirals, all *correlated* materials in the sense that they exist only because electrons repel each other via the Coulomb interaction. A distinct and more exotic case is the *quantum magnet*, where local moments exist at every site but are entangled with each other so thoroughly that the material has no net magnetization at all, microscopically, and every site forms a singlet with its neighbors. We will return to these at length later in the course.

- **Dirac and topological matter**, with graphene, a single honeycomb layer of carbon, as the prototype: electrons here propagate with a topologically protected phase (a Berry phase). The same family includes quantum spin Hall insulators (helical edge states), Chern insulators (chiral edge states), and chiral superconductors (which host Majorana excitations).

- **Superconductors**, from conventional, phonon-mediated ones like aluminum, to *correlated* superconductors such as the cuprates and pnictides, whose mechanism remains one of the biggest open problems in the field.

- **Fractional quantum matter**, where the fundamental excitations are literal fractions of an electron. The fractional quantum Hall state is the best-established example, observed for decades in materials like gallium arsenide; fractional Chern insulators are a newer, more accessible route to the same physics, and a promising building block for new quantum devices.

1.3 Emergent quasiparticles: what putting atoms in a lattice actually buys you

Here is the puzzle that motivates everything that follows. All of this phenomenology is built from the same four ingredients, electrons, protons, neutrons, and photons, and a single, isolated atom shows none of it: no fractionalization, no vortices, no magnons, no spinons. So how does arranging identical atoms into a lattice produce such a rich zoo of collective behavior? Before we build the formalism to answer that question properly, it helps to know what we are looking for. A few of the emergent quasiparticles we will meet along the way:

- **Phonons**: lattice vibrations, the simplest quasiparticle of all, since they exist even classically.
- **Magnons**: spin-1 excitations of a magnetically ordered state, a small precession of the local moment that propagates energy (and spin) through the material. Notably, magnons can propagate even in a Mott insulator, where charge cannot: a magnetic insulator can still carry spin and heat.
- **Spinons**: spin-1/2, charge-neutral excitations that live in quantum magnets, where time-reversal symmetry is *not* broken. A spinon is, in a sense, only the spin half of an electron, with the charge half frozen out, an example of electron fractionalization known as spin-charge separation.
- **Dirac fermions**: massless-looking electrons with a linear dispersion, arising not from interactions but from lattice geometry (graphene's honeycomb lattice is the standard example).
- **Vortices**: topological defects in a superconductor's phase, themselves dynamical quasiparticles, and useful enough that they underlie some of our most precise measurements of the electron charge and Planck's constant.

The course's own summary slide for this list also shows Majorana fermions (which we will meet again shortly, tied to chiral superconductors) and anyons alongside these five. Anyons in particular are not discussed any further in this session, slides or speech; they belong properly to the fractional quantum Hall material in a later session, and are mentioned here only as a name to recognize.

1.4 Setting up the formalism: the Schrödinger equation, two ways

Every quantum mechanical system is governed by the Schrödinger equation, built from a Hamiltonian (think of it, for our purposes, as a matrix) acting on a wave function. Given the wave function at time zero and the Hamiltonian, we can in principle integrate this equation forward and know the state at any later time. This statement is completely general: it holds whether the wave function describes one particle or an entire many-body system. The practical difference is enormous, though. Single-particle problems can always be solved exactly, in every case that matters to us. Many-body problems generically cannot, and the difficulty grows extremely fast with system size, as we will see concretely in a moment. Since most of the quantum matter we care about is genuinely many-body, we need a systematic (if often approximate) way of handling that difficulty, and that is what today's session builds.

1.4.1 First quantization

When electrons can be treated as independent of one another, we can describe the system with a single-particle wave function living in an ordinary Hilbert space: this is *first quantization*, the language you already know from a standard quantum mechanics course. It is a remarkably capable tool, sufficient to describe metals, semiconductors, insulators, and even topological insulators.

1.4.2 Second quantization

First quantization assumes a fixed number of particles, and that assumption fails for a whole class of quantum materials: any system where electrons are not independent, or, more strikingly, where the *number* of electrons itself is not fixed. For those systems we need *second quantization*, a formalism built not around "how many electrons are present" but around how electrons are created and destroyed. It turns out to be essential for superconductors, superfluids, and correlated materials generally, and it is what makes phenomena like spinon fractionalization straightforward to describe at all.

1.5 A warm-up in first quantization, and the puzzle that breaks it

Consider the simplest possible single-particle system: a lattice of sites, one level and one electron each, forming some finite island (a quantum dot, say). With no many-body interactions, this is an ordinary diagonalization problem, exactly analogous to a particle in a box: a lowest energy state, a first excited state, and so on. In this language, "one electron in the ground level" and "one electron in the first excited level" are simply two different single-particle states, or superpositions of the two, but in every case there is exactly one electron in the system, full stop.

Now here is the puzzle. Suppose we want to describe a quantum dot in a coherent superposition of having zero electrons and having two electrons, each with 50% probability. We cannot write this down as an ordinary wave function, because a wave function needs a fixed number of coordinates, and there is no sensible way to have zero coordinates and two coordinates "at the same time." This is not a contrived example either: exactly this kind of reasoning, about a fluctuating number of electrons, is what lets us understand how local magnetic moments couple to each other in real oxides (the Goodenough-Kanamori rules for predicting ferro- versus antiferromagnetism), and it is built into the very definition of a superconductor, whose mean-field order parameter conserves only the *parity* of the electron number, not the number itself. This particular non-particle- conserving description of superconductors has a name of its own, the Bogoliubov-de Gennes formalism, that the slides introduce here but that we will only properly use once we reach the superconductivity session.

1.6 Building second quantization: creation and annihilation operators

To describe systems like this we introduce two operators at every site i: the annihilation operator c_i, which destroys an electron at site i, and the creation operator $c_i^\dagger$, which creates one. The Hamiltonian, and all of the system's dynamics, gets written entirely in terms of these operators.

We define the vacuum (empty) state $|0\rangle$ by

$$c_i|0\rangle = 0 \quad \text{for every site } i. \tag{1.1}$$

A two-site hopping term, an electron able to jump between sites 0 and 1, is then written

$$H_{\text{hop}} = t\left(c_0^\dagger c_1 + c_1^\dagger c_0\right), \tag{1.2}$$

read as "destroy an electron at 1, create one at 0" plus the reverse process. Building up from the vacuum by applying creation operators gives us every state of the two-site system: $|0\rangle$, $c_0^\dagger|0\rangle$ (one electron at site 0), $c_1^\dagger|0\rangle$ (one electron at site 1), and $c_0^\dagger c_1^\dagger|0\rangle$ (both sites occupied).

Notebook check-in: `session1.ipynb`, "The spectra of a dimer" (Core)

Build exactly this two-site system and diagonalize it. Check that the Fock states above really do reproduce the spectrum you'd get from an ordinary two-level first-quantization treatment, before we add any interactions.

1.7 Enforcing fermionic statistics

Nothing so far distinguishes fermions from bosons: everything above applies equally to both. The distinction is imposed through the operator algebra. Fermi statistics requires the many-body wave function to pick up a minus sign when we exchange two particles, and as a direct consequence, a given single-particle level can hold zero or one fermion, never two. We enforce this with anticommutation relations,

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \qquad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \tag{1.3}$$

The second relation is doing the real work: setting $i = j$ makes $(c_i^\dagger)^2 = 0$, which is exactly the statement that you cannot create two electrons in the same level. It simultaneously enforces the wave function's antisymmetry, so Pauli exclusion is built into the algebra itself rather than added by hand.

1.8 First versus second quantization: the same problem, viewed two ways

For a purely single-particle tight-binding Hamiltonian, first and second quantization are, in practice, the same amount of work. Write the first-quantized Hamiltonian with hopping matrix elements t_{ij} between orbitals i, j; the second-quantized version has the very same t_{ij} multiplying $c_i^\dagger c_j$. Diagonalizing either one uses the same unitary transformation U and produces the same eigenvalues. The only formal difference is that in first quantization the diagonalizing coefficients form a vector, while in second quantization they become operators; the physics is identical either way.

1.9 Where the two formalisms diverge: the exponential wall

The gap between the two languages opens up the moment we add genuine many-body interactions, and it opens up fast. Consider just three sites with a generic quartic (four-operator) interaction term added to the kinetic Hamiltonian. We are no longer allowed to restrict ourselves to single-particle states; we need the full Fock space: the empty state, three states with one electron, three states with two electrons, and one state with all three sites filled, for a total of $2^3 = 8$ many-body states; figure 1.1 draws both sides of the comparison explicitly.

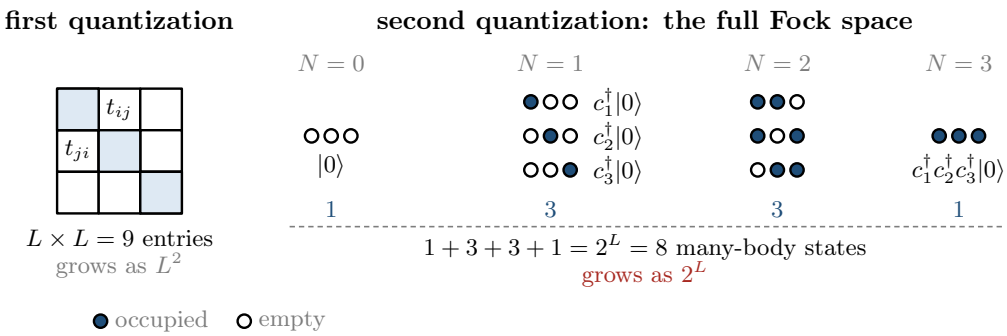


Figure 1.1: The exponential wall, for three sites. A single-particle tight-binding problem is an $L \times L$ matrix (left). The interacting problem needs the full Fock space (right): every site independently empty or occupied, 2^L states in all, organised here by particle number. For $L = 3$ this is a modest 8; by $L \approx 20$ it is already about a million, which is where brute-force diagonalization stops.

In general, for L sites, the single-particle problem is an $L \times L$ matrix, while the many-body problem is $2^L \times 2^L$, and the ground-state wave function itself needs 2^L coefficients to write down. This is why $L \approx 20$ already puts us near the practical limit of what can be handled by brute force. The exact rate of growth depends on what kind of site we are dealing with: spinful fermions give four states per site (empty, up, down, doubly occupied), spinless fermions give two (relevant once a single spin species is isolated, by strong magnetism or a strong field), and pure spin sites, which we will need once we get to quantum magnets and Kondo systems later in the course, also grow exponentially but with a different underlying algebra.

1.10 A first look at eigenstates: finite tight-binding islands

Before turning to interactions, it is worth pausing on what a purely single-particle Hamiltonian already gives us, because it is something we can directly compare to experiment. Take a tight-binding Hamiltonian $H = \sum_{ij} t_{ij}\, c_i^\dagger c_j$ on a honeycomb lattice cut into a finite island, triangular, hexagonal, or rectangular, and diagonalize it. Plotting the probability density of a given eigenvector in real space is directly comparable to what a scanning tunneling microscope measures. Qualitatively, these look exactly like finite-size "particle in a box" states: the lowest-energy state is peaked near the island's center, and higher states develop progressively more nodes. This is a nice first checkpoint: single-particle diagonalization, nothing more, already reproduces a recognizable and experimentally measurable structure.

Notebook check-in: `session1.ipynb`, "Finite sized chains" (Core), and "Density of states of a finite system as a function of the system size" (Challenge)

Diagonalize a finite chain (or island) and look at how the eigenstates' spatial profiles and the density of states change as you grow the system. This is the numerical version of the particle-in-a-box argument above, and a good place to build intuition for how a genuinely infinite, periodic system (which session 2 gets to) is approached in the finite-size limit.

Two software tools will carry us through this and every following session: **PyQula**, a Python library for tight-binding and mean-field calculations, topological properties, and spectral functions, used through the course's Jupyter notebooks, and **Quantum Lattice**, a GUI front end built on PyQula that needs no coding at all. Both are open source, and both are worth having open alongside these notes.

1.11 Spontaneous symmetry breaking: why a symmetric Hamiltonian can have an asymmetric ground state

Before we set up interactions in earnest, it is worth stepping back and asking a more general question, because everything we do for the rest of the session is really an instance of it. Physical theories tend to have a lot of symmetry built in: space is symmetric, time is symmetric, and so on. But reality is not: an object sits in one place and not another, a system evolves forward and not (usually) backward. So why do symmetries present in the Hamiltonian so often fail to survive into the ground state?

Let us start with a very down-to-earth example. Imagine sitting down at a table for a meal with colleagues, bread in front of you on both sides. You have to choose: bread on the left, or bread on the right. There is, of course, a "correct" etiquette answer, but suppose neither of us knows it. Some people will pick left, others right, and once the choice is made, everyone at the table settles into one of two equally good arrangements: everybody-left, or everybody-right. The setup you started from, bread symmetrically placed on both sides, was perfectly symmetric. The outcome, once chosen, is not. That mismatch, a symmetric problem with a non-symmetric solution, is spontaneous symmetry breaking in miniature.

The same logic, made rigorous, is exactly what happens in a classical Ising ferromagnet. The Hamiltonian's set of ground states is symmetric: all spins up is just as good a solution as all spins down. But take a *very* large system, formally infinitely large, the thermodynamic limit, and the system has to commit to one of the two: all spins up, or all spins down. Which one you measure is decided by fluctuations, not by the Hamiltonian, and once decided, the state you observe is no longer symmetric even though the Hamiltonian that produced it always was. This thermodynamic-limit requirement is essential: spontaneous symmetry breaking, strictly speaking, is a statement about infinite systems, and it is a generic feature of many-body interactions and correlations, not something that shows up for a handful of particles. Figure 1.2 collects all of this into one picture: a symmetric energy landscape whose minima are not symmetric, with a barrier between them that only becomes insurmountable in the thermodynamic limit.

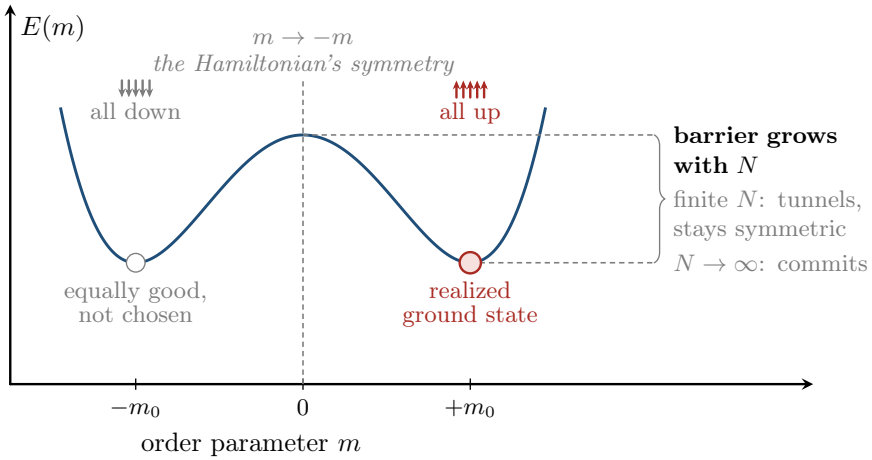


Figure 1.2: Spontaneous symmetry breaking. The energy as a function of the order parameter is symmetric under $m \to -m$, exactly as the Hamiltonian is, yet its minima are not: the system must sit at $+m_0$ or $-m_0$. Which one is chosen by fluctuations, not by the Hamiltonian. The barrier between the two grows with system size, which is why spontaneous symmetry breaking is strictly a statement about the thermodynamic limit.

This gives us a genuinely useful classification scheme: sort states of matter by *which* symmetry they spontaneously break.

- Broken **time-reversal symmetry** gives classical magnets: the magnetization is odd under time reversal, so a magnetic ground state is simply not invariant when you run the movie backward.
- Broken **rotational (inversion) symmetry** gives ferroelectrics: materials that spontaneously develop an electric polarization, the electric analog of a magnetization, breaking the rotational symmetry of the underlying crystal.
- Broken **gauge symmetry** gives superconductors: the phase of the pairing order parameter (or, equivalently, of the creation and annihilation operators) becomes physical once the order sets in. Many observable superconducting phenomena, notably the Josephson effect, are direct consequences of fluctuations in this broken phase; we will develop that properly in the superconductivity session.

It is worth being explicit that this symmetry classification, useful as it is, does not capture everything. Topological materials, topological insulators and fractional Chern insulators among them, can share *exactly* the same broken (or unbroken) symmetries as an ordinary, trivial material and yet behave completely differently, because they differ in their topological classification rather than their symmetry classification. That distinction is the subject of its own dedicated session later in the course; for now, just keep in mind that "which symmetry is broken" is not the whole story of quantum matter.

Every instance of spontaneous symmetry breaking also comes with its own emergent quasiparticle, the low-energy fluctuation of the order that formed:

Broken symmetry	**Realized in**	**Quasiparticle**	**Spin**	**Charge**	**Gap**
Continuous translational	Crystals	Phonon	0	0	Gapless (always)
Spin-rotational $SO(3)$	Magnets	Magnon	1	0	Gapless or gapped
Gauge	Superconductors	Higgs mode	0	0	Gapped

Notice that magnons and the Higgs mode both carry zero charge, but very different spin, and that only the phonon is guaranteed to be gapless in every material; whether a magnon or the Higgs mode is gapped depends on the details of the electronic structure.

Since we cannot literally run time backward in the lab to check whether a material breaks time-reversal symmetry, we instead measure quantities that are themselves odd under time reversal. Two are especially useful: the **Hall conductivity** (a transverse current generated by a longitudinal voltage) and the **magneto-optical Kerr effect** (a different optical reflection for left- versus right-handed circularly polarized light). A nonzero signal in either one is direct evidence of broken time-reversal symmetry.

That gives us, finally, a precise, formal definition of a magnet, beyond "the thing that sticks to your fridge": a magnet is a many-body state whose wave function is not invariant under the time-reversal operator $\hat{T}$,

$$\hat{T}|\Psi\rangle = |\Psi\rangle \;\Rightarrow\; \text{not magnetic}, \qquad \hat{T}|\Psi\rangle \neq |\Psi\rangle \;\Rightarrow\; \text{magnetic}. \tag{1.4}$$

A bar magnet's north and south poles swap under time reversal (since a magnetic pole is tied to an orbital angular momentum, and angular momenta flip sign under $\hat{T}$), so a bar magnet fails the invariance test in (1.4) and is magnetic by this definition. A plain, non-magnetic stick, by contrast, looks the same under time reversal and passes the test.

With this general picture in hand, let us now go build it explicitly, three times over.

1.12 The general interacting Hamiltonian, and the tool we'll use on it

We are now ready to state the problem this whole session is building toward. The Hamiltonians we care about take the generic form

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j \;+\; \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j^\dagger c_k c_l, \tag{1.5}$$

a free (single-particle, exactly solvable) part plus a quartic interaction. The interaction itself can come from plain electrostatic (repulsive) Coulomb repulsion, or from an effective interaction mediated by some other quasiparticle (phonons, magnons, plasmons), which can render the net interaction attractive rather than repulsive. We will treat these interactions phenomenologically, as given effective models, rather than deriving them microscopically from first principles.

Solving (1.5) exactly is generically hopeless, for exactly the exponential reason in Section 1.9. **Mean-field theory** is the tool that lets us extract systematic, if approximate, solutions from it anyway, and it is capable of predicting a wide range of symmetry-broken states. We are going to work through three of them today, the same three cases classified in Section 1.11:

1. **Broken time-reversal symmetry**, magnetism: run the system's movie backwards and it looks different from the original.
2. **Broken (lattice translational) symmetry**, the charge density wave: a Hamiltonian that starts out the same at every site ends up with charge on some sites and not others.
3. **Broken gauge symmetry**, superconductivity: the system develops an order parameter with a physical phase, something the original Hamiltonian's symmetry forbids.

Figure 1.3 sets these three branches side by side, together with the equation each one is developed from and the session it returns in.

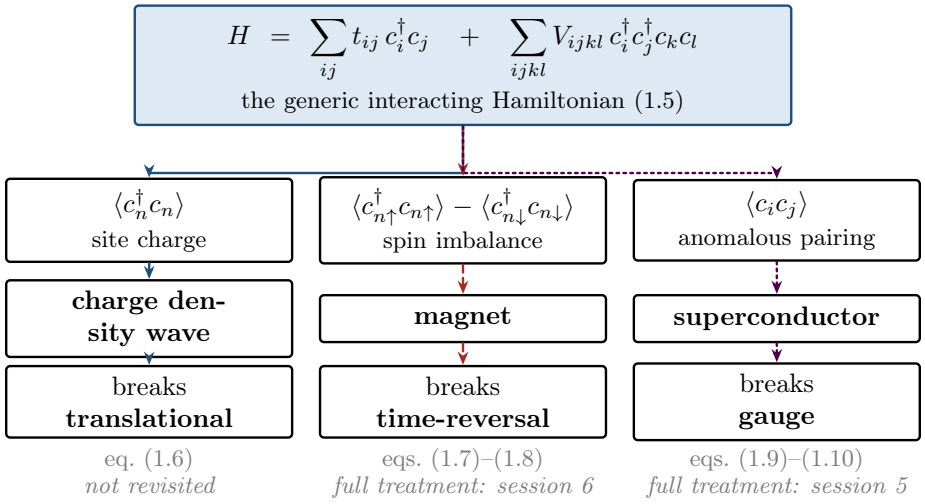


Figure 1.3: One Hamiltonian, three ordered states. The same generic interacting Hamiltonian (1.5), decoupled around three different bilinear expectation values, produces a charge density wave, a magnet, or a superconductor, each breaking a different symmetry. This branching is the organising skeleton of the rest of the course.

Two of today's three cases (magnetism and superconductivity) will come back for a full, much deeper treatment in their own dedicated sessions later in the course; think of what follows as the first pass, establishing the mean-field machinery itself, not the last word on either one.

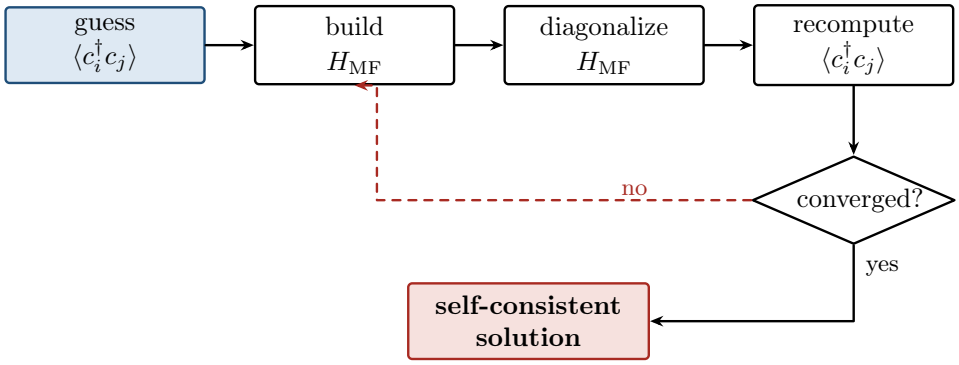
1.13 Worked example 1: the charge density wave

Take the simplest possible interacting case: a spinless one-dimensional chain, nearest-neighbor hopping t, plus a nearest-neighbor Coulomb repulsion V,

$$H = -t \sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.} \right) + V \sum_n c_n^\dagger c_n \, c_{n+1}^\dagger c_{n+1}. \tag{1.6}$$

As written, every site is equivalent by translational symmetry. The quartic term is the troublesome one, so mean-field theory factorizes it: replace one pair of operators by its expectation value, leaving the rest untouched. This is, to be clear, an *uncontrolled* approximation (a more rigorous route exists via a path integral and a saddle-point approximation), but it is the practical workhorse we will use throughout the course. The result is that the interaction term becomes a spatially dependent single-particle on-site energy: nothing forces the expectation value $\langle c_n^\dagger c_n \rangle$ to be the same at every site, so even though the bare Hamiltonian (1.6) is translationally symmetric, the mean-field solution need not be. That mismatch, a symmetric Hamiltonian with an asymmetric ground state, *is* the charge density wave, and it is exactly the same mismatch Section 1.11 described in the abstract.

More rigorously, mean-field theory is a variational search over single-particle product states, minimizing $\langle H \rangle$ with respect to the bilinear expectation values $\langle c_i^\dagger c_j \rangle$ that parameterize the mean-field Hamiltonian. Since the Hamiltonian depends on these expectation values and the expectation values depend on the Hamiltonian, we solve it self-consistently: guess a Hamiltonian, compute the expectation values it implies, build a new Hamiltonian from those, and repeat until the process converges. Figure 1.4 draws that loop; every worked example below, and every notebook callout in this chapter, runs it.



H_{MF} depends on the expectation values, and the expectation values depend on H_{MF}: the loop is what resolves that circularity

Figure 1.4: The mean-field self-consistency loop. Guess the bilinear expectation values, build the single-particle Hamiltonian they imply, diagonalize it, recompute the expectation values that result, and repeat until they stop changing. All three worked examples of this session, and every notebook callout in this chapter, run this identical loop.

Numerically, this shows up exactly as you would hope: at $V = 0$ the spectrum is gapless (a metal); as V grows, a gap opens in the single-particle spectrum and widens, while the charge imbalance between even and odd sites grows from zero toward its maximal, fully polarized value. Figure 1.5 shows the two limits in real space, alongside the spectrum each one has.

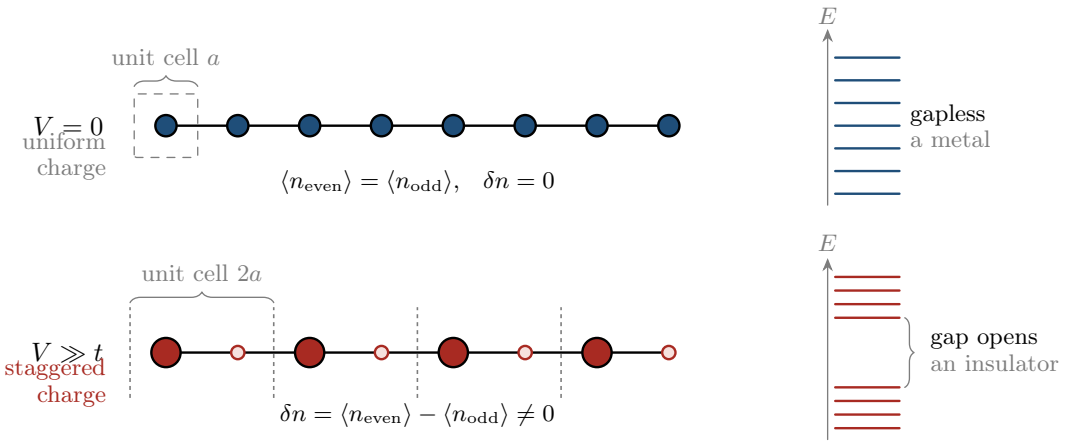


Figure 1.5: The charge density wave. At $V = 0$ every site of the chain (1.6) carries the same charge and the single-particle spectrum is gapless (top). Once the nearest-neighbour repulsion drives the mean-field instability, charge piles up on alternating sites, doubling the unit cell and opening a gap in the spectrum (bottom). The Hamiltonian is translationally symmetric in both cases; its ground state is not.

Notebook check-in: `session1.ipynb`, "Symmetry breaking in the spinless fermionic chain" (Core), and "Gap opening in a spinless interacting model" (Challenge)

Run the self-consistency loop of Section 1.13 yourself: start from an unbiased guess, iterate, and watch the site-resolved charge imbalance grow with V, alongside the gap opening in the spectrum and density of states.

1.14 Worked example 2: magnetism

Now restore the spin degree of freedom and switch the interaction to an on-site (Hubbard) term between spin-up and spin-down electrons,

$$H = -t \sum_{n,\sigma} \left(c^\dagger_{n\sigma} c_{n+1,\sigma} + \text{h.c.} \right) + U \sum_n c^\dagger_{n\uparrow} c_{n\uparrow}\, c^\dagger_{n\downarrow} c_{n\downarrow}. \tag{1.7}$$

The sign of U alone decides the physics: $U < 0$ (attractive) drives superconductivity, which we will get to in Section 1.15; $U > 0$ (repulsive) drives magnetism, our focus here.

The mean-field decoupling proceeds exactly as before, now applied to $\langle c^\dagger_\uparrow c_\uparrow \rangle$ and $\langle c^\dagger_\downarrow c_\downarrow \rangle$ separately, giving a bilinear Hamiltonian with spin-dependent on-site energies. Define the magnetization

$$m = \langle c^\dagger_\uparrow c_\uparrow \rangle - \langle c^\dagger_\downarrow c_\downarrow \rangle. \tag{1.8}$$

Self-consistency gives $m = 0$ automatically for $U < 0$, and a finite m for $U > 0$: a genuinely magnetic solution.

It is worth being precise about *which* symmetry this breaks, tying back to the formal definition in (1.4). Time-reversal symmetry reverses the arrow of time and, with it, every angular momentum in the system, including spin; on the fermionic operators this means it swaps $c^\dagger_\uparrow c_\uparrow \leftrightarrow c^\dagger_\downarrow c_\downarrow$. Since m in (1.8) flips sign under exactly that swap, a nonzero m means the ground state fails the invariance test in (1.4): a magnet, by definition. More generally this is a breaking of the *continuous* spin-rotational symmetry, and it is that continuous symmetry breaking which produces the gapless magnon excitations named in Section 1.11's table; we will study them properly in a later session, and only need the mean-field ground state itself today.

Numerically, ramping up U opens a **Mott gap** in the spectrum, growing linearly with U at large U and associated with the formation of a genuine local moment on every site.

Notebook check-in: `session1.ipynb`, "The interacting Hubbard dimer" (Core), "Magnetic order driven by interactions in a chain" (Core), "Gap opening of the Hubbard model in a chain" (Challenge), and "Gap opening in the Hubbard model on the honeycomb lattice" (Challenge)

Work through the two-site (dimer) case first to get a feel for how the self-consistency loop converges on a small, hand-checkable system, then move to the extended chain and finally the honeycomb lattice (the same finite-island geometries from Section 1.10, now with interactions switched on) to watch the Mott gap open and grow with U.

1.15 Worked example 3: superconductivity and broken gauge symmetry

Return to (1.7) with $U < 0$. The mean-field decoupling we used above always keeps one creation and one annihilation operator together, but nothing forces that choice: we are equally free to take an expectation value between two creation operators (or two annihilation operators),

$$\Delta = \langle c_i c_j \rangle. \tag{1.9}$$

This anomalous, "pairing" term looks qualitatively different from anything we have written so far, but it is a perfectly legitimate mean-field term. Self-consistently, Δ comes out zero for repulsive interactions and can become nonzero for attractive ones: precisely the superconducting order parameter. This worked example is slides-only, with no accompanying notebook section this session; the pairing (Bogoliubov-de Gennes) machinery needed to solve it numerically is left for the dedicated superconductivity session.

Why does this order break gauge symmetry? A gauge transformation acts on the operators as

$$c \to c\, e^{i\varphi}, \qquad c^\dagger \to c^\dagger e^{-i\varphi}. \tag{1.10}$$

Any term built from one creation and one annihilation operator, $c^\dagger c$, is invariant under (1.10), since the two phases cancel. But the pairing term in (1.9) is built from *two* annihilation operators (or two creation operators), so it picks up a phase $e^{2i\varphi}$ (or $e^{-2i\varphi}$) under the very same transformation, and does not cancel. A superconductor with $\Delta \neq 0$ therefore spontaneously breaks the Hamiltonian's gauge symmetry, exactly the "broken gauge symmetry" entry from Section 1.11's classification. This is tied directly back to Section 1.5's fluctuating-electron-number puzzle: breaking gauge symmetry and losing a fixed particle number are, physically, the same statement. A mean-field superconductor genuinely does not have a fixed number of electrons, only a fixed parity.

1.16 Closing summary

- Quantum materials realize an enormous range of phenomena (topological order, fractionalization, correlated superconductivity) that no single atom shows on its own; understanding how they emerge from electrons in a lattice is this course's central question.
- **Second quantization** is the language that lets us treat both fixed-particle-number (single-particle) and fluctuating-particle-number (many-body) systems on the same foot-

ing, built from creation/annihilation operators satisfying the anticommutation relations (1.3).

- The many-body Hilbert space grows as 2^L (Section 1.9), which is why interacting problems need a systematic approximation scheme rather than brute-force diagonalization.
- **Spontaneous symmetry breaking** is the general phenomenon of a symmetric Hamiltonian producing an asymmetric ground state in the thermodynamic limit; states of matter can be classified by which symmetry they break (time-reversal for magnets, rotational for ferroelectrics, gauge for superconductors), each with its own emergent quasiparticle, and a magnet has the precise, formal definition given in (1.4). Topological materials are a reminder that this classification is not exhaustive.
- **Mean-field theory** supplies the practical machinery: factorize the interacting Hamiltonian (1.5) by taking an expectation value over part of the quartic term, then solve the resulting single-particle problem self-consistently.
- The same recipe, applied to three different interaction channels, produces three different symmetry-broken states: a charge density wave via (1.6) (broken translational symmetry), a magnet via (1.7)-(1.8) (broken time-reversal symmetry), and a superconductor via (1.9)-(1.10) (broken gauge symmetry). Magnetism and superconductivity each get a full session of their own later in the course; today's versions are the first pass.

Next session, we leave interactions aside for a while and go back to the single-particle side of today's Hamiltonian, to see how *symmetry* (translational symmetry, specifically) lets us diagonalize it exactly, no approximation required, via Bloch's theorem. The general symmetry-breaking framework built in Section 1.11 will not be revisited there, so it is worth carrying forward from here rather than expecting a recap.

Session 2

Symmetries, Tight-Binding Band Structure Theory, and Effective Models

Last time we built second quantization and put it to work: mean-field theory turned an interacting, quartic Hamiltonian into an effective single-particle one, and we watched that same recipe produce a charge density wave, a magnet, and a superconductor, one for each of the three symmetries a many-body ground state can spontaneously break. We spent real time on what that breaking actually means: the bread-on-the-table picture of a set of equally good solutions with no individual realization respecting the full symmetry, the requirement that this only becomes sharp in the thermodynamic limit, the ferroelectric as a fourth worked example alongside magnets and superconductors, and the contrast with topological order, which can tell two materials apart with identical broken symmetries. That is a complete story in itself, and today's session is not going to repeat any of it.

Instead, today asks a different, more structural question. Once interactions have been packaged away into a mean-field, single-particle Hamiltonian, how do we actually solve it? For a handful of sites we can diagonalize a matrix by brute force, but a real material has of order 10^{23} sites, and no computer will ever diagonalize that directly. The answer, once again, is symmetry, but used in a different role than last time: not to classify which order a ground state breaks, but to simplify the diagonalization itself. When the ground state of a Hamiltonian actually *shares* the Hamiltonian's symmetry, that symmetry stops being a classification tool and becomes a genuine shortcut, one that turns an impossible, infinite-dimensional problem into infinitely many easy, finite-dimensional ones. For a periodic system, this shortcut has a name, Bloch's theorem, and by the end of today we will have used it to derive several real band structures, including the Dirac cone that gives graphene its name and that the topology lecture right after this one will pick up and gap open.

2.1 Motivation and recap

Recall the shape a many-body Hamiltonian takes once it has been written in second-quantized, tight-binding form,

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j,$$

with t_{ij} the hopping matrix elements between sites i and j. Solving a quantum material, in this language, means nothing more than diagonalizing this Hamiltonian: finding the unitary transformation that brings it into the diagonal form

$$H = \sum_n \epsilon_n \, \Psi_n^\dagger \Psi_n, \qquad \Psi_n^\dagger = \sum_i a_{n,i} \, c_i^\dagger.$$

Symmetries let us find this diagonal form, or at least strong qualitative and quantitative properties of it, without ever doing the diagonalization explicitly. That is the motivation for everything that follows today: using symmetry not to classify order, but to simplify the actual solving of a quantum many-body problem.

2.2 Simplifying problems using symmetry: intuitive examples

The trick is one you have already used, even outside quantum mechanics. Take an infinite plane carrying a uniform positive charge, and ask which direction the electric field points at some location off the plane: straight out, or tilted off at some angle? Apply a mirror symmetry, reflecting left and right: the charge distribution is unchanged by the reflection, so whatever the true field is, it has to map to itself under the same reflection. A tilted field fails this test (its mirror image tilts the other way), while a field pointing straight out passes it. Since the electric field at that point is unique, symmetry alone rules out every answer except the one perpendicular to the plane, without ever doing the Coulomb-law integral.

The same argument works on a less trivial shape. Replace the infinite plane with a uniformly charged ellipse ("egg-shaped" charge distribution) and ask the same question about the field at some point off its axis. Mirror symmetry again forces the untilted answer, this time using the ellipse's own reflection axis, and again without any explicit integration.

Mirrors are only the simplest case. A circle is left invariant by an arbitrary rotation; a square is left invariant only by 90°, 180°, and 270° rotations; a less symmetric shape might be left invariant only by a single mirror. Quantum systems can carry much richer, far less obvious symmetries than any of these, and it is exactly this richer structure that we now turn into a systematic tool.

2.3 The formal machinery of symmetry operators

To make the electromagnetism-style argument precise for a quantum system, we need three objects: a wavefunction ψ, a transformed wavefunction $\phi = S\psi$ obtained by applying a symmetry operation, and the operator S itself, a linear (in a finite-dimensional representation, matrix) map from one to the other. A concrete case fixes the idea: if ψ is a p_x orbital and S is a 90° rotation, the transformed state $\phi = S\psi$ is a p_y orbital, two lobes on the x-axis rotated into two lobes on the y-axis. Notice this p_x orbital is *not* an eigenstate of the rotation: it gets mapped to something genuinely different.

Symmetry operators are always unitary. Physically, a symmetry is just a relabelling of our reference frame, a rotation of axes, say, and any such relabelling has to preserve the norm of every state in the Hilbert space. Concretely,

$$S^\dagger S = S S^\dagger = \mathbb{1}, \qquad S^\dagger = S^{-1}. \tag{2.1}$$

A direct consequence is that the eigenvalues of S all have unit modulus: $\langle\psi|S^\dagger S|\psi\rangle = \langle\psi|\psi\rangle$ forces $|\lambda|^2 = 1$ for any eigenvalue λ. So a symmetry eigenvalue can always be written as a pure phase,

$$S|\psi\rangle = e^{i\varphi}|\psi\rangle, \qquad \varphi \in (-\pi, \pi]. \tag{2.2}$$

Two small examples fix this before we bring in a Hamiltonian at all. Take a wavefunction with real, staggered values $(+1, -1, +1, -1)$ on four sites arranged in a ring, and apply a 90° rotation that cyclically shifts the sites by one. The transformed wavefunction is $(-1, +1, -1, +1)$,

exactly -1 times the original: this state is an eigenstate of the rotation with eigenvalue -1, i.e. $\varphi = \pi$ in (2.2). Now take a complex wavefunction $(+1, -i, -1, i)$ on the same ring and apply the same rotation: the transformed wavefunction comes out as i times the original, an eigenstate with eigenvalue i, i.e. $\varphi = \pi/2$.

A less pictorial but more directly useful example: a two-site tight-binding Hamiltonian with a single hopping t between sites 1 and 2,

$$H = t\left(c_1^\dagger c_2 + c_2^\dagger c_1\right),$$

has two eigenstates, the bonding and antibonding combinations $c_1^\dagger + c_2^\dagger$ and $c_1^\dagger - c_2^\dagger$. Under the mirror symmetry that swaps sites 1 and 2, the bonding state picks up eigenvalue $+1$ and the antibonding state picks up eigenvalue -1: different eigenstates of the same Hamiltonian can carry different eigenvalues under the same symmetry.

Now for the central fact of the whole session. Suppose a Hamiltonian is invariant under a symmetry operation, $SHS^{-1} = SHS^\dagger = H$, which is exactly the statement

$$[H, S] = 0. \tag{2.3}$$

By a standard fact from linear algebra, if two operators commute, they share a common eigenbasis. So whenever (2.3) holds, every eigenstate of H can be chosen to also be an eigenstate of S. This is, in the lecturer's own words, perhaps the single most important idea of today, because it is very often far easier to classify the eigenstates of a symmetry operator, a simple phase by (2.2), than to classify the eigenstates of the Hamiltonian directly. In practice, this turns "find all the symmetries of the Hamiltonian" into the very first, and most productive, step of solving it, rather than a side observation: once we know a matrix S commutes with H, classifying S's eigenstates is the same job as classifying H's.

2.4 Worked exercises: guessing eigenstates from symmetry alone

A sequence of small examples shows how much mileage (2.3) gives us, each one adding sites until we finally take the infinite limit.

Three sites on a ring. Take a tight-binding Hamiltonian on three sites arranged in a triangle, with a symmetry R that rotates $1 \to 2 \to 3 \to 1$ by 120°. Since $R^3 = \mathbb{1}$, any eigenvalue $\lambda = e^{i\varphi}$ of R must satisfy $\lambda^3 = 1$, so

$$\varphi_n = \frac{2\pi n}{3}, \qquad n = 0, 1, 2. \tag{2.4}$$

We already know a great deal about H's eigenstates before ever writing H down explicitly. They must take the form $\alpha_1 c_1^\dagger + \alpha_2 c_2^\dagger + \alpha_3 c_3^\dagger$, and because R cyclically permutes the sites while multiplying the whole state by $e^{i\varphi_n}$, the coefficients themselves must carry that same phase from site to site, $\alpha_j \propto e^{i\varphi_n j}$. The eigenstates of any Hamiltonian with this rotational symmetry are exactly the three discrete Fourier modes on the ring, regardless of the specific hopping amplitudes chosen. Figure 2.1 draws the three modes explicitly, and shows what happens to the set of allowed phases as the ring is made larger.

Five sites on a ring. The same argument on a five-site ring with nearest-neighbour hopping, $H = \sum_{\langle ij \rangle} c_i^\dagger c_j$, gives $\varphi_n = 2\pi n/5$, $n = 0, \ldots, 4$, and eigenstates $|\Psi_n\rangle \propto \sum_{j=0}^{4} e^{i\varphi_n j}\, c_j^\dagger |0\rangle$. This already answers a quantitative question without diagonalizing anything: what is the weight of site 1, in the sense of the overlap $\left|\langle\Omega| c_1 \Psi_{\mathrm{GS}}^\dagger |\Omega\rangle\right|^2$, in the lowest-energy eigenstate built from the vacuum $|\Omega\rangle$? Since every $|\Psi_n\rangle$ is a uniform-modulus superposition over all five sites, this overlap equals $1/5$ for every n, lowest energy or not, purely from the symmetry and with no reference to the specific hopping amplitudes at all. The same argument survives

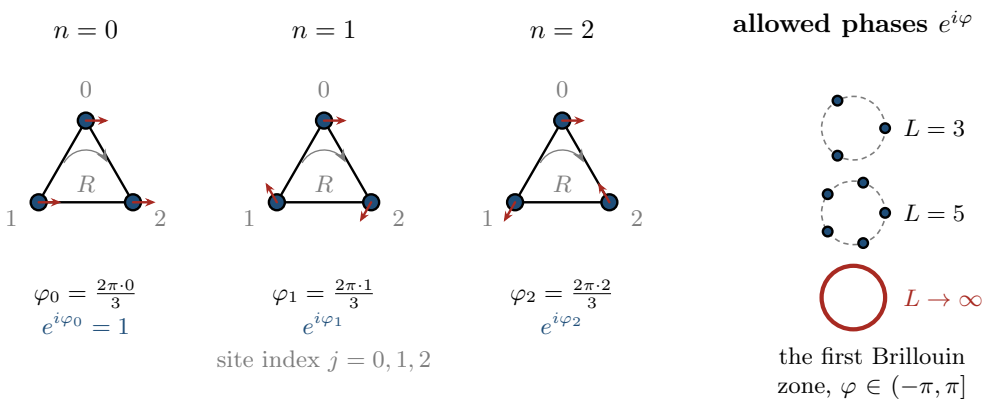


Figure 2.1: Symmetry fixes the eigenstates before the Hamiltonian is ever written down. On a three-site ring the rotation R can only have eigenvalues $e^{2\pi i n/3}$ by (2.4), and the state's coefficients must therefore advance by that same phase from site to site: the eigenstates are the discrete Fourier modes, whatever the hopping amplitudes happen to be. Each site's arrow is the complex coefficient $\alpha_j \propto e^{i\varphi_n j}$. Right: the allowed phases for $L = 3$, $L = 5$ and $L \to \infty$ – in the limit they fill the circle, and that circle is the first Brillouin zone.

unchanged even if nearest-neighbour hopping is replaced by arbitrary long-range hopping (a six-site ring with extra long-range couplings makes the same point): as long as the ring's rotational symmetry is intact, the symmetry eigenvalues, the eigenstates, and expectation values like this one are all fixed by symmetry alone, however complicated the Hamiltonian gets.

It is worth a brief aside that such few-site models are not purely academic. Scanning tunnelling microscopy can position individual atoms with essentially arbitrary precision, and few-site tight-binding rings like these have actually been built and measured this way. The famous stop-motion film built entirely from STM-manipulated atoms, sometimes called "the smallest film ever made," in which every single dot on screen is an individual atom, is a playful reminder of just how precisely such models can now be realized in the lab.

The infinite chain. Now let the ring grow without bound, into an infinite one-dimensional chain with nearest-neighbour hopping. The relevant symmetry is translation by one site, $T : c_n \to c_{n+1}$, and $[H, T] = 0$ for any Hamiltonian built only from hoppings that depend on the distance between sites, not on absolute position. By Section 2.3, H and T share an eigenbasis, and the eigenvalue of T on that shared eigenbasis is, by (2.2), a pure phase $e^{i\varphi}$. This φ is the **Bloch phase**, and the statement that H and T share an eigenbasis, when applied to a system with translational symmetry, *is* Bloch's theorem: if two operators commute, there is a common basis for both of them. The mapping between this phase and the corresponding energy of the Hamiltonian is exactly what we mean by the **dispersion relation** of a periodic system.

2.5 Bloch's theorem and the first band structure

Let us make this fully explicit for the one-dimensional tight-binding chain,

$$H = -t \sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.} \right). \tag{2.5}$$

We take $t > 0$ throughout, matching the sign convention of (1.6) in the previous session.

Because $[H, T] = 0$, we look for eigenstates built as a translationally-covariant superposition over all sites,

$$|\Psi_\varphi\rangle = \sum_n e^{in\varphi}\, c_n^\dagger |0\rangle. \tag{2.6}$$

It is cleanest to invert this relation and define new operators labelled by φ instead of by site,

$$\Psi_\varphi^\dagger = \sum_n e^{in\varphi}\, c_n^\dagger, \qquad c_n^\dagger = \sum_\varphi e^{-in\varphi}\, \Psi_\varphi^\dagger, \tag{2.7}$$

an ordinary Fourier transform between real space and the space of Bloch phases. Substituting (2.7) into (2.5),

$$\sum_n c_n^\dagger c_{n+1} = \sum_{n,\varphi,\varphi'} e^{-in\varphi}\, e^{i\varphi'(n+1)}\, \Psi_\varphi^\dagger \Psi_{\varphi'} = \sum_{\varphi,\varphi'} e^{i\varphi'}\, \delta_{\varphi\varphi'}\, \Psi_\varphi^\dagger \Psi_{\varphi'} = \sum_\varphi e^{i\varphi}\, \Psi_\varphi^\dagger \Psi_\varphi,$$

where the sum over n produced the orthogonality relation $\sum_n e^{in(\varphi'-\varphi)} = \delta_{\varphi\varphi'}$. The Hermitian conjugate term contributes the complex conjugate phase, so

$$H = -t \sum_\varphi \left(e^{i\varphi} + e^{-i\varphi}\right) \Psi_\varphi^\dagger \Psi_\varphi = \sum_\varphi \epsilon_\varphi\, \Psi_\varphi^\dagger \Psi_\varphi, \qquad \boxed{\epsilon_\varphi = -2t\cos\varphi}. \tag{2.8}$$

This is our first band structure, obtained without ever writing down, let alone diagonalizing, an infinite matrix. All we needed was the observation that the eigenstates must carry a definite phase under translation, and then a single Fourier transform did the rest.

A few features of this dispersion are worth drawing out explicitly, since they recur throughout the course. Near the band bottom ($\varphi = 0$), $\epsilon_\varphi \approx -2t + t\varphi^2$, an upward-opening parabola; near the band top ($\varphi = \pi$), writing $\varphi = \pi + \delta$, $\epsilon_\varphi \approx 2t - t\delta^2$, a downward-opening parabola. The two curvatures have *opposite* sign, which (once effective mass is defined properly in Section 2.9) means low-energy quasiparticles near the bottom of the band behave like ordinary, positive-mass electrons, while those near the top behave like negative-mass holes. Near the middle of the band, at $\varphi = \pi/2$ where ϵ_φ crosses zero, the dispersion looks closer to linear than quadratic, a first hint of the Dirac-like physics we meet properly in Section 2.11. All of this is collected in figure 2.2, which also shows the density of states the band produces.

Notebook check-in: `session2.ipynb`, "One dimensional chain" (Core)

Build exactly the chain of (2.5), diagonalize it numerically, and check that the numerical spectrum traces out $\epsilon_\varphi = -2t\cos\varphi$ from (2.8) as you scan φ across the Brillouin zone.

A natural in-class extension checks that the same machinery handles more than nearest-neighbour hopping. Add a second-neighbour hopping term $-\eta\sum_n(c_n^\dagger c_{n+2} + \text{h.c.})$ to (2.5). This term has exactly the same translational symmetry as before, so the eigenstates (2.6) are completely unchanged; only the eigenvalues need recomputing. Repeating the Fourier substitution for the new term (which pairs $c_n^\dagger$ with c_{n+2}, contributing a phase $e^{\pm 2i\varphi}$ instead of $e^{\pm i\varphi}$) gives

$$\epsilon_\varphi = -2t\cos\varphi - 2\eta\cos(2\varphi). \tag{2.9}$$

Notebook check-in: `session2.ipynb`, "One dimensional chain, with longer range hopping" (Core)

Add the second-neighbour hopping η to the chain from the previous callout, and check that the numerically computed band matches (2.9). It is worth also checking the density of states here (Section 2.9), since η can create a second local extremum in ϵ_φ and, with it, an extra Van Hove-type feature in the spectrum.

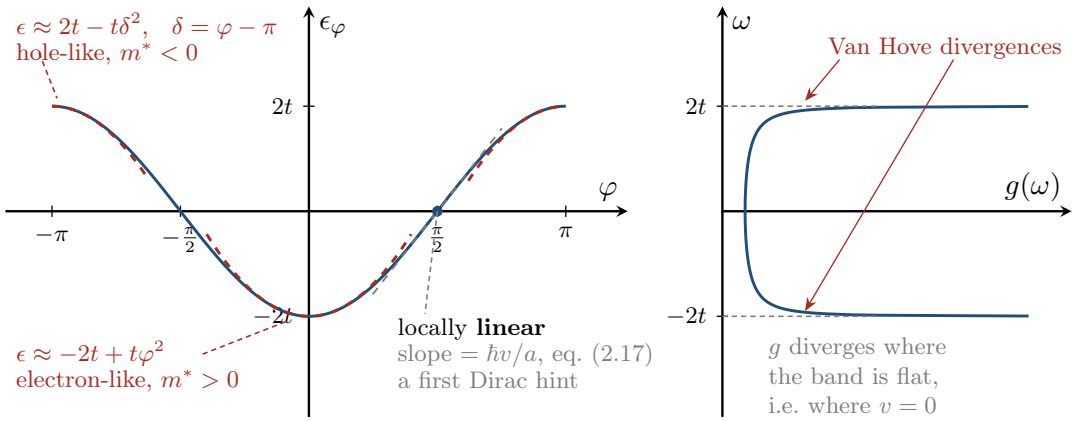


Figure 2.2: The one-dimensional band $\epsilon_\varphi = -2t\cos\varphi$ of (2.8), drawn for $t = 1$. Near the band bottom the dispersion is an upward parabola and quasiparticles behave as ordinary positive-mass electrons; near the top the curvature has the opposite sign and they behave as holes; near midband it is locally linear, a first glimpse of the Dirac physics of Section 2.11. The slope is the group velocity (2.17) and the curvature the inverse effective mass (2.18) – both are read straight off this one curve. Right: the resulting density of states (2.16), which diverges at both band edges, exactly where the group velocity vanishes.

Also worth flagging by name: `pyqula`, a Python package that computes band structures like this one in 1D, 2D, 3D, and beyond, is the computational tool used throughout the course. It appears here only by name; the slides work through an explicit installation walkthrough (`pip install pyqula`) and example scripts for a 1D chain and a 1D two-site-per-cell "ladder," with the resulting band plots shown on screen. That specific installation walkthrough and its code screenshots are slides-only material, not narrated in the transcript and not reproduced here; the notebook is where you actually run the equivalent code yourself.

2.6 What the phase means: quasi-momentum

Let us stop and ask what the phase φ in (2.2) physically means, not just formally. Go back to the continuum Schrödinger equation and ask how translating a wavefunction by the lattice constant a, $\psi(x + a)$, relates to $\psi(x)$. Taylor expanding,

$$\psi(x+a) = \psi(x) + a\,\partial_x\psi(x) + \frac{a^2}{2}\partial_x^2\psi(x) + \cdots = e^{a\partial_x}\psi(x) = e^{\frac{i}{\hbar}ap}\,\psi(x),$$

where $p = -i\hbar\partial_x$ is the momentum operator. If ψ is an eigenstate of the translation operator with eigenvalue $e^{i\varphi}$, comparing the two expressions identifies

$$\varphi = \frac{a\,p}{\hbar} = k\,a, \tag{2.10}$$

the symmetry eigenvalue is literally the crystal momentum $k = p/\hbar$ times the lattice constant. This is why φ is called the *quasi-momentum*: it is directly tied to the momentum operator of the original Schrödinger equation, just measured in units where the lattice constant is absorbed, and it ranges over $\varphi \in (-\pi, \pi]$, the first Brillouin zone.

This identification also tells us where the orthogonal single-orbital basis we have implicitly been using comes from. A real material's atomic orbitals, s, p, d, and so on, are not orthogonal between neighbouring sites; before writing a clean tight-binding Hamiltonian we first pass to an orthogonalized local basis, one orbital per site, obtained systematically via **Löwdin orthogonalization** (a construction the contact session works through properly, and that we simply assume here; the slides also label this basis "Wannier states"). In this basis, symmetry

alone already tells us the eigenstates take the form $e^{in\varphi}c_n^\dagger|0\rangle$, exactly the ansatz (2.6), whether we imagine a finite ring of 7 sites, 11 sites, or the infinite chain we actually want to describe a solid: we can build the same intuition by working through progressively larger finite rings before taking the $L \to \infty$ limit.

2.7 Multi-orbital band structures

Real unit cells almost never contain a single orbital: graphene has two (one per sublattice), transition-metal compounds routinely have several d-orbitals per site, and so on. Let α label the orbital within a unit cell and n label the unit cell itself. The most general bilinear Hamiltonian built from these operators, keeping couplings only within a cell and to the nearest-neighbouring cell (longer-range couplings work by exactly the same trick, just with more terms), is

$$H = \sum_{n,\alpha,\beta} t_{\alpha\beta}\, c_{n,\alpha}^\dagger c_{n,\beta} + \sum_{n,\alpha,\beta} \gamma_{\alpha\beta}\, c_{n,\alpha}^\dagger c_{n+1,\beta} + \text{h.c.}, \tag{2.11}$$

with $t_{\alpha\beta}$ the intracell hoppings and $\gamma_{\alpha\beta}$ the intercell hoppings.

The symmetry that diagonalizes (2.11) is a slightly richer version of pure translation: it shifts the unit-cell index $n \to n+1$ *and* performs a unitary rotation among the orbitals within each cell. The corresponding eigenstates are

$$\Psi_{\varphi,\alpha}^\dagger \sim \sum_{n,\beta} e^{in\varphi}\, U_{\alpha\beta}\, c_{n,\beta}^\dagger,$$

for some unitary U. Substituting this ansatz into (2.11) and repeating the Fourier argument of Section 2.5 orbital-by-orbital shows that the required U is exactly the one that diagonalizes the **Bloch Hamiltonian**

$$H(\varphi) = t + \gamma\, e^{i\varphi} + \gamma^\dagger e^{-i\varphi}, \tag{2.12}$$

a matrix whose size equals the number of orbitals per unit cell. Each entry is indexed by the pair of orbitals (α, β) it couples: the intracell hoppings $t_{\alpha\beta}$ enter that entry directly, with no phase, while the intercell hoppings $\gamma_{\alpha\beta}$ enter dressed with the phase $e^{\pm i\varphi}$ the symmetry eigenvalue provides. This is the general recipe to use again and again: count orbitals per cell to get the matrix size, place each hopping in the entry its orbital pair indexes, multiply the intercell ones by $e^{\pm i\varphi}$, and diagonalize. Note that an intracell hopping between *different* orbitals lands off the diagonal, exactly as the worked example below shows.

A concrete worked example fixes the idea and will matter again soon. Take a two-orbital (A, B) chain, with strong intracell coupling of magnitude 1 between A and B, and a weaker intercell coupling η linking B in one cell to A in the next. The Bloch Hamiltonian (2.12) is purely off-diagonal,

$$H(\varphi) = \begin{pmatrix} 0 & 1+\eta\, e^{-i\varphi} \\ 1+\eta\, e^{i\varphi} & 0 \end{pmatrix}, \qquad \epsilon_\pm(\varphi) = \pm\left|1+\eta\, e^{-i\varphi}\right|. \tag{2.13}$$

This particular alternating-hopping chain is worth remembering by name: it is exactly the Su-Schrieffer-Heeger-type Hamiltonian that reappears, with a topological twist, in the topology lecture right after this one. Figure 2.3 shows the recipe applied to it, matching each bond of the lattice to the matrix entry it produces, and what happens to the two bands as η is tuned through 1.

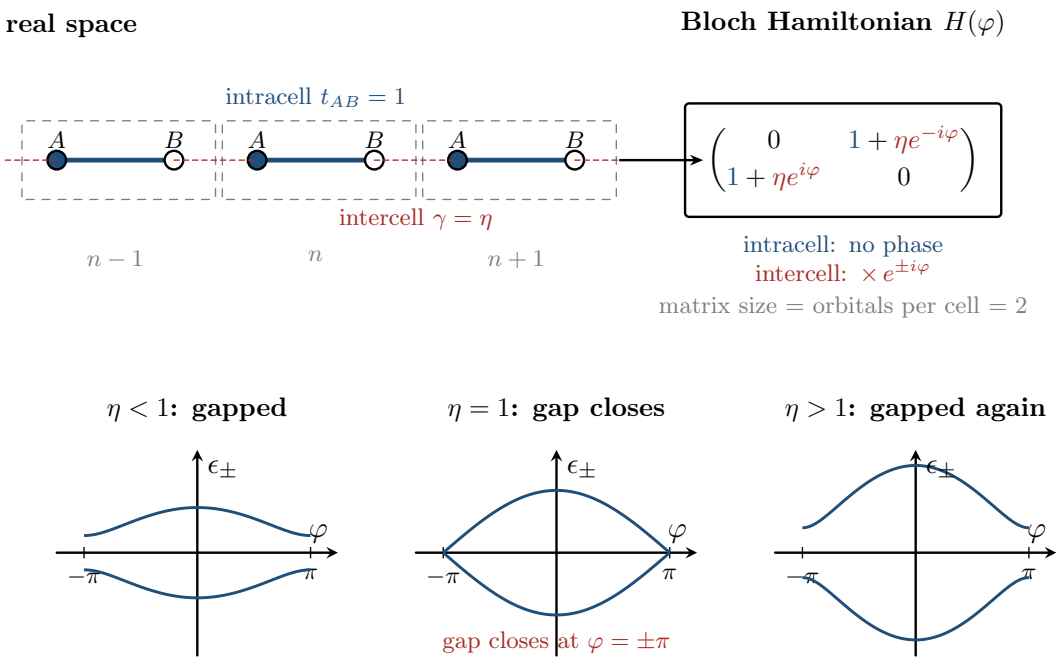


Figure 2.3: Reading a Bloch Hamiltonian off a lattice. The number of orbitals per unit cell sets the matrix size; an intracell hopping enters the entry its orbital pair indexes with no phase, an intercell hopping enters the corresponding entry multiplied by $e^{\pm i\varphi}$. Note that the intracell A–B hopping lands *off* the diagonal. For the alternating-hopping chain (2.13) this gives the two bands $\epsilon_\pm = \pm|1+\eta e^{-i\varphi}|$: gapped for $\eta \neq 1$, and gap-closing exactly at $\eta = 1$ – the Su-Schrieffer-Heeger chain the topology session picks up.

Notebook check-in: `session2.ipynb`, "One dimensional chain, band folding" (Challenge), and "One dimensional ladder, band folding" (Challenge)

The "band folding" section builds exactly this picture from the other direction: take a single-orbital chain and re-describe it using a larger unit cell (equivalently, more than one site per cell), and watch a single band from (2.8) fold into multiple bands, as (2.12) predicts. The ladder section then works through a genuine two-orbital-per-cell model along the lines of (2.13); check that the numerical bands match $\epsilon_\pm(\varphi)$ there.

2.8 Higher-dimensional band structures

A two-dimensional lattice has two independent, mutually commuting translations, T_x and T_y, so Bloch eigenstates carry two independent phases, φ_x and φ_y, one per direction, tied to the crystal momentum along the two lattice vectors by the same Taylor-expansion argument as (2.10), applied once per direction. The space of allowed (φ_x, φ_y) pairs is the **Brillouin zone**. Because $\varphi = 0$ and $\varphi = 2\pi$ label the same translation eigenvalue in both directions, the Brillouin zone is not an open square but a periodic one: identifying the left and right edges, and the top and bottom edges, turns it into a torus (a circle, in the one-dimensional case of Section 2.5). This compactness of the Brillouin zone will matter a great deal once topological invariants enter the next session.

Applying the Fourier trick of Section 2.5 once per spatial direction gives, for a single-orbital square lattice with nearest-neighbour hopping t,

$$\epsilon(\varphi_x, \varphi_y) = 2t\,(\cos\varphi_x + \cos\varphi_y)\,, \tag{2.14}$$

and, applying the identical argument a third time, for a single-orbital cubic lattice,

$$\epsilon(\varphi_x, \varphi_y, \varphi_z) = 2t\left(\cos\varphi_x + \cos\varphi_y + \cos\varphi_z\right). \tag{2.15}$$

The same trick extends systematically to four, five, or more dimensions; nothing in the argument actually depends on being in three-dimensional space. Equations (2.14) and (2.15) are slides-and-transcript material with no notebook counterpart in `session2.ipynb`: the notebook's own two-dimensional worked example goes straight to the honeycomb lattice of Section 2.11 rather than the plain square lattice, and there is no three-dimensional cubic-lattice section at all. If you want to check (2.14) or (2.15) numerically, the same construction used for the honeycomb-lattice callout below extends directly, just with a single-orbital, square (or cubic) Bravais lattice instead.

It is worth being explicit, before moving on, about a point that has been implicit throughout: everything above applies far beyond electrons. The same Bloch-Hamiltonian trick diagonalizes phonon bands, magnon bands, and Bogoliubov-de Gennes quasiparticle bands in a superconductor, and it applies equally well to artificial, engineered lattices, such as cold atoms trapped in optical lattices, that have nothing to do with real crystals at all. We will focus on electronic bands for concreteness, but this generality is exactly why the same machinery reappears for magnons, and for Bogoliubov quasiparticles, later in the course. The physics that comes out of single-particle band theory this way is itself extremely rich: Dirac fermions with zero effective mass, Weyl semimetals (Dirac points generalized to higher dimensions), quantum spin Hall insulators, Chern insulators, crystalline topological phases, and topological superconductors, all of it building on nothing more than the Bloch Hamiltonian $H(\mathbf{k})$ constructed here.

2.9 Properties of an electronic dispersion

Once we have a dispersion $E(\mathbf{k})$, whether from (2.8), (2.13), (2.14), or something more elaborate, several derived quantities carry direct physical meaning.

The **density of states**,

$$g(\omega) = \int_{\mathrm{BZ}} \frac{d^d k}{(2\pi)^d}\, \delta(\omega - E(\mathbf{k}))\,, \tag{2.16}$$

counts how many states sit at energy ω. It governs how strongly a material responds to interactions: a large or divergent density of states means even a weak interaction can drive an instability toward magnetism, a charge density wave, or superconductivity, exactly the three cases from last session, while a vanishing density of states means interactions struggle to have any effect at all, however strong.

The **group velocity**,

$$v(\mathbf{k}) = \frac{1}{\hbar}\nabla_{\mathbf{k}} E(\mathbf{k}), \tag{2.17}$$

is the local slope of the dispersion. Where $v(\mathbf{k}) = 0$, electrons at that $\mathbf{k}$ are effectively immobile; where $v(\mathbf{k})$ is large, they propagate through the lattice largely unaffected by scattering.

The **effective mass**,

$$\left(\frac{1}{m^*}\right)_{ij} = \frac{1}{\hbar^2}\,\frac{\partial^2 E(\mathbf{k})}{\partial k_i\, \partial k_j}, \tag{2.18}$$

is the curvature of the dispersion, and it is the mass actually measured in essentially every transport or cyclotron-resonance experiment; the bare electron mass is not directly observable inside a solid. Depending on the band, m^* can range from nearly zero to extremely large, or,

as we will see shortly for a Dirac dispersion, formally undefined at the point where the bands touch.

Finally, the **Fermi surface** is the locus of **k**-points sharing the energy of the highest occupied state. Its shape controls which electronic instabilities are favoured, how a material responds to an external magnetic field, and it is one of the quantities ARPES (Section 2.12) can map out directly.

Notebook check-in: `session2.ipynb`, "Density of states of the honeycomb lattice" (Core), and "Fermi surface of the honeycomb lattice" (Challenge)

Once the honeycomb dispersion of Section 2.11 is in hand, these two sections let you compute (2.16) and the Fermi surface directly and see how both change as the Fermi level is tuned through the Dirac point.

2.10 Three archetypes of dispersion: parabolic, Dirac, and flat

Real dispersions are complicated functions of **k**, but almost every band structure in this course is locally built out of three limiting archetypes.

1. **Parabolic bands**, $E \propto k^2$, the free-electron-like case, describe most semiconductors and conventional metals. They arise essentially by Taylor expansion near any extremum of a band: we already saw this explicitly near $\varphi = 0$ and $\varphi = \pi$ in (2.8), with opposite-sign effective masses at the two extrema.

2. **Dirac (linear) dispersions**, $E \propto |k|$, are realized in graphene and in a range of topological materials (quantum spin Hall insulators, Chern insulators, and other symmetry-protected topological phases). They carry finite topological invariants (a Berry phase) and produce phenomenology that echoes relativistic physics, even though the underlying electrons are entirely non-relativistic.

3. **Flat bands**, $E(\mathbf{k}) = \text{const}$, might sound like the least interesting case, but by (2.16) a flat band produces a *divergent* density of states, so even weak interactions have an enormous effect wherever a flat band sits near the Fermi level. When the flatness arises for a nontrivial reason, real-space destructive interference rather than simply setting a hopping to zero, it can also carry a finite Chern number, central to the quantum Hall effect and to moiré materials.

Figure 2.4 puts the three side by side with the density of states each one produces, which is what decides whether interactions can do anything with them.

We derive the Dirac case explicitly next, since it is graphene's defining feature and the direct link forward to the topology lecture, then look at two lattices that realize flat bands. Both `pyqula`, introduced by name in Section 2.5, and its GUI companion, a second computational tool shown on the slides under the name **Quantum Lattice** (interface-based, no scripting required, and not named explicitly in speech, which only gestures at "a user interface we will use at some points"), are the tools used throughout the course to compute dispersions like these.

2.11 The Dirac cone: graphene's honeycomb lattice, and flat-band lattices

Graphene's honeycomb lattice has two orbitals per unit cell, one on each of two triangular sublattices, conventionally labelled A and B, sitting on an underlying triangular Bravais lattice.

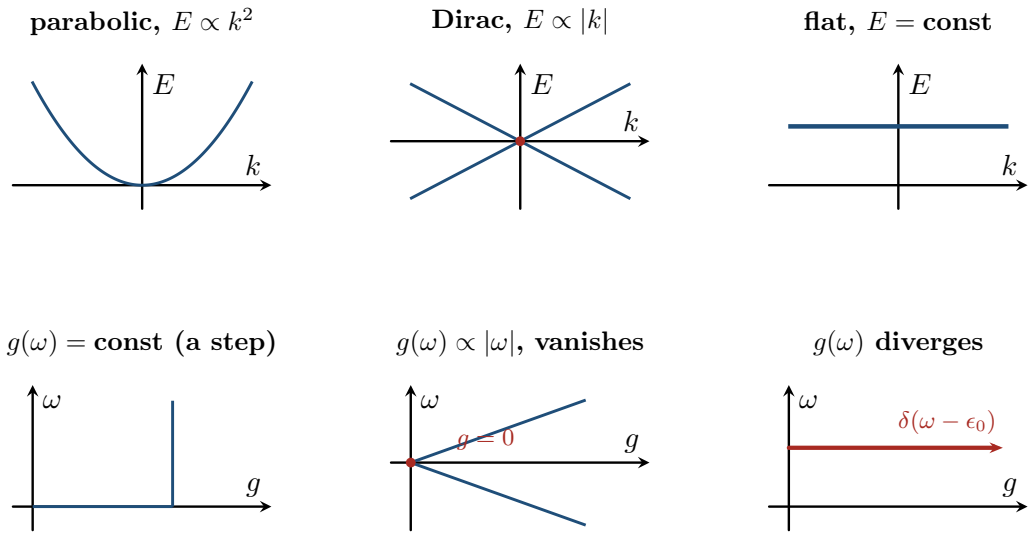


Figure 2.4: The three dispersion archetypes and the densities of states they produce, all in *two* dimensions. A parabolic band gives a constant (step) density of states; a Dirac point makes it vanish linearly, so a finite interaction strength is needed before anything happens; a flat band makes it diverge, so an arbitrarily weak interaction can drive the symmetry-broken states of Session 1. Almost every band structure in this course is, locally, one of these three.

Nearest-neighbour hopping only ever connects A to B (never A to A, nor B to B), so by the recipe of Section 2.7 the Bloch Hamiltonian (2.12) is purely off-diagonal,

$$H(\mathbf{k}) = \begin{pmatrix} 0 & f(\mathbf{k}) \\ f^*(\mathbf{k}) & 0 \end{pmatrix}, \qquad f(\mathbf{k}) = t \sum_{j=1}^{3} e^{i\mathbf{k}\cdot\boldsymbol{\delta}_j}, \tag{2.19}$$

where $\boldsymbol{\delta}_j$, $j = 1, 2, 3$, are the three vectors connecting an A site to its nearest B neighbours. Diagonalizing (2.19) gives $E_\pm(\mathbf{k}) = \pm|f(\mathbf{k})|$. The function $f(\mathbf{k})$ vanishes at two inequivalent points of the Brillouin zone, conventionally called K and K', since the three phases in the sum cancel exactly there, and this is why both bands touch at exactly zero energy at K and K'. That cancellation is worth seeing rather than taking on trust: figure 2.5 shows the three phase factors at $\mathbf{K}$ as three unit vectors 120° apart, which close a triangle and therefore sum to zero.

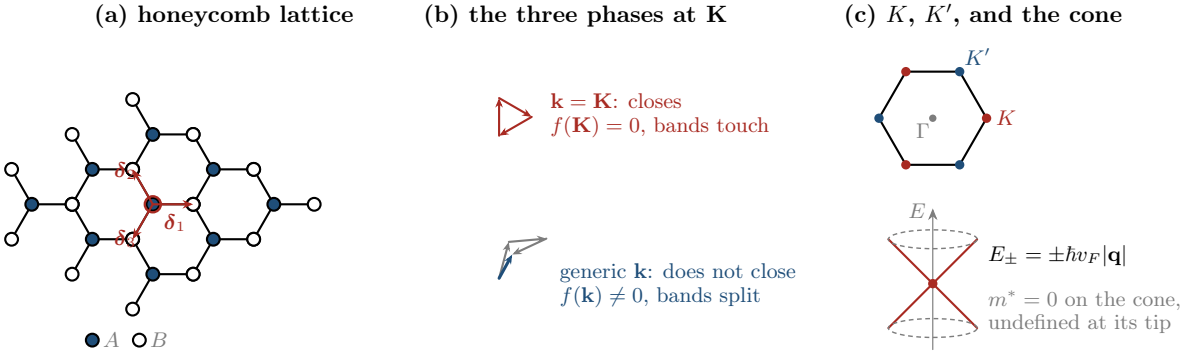


Figure 2.5: Graphene's Dirac cones. **(a)** The honeycomb lattice: two sublattices A, B per unit cell, with the three nearest-neighbour vectors $\boldsymbol{\delta}_{1,2,3}$ from an A site. **(b)** Why the bands touch. At $\mathbf{k} = \mathbf{K}$ the three phase factors $e^{i\mathbf{K}\cdot\boldsymbol{\delta}_j}$ are three unit vectors 120° apart in the complex plane; laid head to tail they close a triangle, so $f(\mathbf{K}) = 0$ in (2.19) and both bands meet at zero energy. At a generic $\mathbf{k}$ they do not close, and the bands are split. **(c)** The two inequivalent zero-energy points K and K' of the Brillouin zone, around which the dispersion is the linear cone $E_\pm = \pm\hbar v_F|\mathbf{q}|$ of (2.21).

Expanding $f(\mathbf{k})$ to linear order around K, writing $\mathbf{k} = \mathbf{K} + \mathbf{q}$ for small $\mathbf{q}$, the off-diagonal entry becomes a linear function of $\mathbf{q}$ (one linear term for its real part, one for its imaginary part), and (2.19) reduces to the standard two-dimensional **Dirac Hamiltonian**,

$$H(\mathbf{q}) \approx \hbar v_F \left(q_x \sigma_x + q_y \sigma_y\right), \qquad v_F = \frac{3ta}{2\hbar}, \tag{2.20}$$

with σ_x, σ_y the Pauli matrices acting on the A/B sublattice space and a the nearest-neighbour bond length. Diagonalizing (2.20) gives

$$E_\pm(\mathbf{q}) = \pm\hbar v_F |\mathbf{q}|, \tag{2.21}$$

a strictly linear, cone-shaped dispersion around K, and by an identical calculation around K': the **Dirac cone**. Notice the effective mass from (2.18) is formally zero everywhere on the cone except exactly at its tip, where it is ill-defined altogether, the "massless-looking electron" phenomenology graphene is famous for. This is the low-energy physics of every honeycomb-lattice material, graphene above all, and there is a great deal more structure hiding at K and K' (a Berry phase, and with it a topological classification) that we leave for the topology lecture right after this one.

Notebook check-in: `session2.ipynb`, "Infinite two-dimensional graphene" (Core)

Build the two-dimensional honeycomb Bloch Hamiltonian (2.19) directly and diagonalize it across the Brillouin zone. Zoom in on K and K' and check numerically that the dispersion there really is linear, i.e. that it matches (2.21) once you are close enough to the Dirac point.

Two standard lattices realize genuinely flat bands from real-space geometry alone, rather than from any fine-tuning of hopping parameters to zero. The **Lieb lattice** is built from a square lattice with one site out of every four removed in a regular pattern; its dispersion has two parabolic bands, much like the ordinary square lattice (2.14), plus one exactly flat band sitting at zero energy. The **kagome lattice** is built instead from a triangular lattice with one site out of every four removed; it has a dispersion closer to the honeycomb lattice's, hosting Dirac-cone-like linear crossings, plus, in addition, a flat band at the edge of the spectrum: with the sign convention of (2.5) it sits at the top, at $\epsilon = 2t$, and it would sit at the bottom for the opposite sign of t. In both cases the flat band arises from destructive interference between hopping paths in real space, a purely geometric effect rather than an interaction effect (the strong-correlation physics only enters once electrons are actually placed into these already-flat bands). Figure 2.6 shows both lattices, both dispersions, and the localized state whose hopping paths cancel.

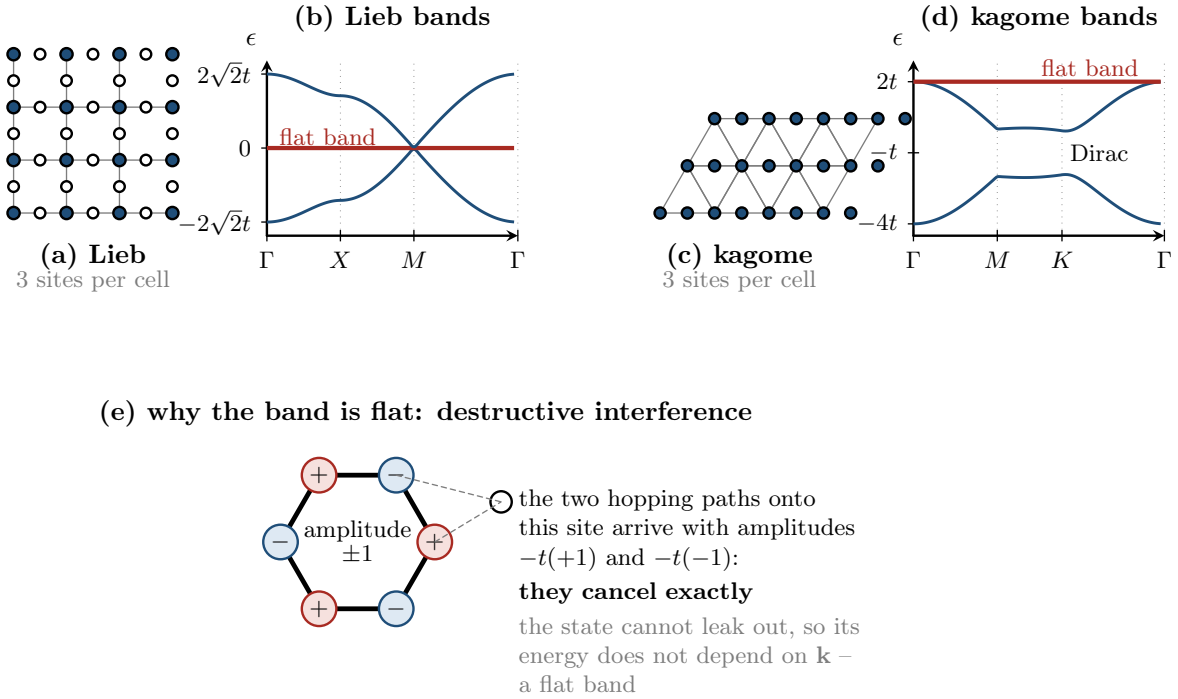


Figure 2.6: Flat bands from geometry alone. **(a)** The Lieb lattice (a square lattice with one site in four removed) and **(b)** its bands: two dispersive bands plus a band that is exactly flat at $\epsilon = 0$, all three touching at M. **(c)** The kagome lattice (corner-sharing triangles) and **(d)** its bands: a Dirac-like crossing at K, plus a flat band at $\epsilon = 2t$. **(e)** Why the flat band exists. An eigenstate can be confined to a single kagome hexagon with amplitudes alternating in sign around it; every path by which it could hop onto an outside site arrives twice with opposite sign and cancels. Nothing has been fine-tuned to zero – the flatness is a property of the lattice geometry, and by (2.16) it makes the density of states diverge.

This is slides-only material with no notebook counterpart in `session2.ipynb`: the notebook works through the honeycomb lattice in depth but has no Lieb-lattice or kagome-lattice section of its own. The band-structure plots for both lattices are shown directly on the slides; the description above captures their qualitative structure, but seeing the actual flat-band

dispersion requires the figure itself.

2.12 Measuring band structures: ARPES

A natural question, once all this formalism is in hand, is whether any of it is actually observable. The energy eigenvalue $E(\mathbf{k})$ is clearly physical, but the quasi-momentum $\mathbf{k}$ itself started life as a symmetry eigenvalue in Section 2.6; is it something we can measure directly, and can we measure the *relation* between the two?

The answer is yes, via **angle-resolved photoemission spectroscopy (ARPES)**. The idea is conceptually simple: shoot photons of known energy and momentum at a material; each photon can eject an electron from the material entirely (the photoelectric effect), and momentum and energy conservation relate the electron's momentum and kinetic energy, both measured directly outside the material, back to its momentum and energy while still inside. Scanning the emission angle and kinetic energy of the ejected electrons therefore maps out $E(\mathbf{k})$ directly. This is exactly how the graphene band structure and Fermi surface are measured in practice, and the measured Dirac cone matches the tight-binding derivation of Section 2.11 closely. This is slides-only material with no notebook counterpart: the actual ARPES images of graphene's measured band structure and Fermi surface are shown on the slides, and there is no way to reproduce a real photoemission experiment in a tight-binding notebook; the description above captures the physical principle, but seeing the measured data itself requires the figure.

2.13 Closing summary

- **Symmetry as a simplification tool.** When the ground state shares the Hamiltonian's symmetry, we can use a commuting symmetry operator, $[H, S] = 0$ in (2.3), to find H's eigenbasis by finding S's eigenbasis instead, a strategy worked through explicitly for rings of three and five sites in Section 2.4.

- **Bloch's theorem** is exactly this idea applied to translational symmetry on an infinite lattice: eigenstates carry a definite phase φ (the quasi-momentum, (2.10)) under translation, turning one infinite-dimensional diagonalization into a continuous family of finite ones, the **Bloch Hamiltonian** $H(\varphi)$.

- We derived the one-dimensional chain's band structure explicitly, $\epsilon_\varphi = -2t\cos\varphi$ in (2.8), and generalized the same machinery to second-neighbour hopping (2.9), multiple orbitals per unit cell via the matrix Bloch Hamiltonian $H(\varphi) = t + \gamma e^{i\varphi} + \text{h.c.}$ in (2.12)-(2.13), and higher spatial dimensions, (2.14) and (2.15).

- A dispersion $E(\mathbf{k})$ determines a material's measurable properties directly: density of states (2.16), group velocity (2.17), effective mass (2.18), and Fermi surface, and every real dispersion is locally built from three archetypes: parabolic, Dirac, and flat.

- The **Dirac cone** of graphene's honeycomb lattice, $H(\mathbf{q}) \approx \hbar v_F(q_x\sigma_x + q_y\sigma_y)$ with $E_\pm = \pm\hbar v_F|\mathbf{q}|$ in (2.20)-(2.21), is this session's single most important worked example, derived directly from the vanishing of the off-diagonal Bloch Hamiltonian (2.19) at the K, K' points; the Lieb and kagome lattices realize flat bands by the same kind of real-space interference, with no fine-tuning required.

- All of this band-structure machinery generalizes directly to phonons, magnons, and Bogoliubov-de Gennes quasiparticles, not just electrons, and is checked against real materials experimentally via ARPES.

If there is one thing to take away: periodic-system spectra are computed via band-structure theory by choosing the shared eigenbasis of the symmetry operator and the Hamiltonian, and the resulting $E(\mathbf{k})$ dispersion relation determines a great deal about a material's electronic properties on its own, before interactions ever re-enter the picture. The Dirac cone derived in Section 2.11 is not just graphene trivia: it is exactly the object the next session picks up and asks what happens when you gap it open, which is where topological band structure, Chern numbers, and the SSH model (already glimpsed in the two-orbital chain of (2.13)) all enter the course in earnest. Flat bands, too, are not the end of the story: the kagome and Lieb constructions here are the direct predecessor of the Landau-level flat bands the quantum Hall session builds several sessions from now.

Session 3

Topological Band Structure Theory

Last session closed on the honeycomb lattice's most striking feature: near the corners of its Brillouin zone, the tight-binding band structure stops looking parabolic and becomes linear, the Dirac cone, one of the three dispersion archetypes (parabolic, Dirac, flat) that we singled out as organizing the rest of the course. We even saw that this is not just a theorist's convenience: ARPES measures that linear dispersion directly in real graphene. So here is the question that organizes today's entire session: what happens if we open a gap at that Dirac point? A gapped Dirac cone turns a metal into an insulator, but not always an ordinary one. Depending on exactly *how* the gap opens, the resulting insulator can carry a hidden invariant, a topological invariant, that shows up nowhere in the bulk spectrum itself but forces the material to have conducting states at its edge, states that no local disorder can remove. Getting to that statement rigorously is going to take some work: we need linear response theory as a tool, the Hall conductivity as the observable that first reveals the invariant, the Chern number as the invariant itself, and the general bulk-boundary correspondence that ties the invariant to edge physics. We will build all of this using the quantum Hall state as the simplest possible example, then bring the same machinery back to a purely one-dimensional model (the SSH chain) before finally returning, in Section 3.10, to gap open the honeycomb Dirac cone itself and see a genuine Chern insulator emerge with no magnetic field anywhere in sight.

3.1 What topological materials are

Before diving into formalism it is worth having a rough taxonomy in view, so that the abstract machinery we are about to build has concrete targets to attach to. Topological materials, as a class, are systems with an excitation gap in the bulk that simultaneously host protected excitations at their surface or edge. That combination, gapped bulk plus gapless boundary, is going to recur throughout the session, and it is genuinely the defining feature: it is not that the edge happens to have low-energy states, it is that the bulk topology *forces* them to be there.

Within that class we will meet, over today and the following session:

- The **quantum Hall effect**, realized in two-dimensional electron gases and in graphene under a strong perpendicular magnetic field; the simplest topological state, and our starting point.
- The **quantum anomalous Hall effect**, the same physics without any external magnetic field at all, which we will construct explicitly in Section 3.10.
- **Quantum spin Hall insulators**, effectively two time-reversed copies of a quantum Hall state glued together (Section 3.11).

- **Topological superconductors**, whose surface excitations are Majorana modes, and which underlie one route to topological quantum computing (we return to these in full in session 5).
- **Topological semimetals**, including Weyl semimetals, whose low-energy quasiparticles are Weyl fermions.
- **Topological Kondo insulators**, this year's spoken example of topology surviving in a strongly correlated, many-body setting, "and many others that do not fit on this slide."

This is slide-only material with no notebook counterpart: the slides name specific material realizations for each entry (graphene for the quantum Hall effect, Cr-doped $(Bi,Sb)_2Te_3$ for the quantum anomalous Hall effect, Bi_2Se_3 for the quantum spin Hall effect, Fe chains on Pb for a topological superconductor, TaAs for a Weyl semimetal, and SmB_6 for its many-body example) that were shown but not narrated in detail, and the course's notebook does not simulate any of these specific compounds: the taxonomy slide's many-body example is the topological Kondo insulator, with SmB_6 as the material realization. What we will simulate, in the models built up over this session, are minimal tight-binding versions of the first three items on the list above.

3.2 Linear response theory: how a measurement reads off the wavefunctions

To see why a bulk band structure can carry an invariant at all, we need a tool for connecting the wavefunctions of a Hamiltonian to something we can actually measure. That tool is linear response theory, and the idea behind it is completely general: perturb the system at some position x and time t, then measure some observable, charge, current, magnetization, heat, at another position x' and time t'. Linear response theory tells us that this measured response depends only on the electronic structure of the unperturbed material.

Two examples make this concrete before we specialize to the Hall conductivity. Put an impurity into a two-dimensional metal and look at how the electron density, or the local density of states, gets modulated around it: linear response theory shows that this modulation pattern (the Friedel oscillation) directly encodes the Fermi wavelength of the electron gas, which is exactly what scanning tunneling microscopy exploits in practice. Similarly, scattering neutrons off a magnetic material lets the neutron's magnetic moment couple to the local moments in the sample; the resulting scattering pattern reconstructs both the magnetic structure and its excitations, in particular magnons. Both of these are slide-only illustrations in the sense that the labeled Friedel-oscillation and neutron-scattering figures were shown on screen without being narrated in full; the physical content of both examples, however, was explained in speech and is worth keeping in mind as the general payoff of the formalism we build next.

Formally, we start from an unperturbed Hamiltonian H_0 and add a perturbation controlled by a dimensionless parameter λ,

$$H(\lambda) = H_0 + \lambda V,$$

so that $\lambda = 0$ recovers the original system and $\lambda = 1$ is the fully perturbed one. Any observable, the energy, the density, the magnetic moment, the current, or the wavefunctions themselves, can then be Taylor-expanded in λ; the derivative of an observable with respect to λ at $\lambda = 0$ is what we call the response of the system (its susceptibility, in the case of a magnetic perturbation). Since every observable can be built from the wavefunctions, the most useful object to expand is the ground-state wavefunction itself. For a state $|n\rangle$ of the unperturbed Hamiltonian H_0 with energy ε_n, standard first-order perturbation theory gives

$$|n\rangle_\lambda = |n\rangle + \lambda \sum_{m\neq n} \alpha_m |m\rangle + \mathcal{O}(\lambda^2), \qquad \alpha_m = \frac{\langle m|V|n\rangle}{\varepsilon_n - \varepsilon_m}. \tag{3.1}$$

Everything that follows today, the Hall conductivity, the Berry curvature, the Chern number, comes directly from plugging (3.1) into the definition of some observable.

3.3 Where perturbation theory breaks down

Before using (3.1), it is worth being honest about when it fails, because several of the most important phenomena in this course live exactly in that failure. The expansion (3.1) implicitly assumes we are in the *perturbative regime*: a small change in the input (the perturbation strength) produces a small change in the output. This is not automatic. Three cases are worth naming now, because we will meet each one properly later in the course:

- An impurity placed in a metal produces the **orthogonality catastrophe**: the many-body ground state after the perturbation becomes exactly orthogonal to the original ground state, however small the perturbation, so no finite-order Taylor expansion in λ can connect the two.
- A quantum spin (a local magnetic moment) coupled to a metal produces the **Kondo effect**: the relevant energy scale, the Kondo temperature, is a non-perturbative function of the coupling constant, and no order of perturbation theory in that coupling reproduces it. We will meet the Kondo lattice model properly in session 8; today's mention is a first flag that ordinary perturbation theory has a hard limit.
- **Superconductivity** is likewise non-perturbative: the BCS gap does not appear at any finite order in the interaction strength starting from the normal state. Session 5 works this out in full.

For today's lecture, and for every model we build in this session, we stay safely inside the regime where (3.1) holds. It is enough, for now, to know that it does not always.

3.4 The Hall conductivity from linear response

Now we specialize (3.1) to the measurement that is going to carry us to topology: the transverse, or Hall, conductivity. Take a two-dimensional material, apply a voltage in the y direction, and ask what current flows in the x direction. In momentum space, a voltage in y enters as a perturbation proportional to $\partial H/\partial k_y$ (it shifts k_y via the Peierls substitution), and the current operator in the x direction is, up to the electron charge, the group-velocity operator $\partial H/\partial k_x$. Both are exactly the kind of matrix element that (3.1) hands us.

Substituting (3.1) into the definition of the Hall conductivity gives an expression built from two velocity operators and an energy denominator, the standard Kubo formula for a filled band n and empty bands m,

$$\sigma_{xy} \;\propto\; \sum_{n \text{ filled}} \sum_{m \text{ empty}} \int_{\mathrm{BZ}} \frac{d^2k}{(2\pi)^2} \, \frac{\langle n|\partial_{k_x} H|m\rangle\langle m|\partial_{k_y} H|n\rangle - (x \leftrightarrow y)}{(\varepsilon_n - \varepsilon_m)^2}. \tag{3.2}$$

At this point the expression still depends explicitly on the band energies through the denominator, which looks like it should depend on all the messy details of the band structure. It does not, and seeing why is the whole point of the next section.

3.5 The Berry curvature, and why the Hall conductivity is an integer

Differentiate the Schrödinger equation $H(k)|u_n(k)\rangle = \varepsilon_n(k)|u_n(k)\rangle$ with respect to k_μ and project onto a different band $\langle u_m(k)|$, $m \neq n$:

$$\langle u_m|\partial_{k_\mu} H|u_n\rangle = \big(\varepsilon_n(k) - \varepsilon_m(k)\big)\,\langle u_m|\partial_{k_\mu} u_n\rangle. \tag{3.3}$$

This identity is exactly the piece we need: it trades a matrix element of $\partial_k H$ for a matrix element of the wavefunction's own k-derivative, divided by the energy difference. Using (3.3) to rewrite the numerator and denominator of (3.2) together, the explicit energy dependence cancels completely, and what remains is an object built purely from wavefunction overlaps, the **Berry curvature** of band n,

$$\Omega_n(k) = -2\,\mathrm{Im}\,\langle \partial_{k_x} u_n(k)|\partial_{k_y} u_n(k)\rangle,$$

so that the Hall conductivity of a set of filled bands is

$$\sigma_{xy} = \frac{e^2}{h}\,C, \qquad C = \sum_{n\ \text{filled}} \frac{1}{2\pi}\int_{\mathrm{BZ}} \Omega_n(k)\,d^2k. \tag{3.4}$$

This is the central result of the session. Two things about (3.4) deserve emphasis. First, σ_{xy} depends only on how the wavefunctions vary across the Brillouin zone, not on the energies at all: you can deform the band structure however you like, and as long as the wavefunctions do not change, the Hall conductivity is untouched. Second, and more strikingly, for a gapped (insulating) system C is always an integer, regardless of how many bands are involved or what the dispersion looks like. This is a genuinely topological statement: the Brillouin zone of a two-dimensional system is a compact manifold (a torus, since opposite edges of the zone are identified), and integrating a curvature over a closed surface always produces an integer, exactly as it does for the Gaussian curvature of an ordinary closed surface in differential geometry. This integer C is the **Chern number**, and (3.4) is the first place in the course where a measurable quantity, the Hall conductivity, is pinned to a topological invariant.

3.6 The quantum Hall state: Landau levels and chiral edge states

The Chern number appears in its cleanest form in the quantum Hall state itself: a two-dimensional electron gas in a strong perpendicular magnetic field. The resulting band structure is exactly the flat-band archetype from session 2, a set of dispersionless Landau levels, and each Landau level individually carries Chern number $+1$. Filling the first Landau level gives total Chern number $C = 1$; filling the first two gives $C = 2$; and so on, so that $\sigma_{xy} = C\,e^2/h$ steps through successive integer plateaus as the chemical potential is raised through successive Landau levels. This is the integer quantum Hall effect, and session 4 works out its full phenomenology (including the fractional generalization); today we only need the Chern-number bookkeeping.

Here is the puzzle this raises. We have an insulator (a gap between Landau levels), yet σ_{xy} is a *conductivity*, meaning some electron somewhere has to be moving. The resolution is that any real sample has an edge, and at that edge the bulk gap closes: gapless edge states appear, carrying exactly the current that (3.4) predicts. These edge states have a further, crucial property: on a given edge, states propagate in only *one* direction. The top edge of a sample carries current one way and only that way; the bottom edge carries it the opposite way and only that way. A mode that propagates in a single direction is called **chiral**, and

chirality is precisely what protects the quantum Hall state against disorder: an electron in a chiral edge mode has nowhere to backscatter into, since there is no counter-propagating state at the same edge and energy to scatter into. No backscattering channel means no resistance in that channel. Figure 3.1 shows both halves of this picture: the closed bulk orbits that carry nothing, the skipping orbits that carry everything, and what happens when an impurity is put in their way.

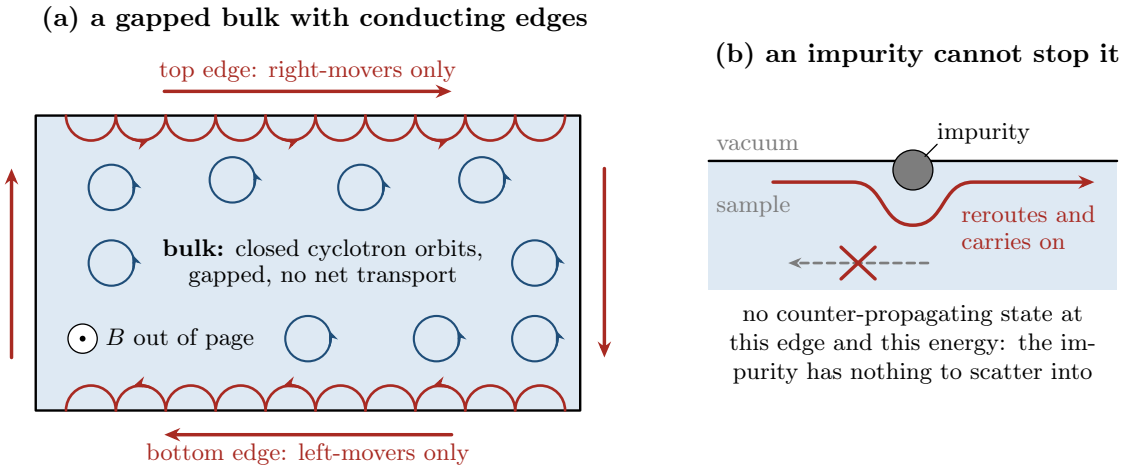


Figure 3.1: Chiral edge states, and why they cannot be backscattered. (a) A quantum Hall sample. In the bulk the electrons execute closed cyclotron orbits, go nowhere, and the system is an insulator with a gap between Landau levels. At the boundary those orbits cannot close; they skip along the edge, and for a given direction of B they skip in one direction only, opposite on the two edges. That single-direction propagation is what the word **chiral** means, and it is what carries the current that $\sigma_{xy} = C\,e^2/h$ predicts even though the bulk is gapped. (b) The consequence. An impurity at the edge can deflect the mode, but there is no counter-propagating state at the same edge and the same energy for it to scatter *into*, so the mode simply reroutes around the obstacle and carries on. No backscattering channel means no resistance, which is why the Hall plateaus are quantized as sharply as they are. Reversing the sign of B reverses every arrow at once.

A concrete model makes this vivid: a tight-binding square lattice with a perpendicular magnetic field, cut into a finite-width ribbon (translationally invariant along the ribbon but finite across it). Diagonalizing this ribbon shows exactly the structure just described: flat, Landau-level-like bulk bands, plus dispersive edge branches crossing the gap, one branch confined to each edge and propagating in only one direction. Zooming into the low-energy region near zero energy, the level spacing between the flat bands approaches the familiar equal spacing of a continuum two-dimensional electron gas in a magnetic field, so the tight-binding model smoothly reproduces the textbook Landau problem in the appropriate limit. This particular computed band-structure figure (bulk flat bands in green plus the two counter-propagating edge branches) is shown on the slides but only described qualitatively in speech; reproducing it numerically is exactly what the notebook callout below asks you to do.

Notebook check-in: `session3.ipynb`, "From electrons in a ribbon to the quantum Hall state" (Core), "The quantum Hall effect in graphene" (Core), and "Localization of the edge states in the quantum Hall effect" (Challenge)

Build a square-lattice (then honeycomb-lattice) ribbon with a magnetic field and reproduce the figure described above: flat Landau-level bulk bands plus chiral edge branches. Check the propagation direction of the edge states at positive versus negative energy on each edge, exactly the chirality argument of Section 3.6, then look at how the edge states localize in real space as a function of energy and how that localization responds to a weaker field or a narrower ribbon.

3.7 What "topology" means for a Hamiltonian, and the bulk-boundary correspondence

We have now seen one topological invariant in action; it is worth stepping back and saying what "topological" actually means, in the mathematical sense the word is borrowed from. Topology classifies objects by whether they can be smoothly deformed into one another without a drastic operation, cutting, tearing, gluing. Three-dimensional surfaces are classified by their number of holes: a donut and a coffee cup both have exactly one hole and can be continuously deformed into each other, while a sphere has none and cannot be deformed into either. Closed curves in one dimension have an analogous classification by knots: curves with inequivalent knots cannot be smoothly untangled into one another.

For a Hamiltonian, the analogous "deformation" is changing the band structure continuously while keeping the bulk gap open. Two Hamiltonians related by such a deformation share the same Chern number, precisely because C in (3.4) is quantized and therefore cannot change continuously; it can only jump, and a jump requires the gap to close at some point along the deformation. So the Chern number plays the role that the number of holes plays for a surface: Hamiltonians with the same Chern number are topologically equivalent (deformable into each other without closing the gap), and Hamiltonians with different Chern numbers are not.

This has an immediate and powerful consequence. Suppose we build a spatially varying Hamiltonian that equals H_1 (Chern number C_1) far to the left and H_2 (Chern number $C_2 \neq C_1$) far to the right. Since H_1 and H_2 cannot be deformed into each other without closing the gap, and the interpolating Hamiltonian passes continuously from one to the other, the gap *must* close somewhere in between, irrespective of how the interface is actually constructed microscopically. This is the **bulk-boundary correspondence**: an interface between two bulk phases with different Chern numbers always hosts a gapless state, and that gapless state's existence depends only on C_1 and C_2, not on any detail of the interface. The vacuum itself has $C = 0$ (it is, trivially, topologically trivial), so a quantum Hall state with $C = 1$ automatically has a gapless edge state at every boundary with vacuum, which is exactly the edge physics of Section 3.6 recovered from a completely general argument. Figure 3.2 sets out the three steps of that argument in order.

The same argument explains the robustness we already invoked. Introduce a local impurity at the edge of a quantum Hall sample: this only changes the microscopic details of the vacuum/quantum-Hall interface, and since the interface's topological excitation must exist regardless of those details, the edge state simply reconstructs itself around the impurity rather than being scattered by it. This is the general mechanism, not special to the quantum Hall case, and it will reappear for every topological invariant we meet from here on.

3.8 Three topological material classes

With the general machinery in place, three classes of topological matter are worth naming before we build explicit models for two of them:

- **Chern insulators**: bulk Chern number $C \neq 0$, chiral edge states, and time-reversal symmetry necessarily broken (we will see in Section 3.11 that time-reversal symmetry forces $\sigma_{xy} = 0$). No backscattering channel means no electrical losses at the edge, which makes them an attractive platform for dissipationless electronics.
- **Quantum spin Hall insulators**: two copies of a Chern insulator, one per spin, with opposite Chern number, so the net Chern number vanishes and time-reversal symmetry is preserved, but the edge states are **helical**: spin-up and spin-down propagate in opposite

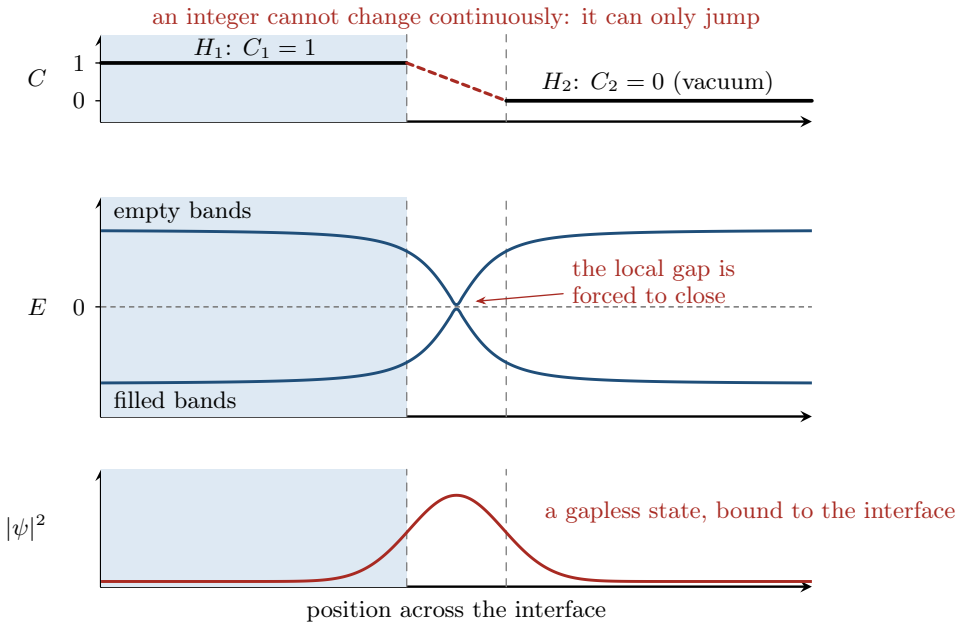


Figure 3.2: The bulk-boundary correspondence, as an argument about an integer. A Hamiltonian is interpolated slowly in space from H_1 on the left to H_2 on the right. Top: the Chern number is an integer, so it cannot slide from $C_1 = 1$ to $C_2 = 0$; it can only jump. Middle: a jump is only allowed where the quantization argument itself fails, that is, where the gap closes, so the local gap is forced down to zero somewhere in the interface region (drawn schematically, in the local-gap approximation). Bottom: what lives there is a gapless state bound to the interface. Nothing in this argument used any detail of how the interface is built, which is why the edge state survives roughened boundaries and impurities: only C_1 and C_2 enter. Taking H_2 to be the vacuum, $C_2 = 0$, this recovers the chiral edge state of figure 3.1 from a completely general argument.

directions along the same edge. This is the natural platform for dissipationless spin transport, i.e. spintronics.

- **Topological superconductors**: a bulk superconducting gap combined with gapless surface excitations that are their own antiparticle, Majorana excitations. Because the corresponding quantum information is stored non-locally, conventional local decoherence channels do not act on it directly, which is the basis of topological quantum computing. We build the minimal model for this case, the Kitaev chain, in session 5, and we will see there that it is literally the same model as the one we are about to construct next.

3.9 The SSH model: a minimal one-dimensional topological insulator

We now build the simplest possible model that shows this physics explicitly: the Su-Schrieffer-Heeger (SSH) chain. It looks almost too simple to be useful, but it is worth taking seriously, not least because the simplest model for a topological superconductor, the Kitaev chain, turns out to be exactly this model written in Majorana language.

Take a one-dimensional lattice with two sites per unit cell, A and B, and two hopping amplitudes: an intracell hopping t (within a unit cell) and an intercell hopping fixed at 1 (between neighboring unit cells). Consider first the extreme limit in which the intracell hopping dominates, $t \gg 1$, so that the chain is fully dimerized with alternating bonds: strong bonds of strength t coupling sites $(0,1)$, $(2,3)$, $(4,5)$, ... and negligible coupling between $(1,2)$, $(3,4)$, ... Each dimer is then an isolated two-level system with bonding and antibonding eigenstates at energies $\pm t$; the whole chain's spectrum is just this pair of levels repeated once per dimer, and there is nothing at zero energy.

Now take the opposite limit, $t \to 0$, in which the strong bonds are the intercell ones and the dimerization pattern is shifted by one site: leave the very first site uncoupled (a "dangling" site), couple $(1,2)$, $(3,4)$, ..., and leave the very last site dangling as well. In the bulk this looks identical to the previous chain (the same alternating dimer pattern, just shifted by one site), but the two ends are qualitatively different: each end now hosts a genuinely uncoupled site. Diagonalizing this chain gives the analogous bonding/antibonding pairs at ± 1 in the bulk, plus **two states pinned exactly at zero energy**, one localized at each end. Figure 3.3 shows the two chains side by side together with the spectra they produce.

This is the minimal example of a topological phase. The fully dimerized chain (first case) is the **trivial phase**: gapped everywhere, including at the ends. The shifted chain (second case) is the **topological phase**: gapped in the bulk, but with gapless, exactly-zero-energy modes at the edges. It looks like a very fine-tuned construction, and it is; real chains have hopping between every pair of neighboring sites. What survives fine-tuning is the important part: turning on a small coupling t between the originally-decoupled sites at the two ends does *not* remove the zero modes, as long as t stays below the bulk hopping strength. Intuitively, for $t \ll 1$ we are weakly coupling an isolated zero-energy level to a gapped bulk, and a weak coupling cannot push a level all the way across a finite gap. More rigorously, this is exactly the deformation argument of Section 3.7: the Hamiltonians at different small t are topologically equivalent, connected without closing the bulk gap, so whatever invariant protects the zero mode at $t = 0$ continues to protect it.

We can make this fully precise using the Bloch Hamiltonian. With intracell hopping t and intercell hopping normalized to 1, the momentum-space Hamiltonian in the (A, B) basis is the 2×2 matrix

$$H(k) = \begin{pmatrix} 0 & d(k) \\ d^*(k) & 0 \end{pmatrix} = (t + \cos k)\,\sigma_x + \sin k\,\sigma_y, \qquad d(k) = t + e^{ik}, \tag{3.5}$$

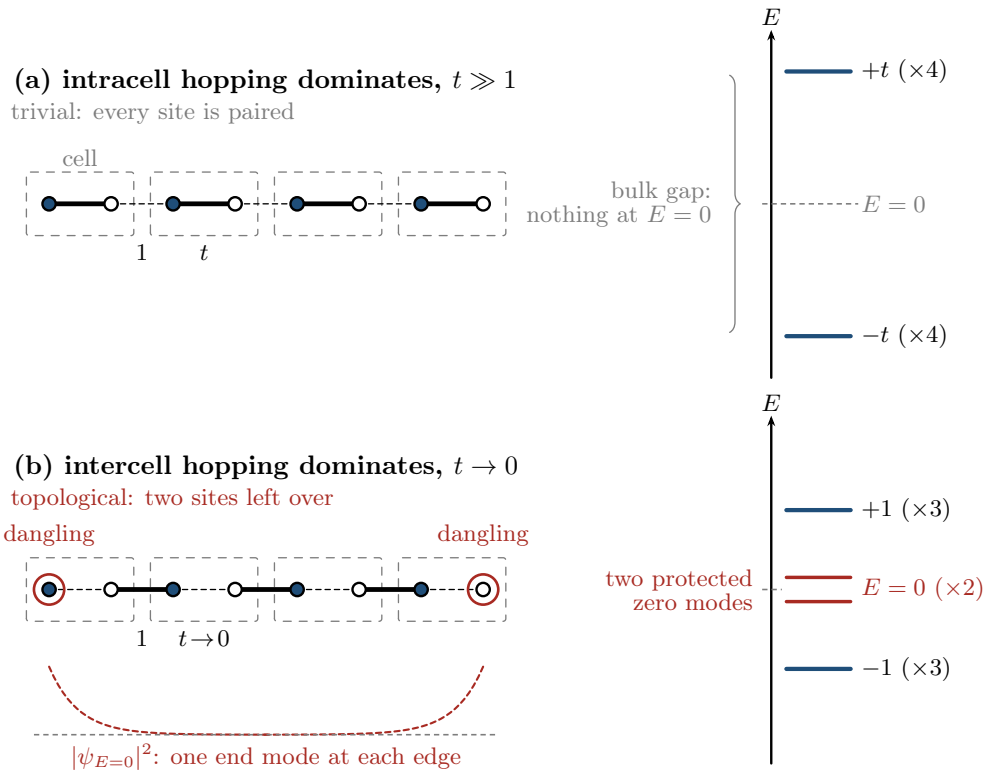


Figure 3.3: The Su-Schrieffer-Heeger chain, drawn with the same eight sites and the same unit-cell convention in both rows, so that the only difference is which of the two bonds is the strong one. (a) When the intracell hopping dominates, $t \gg 1$, every site is paired into a dimer and the spectrum is two levels at $\pm t$ with a clean gap around zero: the trivial phase. (b) When the intercell hopping dominates, $t \to 0$, the dimer pattern is shifted by one site, the first and last sites are left uncoupled, and the same bulk spectrum now carries two extra states pinned at $E = 0$, one localized at each end: the topological phase. Nothing in the bulk distinguishes the two chains; the difference lives entirely at the ends, which is exactly what a bulk invariant has to predict. At small but finite t the two end states hybridize through the chain and split away from zero only exponentially in the chain length.

with eigenvalues $E_\pm(k) = \pm|d(k)| = \pm\sqrt{1 + t^2 + 2t\cos k}$. The gap closes exactly when $d(k) = 0$, which requires $t = 1$ and $k = \pi$: this is the topological phase transition point, and it matches the qualitative $t < 1$ versus $t > 1$ split above, since at $t = 0$ the chain is the fully dangling-site (topological) limit and at $t = 1$ it becomes the fully symmetric, gapless chain.

The topological invariant here is the **Zak phase**, the one-dimensional Berry phase accumulated by the lower-band Bloch state $|u_-(k)\rangle$ as k sweeps once around the Brillouin zone,

$$\gamma = i \oint_{\mathrm{BZ}} \langle u_-(k)|\partial_k u_-(k)\rangle \, dk \pmod{2\pi}. \tag{3.6}$$

Geometrically, γ/π counts how many times $d(k)$ winds around the origin as k runs from 0 to 2π; from (3.5), $d(k) = t + e^{ik}$ traces a unit circle centered at $(t, 0)$ in the complex plane. That circle encloses the origin exactly when $|t| < 1$, and does not when $|t| > 1$: consistent with everything above, $\gamma = \pm\pi$ (one winding, a topological insulator, gapless edge modes) for $t < 1$, and $\gamma = 0$ (no winding, a trivial insulator, no edge modes) for $t > 1$. The value of the Zak phase is in exact one-to-one correspondence with the presence or absence of a zero-energy edge state: this is the one-dimensional version of the bulk-boundary correspondence from Section 3.7. Figure 3.4 draws that winding together with the bands it controls, and makes the point that the two insulating phases are indistinguishable in their spectra alone.

This is lecture and slides-only material with no notebook counterpart: `session3.ipynb` does not include a dedicated SSH-chain section; its numerical exercises start directly from two-dimensional ribbons (the quantum Hall and quantum anomalous Hall constructions of Sections 3.6 and 3.10). If you want to see the zero modes and the Zak-phase winding of (3.6) numerically, the direct way is to build the finite N-site chain from Section 3.9 by hand (following the same tight-binding diagonalization recipe used for the finite islands in session 1) and watch the two near-zero eigenvalues split away from zero as t is increased past 1.

3.10 Gapping the Dirac cone: Chern insulators without Landau levels

Now we return to the question the whole session opened with. A Chern insulator needs a broken time-reversal symmetry (indeed, one can show directly from (3.4) that $\sigma_{xy} \equiv 0$ whenever time-reversal symmetry holds, since time-reversal flips the sign of the Berry curvature). But nothing in the derivation of the Chern number required a magnetic field or Landau levels specifically. So the question becomes: can we engineer a finite Chern number from an ordinary lattice band structure, using only time-reversal-breaking terms that live entirely within the lattice itself?

The recipe is to look for a lattice with a **massive Dirac equation** in its low-energy expansion. Several two-dimensional lattices, the honeycomb lattice, the kagome lattice, a square lattice threaded by π-flux, all host massless Dirac points in their band structure (exactly the Dirac cone session 2 derived for graphene). Near one such point, the low-energy Hamiltonian takes the form

$$H(k) = k_x \, \sigma_x + \lambda \, k_y \, \sigma_y + m \, \sigma_z, \tag{3.7}$$

with $\lambda = \pm 1$ fixed by the lattice's own chirality at that Dirac point and m a mass term that opens a gap, $E_\pm(k) = \pm\sqrt{k_x^2 + k_y^2 + m^2}$. Computing the Berry curvature of the lower band and integrating it (the same procedure as (3.4), now over the small neighborhood of momentum space where the curvature is concentrated) gives the Chern number contributed by this single gapped Dirac cone,

$$C_{\mathrm{Dirac}} = \frac{1}{2} \operatorname{sgn}(m) \operatorname{sgn}(\lambda). \tag{3.8}$$

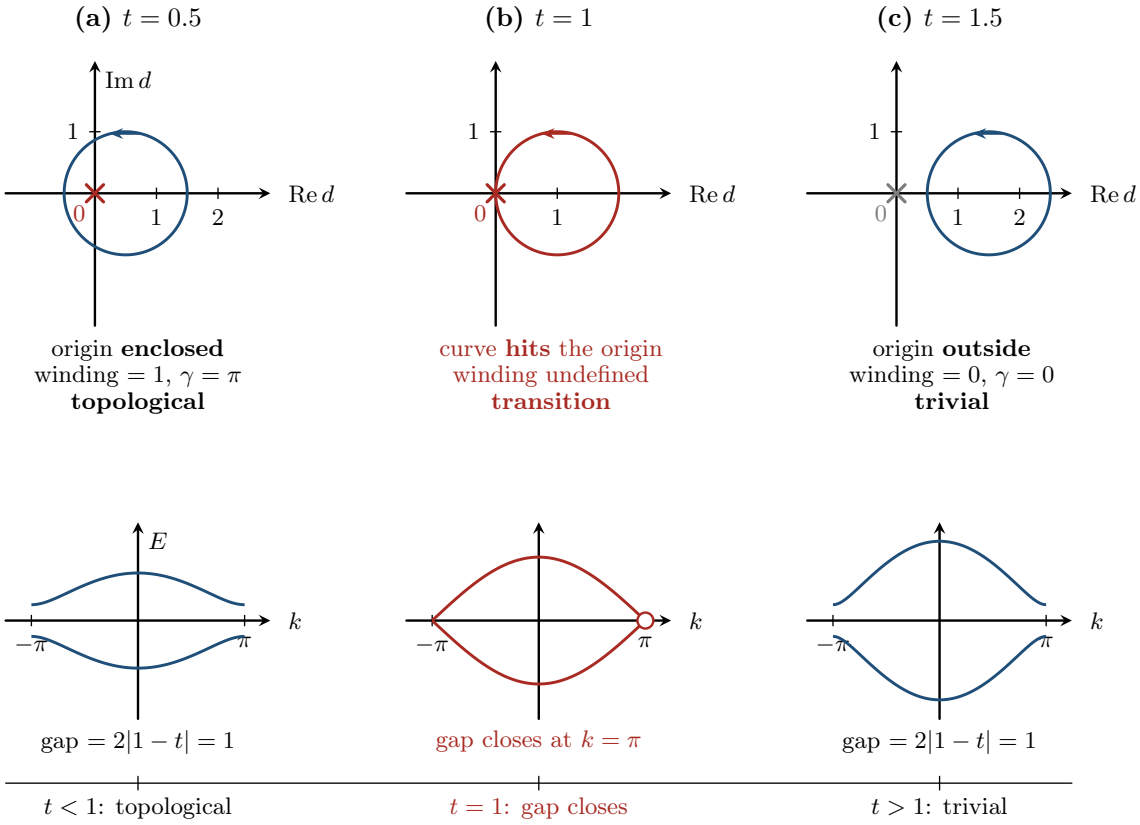


Figure 3.4: The one-dimensional invariant, drawn. Top row: the off-diagonal element $d(k) = t + e^{ik}$ of the SSH Bloch Hamiltonian (3.5), traced out in the complex plane as k sweeps once through the Brillouin zone (arrow: direction of increasing k). It is always the unit circle, but centred at $(t, 0)$, so it encircles the origin for $t < 1$ and misses it for $t > 1$; the winding number of that curve about the origin is exactly γ/π, the Zak phase (3.6) in units of π. Bottom row: the corresponding bands $E_\pm(k) = \pm|d(k)|$, with gap $2|1 - t|$. The two rows say the same thing twice: the only way to unwind the curve is to drag it across the origin, and $d(k) = 0$ is precisely where the gap closes, so the invariant cannot change while the system stays insulating. Note that (a) and (c) have the same gap and look equally unremarkable as spectra – the distinction between them is invisible in the energies and lives entirely in the wavefunctions, which is the whole point of (3.4). By the bulk-boundary correspondence, $\gamma = \pi$ is also exactly the condition for the zero-energy end modes of figure 3.3.

The half-integer in (3.8) is not itself an inconsistency: a real lattice always has an even number of Dirac points related by time-reversal symmetry (the honeycomb lattice has two, conventionally called K and K'), and only the *sum* of their contributions needs to be an integer.

This is exactly where the choice of mass term matters, and it is worth working through the two natural choices explicitly, because they give qualitatively different physics. At K and K' the chirality λ has opposite sign (they are time-reversal partners). Suppose we open the gap with a simple staggered sublattice potential (raising the on-site energy on one honeycomb sublattice, lowering it on the other): this mass term is an ordinary scalar potential, so it has the *same* sign m at both K and K'. From (3.8), the two Dirac points then contribute Chern numbers of opposite sign, $+\frac{1}{2}\,\mathrm{sgn}(m)$ and $-\frac{1}{2}\,\mathrm{sgn}(m)$, which cancel: $C = 0$, an ordinary, topologically trivial insulator (this is the boron-nitride-like route to a gap). Now instead open the gap with a second-neighbor, purely imaginary hopping, the **Haldane coupling**: physically, this amounts to a pattern of local orbital currents circulating within each hexagon, which is odd under time reversal. This mass term flips sign between K and K' *relative to* λ, so the two Dirac points now contribute Chern numbers of the *same* sign, adding rather than cancelling: $C = \pm 1$, a genuine Chern insulator, built entirely from lattice hoppings with no external magnetic field anywhere. This was the first model ever proposed with a finite, quantized Hall conductivity and zero net flux, and it is exactly the gap-opening mechanism session 2's Dirac cone was waiting for. Figure 3.5 lays the two cases out side by side, mechanism and arithmetic together. A gap can also be opened by combining an ordinary magnetization with spin-orbit coupling, which produces the same sign pattern as the Haldane term and gives an equivalent quantum anomalous Hall state. Both routes have material realizations: Haldane-like physics has been realized in twisted graphene bilayers on boron nitride, and the magnetism-plus-spin-orbit route has been realized, and measured, in magnetically doped Bi_2Se_3.

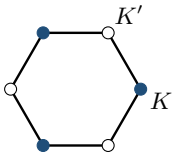
Notebook check-in: `session3.ipynb`, "The quantum anomalous Hall effect" (Core), "Edge states of the quantum anomalous Hall state" (Core), "Topological states and topological invariants" (Challenge), "Topological phase transitions in the bulk" (Challenge), and "Topological phase transitions in the edge" (Core)

Build the Haldane-coupled honeycomb ribbon (no magnetic field) and check that the same chiral edge states from Section 3.6 appear anyway, exactly as the argument above predicts. Then, in the Berry-curvature exercise, integrate the curvature separately for the Haldane coupling and for a sublattice imbalance: you should find a non-zero net Chern number for the Haldane case and a vanishing one for the sublattice-imbalance case, precisely the cancellation-versus-addition distinction worked out around (3.8). Combining both terms and tracking where the bulk gap (and then the edge states) closes as their relative strength changes is the numerical version of the phase transition at $t = 1$ from Section 3.9, now realized in two dimensions. The honeycomb Berry-curvature and edge-dispersion figures on the slides show this same computation; the notebook is where you reproduce it yourself.

3.11 Quantum spin Hall insulators

A Chern insulator needs broken time-reversal symmetry. The quantum spin Hall insulator is what you get if you insist on keeping time-reversal symmetry intact while still wanting a topological, gapped state: take two copies of the massive-Dirac construction (3.7), one for each spin, with **opposite** mass for opposite spins, $m_\uparrow = -m_\downarrow$. From (3.8), the two spin sectors then carry opposite Chern number,

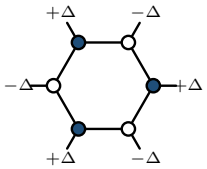
$$C_\uparrow = \tfrac{1}{2}\,\mathrm{sgn}(m_\uparrow)\,\mathrm{sgn}(\lambda), \qquad C_\downarrow = -C_\uparrow, \qquad C = C_\uparrow + C_\downarrow = 0, \qquad C_s \equiv C_\uparrow - C_\downarrow = \pm 2. \tag{3.9}$$



Brillouin zone

The rule (3.8): $C_{\text{Dirac}} = \frac{1}{2}\,\text{sgn}(m)\,\text{sgn}(\lambda)$.
The lattice fixes the chirality: $\lambda = +1$ at K and $\lambda = -1$ at K', because K and K' are time-reversal partners; only the *sum* over the two valleys has to be an integer.

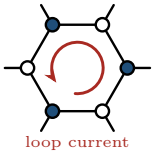
(a) staggered sublattice potential



a scalar on-site potential: **even** under time reversal, so m has the **same** sign at K and K'.

$C_K = \frac{1}{2}(+)(+) = +\frac{1}{2}$
$C_{K'} = \frac{1}{2}(+)(-) = -\frac{1}{2}$
$C = 0$: the two cancel.
An **ordinary insulator**.

(b) Haldane coupling



circulating orbital currents: **odd** under time reversal, so m **flips** sign between K and K'.

$C_K = \frac{1}{2}(+)(+) = +\frac{1}{2}$
$C_{K'} = \frac{1}{2}(-)(-) = +\frac{1}{2}$
$C = +1$: the two add.
A **Chern insulator**, no field.
(Reversing the coupling gives -1.)

Figure 3.5: The whole content of Section 3.10 in one piece of bookkeeping. Equation (3.8) gives each gapped Dirac point a Chern number $\frac{1}{2}\,\text{sgn}(m)\,\text{sgn}(\lambda)$, and the honeycomb lattice's two Dirac points are time-reversal partners with opposite chirality, $\lambda = +1$ at K and $\lambda = -1$ at K'. So everything hinges on how the mass term itself behaves under time reversal. (a) A staggered sublattice potential is an ordinary scalar potential, even under time reversal, so m has the same sign at both valleys, the two half-integers come out opposite, and they cancel: a perfectly ordinary insulator, the boron-nitride-like gap. (b) The Haldane coupling is a pattern of circulating orbital currents, odd under time reversal, so m flips sign between the valleys, the two half-integers come out equal, and they add: $C = \pm 1$, a genuine Chern insulator with quantized σ_{xy} and chiral edge states, built from lattice hoppings alone with no external magnetic field and zero net flux. The two gaps look identical in the bulk spectrum; only the wavefunctions know the difference.

The total Chern number C vanishes, consistent with unbroken time-reversal symmetry, but the **spin** Chern number C_s does not, and it is C_s that controls the edge physics: at each edge, spin-up propagates one way and spin-down propagates the other way, a **helical** rather than chiral edge state.

The natural mechanism that produces exactly this opposite-sign mass is spin-orbit coupling. The relativistic spin-orbit term has the structure $\mathbf{L}\cdot\mathbf{S}$; its z-component, L_zS_z, acts like an effective magnetic field pointing one way for spin-up electrons and the opposite way for spin-down electrons. That is exactly the opposite-sign mass structure (3.9) needs, generated with no net magnetic field and no net time-reversal breaking. Graphene itself, once its (very small) intrinsic spin-orbit coupling is included, is technically a quantum spin Hall insulator by this mechanism; the effect is far more pronounced, and experimentally accessible, in monolayer 1T′-WTe_2, where the quantum spin Hall effect survives up to 100 K.

There is a subtlety worth being precise about. If spin were perfectly conserved, we could simply speak of "the" Chern number for spin-up and "the" Chern number for spin-down independently. Real materials have spin-mixing terms (spin-orbit coupling itself mixes spin to some degree), so a natural worry is that these terms could smoothly close the helical edge states' protection. They cannot, as long as time-reversal symmetry itself survives, and the reason is Kramers' theorem: for spin-1/2 electrons the time-reversal operator squares to -1, which forces an exact twofold degeneracy at every momentum that is its own time-reversal partner (the time-reversal-invariant momenta, Γ and X in this setup). That degeneracy exists whether or not spin-mixing terms are present, so the edge-state crossing at those momenta cannot be gapped out by any perturbation that respects time-reversal symmetry, only by one that breaks it. This is a genuinely different, and stronger, protection mechanism than the "two independent Chern insulators" picture would suggest on its own. Figure 3.6 contrasts the chiral and helical edge dispersions and shows what a time-reversal-breaking term does to the protected crossing.

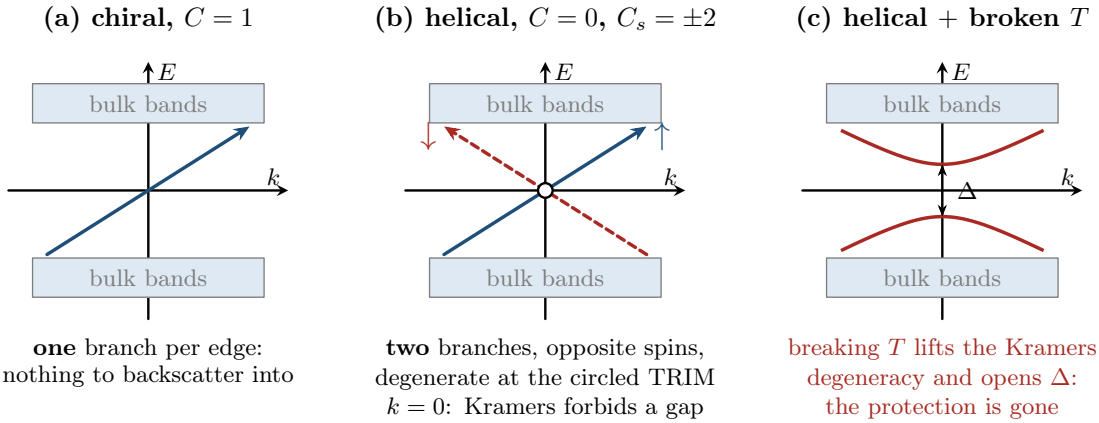


Figure 3.6: Edge dispersions, schematically, for the two topological classes of Section 3.8. Grey blocks are the bulk bands; the branches crossing the bulk gap are the edge states, plotted against the momentum along the edge. (a) A Chern insulator has a single branch per edge: at any energy in the gap there is one state and it moves one way, so a scatterer has nothing to send it back into. (b) A quantum spin Hall insulator has two, one per spin, running in opposite directions on the *same* edge, which is what **helical** means; the total Chern number vanishes but the spin Chern number does not (3.9). (c) What protects the crossing is not that the two spin sectors are separately conserved, but Kramers' theorem: the crossing sits at a time-reversal-invariant momentum, where a twofold degeneracy is guaranteed for spin-1/2 electrons whatever the perturbation, so long as it respects time reversal. Time-reversal-preserving disorder can therefore deform the branches but not gap them; only a time-reversal-breaking perturbation opens the gap Δ in (c), and with it the protection is gone.

Disorder makes the distinction concrete: helical edge states survive disorder that preserves

time-reversal symmetry (the band structure is modified but the gapless crossing at the edge remains), while disorder that breaks time-reversal symmetry opens a genuine gap in the edge spectrum and destroys the protection entirely.

Notebook check-in: `session3.ipynb`, "The quantum spin Hall state" (Core), "The robustness of the quantum spin Hall state" (Core), "Topological invariant of the quantum spin Hall state", and "Finite size effects in topological systems" (Challenge, using exchange coupling plus Rashba spin-orbit coupling)

Build the two-copies-of-a-Chern-insulator model and identify the helical pair of edge states by spin projection and propagation direction, exactly the pattern in (3.9). Then add disorder that preserves versus breaks time-reversal symmetry and confirm the gap only opens in the second case. The "in the lab" note in this notebook section points to the 1T′-WTe_2 measurement mentioned above.

3.12 Topology beyond electrons

We built all of today's machinery, the Chern number, the Zak phase, the bulk-boundary correspondence, around electronic band structures. None of the underlying mathematics actually requires electrons. The same invariants apply to any quantum excitation with a well-defined dispersion relation: **magnons**, the spin-wave excitations of an ordered magnet, and **spinons**, the fractionalized spin excitations of a quantum magnet, both admit exactly the same Berry-curvature-and-Chern-number treatment, and we will carry it out explicitly for magnon bands in session 6 and for spinon bands in session 8. More broadly still, the argument only needed a Hamiltonian, or more generally a matrix describing how a wave propagates, so it applies to any system with a dispersion relation at all: photonic systems (light propagating through a periodic medium), and even purely classical mechanical systems, coupled pendula, vibrating lattices, electronic circuits, have all been shown, in tabletop experiments, to host topologically protected modes. Topology, in the end, is a statement about the internal structure of eigenvectors as a function of momentum, and that structure exists wherever a wave equation does, quantum or not.

If there is one thing to take away from today, it is this: topological systems have protected edge modes, those modes are tied to a non-trivial value of a topological invariant computed entirely from the bulk, and this happens in both one and two dimensions (and beyond), typically protected by a specific symmetry, translational symmetry and gauge invariance for the Chern number, time-reversal symmetry for the quantum spin Hall state.

3.13 Closing summary

- The Hall conductivity, computed via linear response theory from the wavefunction expansion (3.1), reduces to an integral of the Berry curvature over the Brillouin zone and is quantized to an integer, the **Chern number**, for any gapped system: $\sigma_{xy} = C\, e^2/h$ (3.4).
- The quantum Hall state realizes this with flat Landau-level bands, each contributing Chern number 1; the bulk gap forces the current onto **chiral** edge states, one direction per edge, which is why they cannot backscatter and why the plateaus are so sharply quantized (Section 3.6).
- **Topology** classifies Hamiltonians by whether their band structures can be continuously deformed into one another without closing the gap; the Chern number is that classification's invariant, and an interface between regions of different Chern number is forced to

host a gapless state, the general **bulk-boundary correspondence** (Section 3.7).

- The **SSH model** is the minimal one-dimensional realization: a 2×2 Bloch Hamiltonian (3.5) whose lower-band Zak phase (3.6) is 0 (trivial) or π (topological) depending on whether the intracell hopping is larger or smaller than the intercell hopping, with the topological phase hosting protected zero-energy modes at the chain's ends.
- A lattice hosting a **massive Dirac equation** (3.7) contributes Chern number $\frac{1}{2}\,\mathrm{sgn}(m)\,\mathrm{sgn}(\lambda)$ per Dirac point (3.8); whether the contributions from time-reversal-paired Dirac points cancel or add depends entirely on how the mass term transforms under time reversal, which is exactly what separates a trivial gap (staggered sublattice potential) from a genuine **Chern insulator** (the Haldane model), the answer to the session's opening question of what happens when you gap the honeycomb Dirac cone.
- Pairing two oppositely-massed copies of (3.7), one per spin, preserves time-reversal symmetry ($C = 0$) while keeping a non-trivial spin Chern number $C_s = \pm 2$ (3.9): the **quantum spin Hall insulator**, with helical edge states protected by Kramers degeneracy rather than by broken symmetry.
- The same invariants generalize beyond electrons entirely, to magnons and spinons (sessions 6 and 8), and beyond quantum mechanics altogether, to any system with a dispersion relation.

Next session we go back to the flat-band archetype from session 2 and work through it properly: the full quantum Hall effect, Landau levels from first principles, and the fractional quantum Hall state.

Session 4

Electrons in a Magnetic Field, Quantum Hall Effect and Landau Levels

Last time we built the general machinery of topological band structure: the Kubo formula for the Hall conductivity, its rewriting as an integral of Berry curvature, the Chern number as the resulting topological invariant, and the bulk-boundary correspondence that ties a nonzero Chern number to a protected chiral edge state. We worked all of that machinery through two concrete examples, the SSH chain and a gapped Dirac cone on the honeycomb lattice (the Haldane construction), and we did it quite deliberately without ever turning on a magnetic field. That was a promissory note: session 2 had already told us that a Bloch Hamiltonian's dispersion comes in three archetypes, parabolic, Dirac, and flat, and the flat-band case, the one that gives a literal magnetic field its own dedicated topological story, was left for today. So let us pick up exactly where we left off. Today we run session 3's Chern-number machinery through the flat-band case in full: the integer quantum Hall effect, Landau levels as topological flat bands, and then, once we let those flat bands become genuinely many-body objects, the Laughlin wavefunction, fractional charge, and anyonic statistics, the simplest example of fractionalization we will meet in this course. We will close by turning the telescope back around: the Haldane model gives us a quantized Hall conductance with no magnetic field at all, and we will see explicitly that it and the ordinary quantum Hall state are two faces of the very same topological phase.

4.1 Where the quantum Hall state shows up

The first thing worth appreciating about the quantum Hall state is that it is not the property of one particular material; it is a state of matter that a whole family of two-dimensional electron systems can realize, each with its own convenient energy and temperature scale.

- **Semiconductor quantum wells**, historically gallium arsenide (GaAs), were the original platform. There the electrons are essentially parabolic (ordinary, quadratic-dispersion) electrons, the relevant energy scale grows linearly in the magnetic field, and the numbers work out to roughly one millielectronvolt for laboratory-accessible fields, which is why GaAs quantum Hall physics needs temperatures on the order of one kelvin.
- **Two-dimensional materials**, and graphene above all, realize the same physics but with Dirac electrons rather than parabolic ones. As we will derive explicitly below, that changes the energy scale to grow as the square root of the field rather than linearly, and

the slides give the concrete numbers: roughly two orders of magnitude larger, around 100 millielectronvolts, high enough that graphene's quantum Hall plateaus survive up to about 100 K, and in the best samples essentially up to room temperature.

Whatever the platform, the common thread is the one we already met last session: an observable, the Hall conductance, comes out exactly quantized, and that quantization is a statement about topology rather than about any particular material's chemistry.

4.2 Measuring the Hall conductance

The experiment itself is simple to state. Take a two-dimensional system, apply a voltage along one in-plane direction (call it y), and measure the resulting current along the perpendicular in-plane direction (x). The ratio of that transverse current to the applied voltage is the Hall conductance, and it is this object that gets quantized in the quantum Hall regime.

A real device realizing this measurement is a two-dimensional electron gas (2DEG) contacted by several finger gates: a voltage is imposed between a top and a bottom gate, and the current is read off between two gates arranged along x, with everything wired through metallic leads. The device geometry itself carries no new physics beyond what we need conceptually, so we will not dwell on it further; the slide deck's device figures are worth a glance for the real-world picture, but the notes from here on work with the idealized two-dimensional system.

4.3 From the Kubo formula to the Chern number, briefly

We derived this result in full last session, so let us not repeat fifteen lines of algebra that would look almost identical; instead let us recall the logic and confirm that it applies unchanged here. Second-order perturbation theory (the Kubo formula) gives the transverse conductivity as a sum, over all pairs of occupied and empty bands, of matrix elements of $\partial H/\partial k_x$ and $\partial H/\partial k_y$ divided by energy differences. Using the identity that relates $\partial H/\partial k$ to the derivative of the eigenstates themselves (differentiate the eigenvalue equation $H(\mathbf{k})|u_{\mathbf{k}}\rangle = \varepsilon(\mathbf{k})|u_{\mathbf{k}}\rangle$ with respect to $\mathbf{k}$), every explicit energy denominator cancels, and what survives depends only on the wavefunctions: the Berry connection $\mathbf{A}(\mathbf{k}) = i\langle u_{\mathbf{k}}|\nabla_{\mathbf{k}}|u_{\mathbf{k}}\rangle$ and its curl, the Berry curvature $\Omega(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathbf{A}(\mathbf{k})$. Summing over occupied bands and integrating over the Brillouin zone (a closed, compact manifold, a torus) gives an integer,

$$\sigma_{xy} = \frac{e^2}{h}\, C, \qquad C = \frac{1}{2\pi} \sum_{n\,\mathrm{occ.}} \int_{\mathrm{BZ}} \Omega_n(\mathbf{k})\, d^2k \in \mathbb{Z}, \tag{4.1}$$

the Chern number of the occupied bands. This is the same statement as session 3's: the Hall conductance is not just quantized, it is a topological invariant, insensitive to any smooth deformation of the Hamiltonian that does not close the bulk gap. What is new today is not this derivation; it is the specific band structure we are going to plug into it, a magnetic field's Landau levels, and everything that follows once we ask what the many-body ground state of a *partially filled* Landau level looks like.

4.4 Landau levels as Chern-number-one flat bands

A two-dimensional electron gas in a perpendicular magnetic field develops a set of completely flat bands, the Landau levels, indexed $n = 0, 1, 2, \ldots$. "Flat" here means dispersionless: the energy does not depend on momentum at all (we will derive exactly why in Section 4.9). Despite carrying no dispersion, each Landau level carries a finite, and in fact minimal nonzero,

Chern number: $C = 1$. If the chemical potential sits between the N-th and $(N+1)$-th Landau level, the total Chern number of the occupied bands is simply the sum of N individual contributions of 1,

$$C_{\text{tot}} = \sum_{n=1}^{N} C_n = N \quad \Longrightarrow \quad \sigma_{xy} = N \frac{e^2}{h}, \tag{4.2}$$

an integer directly counting the number of filled Landau levels. This is the integer quantum Hall effect: a state built entirely from flat, dispersionless bands, whose Hall conductance is nonetheless an exact, sharply quantized integer multiple of e^2/h.

4.5 Why an insulating bulk still conducts: edge states

Here is the obvious puzzle. If the chemical potential sits in a gap between two Landau levels, the bulk of the sample is, by definition, an insulator: no partially filled band, no bulk states at the Fermi energy. So how can such a system carry a finite, quantized current at all?

The answer, exactly as in session 3's general bulk-boundary argument, is that a nonzero Chern number cannot simply end: at any interface where the Chern number changes (here, the boundary between a Chern-N sample and the vacuum, which has $C = 0$), there must be N protected, gapless, chiral edge modes. A chiral mode propagates in only one direction; the number of edge modes at an interface equals the difference in Chern number across it. Every real sample is finite, so it always has an edge, and the edge states are always there, whether or not we are thinking about them. Applying a voltage between the top and bottom edges of the sample simply creates an imbalance between right-moving and left-moving edge electrons, and that imbalance is exactly the quantized Hall current.

The band structure of a ribbon (a sample infinite in y, finite in x) makes the picture completely explicit: alongside the flat, dispersionless bulk Landau levels, one finds a set of dispersive states localized at the left edge, all propagating in one direction, and a mirror set localized at the right edge, propagating in the opposite direction. Because a given propagation direction is tied to a *location* (left edge or right edge), an impurity sitting on the left edge simply cannot scatter a left-moving electron into a right-moving one: the right edge is macroscopically far away. There is no available final state, so there is no backscattering, and the edge transport is dissipationless. Figure 4.1 puts the ribbon spectrum and its real-space counterpart side by side, and shows the chiral modes crossing the chemical potential.

Notebook check-in: `session4.ipynb`, "Localization of the edge states in the quantum Hall effect in a ribbon" (Challenge), and "Localization of the edge states in the quantum Hall effect in an island" (Challenge)

Build a finite ribbon and a finite island in a magnetic field and look directly at where the edge-state wavefunctions live in real space, confirming the left/right (or perimeter) separation that the no-backscattering argument above depends on.

4.6 The experimental signature: quantized Hall plateaus

Sweeping the magnetic field on a real 2DEG shows this transition directly. At low field, before the quantum regime sets in, the Hall resistance grows linearly with B, exactly the classical Hall effect from ordinary electromagnetism. Once the field reaches the tesla range (the precise scale depends on the material, tracking the Landau-level spacing derived below), a staircase of sharp plateaus appears in the Hall resistance instead. As the field increases further, the cyclotron energy $\hbar\omega_c$ grows, so fewer and fewer Landau levels remain below a fixed chemical

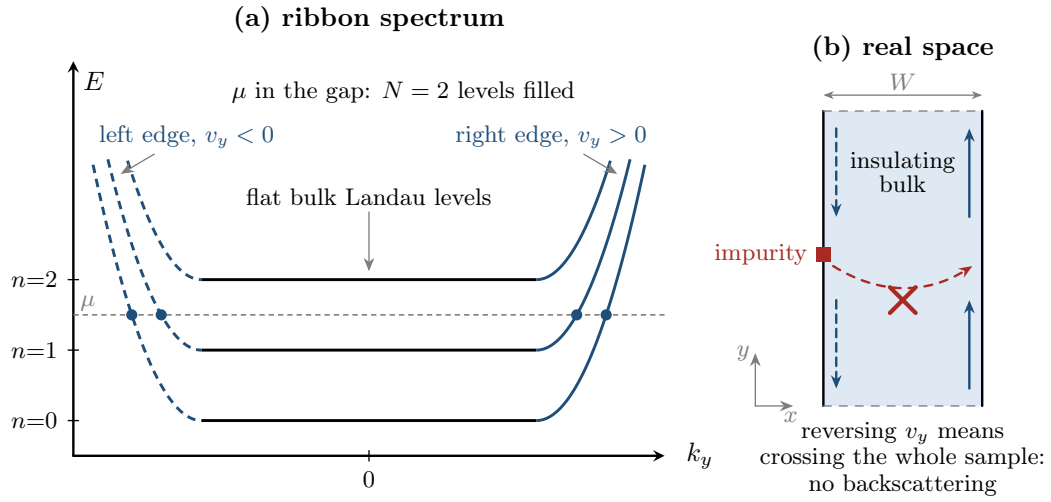


Figure 4.1: The quantum Hall ribbon. (a) The spectrum of a strip that is infinite along y and finite along x: in the bulk the Landau levels (4.6) are exactly flat, so they carry no current at all, but near each boundary the guiding centre $x_0 = \hbar k_y/B$ is pushed into the wall and the level bends up steeply. Those bending branches are the edge states, and each one crosses the chemical potential exactly once per edge, so the number of chiral modes at a given edge equals the number N of filled Landau levels, here $N = 2$. (b) The same statement in real space: the branches with $k_y > 0$ have $v_y = \hbar^{-1}\partial_{k_y}E > 0$ and live on the right edge, the branches with $k_y < 0$ move the opposite way and live on the left edge. Reversing an electron's velocity therefore means moving it to the other side of the sample, which no local impurity can do; that is why the edge transport is dissipationless.

potential; each time the chemical potential sweeps past a Landau level, the number of filled levels N in (4.2) drops by one, and the Hall resistance jumps to the next quantized plateau.

4.7 Coupling electrons to a magnetic field: minimal coupling and the Peierls substitution

To go beyond this phenomenology and actually derive the Landau level spectrum, we need to know how to put a magnetic field into a Hamiltonian at all. Classically, a magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ enters the equations of motion through the canonical momentum,

$$\boldsymbol{\pi} = \mathbf{p} - e\mathbf{A}, \tag{4.3}$$

replacing the ordinary momentum $\mathbf{p}$ everywhere it appears; computing the resulting equations of motion reproduces the familiar Lorentz force law. Throughout this session we set the electron charge magnitude and the speed of light to one (as the slides implicitly do), keeping only $\hbar$ explicit where the Landau level spacing needs it. The same replacement (4.3) works unchanged in quantum mechanics, with $\mathbf{p} \to -i\hbar\nabla$: minimal coupling turns the free Hamiltonian $p^2/2m$ into $\pi^2/2m$, and similarly for the Dirac Hamiltonian below.

One subtlety is unavoidable: $\mathbf{A}$ is never unique. Any $\mathbf{A} \to \mathbf{A} + \nabla\chi$ for a scalar function χ gives the same physical field $\mathbf{B}$, so we always have to pick a gauge. There are two gauges we will actually need, the Landau gauge and the symmetric gauge, and we will meet both in Section 4.8. A related, and slightly counterintuitive, point: applying a uniform out-of-plane field does not itself break the in-plane translational symmetry of the physical system, but *every* choice of $\mathbf{A}$ realizing that field breaks translational symmetry formally, in one direction or in both. Which symmetry survives, and how much of it, is a gauge choice, not a physical one.

The minimal-coupling replacement (4.3) is the natural route for continuum models: Schrödinger or Dirac electrons, where we directly write down and solve a differential equation. For a tight-binding model, written in real-space orbitals, the more natural route puts the field's dependence into the wavefunctions rather than the Hamiltonian, via a local gauge transformation. Concretely, the hopping amplitude t_{ij} is, by definition, an overlap between the orbital wavefunctions at sites i and j; folding the vector potential into those wavefunctions renormalizes every hopping by a complex phase equal to the line integral of $\mathbf{A}$ between the two sites,

$$t_{ij} \;\longrightarrow\; t_{ij}\,\exp\!\left(i\int_{\mathbf{R}_i}^{\mathbf{R}_j}\mathbf{A}\cdot d\boldsymbol{\ell}\right), \tag{4.4}$$

the **Peierls substitution**. It is the tight-binding counterpart of minimal coupling, and the two give the same physics: at low energies, expanding a Peierls-substituted tight-binding model reproduces the continuum minimal-coupling Hamiltonian directly.

4.8 Choosing a gauge: Landau versus symmetric

For a uniform field $\mathbf{B} = B\hat{z}$, two gauges will matter to us.

- The **Landau gauge**, $\mathbf{A} = (0, Bx, 0)$, puts all the spatial dependence into a single coordinate, x. Since $\mathbf{A}$ has no y-dependence, the Hamiltonian built from it still commutes with p_y (translations along y remain a good symmetry), but not with p_x. This gauge preserves *one* translational symmetry, which is exactly what we need to talk about a band structure at all, and it is the gauge behind every ribbon calculation in this session.
- The **symmetric gauge**, $\mathbf{A} = \frac{B}{2}(-y, x, 0)$, breaks both translational symmetries but preserves the system's rotational symmetry instead. It is the natural gauge for writing down many-body wavefunctions with good angular momentum, which is exactly what we need once we get to the fractional quantum Hall state in Section 4.13 onward.

Figure 4.2 makes the difference concrete by writing both gauges as Peierls phases (4.4) on a square lattice: the individual bond phases look nothing alike, but the sum around any plaquette — the only gauge-invariant combination, and the only thing the electrons respond to — is the same flux in both.

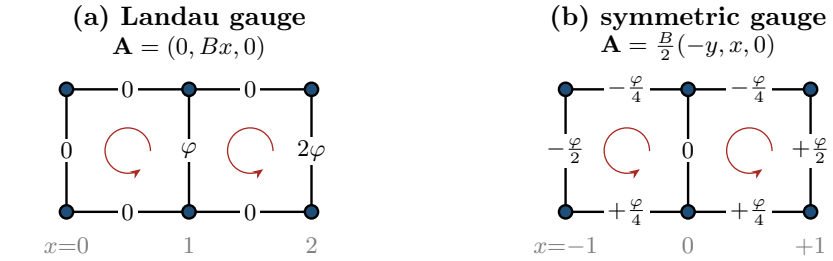
4.9 Landau levels of Schrödinger electrons

Let us now actually solve for the spectrum, starting with the simplest possible case: free electrons, no lattice, no potential beyond the field itself, dispersion $p^2/2m$ before the field is switched on. With minimal coupling and the Landau gauge $\mathbf{A} = (0, Bx, 0)$,

$$H = \frac{\pi_x^2 + \pi_y^2}{2m} = \frac{p_x^2}{2m} + \frac{(p_y - Bx)^2}{2m}. \tag{4.5}$$

We take the carrier charge to be $+e$ in (4.3) and set $e = \hbar = 1$ throughout. This is a bookkeeping choice, but not a free one: it fixes $[\pi_x, \pi_y] = +i\hbar^2/\ell_B^2$, which is the convention the ladder operators of Section 4.10 and the holomorphic wavefunctions of Section 4.13 are built on. For an electron, of charge $-e$, every cyclotron orbit and every edge current below circulates the other way, and nothing else changes.

Because $\mathbf{A}$ has no y-dependence, $[H, p_y] = 0$, so we can look for eigenstates of the form $\psi(x, y) = e^{ik_y y}\phi(x)$. Replacing $p_y \to \hbar k_y$ turns (4.5) into a purely one-dimensional problem in x,



no y-dependence: $[H, p_y] = 0$,
so a band structure in k_y survives

depends on x *and* y: no translation
survives, but the rotation does

going once counterclockwise around *any* plaquette, both gauges give the same $\oint \mathbf{A} \cdot d\boldsymbol{\ell} = \varphi = \Phi_\square$
the bond phases are a gauge choice; the loop sum is the physical flux
(each label is the phase for hopping in the $+x$ or $+y$ direction)

Figure 4.2: Two gauges for the same magnetic field, written as Peierls phases (4.4) on a square lattice. (a) In the Landau gauge the horizontal bonds carry no phase at all and the vertical bond in column x carries $x\varphi$: the phase pattern depends on x but not on y, so translations along y are still an exact symmetry and a band structure in k_y still exists. (b) In the symmetric gauge every bond carries a phase and none of them matches (a), so no translation survives, but the pattern is invariant under rotation about the plaquette centre. Nothing physical distinguishes the two: going once around any plaquette, the individual phases add up to the same φ, which is the flux through that plaquette. The bond phases are a gauge choice, the loop sum is not, and it is the loop sum alone that the electrons respond to.

$$H\phi(x) = \left[\frac{p_x^2}{2m} + \frac{B^2}{2m}\Big(x - \frac{\hbar k_y}{B}\Big)^2\right]\phi(x),$$

which is exactly the Hamiltonian of a quantum harmonic oscillator, centered at $x_0 = \hbar k_y/B$ (a shift absorbed by a change of variable, exactly as the lecture describes) with frequency $\omega_c = B/m$, the cyclotron frequency. Solving it with the same ladder-operator machinery as the ordinary harmonic oscillator gives

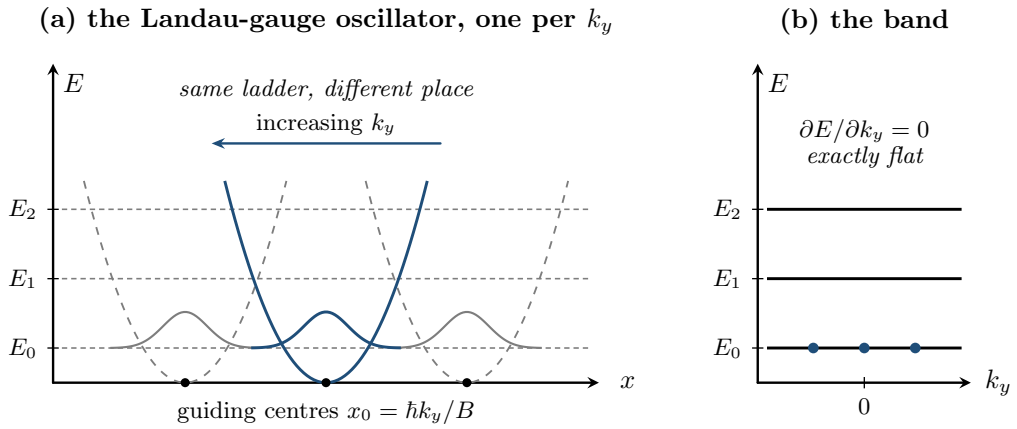
$$E_n = \hbar\omega_c\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots, \tag{4.6}$$

completely independent of k_y: every value of k_y (equivalently, every choice of guiding center x_0) gives the same energy, which is exactly why the Landau levels are flat bands, and why each one carries a macroscopic degeneracy proportional to the sample area. Figure 4.3 draws that mechanism directly: the same well parked in different places, and the degeneracy counted as one orbital per flux quantum through the sample. Plugging in real physical constants for a GaAs-scale field gives the millielectronvolt scale quoted in Section 4.1.

4.10 Landau levels of Dirac electrons

Now repeat the same exercise for a Dirac Hamiltonian, $H_0 = v_F(p_x\sigma_x + p_y\sigma_y)$, the low-energy effective description of graphene near the K point. With the same minimal coupling and Landau gauge, $\pi_x = p_x$, $\pi_y = p_y - Bx$, the Hamiltonian is now an *off-diagonal* 2×2 matrix,

$$H = v_F\begin{pmatrix} 0 & \pi_x - i\pi_y \\ \pi_x + i\pi_y & 0 \end{pmatrix}.$$



on a sample of size $W \times L_y$ the allowed k_y are spaced by $2\pi/L_y$,
so the guiding centres are spaced by $\Delta x_0 = 2\pi\hbar/BL_y$

number of orbitals per Landau level $= W/\Delta x_0 = BWL_y/2\pi\hbar = \Phi/\Phi_0$:
one state per flux quantum through the sample

Figure 4.3: Why the Landau levels are flat. (a) In the Landau gauge each k_y reduces (4.5) to a one-dimensional harmonic oscillator in x, identical in every respect except that it is centred on the guiding centre $x_0 = \hbar k_y/B$: raising k_y slides the whole well to the left without changing its curvature. Every well therefore carries the identical ladder $E_n = \hbar\omega_c(n + \frac{1}{2})$, drawn here as the horizontal guides, and the ground-state wavepacket of each one is the same shape parked in a different place. (b) Plotted against k_y, that is a set of exactly dispersionless bands. The flatness is not an artefact of the gauge: it is the statement that the energy knows nothing about where in the sample the electron sits, which is also where the enormous Landau-level degeneracy comes from.

H itself is not diagonal and cannot be solved by inspection, but H^2 is. Introduce the magnetic length $\ell_B = \sqrt{\hbar/B}$ and ladder operators $a = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x + i\pi_y)$, $a^\dagger = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x - i\pi_y)$; since $[\pi_x, \pi_y] = i\hbar^2/\ell_B^2$, these satisfy the ordinary bosonic commutation relation $[a, a^\dagger] = 1$. In terms of $a, a^\dagger$,

$$H = \frac{\sqrt{2}\hbar v_F}{\ell_B}\begin{pmatrix} 0 & a^\dagger \\ a & 0 \end{pmatrix} \quad \Longrightarrow \quad H^2 = \frac{2\hbar^2 v_F^2}{\ell_B^2}\begin{pmatrix} a^\dagger a & 0 \\ 0 & aa^\dagger \end{pmatrix} = \frac{2\hbar^2 v_F^2}{\ell_B^2}\begin{pmatrix} N & 0 \\ 0 & N+1 \end{pmatrix},$$

with $N = a^\dagger a$ the harmonic-oscillator number operator: exactly the "$p^2 + B^2x^2 + \text{const}$, which looks like a harmonic oscillator" statement from the transcript. Since H^2 is diagonal and harmonic-oscillator-like, its eigenvalues are simply $2\hbar^2 v_F^2 n/\ell_B^2$ for $n = 0, 1, 2, \ldots$, and the energy itself is the square root,

$$E_n = \operatorname{sgn}(n)\, v_F\sqrt{2\hbar B|n|}, \qquad n = 0, \pm 1, \pm 2, \ldots, \tag{4.7}$$

with a genuine, exactly zero-energy mode at $n = 0$ (the lower spinor component vanishes identically there, so there is no partner state to pair it with). This is the origin of graphene's $\sqrt{B}$ scaling: because the spectrum is the *square root* of a harmonic-oscillator spectrum rather than the spectrum itself, both the field-dependence and the level spacing change character, going from linear, equally-spaced levels for Schrödinger electrons to $\sqrt{n}$-spaced levels growing as $\sqrt{B}$ for Dirac electrons, which is exactly why graphene's Landau levels sit roughly two orders of magnitude higher in energy than a GaAs 2DEG's, at fixed field. Figure 4.4 puts the two ladders side by side, including the zero mode that has no counterpart at all in the Schrödinger case.

Notebook check-in: `session4.ipynb`, "The dispersion of the square lattice" (Reminder), "Bands in the conventional quantum Hall state" (Reminder), "Landau levels of conventional electrons" (Core), "Magnetic field dependence of Landau levels" (Core), and "Imaging the magnetic field dependence in the Landau levels" (Core), together with "The dispersion of the honeycomb lattice" (Reminder), "Bands in the Dirac quantum Hall state" (Reminder), "Landau levels of the Dirac equation" (Core), "Magnetic field dependence of Dirac Landau levels" (Core), and "Imaging the magnetic field dependence in the Dirac Landau levels" (Core)

Build a square-lattice ribbon (parabolic band bottom) and a honeycomb-lattice armchair ribbon (Dirac cone) with the Peierls substitution (4.4) switched on, and confirm both spectra directly: equally-spaced flat Landau levels growing linearly in B for the square lattice, reproducing (4.6), against unequally-spaced levels growing as $\sqrt{B}$ for the honeycomb lattice, reproducing (4.7). The "imaging" sections let you track individual level energies as an explicit function of field strength, which is the most direct numerical check of the linear-versus-square-root scaling derived above.

4.11 Numerical confirmation: Landau levels and edge states on the lattice

It is worth being explicit about how the continuum results of Sections 4.9 and 4.10 connect to tight-binding lattices, since this is where the edge states of Section 4.5 first become visible alongside the bulk Landau levels in a single calculation. Take a finite-width ribbon of a square lattice (whose band bottom is parabolic, hence a stand-in for the Schrödinger 2DEG), add a uniform orbital field through the Peierls substitution (4.4), and set the chemical potential near the bottom of the band, so the lattice model is effectively describing a dilute, nearly-free electron gas. The resulting band structure shows exactly what the continuum calculation predicted: a set of flat bulk bands (the Landau levels) sitting alongside dispersive edge bands,

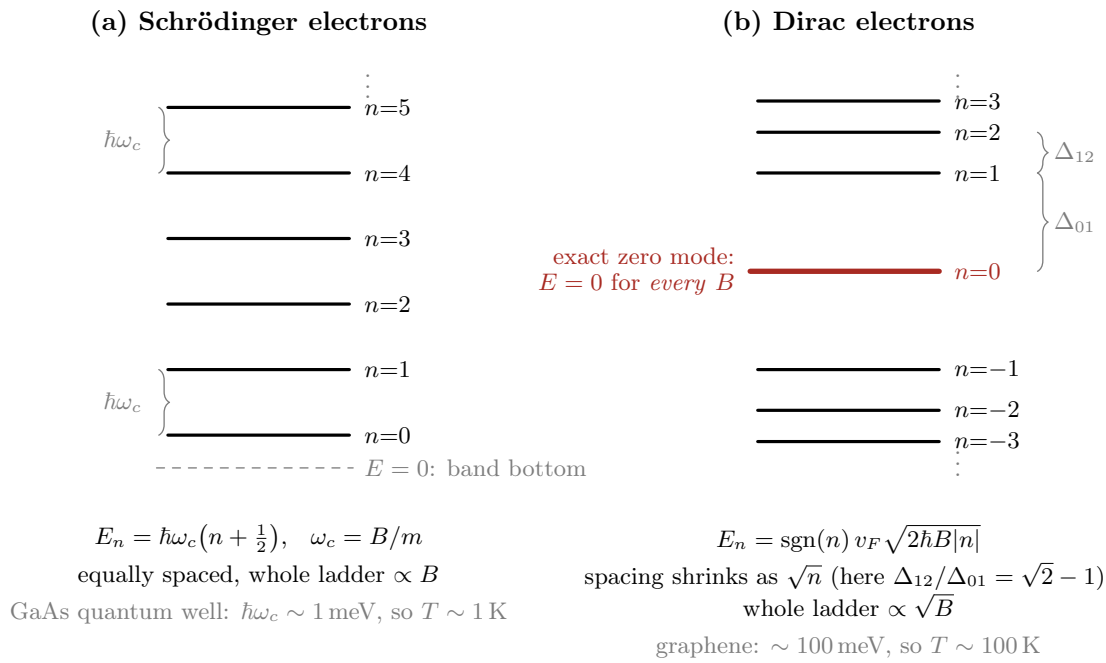


Figure 4.4: The same magnetic field, two dispersions, two completely different ladders. (a) Schrödinger electrons give the harmonic-oscillator spectrum (4.6): every level sits a constant $\hbar\omega_c = \hbar B/m$ above the one below it, and the whole ladder stretches linearly in B. (b) Dirac electrons give the square root of that spectrum, equation (4.7): the levels crowd together as $\sqrt{n}$, they come in a particle-hole symmetric pair for every $|n| \geq 1$, and the overall scale grows only as $\sqrt{B}$ – which is nevertheless enough to put graphene's level spacing about two orders of magnitude above a GaAs quantum well's at the same field. The $n = 0$ level is the qualitative novelty: it is pinned to exactly zero energy for every field strength, because it is the one solution with no partner in the other spinor component. The two vertical scales are independent of each other; it is the pattern of spacings that is being compared, not the absolute energies.

one branch localized at each edge of the ribbon and propagating in opposite directions, precisely the chiral pair described in Section 4.5. Repeating the same construction on a honeycomb-lattice armchair ribbon, whose low-energy physics is a genuine Dirac equation at the K and K' points, reproduces the $\sqrt{B}$ Landau levels of Section 4.10 together with their own chiral edge states. Both ribbon band structures (width-30 examples in the slides) are genuinely slide-only figures in the sense that the transcript only describes them qualitatively; the plots themselves carry the visual evidence, and building the equivalent ribbons yourself is exactly what the notebook callout above asks you to do.

4.12 Ladder operators and the many-body Landau Hamiltonian

Sections 4.9 and 4.10 both found harmonic-oscillator structure by working in a specific gauge. We can make that structure manifest in a gauge-independent way by defining ladder operators directly from the canonical momentum,

$$a = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x + i\pi_y), \qquad a^\dagger = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x - i\pi_y),$$

which, using only $[\pi_x, \pi_y] = i\hbar^2/\ell_B^2$ (a statement purely about the field, not about any particular gauge), satisfy the canonical bosonic commutation relation $[a, a^\dagger] = 1$. In terms of these operators the *full* quantum Hall Hamiltonian for a Schrödinger electron becomes

$$H = \hbar\omega_c \left(a^\dagger a + \frac{1}{2} \right), \tag{4.8}$$

manifestly a harmonic oscillator, with eigenvalues reproducing (4.6) directly and with no reference to a gauge at all. This is the cleanest way to see that "Landau levels behave like a harmonic oscillator" is not an artifact of the Landau gauge; it is a structural property of any electron in a uniform field.

4.13 The symmetric gauge and the lowest-Landau-level wavefunctions

To go from single-particle Landau levels to a genuinely many-body quantum Hall wavefunction, we switch to the symmetric gauge, $\mathbf{A} = \frac{B}{2}(-y, x, 0)$, and to the complex coordinate $z = x + iy$. Writing $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$, the ladder operator of Section 4.12 becomes, in this gauge, proportional to $\partial_{\bar{z}} + z/4\ell_B^2$. The lowest-Landau-level (LLL) condition, $a\psi = 0$, is therefore the first-order differential equation

$$\left(\partial_{\bar{z}} + \frac{z}{4\ell_B^2} \right) \psi(z, \bar{z}) = 0 \quad \Longrightarrow \quad \psi(z, \bar{z}) = f(z)\, e^{-|z|^2/4\ell_B^2}, \tag{4.9}$$

for an *arbitrary holomorphic function* $f(z)$, that is, any function built only from powers of z and never $\bar{z}$. This is the precise statement behind "any power z^m times a Gaussian is a zeroth-Landau-level wavefunction": holomorphicity is not an extra assumption, it is exactly what solving the LLL condition in the symmetric gauge forces on us. The single-particle LLL orbitals are then

$$\psi_m(z) = z^m\, e^{-|z|^2/4\ell_B^2}, \qquad m = 0, 1, 2, \ldots, \tag{4.10}$$

each labeled by an angular momentum m (the eigenvalue of L_z in this gauge), all sharing the same Gaussian envelope but with the polynomial prefactor z^m pushing the orbital's probability

density further from the origin as m grows, since $|\psi_m|^2$ peaks where $\frac{d}{d|z|}\left(|z|^{2m}e^{-|z|^2/2\ell_B^2}\right) = 0$, at radius $r_m = \sqrt{2m}\,\ell_B$.

4.14 The many-body wavefunction of the filled lowest Landau level

Now we ask for the many-body ground state in which *every* LLL orbital is occupied, for N electrons. Two requirements pin this state down completely: it must be built entirely from LLL orbitals (4.10), and it must be antisymmetric under exchange of any two electrons, as fermion statistics demands. Both requirements are satisfied at once by

$$\Psi_1(z_1, \ldots, z_N) = \prod_{i<j}(z_i - z_j)\ \exp\left(-\sum_i \frac{|z_i|^2}{4\ell_B^2}\right). \tag{4.11}$$

The Vandermonde product $\prod_{i<j}(z_i - z_j)$ does two jobs simultaneously. First, it is automatically antisymmetric: exchanging any two coordinates $z_i \leftrightarrow z_j$ flips the sign of exactly the one factor $(z_i - z_j)$, giving the overall minus sign Pauli's principle demands. Second, it is precisely the Slater determinant of the single-particle orbitals $1, z, z^2, \ldots, z^{N-1}$, since $\det\left(z_i^{j-1}\right) = \prod_{i<j}(z_j - z_i)$ is the standard Vandermonde determinant identity, so (4.11) is nothing but the Slater determinant of all N lowest LLL orbitals, times their common Gaussian envelope (the individual Gaussians simply combine into the single collective exponential shown). This is the exact many-body ground state of N electrons filling the lowest Landau level completely: the integer quantum Hall state at filling $\nu = 1$.

4.15 Angular momentum, enclosed flux, and the uniform electron gas

Section 4.13 already told us that larger angular momentum m pushes an orbital's peak further from the origin, to radius $r_m = \sqrt{2m}\,\ell_B$; equivalently, that orbital encloses a larger area, and therefore a larger amount of magnetic flux. This link between angular momentum and enclosed flux is what we will use in Section 4.17 to track how much charge a quasiparticle carries. For now, the important consequence is about (4.11) itself: summing over the full Slater determinant, one finds a density profile that is essentially uniform (flat) out to a radius set by the highest occupied angular momentum $m = N - 1$, and falls sharply to zero beyond it. The filled lowest Landau level, in other words, is a genuinely featureless, uniform two-dimensional electron gas, which is exactly the property we will now ask a *fractionally* filled Landau level to share. Figure 4.5 shows both halves of that statement: the individual orbitals as nested rings of growing radius, and the flat droplet they add up to.

4.16 Fractional filling and the Laughlin wavefunction

Suppose instead we want to occupy only a fraction of the available LLL orbitals, say two out of every three, but in a way that keeps the electron gas uniform rather than simply leaving isolated orbitals empty (which would create an obviously non-uniform, "lumpy" density, since each empty orbital sits at its own fixed radius). What we are looking for is a many-body wavefunction that: still lives entirely in the LLL, so it is built only from the coordinates z_i and never $\bar{z}_i$; is still antisymmetric, since the particles are still electrons; and, heuristically, keeps the electrons as far apart from one another as possible, since that is what minimizing a repulsive Coulomb interaction energy would favor.

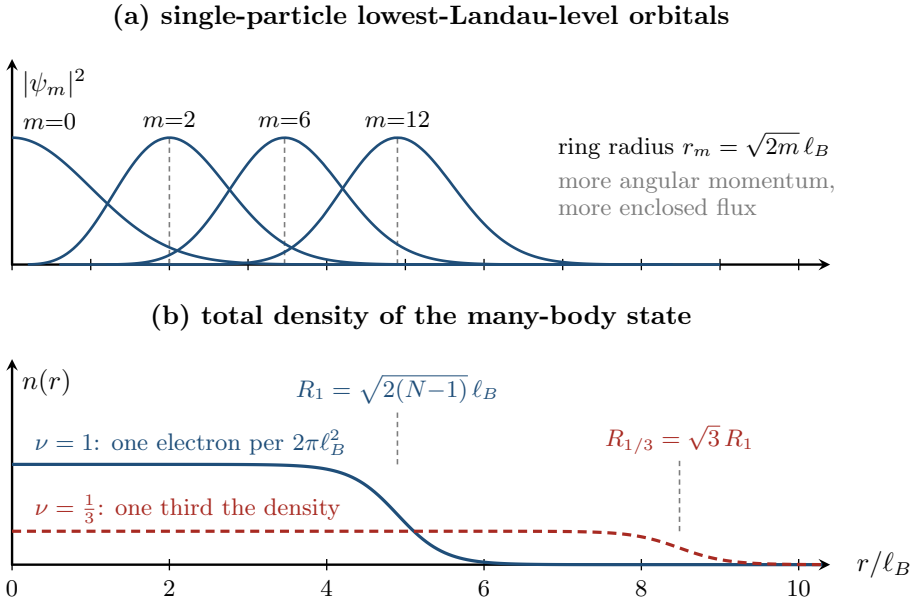


Figure 4.5: Angular momentum is radius. (a) The lowest-Landau-level orbitals (4.10), $|\psi_m|^2 \propto r^{2m}e^{-r^2/2\ell_B^2}$, each renormalized here to the same peak height: they are nested rings, and raising m moves the ring outward, to $r_m = \sqrt{2m}\,\ell_B$, enclosing more flux. (b) Stacking every orbital from $m = 0$ to $m = N - 1$ therefore fills a disc, and the rings overlap in exactly such a way that the total density is flat inside it and falls off sharply at the edge: the filled lowest Landau level is a featureless, uniform electron droplet, one electron per flux quantum. The Laughlin state (4.12) at $\nu = 1/3$ is the second profile: the same N electrons now use orbitals up to $m = 3(N-1)$, so the droplet swells by $\sqrt{3}$ in radius and thins to a third of the density, while staying just as uniform. That uniformity is the property the Laughlin construction is built to preserve; the edge profiles are schematic.

The construction that achieves all three requirements at once is remarkably simple: take the filled-LLL wavefunction (4.11) and raise the Vandermonde factor to a power,

$$\Psi_m(z_1, \ldots, z_N) = \prod_{i<j} (z_i - z_j)^m \; \exp\left(- \sum_i \frac{|z_i|^2}{4\ell_B^2} \right), \tag{4.12}$$

the **Laughlin wavefunction**. The exponent m must be an odd integer: an even power would make the wavefunction *symmetric* rather than antisymmetric under exchange, which fermions cannot allow. Raising the exponent directly increases the "avoidance" between electrons (the wavefunction now vanishes to m-th order, not just linearly, whenever two electrons coincide), which is exactly the correlation effect we were after. Working out the resulting electron density gives filling factor

$$\nu = \frac{1}{m}, \tag{4.13}$$

which one can see directly by counting: the maximum power of any single coordinate appearing in the fully expanded polynomial (4.12) is $m(N-1)$, so the wavefunction uses angular-momentum orbitals up to $m(N-1)$, meaning N electrons are spread over $\approx m(N-1)+1 \approx mN$ available LLL orbitals, a filling fraction of $N/(mN) = 1/m$, matching (4.13). It is worth pausing on how remarkable this construction is: starting from nothing but the single-particle LLL orbitals, Pauli's exclusion principle, and a heuristic correlation argument, we land on the exact ground state of a strongly interacting many-body problem, and it is going to hand us fractional charge in the very next section.

4.17 Quasiholes and quasielectrons

From this point through the end of Section 4.18, the material has no counterpart in `session4.ipynb`*. The notebook's coverage of this session stops at the integer quantum Hall state, its Landau levels, and its edge and Chern-insulator physics (Sections 4.10, 4.11, and 4.19 below); the Laughlin wavefunction, quasiholes and quasielectrons, and the Berry-phase argument for fractional charge and anyonic statistics are transcript-and-slide material with no numerical exercise built around them.*

Excitations above the Laughlin ground state (4.12) turn out to have an equally simple construction in these coordinates. A **quasihole** at the origin is obtained by multiplying the wavefunction by an extra overall factor of z_i for every electron; a quasihole at an arbitrary location w is obtained instead by inserting a factor $(z_i - w)$ for each electron,

$$\Psi_{\mathrm{qh}}(w) = \prod_i (z_i - w)\, \Psi_m(z_1, \ldots, z_N). \tag{4.14}$$

Multiplying by this extra factor raises the effective power associated with every orbital by one, which, by the angular-momentum-versus-radius relation of Section 4.15, pushes charge density outward, away from w, leaving a depleted region there: a quasihole. Repeating this construction m times, at m different locations, removes exactly one full electron's worth of charge overall, consistent with each individual quasihole carrying only a fraction $1/m$ of an electron charge (we will confirm this fraction directly via the Berry phase in Section 4.18). A **quasielectron** is, in this same language, the inverse operation: instead of raising the power (multiplying by $z_i - w$), we lower it, implemented by a derivative with respect to $z_i - w$ rather than a multiplication; this pulls charge inward instead of pushing it outward, giving an excitation of opposite (negative) fractional charge.

4.18 Berry phase, fractional charge, and anyonic statistics

We now use the Berry phase, the same object session 3 built the Chern number from, to show directly that the quasihole excitation of Section 4.17 really does carry a fraction of an electron charge, and that a pair of them obey neither fermionic nor bosonic exchange statistics.

Fractional charge. Take the single-quasihole wavefunction (4.14) and adiabatically move w around a closed loop, parameterizing $w = |w|e^{i\theta}$ and computing the resulting Berry phase as θ runs from 0 to 2π. The result comes out as an Aharonov-Bohm phase, proportional to the flux enclosed by the loop, but carrying an extra prefactor of $1/m$ relative to what an ordinary charge-e particle would pick up around the same loop. Since the Aharonov-Bohm phase for a particle of charge q around enclosed flux Φ is $2\pi q\Phi/\Phi_0$ (with Φ_0 the flux quantum), the extra $1/m$ we find is a direct, unambiguous statement about the *charge* of the excitation:

$$q_{\text{qh}} = \frac{e}{m}. \tag{4.15}$$

The electron is no longer the fundamental unit of charge for these excitations; a genuine fraction of it is.

Anyonic statistics. Now take a wavefunction with *two* quasiholes, one fixed at the origin and one at w, and move the second one in a loop around the first, again parameterizing $w = |w|e^{i\theta}$. The resulting Berry phase splits into two pieces: the same Aharonov-Bohm contribution as before (charge times enclosed flux), plus an additional **statistical phase** picked up purely from winding one quasiparticle around the other,

$$\theta_{\text{stat}} = \frac{2\pi}{m}. \tag{4.16}$$

One bookkeeping point matters when comparing this to the familiar cases: taking one quasiparticle all the way around another is *two* successive exchanges, so the exchange phase is half of (4.16), namely π/m. For ordinary fermions, exchanging two particles gives a phase of π; for bosons, a phase of 0. The exchange phase π/m is neither, and it interpolates between the two familiar cases as m grows (see figure 4.6): these quasiparticles are **anyons**, excitations whose exchange statistics interpolate continuously between the fermionic and bosonic cases, labeled by the same integer m that set the filling fraction. We started from a many-body wavefunction built entirely out of ordinary electrons, genuine fermions, and out of that wavefunction's own correlations we have produced excitations that are neither fermions nor bosons: this is the simplest example of fractionalization in a quantum many-body system that we will meet in this course.

4.19 The quantum Hall effect without Landau levels: the Haldane model, revisited

This closing section is a deliberate callback, not new material: session 3 already built the Haldane model, as its own worked example of a Chern insulator obtained by gapping a Dirac cone without any external field. We revisit it here specifically because it closes the loop this session opened: everything above required an actual magnetic field to produce Landau levels, so it is worth confirming explicitly that the quantized Hall conductance itself never actually required one.

As a quick reminder of the construction: start from the honeycomb lattice with first-neighbor hopping only (an ungapped Dirac cone at K and K'), and add a second-neighbor hopping that is purely imaginary, which breaks time-reversal symmetry. In the Bloch Hamiltonian this term appears as a valley-dependent mass, with opposite sign at K and K' for an ordinary staggered on-site potential, but with the *same* sign at both valleys for the Haldane

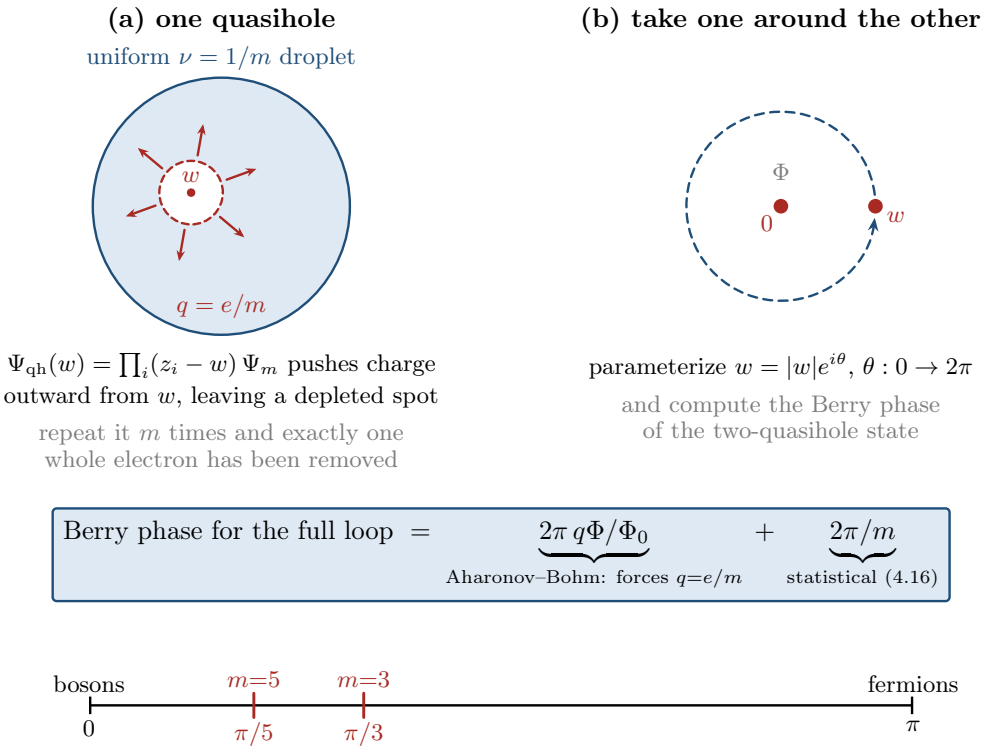


Figure 4.6: Fractional charge and fractional statistics, both read off a Berry phase. (a) Multiplying the Laughlin state by $\prod_i (z_i - w)$, equation (4.14), raises every electron's angular momentum about w by one, which by figure 4.5 pushes charge radially outward and leaves a depleted spot behind. Doing it m times removes exactly one electron's worth of charge, so one quasihole is short of an electron by a factor m: it carries $q = e/m$. (b) Carrying one quasihole once around another and tracking the Berry phase separates that statement into two pieces: an Aharonov-Bohm piece, proportional to the enclosed flux, whose prefactor is what fixes $q = e/m$ in (4.15), and a leftover piece $2\pi/m$ that depends on nothing but the fact that one quasiparticle went around the other. A full loop is two exchanges, so the exchange phase itself is π/m, drawn at the bottom: for every odd $m > 1$ it lands strictly between the bosonic 0 and the fermionic π, which is what makes these excitations anyons.

(imaginary second-neighbor) term. Equal-sign masses at both valleys give equal-sign Berry curvature at both valleys too, so the two valley contributions add rather than cancel when integrated over the Brillouin zone, leaving a net, nonzero Chern number, an insulating bulk gap, and a chiral edge state, all with $B = 0$ throughout.

What is genuinely new in today's version of this story is the explicit **adiabatic connection** between the Haldane model and the ordinary field-driven quantum Hall state of this session. Start in the Haldane model (imaginary second-neighbor hopping, no field); switch on a real magnetic field while keeping the Haldane coupling, and Landau levels develop out of the previously dispersive bands; then switch the Haldane coupling back off, leaving the system in the ordinary quantum Hall state built from Landau levels alone. At every step of this path the bulk gap stays open and the chiral edge state never disappears. Since the two end points can be connected without ever closing the gap, the Haldane-model Chern insulator and the Landau-level quantum Hall state are *topologically equivalent*: the very same phase of matter, reached by two completely different physical routes, one that needs a magnetic field and one that does not.

Notebook check-in: `session4.ipynb`, "Topological equivalence between the Quantum Hall state and a Chern insulator" (Challenge), and "Topological domains in the quantum Hall state" (Challenge)

Build the Haldane model, turn a magnetic field on and off numerically alongside the Haldane coupling, and track the bulk gap and Chern number through the adiabatic path described above. The "topological domains" exercise lets you probe spatial regions where the two couplings compete directly, a hands-on version of the equivalence argument just given.

4.20 Real materials realizing Chern and fractional Chern physics

This survey, like Section 4.17-4.18, is transcript-and-slide content with no notebook counterpart; there is no numerical exercise in `session4.ipynb` built around any of the specific compounds below.

The zero-field Chern insulator is not just a theoretical curiosity. Chromium-doped $(Bi,Sb)_2Te_3$, a magnetically doped topological insulator, breaks time-reversal symmetry through genuine magnetism and realizes an integer quantized Hall conductance with no applied field. Twisted graphene bilayers aligned with a boron nitride substrate realize the same physics through a different mechanism, orbital currents rather than magnetic moments. On the fractional side, twisted two-dimensional materials can host topological flat bands whose mathematical structure mimics an artificial, pseudo-magnetic field tied to elastic lattice reconstruction rather than any real field; twisted bilayers of molybdenum ditelluride ($MoTe_2$) are a recent, striking example, showing genuine fractional quantum Hall physics at zero magnetic field.

4.21 Closing summary

- The Hall conductivity is a topological invariant, the Chern number of the occupied bands, exactly as derived (more fully) in session 3; equation (4.1) is the same result confirmed once more here rather than re-derived from scratch.
- A magnetic field turns a two-dimensional electron gas into a set of flat Landau levels, each carrying Chern number one, so the Hall conductivity is quantized to Ne^2/h for N filled Landau levels, equation (4.2).
- An insulating bulk still conducts because a nonzero Chern number forces chiral, backscattering-free edge states at the sample boundary (Section 4.5), directly visible in ribbon band

structures.

- Landau levels arise from minimal coupling (4.3) or, on a lattice, the Peierls substitution (4.4); Schrödinger electrons give equally spaced levels $E_n = \hbar\omega_c(n + \frac{1}{2})$, equation (4.6), while Dirac electrons give $E_n \propto \sqrt{nB}$, equation (4.7), explaining graphene's much higher quantum Hall temperature scale.

- In the symmetric gauge, lowest-Landau-level wavefunctions are holomorphic, $\psi_m(z) \propto z^m e^{-|z|^2/4\ell_B^2}$, equation (4.10); the filled-LLL many-body state is the Vandermonde determinant times a collective Gaussian, equation (4.11).

- Raising the Vandermonde factor to an odd power m gives the Laughlin wavefunction, equation (4.12), with filling factor $\nu = 1/m$, equation (4.13); its quasihole excitations, equation (4.14), carry fractional charge e/m, equation (4.15), and obey anyonic statistics, picking up $2\pi/m$ on a full braid, equation (4.16), that is an exchange phase π/m strictly between the bosonic 0 and the fermionic π: the simplest instance of fractionalization we will meet.

- The Haldane model shows the quantized Hall conductance never strictly required a magnetic field at all, and is adiabatically, topologically equivalent to the field-driven quantum Hall state, a callback to session 3 rather than new machinery.

Today's order was driven entirely by a magnetic field, an external, symmetry-breaking knob we simply turned on by hand. Next session we meet a completely different route to order: attractive interactions between electrons themselves, driving the system to superconduct without any external field at all, and we will see that the very same trivial-versus-chiral topological classification built here and in session 3 comes right back to sort superconducting order parameters into conventional and topological classes. It is also worth holding on to the flavor of fractionalization we just built, since we will meet a second, quite different-looking instance of it very soon: here, correlations inside a partially filled topological flat band split the electron's charge and turned its exchange statistics anyonic; next time, pairing correlations in a superconductor will split the electron into Majorana halves instead. Same underlying idea, a correlated topological state hosting excitations that are not the electron itself, reached by two genuinely different mechanisms.

Session 5

Superconductivity, Nambu Representation and Majorana Physics

Session 4 closed by holding on to the flavor of fractionalization we had just built there, and by pointing forward to a second, quite different-looking instance of it we were about to meet: correlations inside a partially filled topological flat band split the electron's charge and turned its exchange statistics anyonic; here, we were told, pairing correlations in a superconductor would split the electron into Majorana halves instead. Today is where that promissory note gets cashed in. Superconductors and fractional quantum Hall states are, as we will restate carefully below, two correlated states that both host fractional excitations via opposite mechanisms: one has a mean-field description and needs an attractive interaction, the other has none and needs a strong repulsive one. Same underlying idea, a correlated state hosting excitations that are not the electron itself, reached by two genuinely different routes; today we build the machinery for the superconducting route in full.

That machinery has one further piece of unfinished business behind it. Back in session 1 we built mean-field theory as a general tool and then ran it through three symmetry-broken states in quick succession: a charge density wave, a magnet, and, as our third and briefest worked example, a superconductor. There, starting from an attractive Hubbard interaction, we let ourselves take an expectation value between two annihilation operators instead of one creation and one annihilation operator, defined the pairing order $\Delta = \langle c_i c_j \rangle$ (session 1, eq. 1.9), and showed that this term picks up a phase $e^{2i\varphi}$ under a gauge transformation while every ordinary bilinear term stays invariant: a superconductor spontaneously breaks gauge symmetry, and with it, a fixed electron number. That was deliberately a first pass. We said at the time that magnetism and superconductivity would each get "a full, much deeper treatment in their own dedicated sessions later in the course," and today is superconductivity's turn (magnetism is deferred once more, to the next recording, which in this course's arc is session 6).

So today we pick that thread back up and build the actual machinery around it. We will need a new representation, the Nambu spinor, to even diagonalize a Hamiltonian that mixes creation and annihilation operators the way a pairing term does; that representation hands us the Bogoliubov-de Gennes Hamiltonian, the central object of the whole session. We will use it to classify superconducting order by symmetry (singlet versus triplet, s-, p- and d-wave), to show that two superconductors can look spectrally identical and yet carry completely different topology, reusing exactly the Chern-number machinery from the topological-band-structure and quantum-Hall sessions to tell them apart. And we will follow that topological distinction all the way down to its most dramatic consequence: the Kitaev chain, a model we will recognize as nothing more than the SSH model from the topological-band-structure session, rewritten in the language of Majorana operators, whose topological zero modes turn out to be their own antiparticles. We will close by asking what these Majorana modes are good for (topological

quantum computing), and by handing off to session 6, which returns to do for magnetism exactly what we are about to do for superconductivity, the course's one genuinely spoken cliffhanger: the same interacting Hamiltonian can mean-field into either a superconductor or a magnet, and magnetism is what got deferred to the next recording.

5.1 Weakly versus strongly correlated matter, revisited

Let us restate the general setting, because it is worth having sharply in view before we specialize. Every interacting Hamiltonian we care about in this course has the same generic shape,

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j \;+\; \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j^\dagger c_k c_l,$$

a free bilinear part plus a quartic interaction (this is exactly session 1's eq. 1.5). When mean-field theory can turn the quartic term into a good effective bilinear Hamiltonian, we call the resulting state *weakly correlated matter*: this covers magnets, charge density waves, and superconductors, all of the states session 1 built. When no such single-particle mean-field Hamiltonian captures the physics, we are dealing with *strongly correlated matter*, and we need exact or numerical solutions instead; the fractional quantum Hall state from the quantum-Hall session is our standing example of that harder case.

It is worth noticing that superconductors and the fractional quantum Hall state actually sit at opposite corners of this classification, even though both eventually host "fractional" excitations. A superconductor has a mean-field description, arises from an *attractive* interaction, is destroyed by a magnetic field, and (as we are about to see) hosts Majorana states. The fractional quantum Hall state has no mean-field description, arises from strong *repulsive* interactions, typically requires a strong magnetic field, and hosts fractionally charged excitations. Majorana excitations and fractional-quantum-Hall excitations are both, loosely speaking, "a fraction of an electron," but they are fractions produced by entirely different mechanisms; it is one of the nicer illustrations in this course of how very different microscopic physics can converge on a shared qualitative phenomenon, emergent fractionalization, without the underlying mechanism being at all the same.

5.2 Superconductivity as a macroscopic quantum phenomenon

Before diving into the Hamiltonian, it is worth pausing on why superconductivity gets singled out as one of the cleanest windows we have onto quantum mechanics at macroscopic scale. A superconductor is built from trillions of atoms, and yet a measurement on the bulk sample can read off, essentially exactly, the microscopic constants $\hbar$ and e. This works because gauge symmetry breaking gives the many-body wave function a single, well-defined phase $\theta(\mathbf{r})$, and single-valuedness of that phase around a closed loop is a very strong constraint.

Consider a superconducting ring (or, equivalently, a hole in a bulk superconductor) threaded by magnetic flux, and take the loop deep enough inside the superconductor that the supercurrent density vanishes there (Meissner screening). The supercurrent of Cooper pairs (charge $2e$, since a pair is made of two electrons) is proportional to the gauge-invariant combination $\hbar\nabla\theta - 2e\mathbf{A}$; setting the current to zero on our loop forces

$$\oint \nabla\theta \cdot d\boldsymbol{\ell} = \frac{2e}{\hbar} \oint \mathbf{A} \cdot d\boldsymbol{\ell} = \frac{2e}{\hbar}\Phi.$$

But θ is the phase of a single-valued wave function, so going once around the loop it can only change by an integer multiple of 2π: $\oint \nabla\theta \cdot d\boldsymbol{\ell} = 2\pi n$ for integer n. Combining the two

gives flux quantization,

$$\Phi = n\,\Phi_0, \qquad \Phi_0 = \frac{h}{2e}. \tag{5.1}$$

The geometry and the phase-winding argument are collected in figure 5.1. This is exactly the mechanism behind the precision of Josephson-junction critical-current measurements: it is a direct, macroscopic readout of h/e, made possible entirely by gauge symmetry breaking. It is worth noticing the parallel with the quantum Hall state from earlier in the course: there too, a macroscopic measurement (the Hall conductance, quantized in units of e^2/h by the Chern number) reads off a ratio of the same two constants, but through a completely different mechanism, the quantization of a single-particle topological invariant rather than the breaking of a continuous symmetry. Combining the two measurements, the flux quantum and the Hall conductance quantum, is in fact one of the most precise ways we have of extracting h and e separately; it is a nice, concrete example of the emergence this course keeps returning to; macroscopic measurements on billions of atoms handing you exact microscopic constants.

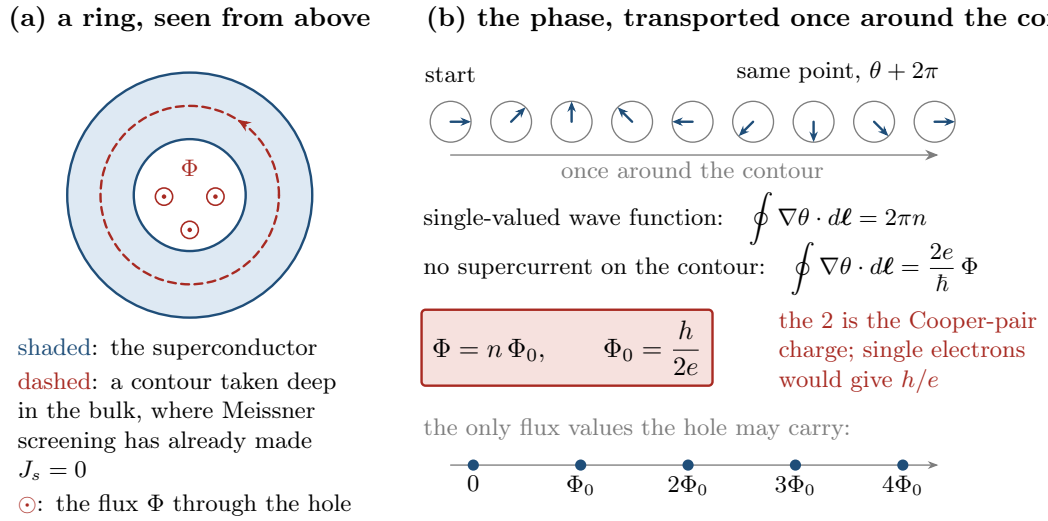


Figure 5.1: Flux quantization, drawn. Left: a superconducting ring; deep in the bulk the Meissner effect has already killed the supercurrent, so on the dashed contour the gauge-invariant combination $\hbar\nabla\theta - 2e\mathbf{A}$ vanishes and the phase gradient is locked to the vector potential. Right: transporting the condensate phase θ once around that contour must return it to itself, so it can only wind by a whole number of turns. The two statements together give $\Phi = n\,h/2e$: the flux through the hole is quantized, and the 2 in the denominator is nothing but the charge $2e$ of a Cooper pair. Nothing here is a property of one electron; it is a macroscopic sample handing back a ratio of microscopic constants, purely because gauge symmetry is broken.

5.3 What a superconductor looks like: resistance and the Meissner effect

Historically, superconductivity was discovered through its transport signature: as you cool an ordinary metal, resistivity falls smoothly because there are fewer phonons for electrons to scatter off. A superconductor instead shows a *sudden* drop of the resistance to exactly zero at a critical temperature T_c; below T_c the resistance stays at zero, not just small. T_c is precisely the temperature at which gauge symmetry breaking sets in. This is slides-only material with no notebook counterpart (the resistance-versus-temperature schematic and the field-expulsion figure are shown and described on the slides but there is no accompanying notebook cell): our companion notebook starts from the Hamiltonian side, not from transport measurements.

The second hallmark is the **Meissner effect**: above T_c a metal tolerates an external magnetic field just fine, but below T_c the superconductor actively expels the field from its bulk. In type-II superconductors, the only field that survives inside the material is confined to quantized vortices, themselves dynamical topological defects of the broken-gauge-symmetry order parameter. Both signatures, zero resistance and flux expulsion, are direct consequences of the same underlying event: the condensation of Cooper pairs into a single macroscopic quantum state with a rigid, spontaneously chosen phase.

5.4 The interacting Hamiltonian and the origin of attractive interactions

We start, as in session 1, from the generic interacting Hamiltonian $H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l$. The interaction V_{ijkl} can come from plain electrostatic (repulsive) Coulomb repulsion, but it can equally well be an *effective* interaction mediated by some other quasiparticle in the material: phonons (lattice vibrations), magnons (fluctuations of a magnetic moment), plasmons (collective charge fluctuations), or even more exotic mediators like skyrmion fluctuations or valence fluctuations. What matters for superconductivity is not the microscopic origin of V_{ijkl} but its net sign in whichever channel turns out to matter: superconductivity requires that channel to be effectively *attractive*.

This is the natural place to draw the distinction between the two families of superconductors:

- **Conventional superconductors**, where the attractive channel is mediated by phonons. This is the best-understood case: mean-field (BCS) theory works quantitatively well, and critical temperatures of simple conventional superconductors like aluminum or lead can actually be predicted from theory.
- **Unconventional superconductors**, mediated by anything that is not a phonon: antiferromagnetic or ferromagnetic magnons, plasmons, valence fluctuations, charge fluctuations, and so on. Here we have no comparably reliable predictive theory; identifying even *which* quasiparticle mediates the attraction in a given unconventional superconductor is often still an open research question, and it remains one of the major unsolved problems the field is currently working on.

5.5 The mean-field pairing Hamiltonian, done properly

Let us now build the pairing mean-field Hamiltonian with the same level of care session 1 used for its charge-density-wave and magnetic examples, rather than just gesturing at $\Delta = \langle cc \rangle$. Take the simplest interacting Hamiltonian that can produce superconductivity, a one-dimensional tight-binding chain with an attractive on-site interaction between opposite spins,

$$H = -t \sum_{n,\sigma} \left(c_{n\sigma}^\dagger c_{n+1,\sigma} + \text{h.c.} \right) - \mu \sum_{n\sigma} c_{n\sigma}^\dagger c_{n\sigma} + U \sum_n c_{n\uparrow}^\dagger c_{n\downarrow}^\dagger c_{n\downarrow} c_{n\uparrow}, \qquad U < 0. \tag{5.2}$$

This is exactly session 1's eq. 1.7, now with $U = -|U| < 0$: the attractive branch of that same Hamiltonian, whose repulsive branch instead drives magnetism (session 6's subject). We mean-field decouple the quartic term in the *pairing* (particle-particle) channel this time rather than the density channel: define the order parameter

$$\Delta_n \equiv U \langle c_{n\downarrow} c_{n\uparrow} \rangle, \tag{5.3}$$

and factorize $c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}c_{n\downarrow}c_{n\uparrow} \approx \langle c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}\rangle c_{n\downarrow}c_{n\uparrow} + c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}\langle c_{n\downarrow}c_{n\uparrow}\rangle - \langle c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}\rangle\langle c_{n\downarrow}c_{n\uparrow}\rangle$, the same Wick-type factorization used throughout session 1, just applied to a different pair of operators. The resulting mean-field Hamiltonian is

$$H_{\mathrm{MF}} = -t\sum_{n\sigma}\left(c^\dagger_{n\sigma}c_{n+1,\sigma} + \mathrm{h.c.}\right) - \mu\sum_{n\sigma}c^\dagger_{n\sigma}c_{n\sigma} + \sum_n\left(\Delta_n\, c^\dagger_{n\uparrow}c^\dagger_{n\downarrow} + \Delta_n^*\, c_{n\downarrow}c_{n\uparrow}\right) - \sum_n\frac{|\Delta_n|^2}{U}. \tag{5.4}$$

Self-consistently, Δ_n comes out exactly zero for $U > 0$ (repulsive) and can become nonzero for $U < 0$ (attractive): this is precisely the superconducting order parameter, and it is exactly the term that breaks gauge symmetry.

Notebook check-in: `session5.ipynb`, "s-wave superconductivity driven by attractive interactions" (Core), and "Dependence on the strength of the attractive interaction" (Challenge)

Build the attractive-Hubbard chain of (5.2), run the pairing self-consistency loop that produces Δ in (5.3)-(5.4), and watch Δ grow from zero as you make U more negative, exactly paralleling how session 1's charge imbalance and magnetization grew with V and $U > 0$ respectively.

5.6 Why pairing kills a fixed particle number

Now let us make the "fluctuating electron number" claim from session 1 fully rigorous, rather than just asserting it. Look at the eigenvalue equation $H_{\mathrm{MF}}|G\rangle = E|G\rangle$ for the Hamiltonian in (5.4), and expand the ground state in the occupation-number (Fock) basis, sorted by total particle number N: $|G\rangle = \sum_N a_N|N\rangle$. The hopping and chemical-potential terms in (5.4) conserve N exactly (one $c^\dagger$ and one c each). The pairing terms $\Delta_n c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}$ and $\Delta^*_n c_{n\downarrow}c_{n\uparrow}$ change N by $+2$ or -2 respectively; crucially, *no term in H_{MF} ever changes N by ± 1*. So H_{MF} only ever connects Fock states whose particle numbers differ by an even integer. For $|G\rangle$ to be an eigenstate, this forces a_N to be nonzero only within one fixed parity class of N (all-even or all-odd), but says nothing about which particular even (or odd) N's appear: generically $a_N \neq 0$ for several different values of N within that class, e.g. $N = 0, 2, 4, \ldots$ simultaneously. The mean-field ground state is therefore *not* an eigenstate of the total number operator $\hat{N}$, even though it does have a well-defined fermion parity. This is the precise sense in which "a superconductor has a fluctuating electron number, but a fixed parity," exactly the statement session 1 flagged back in its discussion of the quantum-dot superposition puzzle (session 1, section 1.5).

5.7 The gauge transformation, made explicit

We already wrote the second-quantized gauge transformation in session 1 (eq. 1.10): $c \to c\,e^{i\varphi}$, $c^\dagger \to c^\dagger e^{-i\varphi}$, the second-quantized shadow of the statement that the phase of a first-quantized wave function carries no physical meaning. Let us now check explicitly which terms in a realistic Hamiltonian survive this transformation and which do not. Ordinary hopping $c^\dagger_i c_j$ is invariant, since the two phases cancel: $c^\dagger_i c_j \to e^{-i\varphi}c^\dagger_i \cdot e^{i\varphi}c_j = c^\dagger_i c_j$. The same cancellation protects essentially every other bilinear term we have met so far: magnetic (spin-dependent) terms, spin-orbit coupling, and any density-density interaction term $c^\dagger_i c_i c^\dagger_j c_j$, since all of these are built from an equal number of creation and annihilation operators pairing up index by index.

The pairing term is the one exception. Under the same transformation,

$$\Delta\, c^\dagger_{n\uparrow} c^\dagger_{n\downarrow} \;\longrightarrow\; \Delta\, e^{-2i\varphi}\, c^\dagger_{n\uparrow} c^\dagger_{n\downarrow}, \tag{5.5}$$

it picks up the phase *twice*, because both operators are creation operators rather than one creation and one annihilation operator. There is no way to choose $\varphi \neq 0 \pmod{\pi}$ that leaves this term invariant, so a Hamiltonian containing a nonzero Δ is not gauge invariant term by term; equivalently, if we insist on writing the transformation as acting on the order parameter itself, $\Delta \to \Delta\, e^{2i\varphi}$, then any particular fixed phase of Δ is a choice the ground state makes spontaneously, exactly the hallmark of continuous symmetry breaking. This is the same logic, applied with full explicit rigor, that session 1 used to establish the gauge-symmetry-breaking of eq. 1.9 there; we have now seen precisely why the factor is 2φ and not φ.

5.8 The Nambu spinor and the Bogoliubov-de Gennes Hamiltonian

Here is the technical obstacle the pairing term creates: we cannot diagonalize (5.4) with an ordinary vector of orbital operators, because the anomalous term mixes creation and annihilation operators rather than pairing them up the way every Hamiltonian we diagonalized in sessions 1-4 did. The fix is to *enlarge* the operator basis so that creation and annihilation operators of the original problem both become "annihilation-like" entries of a bigger spinor. Working in momentum space, with a single orbital and both spins, define the **Nambu spinor**

$$\Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{\mathbf{k}\downarrow} \\ c^\dagger_{-\mathbf{k}\downarrow} \\ -c^\dagger_{-\mathbf{k}\uparrow} \end{pmatrix}, \tag{5.6}$$

an "electron sector" (the first two components, ordinary annihilation operators) glued to a "hole sector" (the last two components, creation operators at $-\mathbf{k}$, spins swapped, with an extra minus sign planted on the very last entry). That sign is not decorative: it is exactly what is needed to make the pairing term come out looking like two identical, decoupled copies of the same 2×2 problem once we write H_{MF} in terms of $\Psi_{\mathbf{k}}$, rather than mixing the two spin blocks together through a Pauli matrix. For a spin-singlet pairing $\Delta(\mathbf{k})$ (momentum-independent and equal to Δ for the on-site case of (5.4) Fourier-transformed), one checks directly, using the fermionic anticommutation relations to reorder operators where needed, that the mean-field Hamiltonian can be written exactly as

$$H_{\mathrm{MF}} = \frac{1}{2}\sum_{\mathbf{k}} \Psi^\dagger_{\mathbf{k}}\, H_{\mathrm{BdG}}(\mathbf{k})\, \Psi_{\mathbf{k}} + \text{const.}, \qquad H_{\mathrm{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) & 0 & \Delta(\mathbf{k}) & 0 \\ 0 & \varepsilon(\mathbf{k}) & 0 & \Delta(\mathbf{k}) \\ \Delta(\mathbf{k})^* & 0 & -\varepsilon(\mathbf{k}) & 0 \\ 0 & \Delta(\mathbf{k})^* & 0 & -\varepsilon(\mathbf{k}) \end{pmatrix}, \tag{5.7}$$

where $\varepsilon(\mathbf{k}) = -2t(\cos k_x + \cos k_y) - \mu$ is the ordinary band dispersion (assumed inversion symmetric, $\varepsilon(-\mathbf{k}) = \varepsilon(\mathbf{k})$). This 4×4 matrix is the **Bogoliubov-de Gennes (BdG) Hamiltonian**: it is, in a very concrete sense, the single most important object for describing excitations of a superconductor. Note that the two diagonal entries in the lower-right carry $-\varepsilon(\mathbf{k})$, not $+\varepsilon(\mathbf{k})$: this is a direct consequence of rewriting $c^\dagger_{-\mathbf{k}\sigma} c_{-\mathbf{k}\sigma}$ in terms of $c_{-\mathbf{k}\sigma} c^\dagger_{-\mathbf{k}\sigma} = 1 - c^\dagger_{-\mathbf{k}\sigma} c_{-\mathbf{k}\sigma}$ via the anticommutation relations (this is exactly where the "hole" language comes from: a missing electron at $-\mathbf{k}$ behaves like a particle of opposite energy).

Because the spinor (5.6) was built with exactly that extra minus sign, the 4×4 problem in (5.7) already comes out block-diagonal: one 2×2 block built from $(c_{\mathbf{k}\uparrow}, c^{\dagger}_{-\mathbf{k}\downarrow})$ and an identical one from $(c_{\mathbf{k}\downarrow}, -c^{\dagger}_{-\mathbf{k}\uparrow})$, each with reduced BdG matrix

$$h_{\mathrm{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) & \Delta(\mathbf{k}) \\ \Delta(\mathbf{k})^* & -\varepsilon(\mathbf{k}) \end{pmatrix}. \tag{5.8}$$

This is exactly the "4×4 matrix, single orbital with two spins, diagonal entries $\varepsilon(\mathbf{k})$, $\varepsilon(\mathbf{k})$, $-\varepsilon(\mathbf{k})$, $-\varepsilon(\mathbf{k})$, off-diagonal entries Δ" structure read directly off the slides and quoted in the lecture, now derived rather than merely asserted. Diagonalizing (5.8) gives the celebrated BCS quasiparticle dispersion

$$E(\mathbf{k}) = \pm\sqrt{\varepsilon(\mathbf{k})^2 + |\Delta(\mathbf{k})|^2}. \tag{5.9}$$

Notebook check-in: `session5.ipynb`, "Nambu electronic structure without superconductivity" and "Nambu electronic structure with superconductivity" (both Reminder)

Build the Nambu-doubled Hamiltonian of (5.6)-(5.7) with $\Delta = 0$ first, and confirm you get exactly two copies of the same band structure, one "physical" (electron sector) and one "spurious" mirror copy (hole sector) with dispersion $-\varepsilon(\mathbf{k})$. Then switch Δ on and watch the gap in (5.9) open at every crossing between the two copies.

5.9 A worked example: gap opening independent of dispersion

Equation (5.9) already tells us something the transcript's slides illustrate with explicit band plots: the gap opens at the minimum of $E(\mathbf{k})$, which happens wherever $\varepsilon(\mathbf{k}) = 0$, i.e. exactly on the original Fermi surface, and the minimum gap there is $|\Delta(\mathbf{k})|$ evaluated on that Fermi surface. Nothing in this argument depends on the details of $\varepsilon(\mathbf{k})$: not the range of hopping, not the lattice geometry, not the position of the chemical potential. Concretely, take the single-orbital square lattice with $\varepsilon(\mathbf{k}) = \cos k_x + \cos k_y$ (units where $t = 1$) and a chemical potential tuned so that the "physical" red band and the "spurious" blue mirror band cross; turning on a constant Δ opens a gap of size $2|\Delta|$ at every such crossing, for *any* choice of chemical potential along that dispersion. Exactly the same statement holds if we redo the calculation on the triangular lattice instead of the square lattice: as we move the chemical potential across the triangular-lattice band, a uniform Δ still gaps the spectrum everywhere the physical and spurious bands cross. This is the generic signature of *conventional* (fully gapped) superconductivity: a uniform, momentum-independent $\Delta(\mathbf{k})$ always fully gaps the Fermi surface, regardless of microscopic detail.

Notebook check-in: `session5.ipynb`, "Superconductivity, full spectral function and electron spectral function" (Core), and "The impact of superconductivity in an electronic structure" (Challenge)

Reproduce the square-lattice example above: plot the electron-sector spectral function with $\Delta = 0$ (two crossing copies) and then with $\Delta \neq 0$ (gap of size $2|\Delta|$ opening at the crossing, per eq. 5.9). Repeat for the triangular-lattice dispersion and confirm the gap opens regardless of where you put the chemical potential.

5.10 Three scenarios for one Fermi surface: metal, gapped SC, nodal SC

Not every pairing function $\Delta(\mathbf{k})$ is momentum-independent, and this is where things get more interesting. Given one electronic dispersion $\varepsilon(\mathbf{k})$, there are, in general, three qualitatively

different outcomes:

1. **No superconductivity** ($\Delta = 0$): a metal, gapless everywhere the Fermi surface sits.
2. **Conventional (s-wave) superconductivity**: $\Delta(\mathbf{k}) \neq 0$ everywhere on the Fermi surface, so $E(\mathbf{k})$ in (5.9) is bounded away from zero everywhere; a full gap.
3. **Unconventional, nodal superconductivity**: $\Delta(\mathbf{k})$ vanishes along particular lines or points of the Brillouin zone. Wherever the Fermi surface crosses one of those nodal lines, $E(\mathbf{k}) = 0$ even though $\Delta \neq 0$ nearby; the system is gapped in most of reciprocal space but remains gapless (metallic) at isolated momenta or along nodal lines.

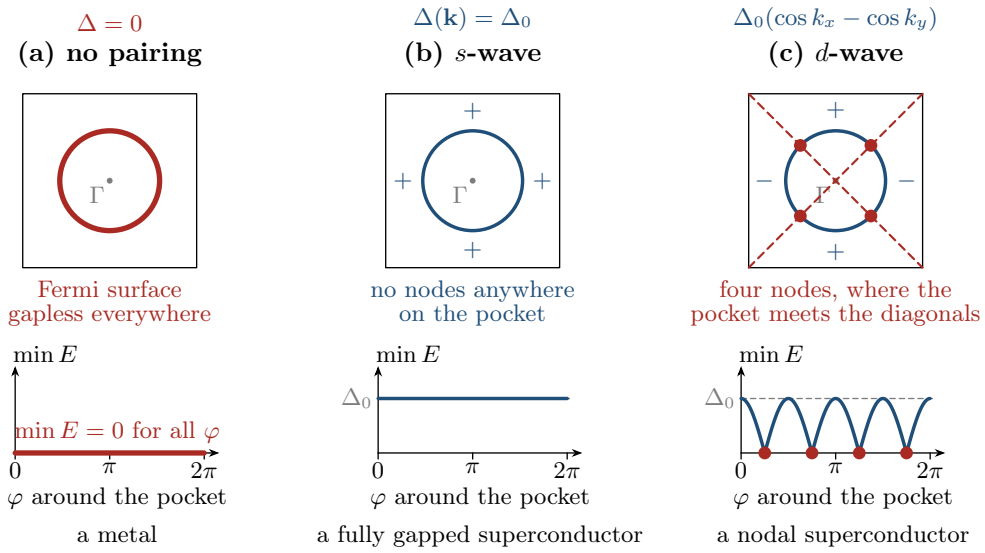


Figure 5.2: One Fermi surface, three outcomes. Top: the Brillouin zone with a small Fermi pocket around Γ; bottom: the minimum of the quasiparticle energy $E(\mathbf{k}) = \sqrt{\varepsilon(\mathbf{k})^2 + |\Delta(\mathbf{k})|^2}$ as we walk once around that pocket. Since $\varepsilon = 0$ on the Fermi surface, that minimum is simply $|\Delta|$ evaluated there, which is why the pairing function only ever matters *on* the Fermi surface. (a) With no pairing the whole pocket sits at zero energy: a metal. (b) A constant Δ_0 lifts every point of it by the same amount: a full gap. (c) The d-wave gap (5.13) vanishes on the zone diagonals, and the pocket crosses them in four places, so four gapless points survive inside an otherwise gapped superconductor. Nodes elsewhere in the zone are invisible to the spectrum; only nodes that intersect the Fermi surface produce gapless quasiparticles.

Figure 5.2 puts the three side by side, in the Brillouin zone and again as the gap read off around the Fermi surface itself, which is the only place it can be read off at all. The high-temperature cuprate superconductors are the standard real-world example of scenario 3: they are genuine superconductors, but with a gap function that has nodes, so parts of their Fermi surface remain gapless even in the superconducting state. This three-way split holds for any lattice geometry; the lecture works through it explicitly for both the square and triangular lattice Fermi surfaces.

Notebook check-in: `session5.ipynb`, "Unconventional singlet superconducting orders" (Challenge), and "Gapped and nodal superconducting orders" (Core)

Build a momentum-dependent $\Delta(\mathbf{k})$ that vanishes along a line in the Brillouin zone (e.g. the d-wave form of section 5.12 below) and confirm numerically, via (5.9), that the gap closes exactly where the Fermi surface crosses that nodal line, distinguishing this from the uniformly-gapped case of section 5.9.

5.11 The general pairing Hamiltonian and its symmetry classification

So far Δ has been a single number. In general the pairing order can depend on momentum and carry spin structure: $\Delta_{\mathbf{k},\sigma_1\sigma_2}$, with $\sigma_1, \sigma_2 \in \{\uparrow, \downarrow\}$, conveniently packaged into the 2×2 pairing matrix $\Delta_{\mathbf{k}} = \begin{pmatrix} \Delta_{\mathbf{k},\uparrow\uparrow} & \Delta_{\mathbf{k},\uparrow\downarrow} \\ \Delta_{\mathbf{k},\downarrow\uparrow} & \Delta_{\mathbf{k},\downarrow\downarrow} \end{pmatrix}$, and built from a generic mean-field Hamiltonian

$$H_{\text{pair}} = -\frac{1}{2}{\sum_{\mathbf{k}}}' \sum_{\sigma_1\sigma_2} \left[\Delta_{\mathbf{k},\sigma_1\sigma_2}\, c^\dagger_{\mathbf{k}\sigma_1} c^\dagger_{-\mathbf{k}\sigma_2} + \Delta^*_{\mathbf{k},\sigma_1\sigma_2}\, c_{\mathbf{k}\sigma_1} c_{-\mathbf{k}\sigma_2} \right], \tag{5.10}$$

Real-space and reciprocal-space pictures of Δ are Fourier-transform equivalent, so we can equally well think of this order in terms of how the pair wave function looks as a function of the separation between the two paired electrons in real space. The key move is to classify superconducting order purely by the *symmetry* of $\Delta_{\sigma\sigma'}(\mathbf{k})$, because that symmetry alone controls how the superconductor is going to respond to perturbations (disorder, magnetic fields), independent of microscopic detail.

Fermionic antisymmetry (the operators $c^\dagger_{\mathbf{k}\sigma} c^\dagger_{-\mathbf{k}\sigma'}$ anticommute) forces

$$\Delta_{\sigma\sigma'}(\mathbf{k}) = -\Delta_{\sigma'\sigma}(-\mathbf{k}), \tag{5.11}$$

a minus sign under the *combined* operation of exchanging spin indices and sending $\mathbf{k} \to -\mathbf{k}$; this combined statement always holds regardless of the type of pairing. The physically meaningful classification comes from splitting $\Delta_{\sigma\sigma'}(\mathbf{k})$ into its part symmetric and its part antisymmetric under spin exchange alone, which (via (5.11)) is equivalent to splitting it by its parity under $\mathbf{k} \to -\mathbf{k}$ alone:

$$\begin{aligned} &\textbf{Spin singlet (even parity):} \quad \Delta(\mathbf{k}) = \Delta(-\mathbf{k}), \qquad \Delta_{\sigma\sigma'} = -\Delta_{\sigma'\sigma} \text{ (spin-antisymmetric)}, \\ &\textbf{Spin triplet (odd parity):} \quad \Delta(\mathbf{k}) = -\Delta(-\mathbf{k}), \qquad \Delta_{\sigma\sigma'} = +\Delta_{\sigma'\sigma} \text{ (spin-symmetric)}. \end{aligned} \tag{5.12}$$

Concretely, on-site s-wave pairing ($\Delta(\mathbf{k}) = \Delta$, constant, arising from a local attractive interaction as in section 5.5) is even in $\mathbf{k}$, hence spin-singlet. This symmetry classification has an immediate physical payoff: a spin-singlet superconductor is essentially insensitive to non-magnetic disorder (Anderson's theorem, the pair wave function only cares about time-reversed partners, which non-magnetic disorder preserves), while a spin-triplet superconductor is strongly suppressed by even weak non-magnetic disorder, because scattering mixes the different momentum directions that the order parameter needs to keep coherent phase relations between. This is precisely why spin-singlet, s-wave superconductors vastly outnumber spin-triplet ones in nature: essentially any real material has some level of non-magnetic disorder, and that disorder selectively kills the triplet channel while leaving the singlet channel intact.

5.12 Building a spin-triplet superconductor out of a magnet

Given how fragile spin-triplet order is, how would we ever engineer one deliberately? One route is a toy model that combines a ferromagnetic exchange field with an attractive interaction in the same Hamiltonian,

$$H = -t \sum_{\langle ij\rangle,\sigma} c^\dagger_{i\sigma} c_{j\sigma} \underbrace{- h \sum_i \left(c^\dagger_{i\uparrow} c_{i\uparrow} - c^\dagger_{i\downarrow} c_{i\downarrow} \right)}_{\text{exchange field}} + \underbrace{U \sum_{\langle ij\rangle} c^\dagger_{i\sigma} c^\dagger_{j\sigma'} c_{j\sigma'} c_{i\sigma}}_{\text{attractive interaction}}, \qquad U < 0.$$

Solved self-consistently at mean-field level, both order parameters (the ferromagnetic exchange field and the pairing amplitude) can be nonzero simultaneously: the ground state breaks both time-reversal symmetry (it is magnetic) *and* gauge symmetry (it is superconducting) at once, and the pairing that survives is, necessarily, spin-triplet, since the exchange field explicitly splits the $\uparrow$ and $\downarrow$ Fermi surfaces and thereby suppresses the singlet channel that needs them degenerate. Physically, s-wave singlet pairing is normally the leading instability precisely because it can gap the *entire* Fermi surface (maximal condensation energy); once the exchange field quenches that channel, a different pairing channel, here the odd-parity, spin-triplet, effectively p-wave channel, becomes the leading instability instead.

The truly striking feature of this state shows up when we put it on a finite ribbon rather than an infinite plane: the bulk spectrum is gapped, exactly as we would expect from a superconductor, but the edges of the ribbon carry gapless excitations. A superconductor that is fully gapped in the bulk and gapless at the edge; this is, by now, a familiar signature from the topological-band-structure and quantum-Hall sessions, and indeed these edge modes are identified as topological in origin, tied to a finite Chern number of the underlying BdG band structure exactly the way a Chern insulator's edge states were tied to its Chern number.

Notebook check-in: `session5.ipynb`, "The impact of a spin splitting in an electronic structure" (Reminder), "A spin triplet superconductor" (Core), "Majorana modes in a spin triplet superconductor" (Challenge), "Interaction-induced spin-triplet superconductivity" (Challenge), and "Topological modes in interaction-driven spin triplet superconductors"

Start from the plain spin-split (exchange-field-only) band structure to build intuition, then run the self-consistency loop for the model above and confirm a spin-triplet Δ emerges. Move to a ribbon geometry and locate the gapless edge modes described here, then connect them (in the later challenge notebook sections) to the Majorana physics of sections 5.15-5.17.

5.13 Nodal singlet order: d-wave, cuprates, and the angular-momentum expansion

Within spin-singlet, even-parity order there is a further useful distinction: a singlet gap can be uniformly nonzero (the s-wave case of section 5.9) or it can be nodal. The classic example is **d-wave** order on the square lattice,

$$\Delta_d(\mathbf{k}) = \Delta_0 \left(\cos k_x - \cos k_y\right), \tag{5.13}$$

which vanishes exactly along the Brillouin-zone diagonals $k_x = \pm k_y$. Combined with (5.9), the quasiparticle spectrum $E(\mathbf{k}) = \sqrt{\varepsilon(\mathbf{k})^2 + \Delta_d(\mathbf{k})^2}$ is gapless precisely where the Fermi surface crosses these diagonals: a nodal singlet superconductor. This is exactly the pairing symmetry realized in the cuprate high-temperature superconductors, many of which have critical temperatures in the hundreds of kelvin and are spin-singlet, but decidedly *not* s-wave.

There is a clean, general organizing scheme behind this, directly analogous to the spherical-harmonic expansion of an atomic orbital wave function: expand the superconducting gap in an angular-momentum basis on the Fermi surface,

$$\Delta(\mathbf{k}) \;\sim\; \sum_L \Delta_L \, Y_L(\mathbf{k}), \tag{5.14}$$

$L = 0$: s-wave (conventional, typically phonon-driven),

$L = 1$: p-wave (spin-triplet, often ferromagnetic-magnon-driven),

$L = 2$: d-wave (spin-singlet, often antiferromagnetic-magnon-driven; cuprates).

In real/reciprocal space, s-wave order looks flat and constant, with no momentum dependence at all; $d_{x^2-y^2}$ order, as in (5.13), alternates sign around the Fermi surface in a four-lobed $+,-,+,-$ pattern (the same nomenclature and the same lobe structure as an atomic d orbital); and there is a third possibility, *extended s-wave*, which is still $L = 0$ (same overall symmetry as plain s-wave under the lattice point group) but is nodal along the *radial* direction: positive near the zone center Γ, passing through a node, and negative further out. Figure 5.3 draws these sign structures over the Brillouin zone alongside the p- and f-wave cases below.

The lecture repeats this same three-way classification (uniform s-wave, a nodal order, and extended s-wave) on the triangular lattice, where the nodal case shows six alternating-sign lobes around the Fermi surface: an **f-wave order,** $L = 3$ (confirmed by the instructor; the "L equal to two" heard in the raw recording was a slip, since a six-lobe alternating-sign pattern is the standard signature of $L = 3$, not $L = 2$, which is already used above for the square lattice's four-lobed $d_{x^2-y^2}$ state). Note that this one steps outside the present section's singlet family: six lobes alternating in sign means the gap changes sign under a half turn, so $\Delta(-\mathbf{k}) = -\Delta(\mathbf{k})$, and by (5.12) an $L = 3$ order is odd-parity and hence necessarily *spin-triplet*, not singlet like the s-, extended-s- and d-wave cases above.

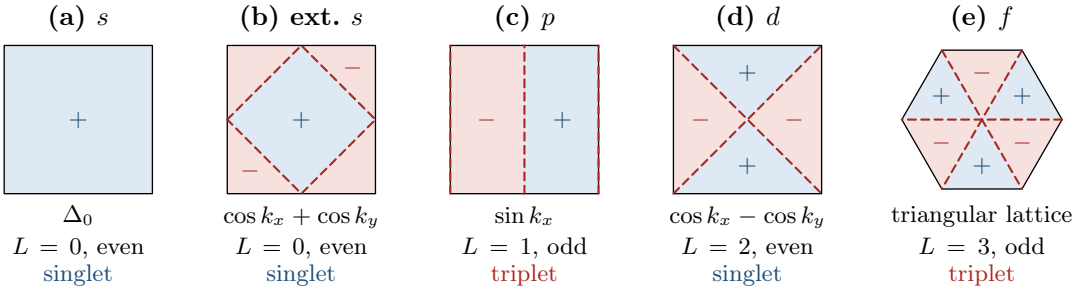


Figure 5.3: The sign structure of $\Delta(\mathbf{k})$ over the Brillouin zone, ordered by angular momentum L as in (5.14); dashed lines are the nodes. Reflecting a panel through its centre sends $\mathbf{k} \to -\mathbf{k}$: panels with even L come back unchanged and are therefore spin singlets, panels with odd L come back with every sign flipped and are spin triplets, which is (5.12) read off a picture. Counting lobes is the quickest diagnostic there is: two lobes means $L = 1$, four means $L = 2$, six means $L = 3$, and any pattern whose lobes alternate an odd number of times in a half turn is necessarily odd in $\mathbf{k}$, hence triplet – which is why the six-lobed f-wave state (e) is a spin triplet even though it sits next to the singlet d-wave state in every other respect. Extended s-wave (b) is worth the second look, since its node is radial rather than angular: it has the same even parity and the same point-group symmetry as plain s-wave, and only the position of its node ring relative to the Fermi surface decides whether it is nodal in the spectrum. The overall sign of Δ is a gauge choice; only the sign *changes* are physical.

Notebook check-in: `session5.ipynb`, "Gapped and nodal superconducting orders" (Core)

Implement the $d_{x^2-y^2}$ gap of (5.13) on the square lattice, plot $\Delta(\mathbf{k})$ over the Brillouin zone and confirm the four-lobe alternating-sign pattern and the location of its nodes relative to a given Fermi surface. The triangular-lattice $L = 3$ nodal (f-wave) case discussed above is also available in this notebook section if you want to see the six-lobe pattern directly.

5.14 Same gap, different topology: trivial s-wave versus chiral p-wave

Here is the pivot point of the whole session. Two superconducting gaps can look completely identical in their *spectral* properties, both fully gapping the Fermi surface with, say, the same

minimum gap size, and yet be topologically inequivalent. Compare plain s-wave pairing, $\Delta(\mathbf{k}) = \Delta_0$, with **chiral p-wave** pairing,

$$\Delta_{p+ip}(\mathbf{k}) = \Delta_0 \left(\sin k_x + i \sin k_y\right), \tag{5.15}$$

an odd-parity, spin-triplet order (consistent with (5.12): $\Delta_{p+ip}(-\mathbf{k}) = -\Delta_{p+ip}(\mathbf{k})$) that winds once around the origin of the complex plane as $\mathbf{k}$ circles the Fermi surface. Both (5.9)-with-Δ_0 and (5.9)-with-Δ_{p+ip} can fully gap the same Fermi surface, so a measurement of the bulk gap alone cannot distinguish them. But exactly as in the topological-band-structure session's Chern-number classification of gapped bands, we can compute the Chern number of the occupied BdG bands for each case, and they come out completely different: the trivial s-wave state has Chern number 0, while the chiral p-wave state has Chern number ± 2 (the sign set by the chirality, $p_x + ip_y$ versus $p_x - ip_y$).

What distinguishes them, drawn in figure 5.4, is the winding of the *phase* of $\Delta(\mathbf{k})$ as $\mathbf{k}$ is carried once around the Fermi surface: constant for s-wave, one full turn for $p + ip$.

This has exactly the consequence the bulk-boundary correspondence would lead us to expect. On a finite system, the s-wave superconductor stays gapped everywhere, bulk and edge alike, just like a trivial (Chern-number-zero) insulator. The chiral p-wave superconductor, by contrast, is gapped in the bulk but carries gapless *chiral* edge modes: on any one edge, $|C|$ branches all propagating the same way (here two, one from each of the two identical spin blocks), with the opposite edge carrying the same number of branches running the opposite way, precisely as a Chern-number-± 2 state must. On a ribbon, then, one sees two counter-propagating sets of branches, but they are localized on opposite edges rather than sharing one. These particular edge modes turn out to be "fractions of an electron" in a very specific and rigorous sense: they are Majorana modes, and understanding exactly what that means is the subject of the rest of this session.

Notebook check-in: `session5.ipynb`, "Trivial and topological superconducting order parameters, bulk spectra" (Core), and "...edge spectra" (Core)

Build the s-wave and chiral p-wave BdG Hamiltonians side by side, confirm their bulk spectra look alike (both gapped, comparable gap size), then switch to a ribbon geometry and confirm only the p-wave case develops gapless chiral edge states, with the two ribbon edges carrying branches of opposite velocity, exactly the Chern-number-driven distinction of section 5.14.

5.15 Majorana bound states: definition and the electron-splitting trick

A **Majorana fermion** is defined algebraically by the condition that its field operator is its own Hermitian conjugate,

$$\gamma^\dagger = \gamma, \tag{5.16}$$

meaning creation and annihilation of the quasiparticle are the same operation: the particle is its own antiparticle. This sounds like it has no place in an ordinary material, since an ordinary electron is emphatically not its own antiparticle ($c^\dagger \neq c$), and a semiconductor, say, never hosts anything of the kind.

But there is a purely algebraic fact that opens the door: *any* ordinary fermionic operator can be written as a combination of two Hermitian, mutually anticommuting Majorana operators,

$$\gamma_A = c + c^\dagger, \qquad \gamma_B = -i\left(c - c^\dagger\right), \tag{5.17}$$

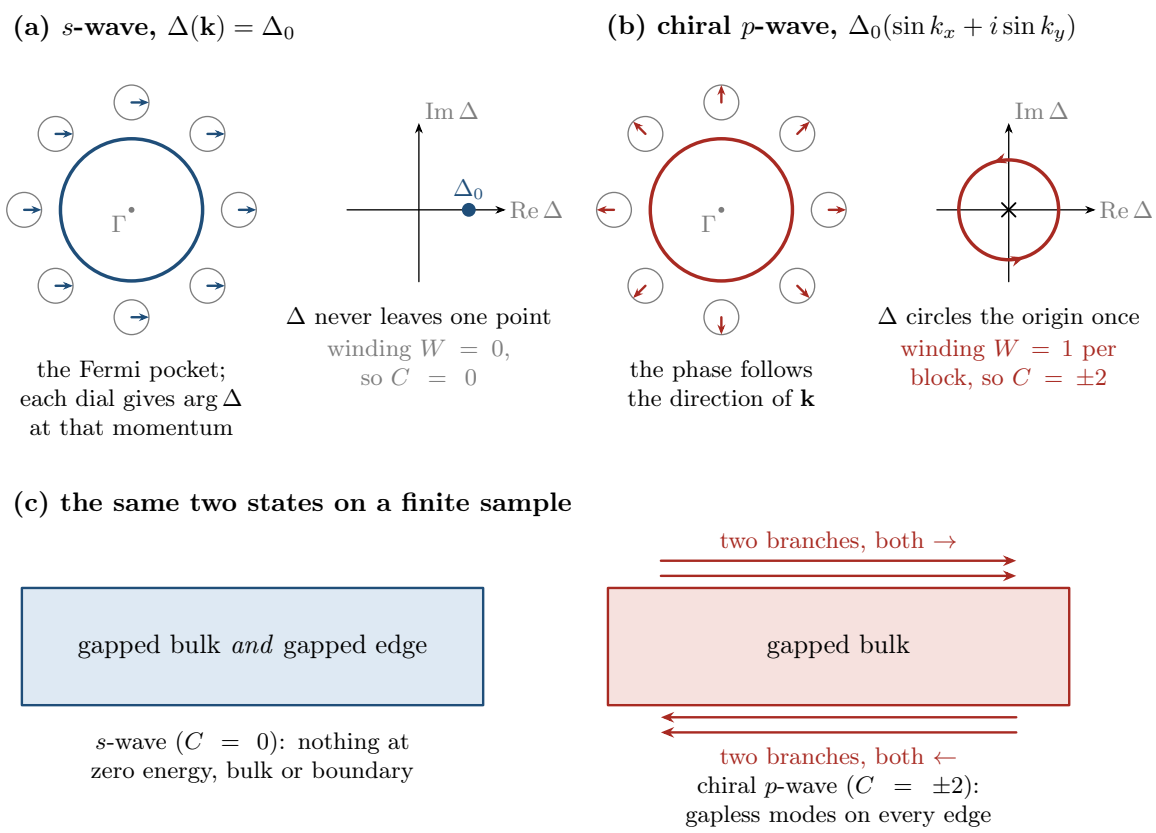


Figure 5.4: Two gaps that a spectroscopy experiment cannot tell apart. Each dial is the phase of $\Delta(\mathbf{k})$, an arrow in the complex plane, drawn at the point of the Fermi pocket where it is evaluated. (a) For s-wave pairing that phase is the same everywhere, so as $\mathbf{k}$ goes once around the pocket the value of Δ never moves off a single point of the complex plane: winding number zero, Chern number zero. (b) For chiral p-wave pairing (5.15) the phase instead tracks the direction of $\mathbf{k}$, so Δ traces a loop that encircles the origin once. The modulus is essentially the same in both cases, so both fully gap the same pocket by the same amount; only the winding differs, and the winding is what the Chern number counts. (c) The consequence on a finite sample: the trivial state is gapped at its boundary as well as in its bulk, while the chiral state must carry $|C|$ gapless branches on each edge, all running the same way, and the opposite way on the opposite edge. Here $C = \pm 2$, one branch from each of the two identical spin blocks of (5.8).

so that $\gamma_A^\dagger = \gamma_A$, $\gamma_B^\dagger = \gamma_B$, $\{\gamma_A, \gamma_B\} = 0$, $\gamma_A^2 = \gamma_B^2 = 1$ (using $\{c, c^\dagger\} = 1$), and, inverting (5.17),

$$c = \frac{1}{2}\left(\gamma_A + i\gamma_B\right), \qquad c^\dagger = \frac{1}{2}\left(\gamma_A - i\gamma_B\right). \tag{5.18}$$

This is initially just a change of variables, a mathematical trick with no new physics in it: every electron is *always* secretly two Majorana operators, tightly bound together into an ordinary fermion, and normally there is no way to observe γ_A or γ_B individually, only their fixed combination c. What makes superconductors special is that BdG eigenstates are already electron-hole mixtures by construction (recall the Nambu spinor of section 5.8 explicitly combines $c_{\mathbf{k}\sigma}$ with $c^\dagger_{-\mathbf{k}\sigma'}$), so a superconductor is exactly the kind of setting where, with the right pairing symmetry and the right geometry, the two Majorana components of an electron can become spatially *separated* (deconfined) rather than staying locked together at one site. That deconfinement is what the rest of this section is about.

5.16 The Kitaev chain: the minimal model with isolated Majoranas

The simplest superconductor capable of hosting isolated Majoranas is, appropriately, about as simple as a superconducting Hamiltonian can be: a one-dimensional, *spinless* chain with nearest-neighbor hopping and nearest-neighbor pairing,

$$H_{\text{Kitaev}} = -t\sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.}\right) - \mu\sum_n c_n^\dagger c_n + \Delta\sum_n \left(c_n c_{n+1} + c_{n+1}^\dagger c_n^\dagger\right). \tag{5.19}$$

Pairing has to sit on a bond here rather than on-site: with spinless fermions, an on-site term $c_n c_n$ vanishes identically by the Pauli principle ($c_n^2 = 0$), so the simplest allowed pairing is exactly the nearest-neighbor term above. Fourier transforming, the dispersion and the pairing become

$$\varepsilon(k) = -2t\cos k - \mu, \qquad \Delta(k) = 2i\Delta\sin k, \tag{5.20}$$

so the anomalous term is manifestly odd in k: this is, as (5.12) already tells us, necessarily a spin-triplet-type (here, spinless p-wave) superconductor, consistent with the Pauli-forbidden on-site channel.

Now set $\mu = 0$ and tune to the special point $\Delta = t$, and substitute the Majorana decomposition (5.17)-(5.18) at every site, $c_n = \frac{1}{2}(\gamma_{2n-1} + i\gamma_{2n})$. Carrying the substitution through both the hopping and the pairing term (the algebra is a direct but slightly lengthy exercise in the anticommutation relations of the γ's, which we encourage you to check yourself), the two terms combine so that every $\gamma_{2n-1}\gamma_{2n+2}$ contribution cancels exactly between hopping and pairing, leaving a remarkably clean result:

$$H_{\text{Kitaev}}\Big|_{\Delta=t,\ \mu=0} = it\sum_{n=1}^{N-1}\gamma_{2n}\,\gamma_{2n+1}. \tag{5.21}$$

Every Majorana operator $\gamma_2, \gamma_3, \gamma_4, \ldots, \gamma_{2N-1}$ appears in this sum, paired up between neighboring *sites*, but the very first Majorana, γ_1, and the very last, γ_{2N}, never appear at all. They drop out of the Hamiltonian entirely: they cost zero energy, and they are completely uncoupled from everything else in the system. This is precisely the same structure we met in the topological-band-structure session's SSH model, in its dimerized strong-coupling limit: there, at full dimerization, the bulk pairs up into bonding/antibonding dimers while a leftover

site at each end decouples and produces a zero-energy edge mode. The Kitaev chain in (5.21) *is* that same structure, just written in Majorana language rather than ordinary fermion language: two decoupled end Majoranas standing in for the SSH model's two decoupled end sites. We are not re-deriving the SSH model here; we are simply recognizing it, dressed up in a new set of operators. Which Majorana is bound to which, in this phase and in the trivial one, is drawn in figure 5.5.

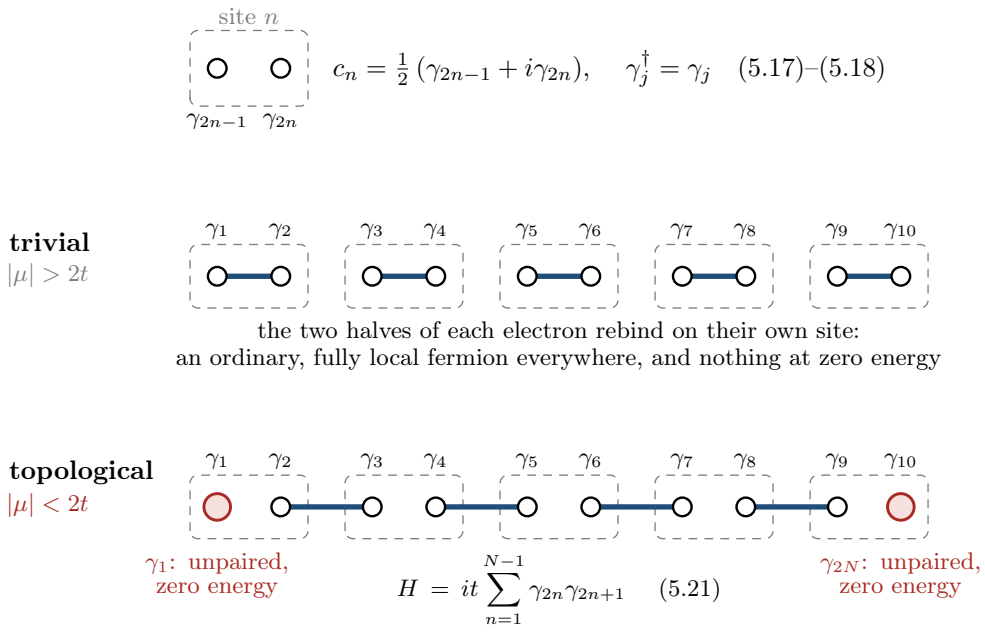


Figure 5.5: The Kitaev chain, drawn in Majorana variables, for $N = 5$ sites (so $\gamma_{2N} = \gamma_{10}$). Splitting every electron into its two Majorana halves (5.17) is free of physical content on its own – what matters is which halves the Hamiltonian binds together. In the trivial phase (top) the binding is intra-site: the two halves of each electron stay locked to one another, exactly as they were in the vacuum, and every mode costs energy. In the topological phase (bottom) the binding is inter-site, γ_{2n} with γ_{2n+1}, and the pattern comes up one bond short at each end: γ_1 and γ_{2N} appear nowhere in (5.21), cost nothing, and sit at opposite ends of the sample. The ordinary fermion they would jointly make up has been split in half and the halves put a chain length apart, which is what *deconfined* means here. This is the fully dimerized SSH chain of the topological-band-structure session with its two leftover end sites, relabelled: same bookkeeping, new operators.

Notebook check-in: `session5.ipynb`, "Emergence of Majorana modes as a function of the size of the superconductor" (Core)

Build the Kitaev chain of (5.19), diagonalize it at the fine-tuned point $\Delta = t$, $\mu = 0$, and confirm directly (as system size grows) that two states pin exactly to zero energy and localize at the two ends, matching the two decoupled Majoranas γ_1 and γ_{2N} predicted by (5.21).

5.17 Periodic versus open boundaries: the topological zero modes

Diagonalizing (5.19) on a *periodic* chain gives a fully gapped spectrum, with a superconducting gap set by Δ and a nontrivial bulk topological invariant (the same kind of $\mathbb{Z}_2$/winding invariant that classified the SSH model's two phases via the Zak phase in the topological-band- structure session, now computed from the BdG bands instead of an ordinary Bloch Hamiltonian). On an *open* chain, say $N = 100$ sites with the two ends disconnected, the bulk spectrum keeps essentially the same gap, but two extra states appear pinned exactly at zero energy, localized

respectively at the left end and the right end of the chain: these are precisely the unpaired Majoranas γ_1 and γ_{2N} from (5.21), now visible as genuine mid-gap states of the open chain.

These zero modes are remarkably robust. Moving away from the fine-tuned point ($\Delta \neq t$, finite μ), the zero modes survive as long as we stay within the same topological phase. The precise, standard criterion for the general Kitaev chain (5.19) with $\Delta \neq 0$ is

$$|\mu| < 2t \implies \text{topological phase (unpaired end Majoranas)}, \qquad |\mu| > 2t \implies \text{trivial phase}. \tag{5.22}$$

In the trivial phase, large chemical potential effectively pins each site to be either empty or doubly occupied in the fermionic language, and the two Majoranas that make up each site's electron recombine locally into an ordinary, confined fermion; there is nothing left to deconfine. In the topological phase, by contrast, p-wave pairing dominates over the trivial on-site physics, and the two Majoranas at each site pair up with their neighbors' Majoranas instead of with each other, leaving the two end Majoranas with no partner at all. This is exactly the same "strong coupling / dimerized limit versus weak coupling / atomic limit" contrast that distinguished the SSH model's topological and trivial phases; the Kitaev chain reproduces it because, as we saw in section 5.16, it literally is the SSH model, just relabeled.

5.18 Engineering a topological superconductor from a quantum spin Hall insulator

Chiral and p-wave superconductors of the kind we need are rare in nature, so in practice they are usually *engineered* rather than found. One instructive recipe reuses the quantum spin Hall (QSH) insulator from the topological-band-structure session directly. Take a QSH insulator, tungsten ditelluride (WTe_2) is the concrete material example used, whose edge carries helical modes: spin-up electrons propagate one way, spin-down electrons propagate the other way, on the same edge. Now place ordinary ferromagnets on two segments of that edge. A magnet locally breaks time-reversal symmetry, which gaps out the helical edge modes underneath it (recall from the topological-band- structure session that the QSH edge's protection relies on time-reversal symmetry via Kramers degeneracy; break that symmetry locally and the protection is gone there). What remains ungapped is a short, isolated segment of helical edge sandwiched between the two magnetic regions.

That isolated segment only has spin-up electrons moving right and spin-down electrons moving left (no spin-down moving right, and no spin-up moving left): counting states purely by propagation direction, this segment is *effectively spinless*, exactly the ingredient the Kitaev-chain construction of section 5.16 needs. Proximity-coupling this segment to an ordinary s-wave superconductor then reproduces, on this effectively spinless one-dimensional system, exactly the physics of (5.19): a one-dimensional p-wave superconductor with Majorana zero modes pinned at the two ends of the isolated segment. The whole stack is drawn in figure 5.6.

Notebook check-in: `session5.ipynb`, "Artificial topological superconductivity" (Challenge), "Edge states of an artificial topological superconductor" (Challenge), and "Topological phase transition as a function of the exchange field"

Build the QSH-edge-plus-magnets-plus-superconductivity construction described above and confirm the emergence of end-localized zero modes as you vary the exchange field strength, watching the system cross the (5.22)-type topological/trivial boundary as the magnetic gapping strength changes.

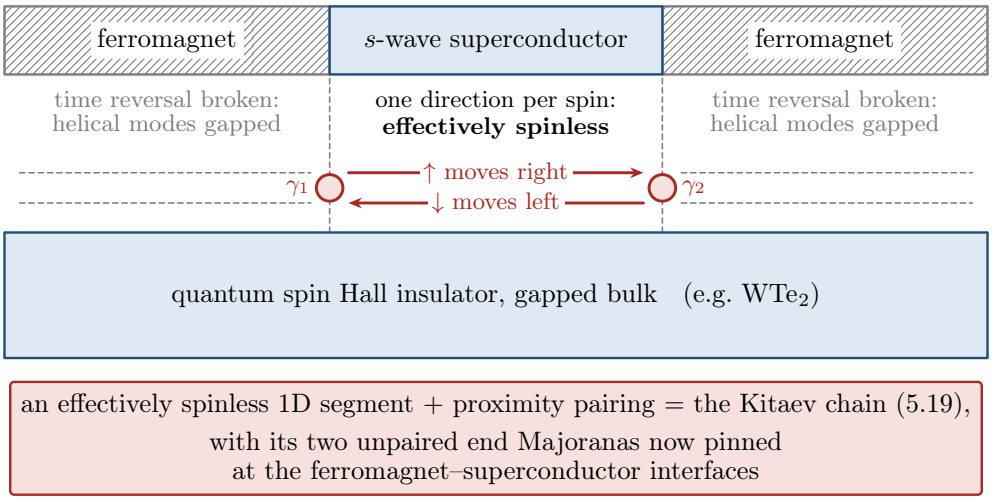


Figure 5.6: Building the Kitaev chain out of parts that exist. The edge of a quantum spin Hall insulator is helical: on one and the same edge, spin-up electrons run one way and spin-down electrons the other. Depositing a ferromagnet on a stretch of that edge breaks time-reversal symmetry locally and destroys the Kramers protection there, gapping the edge underneath it; two ferromagnets leave one short helical segment surviving between them. That segment offers only one propagation direction per spin, so counted as channels it is a *spinless* one-dimensional wire – precisely the ingredient (5.19) demands and the one nature does not hand us directly. Proximity coupling it to an ordinary s-wave superconductor supplies the pairing, and the result is a one-dimensional p-wave superconductor carrying a Majorana zero mode at each end of the segment, that is, at the two ferromagnet–superconductor interfaces. Note what was *not* required: no unconventional superconductor appears anywhere in the recipe, only an s-wave one, a magnet, and a topological insulator.

5.19 Topological quantum computing with Majoranas

The reason Majorana modes attract so much attention beyond their own intrinsic strangeness is their proposed use in fault-tolerant quantum computing. Majoranas are **non-Abelian anyons**: exchanging (braiding) one Majorana around another multiplies the many-body ground state by a phase, and unlike an ordinary phase, the *order* in which different braiding operations are carried out matters, an inherently non-commuting (non-Abelian) algebra. This follows directly from the structure of the BdG Hamiltonian once more than two Majoranas are present.

Concretely, four Majorana operators $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ can be recombined, via (5.18)-type pairings, into two ordinary ("artificial") fermionic operators, so a system with four Majoranas encodes two ordinary fermionic modes. The resulting many-body ground-state manifold, empty, one mode filled, the other mode filled, both filled, is four-fold degenerate (all four states cost zero energy, by the same argument that made γ_1 and γ_{2N} zero-energy in the Kitaev chain), and this degenerate manifold is exactly the proposed computational basis for a topological qubit.

Braiding two Majoranas, say γ_1 around γ_2, acts on the ground-state manifold as a unitary

$$U_{12} = \exp\left(\frac{\pi}{4}\,\gamma_1\gamma_2\right), \tag{5.23}$$

which multiplies states by a relative phase built from $e^{i\pi/4}$ times the Majorana product; the precise form is not the point so much as what it implies: braiding the pair $(1,2)$ and then the pair $(3,4)$ produces a different final state than braiding $(3,4)$ first and $(1,2)$ second, and braiding $(1,4)$ is different again. This non-commuting braid algebra is exactly what lets sequences of purely topological, purely geometric moves (physically moving Majoranas around

one another) implement non-trivial quantum logic gates. Because the resulting qubit is stored non-locally, split between spatially separated Majoranas rather than living in one local degree of freedom, it is intrinsically resistant to local decoherence and to weak, local disorder; this robustness is the whole appeal of the topological-quantum-computing proposal. This braiding-and-quantum-computing material is slides-and-transcript-only content with no counterpart in `session5.ipynb`: none of the notebook's sections implement braiding dynamics or a qubit manifold, which makes sense given how far this is from a static tight-binding diagonalization; it is a genuinely separate, dynamical calculation.

5.20 Closing summary

- **Superconductivity is a mean-field, attractive-interaction phenomenon.** Starting from an attractive Hubbard chain (5.2), pairing mean-field theory produces the order parameter $\Delta_n = U\langle c_{n\downarrow} c_{n\uparrow}\rangle$ (5.3) and mean-field Hamiltonian (5.4).
- **Pairing breaks particle-number conservation down to parity** (section 5.6), and this is exactly the statement that gauge symmetry is spontaneously broken: the pairing term picks up a phase $e^{-2i\varphi}$ under the gauge transformation (5.5), unlike ordinary bilinear terms.
- **Flux quantization**, $\Phi = n\,h/2e$ (5.1), is a direct, measurable macroscopic consequence of that broken gauge symmetry, on par with the quantum Hall conductance quantum from earlier sessions.
- **The Nambu spinor (5.6) and the Bogoliubov-de Gennes Hamiltonian (5.7)-(5.8)** are the tools that let us diagonalize any mean-field superconducting Hamiltonian, giving the quasiparticle dispersion $E(\mathbf{k}) = \pm\sqrt{\varepsilon(\mathbf{k})^2 + |\Delta(\mathbf{k})|^2}$ (5.9); a gap opens at the original Fermi surface regardless of microscopic dispersion details.
- **The symmetry of $\Delta(\mathbf{k})$ classifies the superconductor**: singlet (even in $\mathbf{k}$) versus triplet (odd in $\mathbf{k}$) via (5.11)-(5.12), with an angular-momentum ($L = 0, 1, 2, \dots$) subclassification (5.14) that recovers s-, p-, and d-wave order, the last of these realized by cuprate high-T_c superconductors via the nodal $d_{x^2-y^2}$ gap (5.13).
- **Spectrally identical gaps can carry different topology.** Trivial s-wave ($C = 0$) and chiral p-wave ($C = \pm 2$, eq. 5.15) pairing both fully gap the same Fermi surface, but only the topological one hosts gapless edge modes, exactly the Chern-number bulk-boundary logic from the topological-band-structure and quantum-Hall sessions.
- **The Kitaev chain (5.19) is the SSH model in Majorana language.** At the fine-tuned point it reduces to $H = it\sum_n \gamma_{2n}\gamma_{2n+1}$ (5.21), leaving two unpaired, zero-energy end Majoranas; these survive away from fine-tuning throughout the topological phase $|\mu| < 2t$ (5.22).
- **Majoranas are non-Abelian anyons.** Braiding them implements a non-commuting operator algebra (5.23) on a topologically degenerate ground-state manifold, the basis for proposed topological quantum computing schemes; this last piece is slides-and-transcript material with no notebook counterpart.

Today we followed session 1's superconductivity thread all the way from the bare pairing order parameter to Nambu spinors, Bogoliubov-de Gennes Hamiltonians, singlet/triplet classification, and finally to Majorana zero modes in the Kitaev chain. Session 1's fork had a second branch we have not touched since its first pass: magnetism, broken time-reversal

symmetry rather than broken gauge symmetry. That is exactly where we go next. Session 6 returns to the very same Hubbard Hamiltonian that opened today's session 5.5, now on its repulsive ($U > 0$) side, and gives magnetism the same depth of treatment we have just given superconductivity.

Session 6

Classical Magnetism and Magnons

Last session closed on a sentence worth repeating exactly, because it is the sentence that opens today's lecture too: the same Hubbard-like Hamiltonian can mean-field into either a superconductor or a magnet, and magnetism was deferred to the next recording. Today is that next recording. We already met a first pass at mean-field magnetism all the way back in session 1 (the Hubbard chain, the magnetization m, the Mott gap), but that was deliberately left as an appetizer: a sign flip away from the superconducting case, worked out just far enough to show that magnetism *is* a symmetry-broken mean-field state. Today we go much further. We will solve the Hubbard model's magnetism numerically and see ferromagnetism and antiferromagnetism compete as two genuinely different self-consistent solutions of the same Hamiltonian; we will find the precise threshold interaction strength above which magnetism appears at all; and then, because that threshold sits exactly where mean-field theory itself becomes unreliable, we will leave mean field behind and derive, from first principles, the effective spin Hamiltonian that governs the strongly correlated limit: the Heisenberg model. From there we go one level further still and ask about the *excitations* living on top of a magnetically ordered state, which brings us to magnons, the magnetic analogue of a phonon, built rigorously via the Holstein-Primakoff transformation. And once we have a genuine band structure for magnons, we will find that the topological machinery from sessions 3 and 4, Chern numbers, Berry curvature, chiral edge states, applies to them too, giving us topological magnon bands as a third application of a tool we have now used on electrons twice and on magnons once.

One piece of framing before we start, because it explains how this session ends. Today's recording is a single, self-contained lecture on classical magnetism and magnons only; it does not also carry a second recording on quantum magnetism and the Kondo model the way it might sound from the opening preview of spinons below. That material genuinely belongs to its own session later in the course. What is true, and worth saying plainly rather than papering over, is that today's recording simply stops once it reaches the end of the "topological magnons" material: there is no closing take-home summary slide narrated at the end, the lecture just runs out of planned material. So these notes end the same way, on topological magnons, without inventing a wrap-up sentence the lecture itself never says.

6.1 Two classes of magnets, and the excitations that live on each

Let us start by drawing the map of where we are headed, because today's lecture actually previews material from two future sessions before settling into its own. There are two big classes of magnetic system.

The first class is magnets that break time-reversal symmetry. These are materials with a

net magnetization on every site (or a staggered one), and a finite spin angular momentum is precisely what makes a system look different when its "movie" is run backwards, which is the definition of broken time-reversal symmetry. Ferromagnets, antiferromagnets, and noncollinear spin spirals all belong here. Iron and cobalt are the two materials the lecture actually names, repeatedly, as concrete examples of this family. This is the class we spend all of today on.

The second class is magnets that do *not* break time-reversal symmetry, at least not classically: quantum magnets, among them quantum spin liquids. These have a local moment on every site, but the spins are in quantum superposition rather than settling into a static classical arrangement, so time-reversal symmetry need not be broken overall. A related case, also mentioned today only in passing, is the heavy-fermion Kondo system: a combination of local moments in quantum superposition together with an itinerant metallic electron sea. (The "Classical and quantum magnetism" slide illustrates all three families with material photographs, iron for the ferromagnet/antiferromagnet family, herbertsmithite for quantum spin liquids, samarium hexaboride SmB_6 for heavy-fermion insulators, but none of those three compound names is actually spoken in this year's narration; they are slide-only labels, worth knowing about but not something the lecture itself says out loud.)

Each class comes with its own emergent excitation. Time-reversal-breaking magnets host **magnons**: spin-1, charge-neutral fluctuations of the order parameter, the direct magnetic analogue of a phonon. Because magnons carry no charge, they are completely unaffected by electric fields, and they can propagate in materials that are electronic insulators: a ferromagnetic (or antiferromagnetic) Mott insulator lets no current through, yet a magnon can still carry spin and heat across it. Quantum magnets instead host **spinons**: spin-1/2, charge-neutral excitations, sometimes described informally as "half an electron," since an electron's spin and charge degrees of freedom have effectively separated. We will build magnons carefully over the rest of today's session. Spinons, and the Kondo lattice model, are only previewed here as a name and a spin quantum number; their full construction is a later session's material (session 8), not something today's lecture returns to.

6.2 A recap: symmetry breaking and the mean-field philosophy

We have seen this move twice already in this course, first in session 1 in miniature, then in full depth for superconductivity in session 5, so let us state it once more, briskly, before putting it to work on magnetism specifically. Many-body interactions can spontaneously break a symmetry that the Hamiltonian itself possesses. Three cases run through this entire course: magnets break time-reversal (equivalently, real-space spin-rotation) symmetry; charge density waves break lattice translational symmetry; superconductors break gauge symmetry. All three emerge from the same generic starting point, a kinetic (single-particle, exactly solvable) term plus a quartic many-body interaction, and all three are extracted the same way: mean-field theory factorizes the quartic term by replacing part of it with its expectation value, turning an intractable many-body problem into a solvable single-particle one, solved self-consistently.

Where do the interactions themselves come from? Two sources. Electrostatic (Coulomb) repulsion between electrons is the classical, always-present contribution. Interactions can also be mediated by other quasiparticles (phonons, magnons, plasmons), which can flip the *net* effective interaction from repulsive to attractive. Magnetism specifically needs the net interaction to be repulsive; since bare Coulomb repulsion is already repulsive, this is the easy case to realize, which is one reason magnetism is so common in nature.

6.3 The Hubbard model and mean-field magnetism, revisited

The minimal Hamiltonian for all of this is the one-dimensional Hubbard chain we first met in session 1,

$$H = -t \sum_{n,\sigma} \left(c^\dagger_{n\sigma} c_{n+1,\sigma} + \text{h.c.} \right) + U \sum_n c^\dagger_{n\uparrow} c_{n\uparrow}\, c^\dagger_{n\downarrow} c_{n\downarrow}, \tag{6.1}$$

hopping t plus a purely local (on-site) density-density interaction between opposite spins, the Hubbard term. Negative U (attractive) drives superconductivity, session 5's territory; positive U (repulsive) drives magnetism, ours today.

The mean-field decoupling proceeds exactly as before: replace $c^\dagger_{n\uparrow} c_{n\uparrow}$ and $c^\dagger_{n\downarrow} c_{n\downarrow}$ by their expectation values, leaving a bilinear Hamiltonian with spin-dependent on-site energies. Nothing forces these two expectation values to agree, and the imbalance between them is exactly the magnetization,

$$m = \langle c^\dagger_\uparrow c_\uparrow \rangle - \langle c^\dagger_\downarrow c_\downarrow \rangle, \tag{6.2}$$

which plays the role of an exchange field: the resulting mean-field Hamiltonian looks schematically like $H_{\text{MF}} \sim -\frac{U}{2}\, m\, \boldsymbol{\sigma} \cdot c$, where $\boldsymbol{\sigma}$ is the vector of Pauli matrices. Self-consistency gives $m = 0$ for $U < 0$ and a genuinely magnetic $m \neq 0$ for $U > 0$, exactly as in session 1.

Here is a piece of rigor the slides show but the spoken explanation only gestures at: nothing in this construction singles out the z axis. The Hubbard interaction can be rewritten exactly, for a single site, in terms of the local spin operator $\mathbf{S} = \frac{1}{2} c^\dagger_\alpha \boldsymbol{\sigma}_{\alpha\beta} c_\beta$ and the total occupation $n = n_\uparrow + n_\downarrow$, as $U\, n_\uparrow n_\downarrow = \frac{U}{4} n^2 - U \mathbf{S}^2$ (up to the usual $n^2 = n + 2 n_\uparrow n_\downarrow$ identity for fermionic occupation numbers). Mean-field theory then factorizes the quadratic operator $\mathbf{S}^2$ the same way it factorizes any product of two operators, $\mathbf{S}^2 \approx 2\langle \mathbf{S} \rangle \cdot \mathbf{S} - \langle \mathbf{S} \rangle^2$, and the resulting exchange field points along $\mathbf{m} \equiv \langle \mathbf{S} \rangle$, in *whichever* direction that expectation value happens to be nonzero. Choosing $\langle c^\dagger_\uparrow c_\uparrow \rangle \neq \langle c^\dagger_\downarrow c_\downarrow \rangle$, as in (6.2), picks out $m_z \neq 0$; choosing instead $\langle c^\dagger_\uparrow c_\downarrow \rangle \neq 0$ picks out m_x or m_y. Mean-field magnetism is therefore not restricted to the z axis at all: it produces magnets with magnetization along any direction in space, including genuinely non-collinear arrangements where $\mathbf{m}$ points in different directions on different sites. This is exactly the "non-collinear mean-field Hamiltonian" figure on the slides, which the spoken lecture only names in passing.

Mean field is only trustworthy when $U \ll t$; we will see in Section 6.6 that magnetism typically needs the opposite regime, which is precisely why the strongly localized limit of Section 6.7 onward matters so much.

6.4 Numerical mean-field magnetism: exchange splitting and antiferromagnetic order

Let us see the machinery of Section 6.3 actually run. First take the one-dimensional chain (6.1) with a single-site unit cell at 20% filling and solve it self-consistently. The resulting band structure exchange-splits: the spin-up band (occupied up to the chemical potential) sits at lower energy than the spin-down band (empty at this filling), and the splitting between them, the **exchange splitting**, grows with U. Since only spin-up states are occupied, this is a fully polarized ferromagnetic solution with $m \neq 0$.

Now enlarge the unit cell to two sites and work at half filling instead. At $U = 0$ there are two bands (one per site in the cell), crossing at one point: the system is metallic. Turning on $U > 0$ opens a mean-field gap, and in real space the two sublattices develop opposite spin polarization: one site mostly up, its neighbor mostly down. The *net* magnetization is zero,

but the *staggered* magnetization is not: this is antiferromagnetic order, and (as always with mean-field gap opening) it lowers the total electronic energy relative to the ungapped state, because the states that get pushed down in energy by the gap are exactly the occupied ones.

Notebook check-in: `session6.ipynb`, "(Reminder) Bandstructure of a tight binding chain with two sites" and "(Exercise) Magnetic order driven by interactions in a chain"

Start from the two-site, non-interacting band structure (the metallic crossing case), then switch on U and watch the mean-field gap open and the real-space spin pattern become antiferromagnetic, reproducing (6.2)'s exchange splitting directly. The single-site, 20% filling ferromagnetic case from the same section is the companion calculation.

6.5 Ferromagnet versus antiferromagnet: which one wins?

The Hubbard chain (6.1) admits *both* a ferromagnetic self-consistent solution (single-site cell, metallic, gapless) and an antiferromagnetic one (two-site cell, insulating, gapped) at the same filling and the same U. Both are genuine fixed points of the self-consistency loop; only one can be the true ground state, and we settle that by comparing total energies. The gapped antiferromagnetic solution always wins here: opening a gap pushes the occupied single-particle energies down, so the insulating configuration has lower total energy than the gapless ferromagnetic one at every $U > 0$ we can reach numerically in this half-filled chain.

It is worth tracking how large that energy advantage is as a function of U. At $U = 0$ the two configurations are identical, so the energy difference vanishes. As U grows from zero the gap opens and the energy difference grows with it. But then, surprisingly, as U keeps growing the energy difference starts *shrinking* again, and in the $U \to \infty$ limit (equivalently, the limit where the kinetic energy becomes negligible) the two configurations become degenerate. Numerically this energy difference fits a $1/U$ functional form. We are going to rederive this exact $1/U$ scaling analytically in Section 6.8, from a completely independent calculation, as a genuine cross-check between the mean-field numerics and a many-body perturbative treatment.

6.6 A criterion for magnetism: the Stoner threshold

A natural question: does *any* repulsive $U > 0$ produce magnetism, or is there a threshold? Write the mean-field bands as $E_\sigma(\mathbf{k}) = E(\mathbf{k}) - \sigma \frac{U}{2} m$ for $\sigma = \pm 1$ (spin up/down), with m the self-consistent magnetization of (6.2). At zero temperature, self-consistency reads

$$m = \sum_{\mathbf{k}} \Big[n_F\big(E(\mathbf{k}) - \tfrac{U}{2}m\big) - n_F\big(E(\mathbf{k}) + \tfrac{U}{2}m\big) \Big].$$

Expanding the right-hand side for small m gives $m \approx U\, m\, D(E_F)$, where $D(E_F)$ is the density of states at the Fermi level. This has the trivial solution $m = 0$ always, but a *nontrivial* solution branches off exactly when the linear coefficient reaches one:

$$U\, D(E_F) \geq 1. \tag{6.3}$$

This is the Stoner criterion. Below threshold only $m = 0$ is stable, and the system is nonmagnetic no matter how you perturb it; above threshold the nonmagnetic solution becomes unstable and a finite m develops spontaneously. This is exactly why some metals with sizable Coulomb repulsion, aluminum or lead, stay nonmagnetic: their density of states at the Fermi level is too small for the product in (6.3) to clear one. Iron and cobalt, by contrast, combine a large density of states (from narrow d bands) with a sizable effective U, comfortably clearing the threshold. Numerically, this shows up exactly as (6.3) predicts: the self-consistent

magnetization of the chain stays pinned at zero below a critical U, then grows continuously once U crosses that threshold. Figure 6.1 shows both halves of this statement at once: why a nontrivial solution of the self-consistency condition only exists above threshold, and what the resulting magnetization curve looks like.

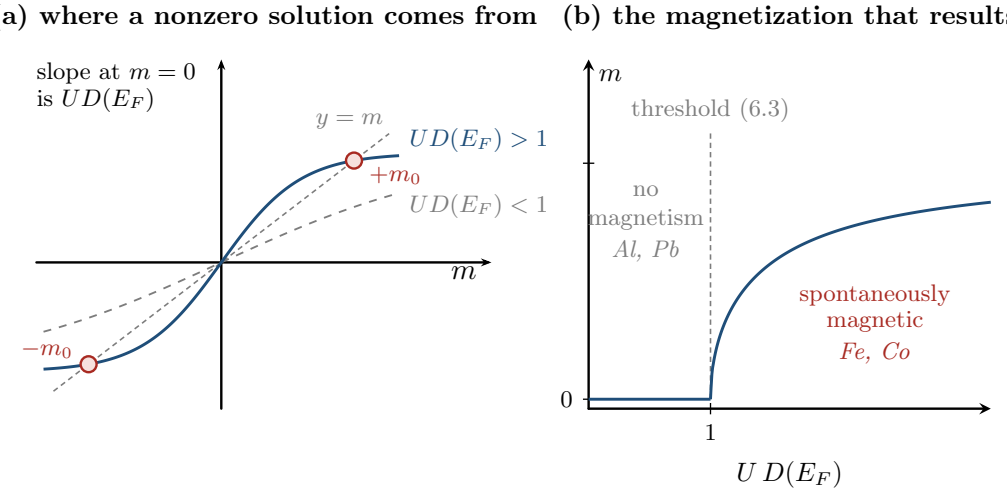


Figure 6.1: The Stoner criterion (6.3), read graphically. (a) The self-consistency condition $m = f(m)$, where f is the right-hand side of the zero-temperature gap equation: an odd function that saturates once the bands are fully polarized, and whose slope at the origin is exactly $UD(E_F)$. If that slope is below one the only crossing with $y = m$ is the trivial $m = 0$; once it exceeds one a pair of nontrivial solutions $\pm m_0$ appears and the nonmagnetic state becomes unstable. (b) The resulting order parameter: pinned at zero until $UD(E_F)$ reaches one, then growing continuously. Whether a metal magnetizes is therefore not decided by U alone but by the *product* of U with the density of states at the Fermi level, which is why aluminium and lead stay nonmagnetic despite sizeable Coulomb repulsion, while the narrow d bands of iron and cobalt clear the threshold comfortably.

Notebook check-in: `session6.ipynb`, "(Exercise) Intermediate coupling regime and Stoner instability"

Reproduce the threshold behavior of (6.3) directly: scan the self-consistent magnetization as a function of U and locate the critical interaction where m first becomes nonzero.

Notebook check-in: `session6.ipynb`, "(Exercise) Magnetism in flat bands"

A flat band has (in principle) a divergent density of states at E_F, so the Stoner product $UD(E_F)$ in (6.3) clears threshold for essentially any $U > 0$. This exercise is not narrated in the transcript, but it is the natural numerical extension of Section 6.6: a flat-band system is the "easiest possible" case for magnetism to appear.

One important consequence closes this section and opens the next: (6.3) tells us that magnetism generically wants $UD(E_F) \gtrsim 1$, and for a reasonable bandwidth that typically means U needs to be comparable to or larger than t, not smaller. But mean-field theory itself is only reliable for $U \ll t$ (Section 6.3). So the regime where magnetism actually shows up is, generically, exactly the regime where the tool we have been using breaks down. We need a genuinely different approach for $U \gg t$, and that is what the rest of this session builds.

6.7 Where mean field breaks down: the strongly localized limit

Let us look again at the half-filled chain from Section 6.4, but now track the magnetization per site as U grows from zero to very large values. At $U = 0$ there is no interaction and $m = 0$. As U increases, m grows, and as $U \to \infty$ it saturates at exactly one Bohr magneton per site: a full, localized spin-1/2 moment on every site, pointing in alternating directions from one sublattice to the next (the "Ising-like" antiferromagnetic state the numerics converge to).

This saturation is telling us something physical: in the large-U limit electrons stop moving altogether. Hopping an electron onto an already-occupied neighboring site costs an energy U, and once $U \gg t$ that cost is prohibitive, so each site simply keeps its one electron, localized, with only its spin free to fluctuate. The right way to organize perturbation theory in this limit is therefore *not* around the kinetic term (as mean field implicitly does), but around localization itself: treat the interacting, localized configuration as the zeroth-order ground state, and treat hopping as a small perturbation on top of it.

6.8 From the Hubbard dimer to the Heisenberg model

Let us make this precise on the smallest possible example: two sites, 0 and 1, hopping t between them, local repulsion U on each, and exactly two electrons total (half filling). The Fock space at this filling has $\binom{4}{2} = 6$ states: four with one electron on each site (each site's electron can independently be up or down) and two with both electrons on the same site (one electron up, one down, by Pauli exclusion).

Group these by energy in the atomic limit ($t = 0$). The four singly-occupied configurations, $|\uparrow,\uparrow\rangle$, $|\downarrow,\downarrow\rangle$, $|\uparrow,\downarrow\rangle$, $|\downarrow,\uparrow\rangle$, cost no interaction energy at all: call this the **low-energy manifold**, at energy 0. The two doubly-occupied configurations, $|\uparrow\downarrow,0\rangle$ and $|0,\uparrow\downarrow\rangle$, each pay the full charging energy U: the **high-energy manifold**. In the low-energy manifold, every state has exactly one electron per site, so the charge degree of freedom is completely frozen; only the spin on each site can still fluctuate. The natural low-energy effective Hamiltonian is therefore a Hamiltonian for spin alone, and rotational invariance fixes its form up to an overall coupling: $H_{\text{eff}} = J\,\mathbf{S}_0 \cdot \mathbf{S}_1$, the two-site Heisenberg Hamiltonian. What remains is to extract J from the microscopic t and U.

Hopping (the perturbation) conserves total spin, so it cannot connect the low-energy manifold to the high-energy one for the *parallel*-spin states at all: applying H_t to $|\uparrow,\uparrow\rangle$ would require moving a spin-up electron onto a site that already holds one, which Pauli exclusion forbids outright. The three parallel/triplet-like states ($|\uparrow,\uparrow\rangle$, $|\downarrow,\downarrow\rangle$, and the symmetric combination $\frac{1}{\sqrt{2}}(|\uparrow,\downarrow\rangle + |\downarrow,\uparrow\rangle)$) are therefore exact eigenstates of the *full* Hamiltonian at energy exactly $E_T = 0$: hopping simply cannot touch them, to any order. The antiparallel/singlet combination $|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow,\downarrow\rangle - |\downarrow,\uparrow\rangle)$ is different: hopping *can* connect it to the (symmetric) doubly-occupied combination $|D\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow,0\rangle + |0,\uparrow\downarrow\rangle)$, with matrix element $\langle D|H_t|S\rangle = -2t$. Restricted to the two-dimensional subspace $\{|S\rangle, |D\rangle\}$, the Hamiltonian is exactly

$$H = \begin{pmatrix} 0 & -2t \\ -2t & U \end{pmatrix},$$

with exact eigenvalues $E_\pm = \frac{U}{2} \pm \sqrt{(U/2)^2 + 4t^2}$. The singlet ground-state energy is the lower root, $E_S = \frac{U}{2} - \sqrt{(U/2)^2 + 4t^2}$; expanding for $U \gg t$,

$$E_S \;\approx\; -\frac{4t^2}{U} + O(t^4/U^3), \qquad E_T = 0 \text{ exactly.}$$

Comparing this to the Heisenberg dimer's exact spectrum, $E_{\text{singlet}} = -\frac{3}{4}J$ and $E_{\text{triplet}} = +\frac{1}{4}J$ (so the singlet-triplet gap is $\Delta E = E_T - E_S = J$), we read off

$$J = \frac{4t^2}{U}. \tag{6.4}$$

$J > 0$: antiferromagnetic, the singlet sits below the triplet. This confirms, with the correct numerical prefactor, the second-order perturbation theory sketched in the lecture (pristine Hamiltonian = the Hubbard term, perturbation = hopping, matrix element t, energy denominator U), and it reproduces exactly the $1/U$ scaling found numerically for the ferromagnet/antiferromagnet energy splitting back in Section 6.5: that mean-field energy difference *is* (twice) the exchange coupling (6.4), so the two completely different calculations, mean-field numerics and many-body perturbation theory, agree. Figure 6.2 collects the whole argument: the Pauli blockade that protects the triplet, the virtual excursion that lowers the singlet, and the exact spectrum whose splitting is J.

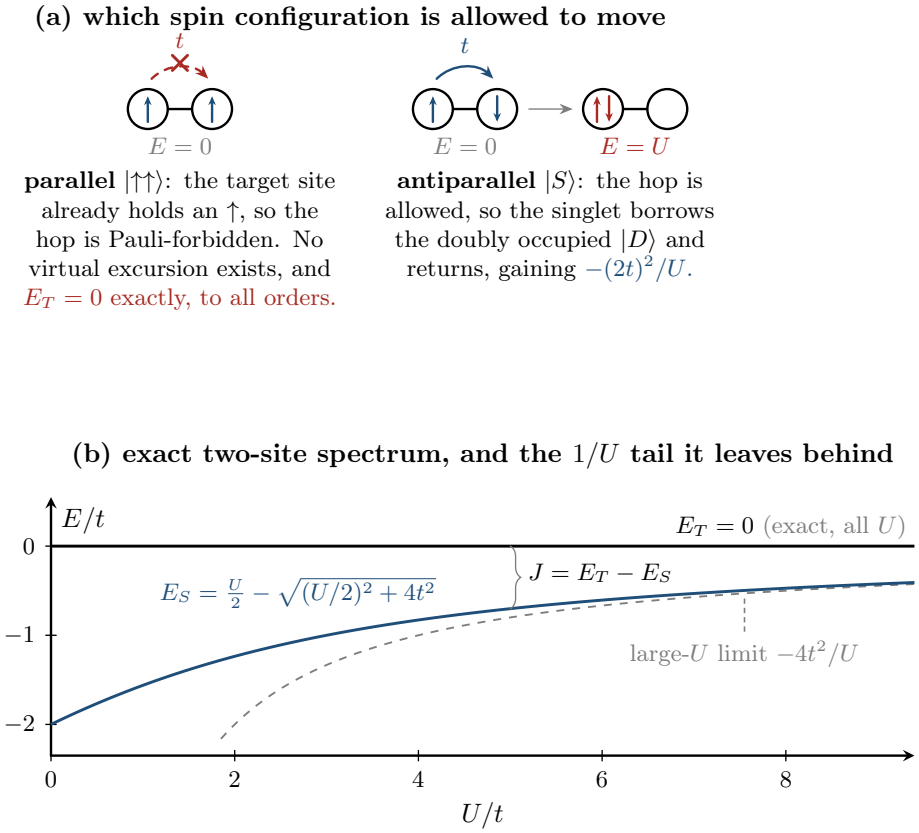


Figure 6.2: Superexchange on the two-site Hubbard model, and why it always comes out antiferromagnetic. (a) With parallel spins the only hop available would put two ↑ electrons on one site, which Pauli exclusion forbids outright: the triplet states cannot lower their energy at any order and sit at exactly $E_T = 0$. With antiparallel spins the hop is allowed, so the singlet borrows a virtual double occupancy of energy U and returns, gaining second-order energy $-4t^2/U$. Magnetic order here is a purely kinetic effect: antiparallel spins are cheaper because they are the only ones allowed to move. (b) The exact singlet energy $E_S = \frac{U}{2} - \sqrt{(U/2)^2 + 4t^2}$ against U/t, with $E_T = 0$ exact for every U. The singlet-triplet splitting *is* the Heisenberg coupling J of (6.4), and it decays as $4t^2/U$: the same $1/U$ law that the mean-field ferromagnet/antiferromagnet energy difference of Section 6.5 was found to obey numerically.

Notebook check-in: `session6.ipynb`, **"(Reminder) The spectra of a dimer," "(Exercise) The interacting Hubbard dimer," "Gap opening in the Hubbard model," and "(Exercise) The strongly interacting limit"**

Diagonalize the six-state Hubbard dimer directly and confirm the singlet-triplet splitting matches (6.4). Then push U/t large and check that the gap and the low-energy spectrum converge to the Heisenberg-model prediction, cross-checking this section's exact result against brute-force diagonalization.

6.9 The Heisenberg model on a general lattice

Nothing in Section 6.8's argument used the fact that we had only two sites. For a generic lattice with hopping t_{ij} between sites i and j and local interaction U, the same second-order argument applies to every bonded pair independently, in the limit $U \gg t_{ij}$, giving the generalized Heisenberg Hamiltonian

$$H_{\text{Heis}} = \sum_{\langle ij \rangle} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j, \qquad J_{ij} = \frac{4t_{ij}^2}{U}. \tag{6.5}$$

Because $J_{ij} \propto t_{ij}^2$, this superexchange coupling derived directly from a single-orbital Hubbard model is always positive, antiferromagnetic, regardless of the sign or geometry of t_{ij}. More general (non-Hubbard, multi-orbital) microscopic models can and do produce couplings of either sign; we will see explicitly in Section 6.11 how ferromagnetic coupling arises once a second, intermediate orbital gets involved. The sign convention throughout is: $J_{ij} > 0$ favors antiparallel neighboring spins (antiferromagnetic), $J_{ij} < 0$ favors parallel neighboring spins (ferromagnetic). Treating the spins in (6.5) as classical vectors already reproduces both classical ground states on a two-site bond: antiparallel for $J > 0$, parallel for $J < 0$.

6.10 Bipartite order versus geometric frustration

On a **bipartite** lattice, one where the sites split cleanly into two sublattices A and B such that every bond connects an A site to a B site (square, honeycomb), the antiferromagnetic Heisenberg model (6.5) has an obvious classical ground state: all spins on A point one way, all spins on B point the opposite way, satisfying every single bond's preference for antiparallel neighbors simultaneously. This is exactly the two-sublattice, collinear antiferromagnetic order we already saw appear at the mean-field level in Section 6.4, and (6.5) explains *why*: in the large-U limit the mean-field Hubbard model on a bipartite lattice literally reduces to this Heisenberg model.

Frustrated lattices are different. On a **non-bipartite** lattice, kagome or triangular being the standard examples, there is no way to two-color the sites so that every bond connects opposite colors; a triangular plaquette, for instance, has three mutually neighboring sites, and no assignment of up/down spins can make all three bonds antiparallel simultaneously. The antiferromagnetic tendency of $J_{ij} > 0$ is still there on every individual bond, but it cannot be satisfied everywhere at once: this is **geometric frustration**. The system's best compromise is typically a noncollinear arrangement (classic case: spins at 120° to each other on a triangular plaquette) that maximizes the total antiferromagnetic character bond by bond, without fully satisfying any single bond. This noncollinear character shows up directly in the mean-field solution of the corresponding electronic Hamiltonian too, not just in the spin-model picture. Figure 6.3 places the bipartite case, the frustrated collinear attempt and the 120° compromise side by side, with the classical bond energies that decide between the last two.

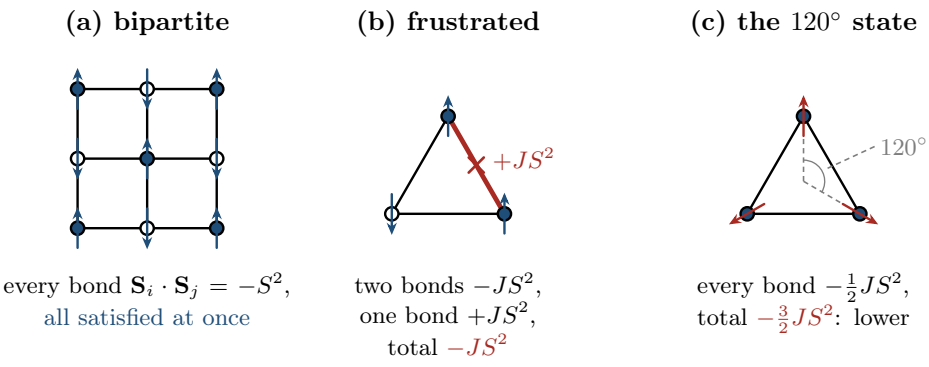


Figure 6.3: Why geometry decides whether an antiferromagnet can order. (a) On a bipartite lattice the sites two-colour, every bond joins the two colours, and the Néel state of Section 6.10 satisfies all of them at once. (b) On a triangle no two-colouring exists: fix two spins antiparallel and the third ends up parallel to one of its neighbours whatever you choose, leaving one bond frustrated. Treating the spins classically, that collinear compromise pays $+JS^2$ on the offending bond and totals $-JS^2$ per triangle. (c) The system does better by giving up on satisfying any single bond fully: three spins at 120° give $\mathbf{S}_i \cdot \mathbf{S}_j = -S^2/2$ on *every* bond and a total of $-\frac{3}{2}JS^2$, half again lower than (b). This is the origin of the noncollinear order that the mean-field solution of the corresponding electronic Hamiltonian also converges to.

Notebook check-in: `session6.ipynb`, "(Exercise) Magnetic instabilities of the square lattice" and "(Exercise) Non-collinear magnetic order from geometric frustration"

Compare a bipartite lattice (square) against a frustrated one (triangular or kagome) at the mean-field level: the square lattice should converge to the collinear two-sublattice order of Section 6.10, while the frustrated lattice should converge to a noncollinear compromise, both traceable back to (6.5)'s sign convention on each bond.

6.11 Where ferromagnetic coupling comes from: superexchange

Section 6.9's direct t^2/U mechanism is always antiferromagnetic, yet plenty of real magnetic materials are ferromagnets. So ferromagnetic coupling needs a different microscopic route: it typically requires an intermediate atom (a ligand) sitting between the two magnetic ions rather than a single direct hop. Picture two transition-metal ions, cobalt, iron, or nickel say, bridged by a ligand such as oxygen. Electrons can, as before, hop directly between the two metal sites, and that channel is antiferromagnetic exactly as in Section 6.9. But there is a second virtual process available: an electron hops from the ligand onto one metal ion, and another electron hops from the ligand onto the other, leaving a virtual state with two electrons (a full shell) on the ligand itself. Depending on which d orbitals are involved and the metal-ligand-metal bond angle, the energy correction from this virtual ligand-double-occupation channel can favor the ferromagnetic configuration over the antiferromagnetic one, flipping the net sign of the effective coupling. The systematic version of this reasoning is the **Goodenough-Kanamori rules**, which predict the sign of the exchange coupling from orbital filling and bond geometry, and which can also be derived systematically from self-consistent multi-orbital calculations (density functional theory, for instance). This is slides-and-transcript material with no dedicated notebook exercise in this session; the notebook's exercises stay within the single-orbital Hubbard model.

6.12 Anisotropic exchange: Dzyaloshinskii-Moriya, Ising, and Kitaev terms

Everything in (6.5) assumed the coupling is an isotropic scalar product, $\mathbf{S}_i \cdot \mathbf{S}_j$, which is what rotational (spin) symmetry forces in the absence of spin-orbit coupling. Once spin-orbit coupling enters the underlying electronic Hamiltonian, additional, less symmetric spin couplings become allowed. Three appear repeatedly in real materials:

Antisymmetric (Dzyaloshinskii-Moriya) exchange,

$$H_{\mathrm{DM}} = \sum_{\langle ij \rangle} \mathbf{D}_{ij} \cdot \left(\mathbf{S}_i \times \mathbf{S}_j \right),$$

a cross product rather than a dot product. It requires broken mirror symmetry (which is exactly why it is prominent at surfaces and interfaces, where the substrate itself breaks that mirror), and it favors neighboring spins rotated roughly 90° relative to each other, promoting noncollinear, often spiral, order.

Anisotropic (Ising-like) exchange, a single-component coupling such as $\Lambda\, S_i^z S_j^z$, which promotes an easy axis (or, with the opposite sign, an easy plane): the system magnetizes more readily along one direction than another.

Kitaev interaction, bond-dependent coupling on a honeycomb-like lattice: an $S_i^x S_j^x$ term on one bond direction, $S_i^y S_j^y$ on a second, $S_i^z S_j^z$ on a third,

$$H_{\mathrm{Kitaev}} = \sum_{\langle ij \rangle \in x\text{-bonds}} K\, S_i^x S_j^x + \sum_{\langle ij \rangle \in y\text{-bonds}} K\, S_i^y S_j^y + \sum_{\langle ij \rangle \in z\text{-bonds}} K\, S_i^z S_j^z .$$

Even on a bipartite honeycomb lattice, where geometry alone would not frustrate anything, this bond-dependence is itself intrinsically frustrating, since no single spin orientation can simultaneously satisfy all three bond types at once (figure 6.4). This is the interaction that makes the Kitaev model a paradigm for quantum spin liquids, which is why it is flagged here explicitly as important groundwork for the frustrated-magnet material picked up in sessions 7 and 8.

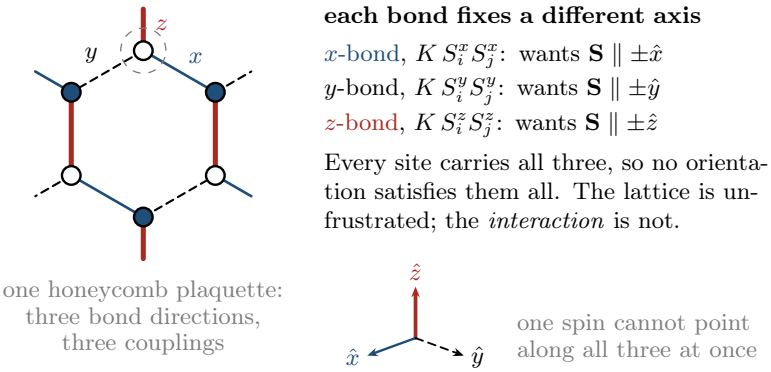


Figure 6.4: The Kitaev interaction and the second kind of frustration. The honeycomb lattice is bipartite: its sites two-colour (filled/open) exactly as in figure 6.3(a), so an ordinary Heisenberg antiferromagnet on it is not frustrated at all. The Kitaev coupling frustrates it anyway, because each of the three bond directions couples only *one* spin component: the x-bond cares only about S^x and so wants both of its spins along $\pm\hat{x}$, the y-bond wants $\pm\hat{y}$, the z-bond wants $\pm\hat{z}$. Every site sits on one bond of each type, and no single orientation can satisfy three mutually orthogonal demands at once. This is frustration built into the interaction rather than into the geometry, and it is what makes the Kitaev model a paradigm for quantum spin liquids in sessions 7 and 8.

Notebook check-in: `session6.ipynb`, "(Exercise) Non-collinear magnetism from spin-orbit coupling"

Add a Dzyaloshinskii-Moriya-type term to the mean-field Hamiltonian and watch the ground state tilt away from collinear order, the electronic counterpart of the spin-model statement above.

6.13 Magnons: fluctuations of the order parameter

Up to this point we have only asked about ground states. Let us now ask about excitations. Take the Heisenberg model (6.5) with $J < 0$ (ferromagnetic), promote the spins to genuine quantum-mechanical operators obeying the angular-momentum algebra $[S_i^a, S_j^b] = i\hbar\,\delta_{ij}\epsilon^{abc}S_i^c$ (we set $\hbar = 1$ from here on, standard for this kind of calculation), and ask for its low-lying excited states. Unlike the interacting electron problem, the ferromagnetic Heisenberg model's exact ground state is known immediately: it is the fully polarized state, every spin aligned, since no other configuration can lower $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$ any further for $J < 0$. A **magnon** is a small, propagating fluctuation of the magnetization away from this fully polarized state: qualitatively, a single spin tipped slightly out of alignment, delocalized coherently over the whole lattice as a wave. The question is how to extract these excitations systematically from a genuinely many-body spin Hamiltonian, and the answer is a change of variables that turns the spin problem into a bosonic tight-binding problem.

6.14 The Holstein-Primakoff transformation

The **Holstein-Primakoff transformation** maps spin operators exactly onto bosonic creation and annihilation operators $a_i^\dagger, a_i$ with $[a_i, a_j^\dagger] = \delta_{ij}$. For a spin of magnitude S on site i,

$$S_i^z = S - a_i^\dagger a_i, \qquad S_i^+ = \sqrt{2S - a_i^\dagger a_i}\; a_i, \qquad S_i^- = a_i^\dagger \sqrt{2S - a_i^\dagger a_i}. \tag{6.6}$$

This is an *exact* rewriting: substituting (6.6) into the bosonic commutation relations reproduces the full $SU(2)$ spin algebra $[S_i^z, S_i^\pm] = \pm S_i^\pm$, $[S_i^+, S_i^-] = 2S_i^z$ exactly, with no approximation yet made. The physical picture behind it is direct: $a_i^\dagger a_i$ counts how many units of spin have been removed from the fully polarized value S on site i, i.e. it counts **magnons** on that site, and $S_i^z = S - a_i^\dagger a_i$ says exactly that. The square-root operator factors are what make (6.6) exact even far from full polarization; they are also what makes the Hamiltonian, written this way, look more complicated rather than less.

The simplification comes from linearizing: whenever the expectation value of $a_i^\dagger a_i$ stays much smaller than S, i.e. whenever we stay close to the fully polarized ground state (precisely the regime we want, since we are looking for low-lying excitations above that ground state), we can expand the square roots and keep only the leading term,

$$S_i^+ \approx \sqrt{2S}\, a_i, \qquad S_i^- \approx \sqrt{2S}\, a_i^\dagger, \qquad S_i^z = S - a_i^\dagger a_i. \tag{6.7}$$

This linearized form is the one used to compute magnon spectra in practice. The bookkeeping it implies is worth stating plainly, since it is the cleanest way to remember the whole transformation: S^+ (which *raises* S^z, i.e. increases the local magnetization back toward saturation) *destroys* a magnon, while S^- (which lowers the magnetization) *creates* one; the total magnetization of the system is the fully saturated value minus the total magnon number, $\sum_i \langle S_i^z \rangle = NS - \sum_i \langle a_i^\dagger a_i \rangle$. Each individual magnon carries exactly one unit of spin, so magnons are $S = 1$ excitations, consistent with what we said about them back in Section 6.1, and they exist only because the ground state they live on top of has already spontaneously broken spin-rotation symmetry. Figure 6.5 draws all three pieces: the fully polarized ground

state, the delocalized precession wave that carries a single magnon, and this raising/lowering dictionary.

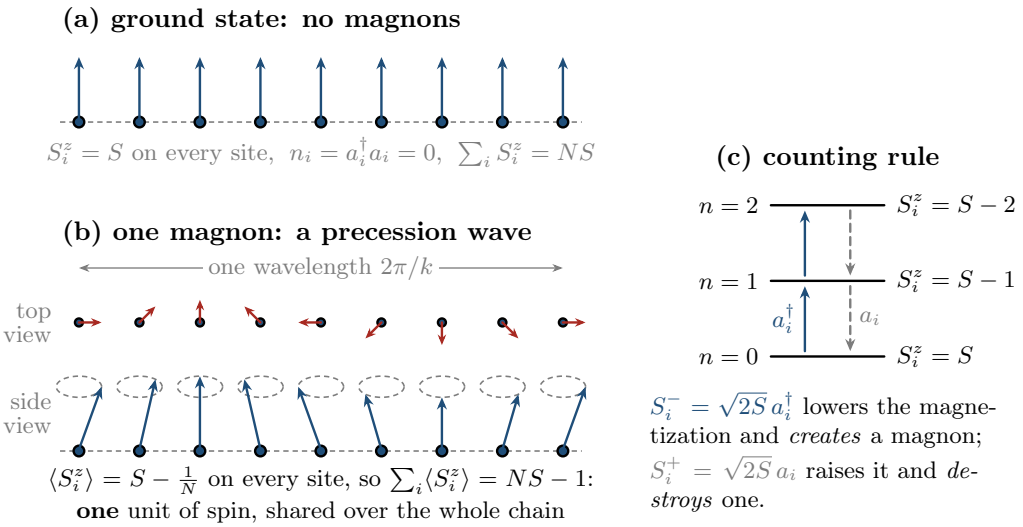


Figure 6.5: What (6.6)–(6.7) actually describe. (a) The ferromagnetic ground state: every spin at its maximal $S^z = S$, and therefore zero magnons. (b) A single magnon is *not* one spin flipped over on one site. It is every spin tipped slightly out of alignment and precessing on its cone, with the precession phase advancing by $\mathbf{k} \cdot \boldsymbol{\delta}$ from site to site (seen most clearly in the top view, where the transverse component turns steadily and closes on itself after one wavelength). The missing unit of spin is therefore shared coherently across the whole lattice: $\sum_i \langle S_i^z \rangle = NS - 1$. Because the tipping angle is small, $\langle a_i^\dagger a_i \rangle \ll S$ everywhere, which is exactly the condition under which the square roots in (6.6) may be expanded; the tipping angle is drawn far larger here than it really is. (c) The counting rule that makes the transformation easy to remember: the boson number counts spin removed from saturation, so the *lowering* operator S^- creates a magnon and the *raising* operator S^+ destroys one.

Substituting (6.7) into the ferromagnetic Heisenberg Hamiltonian, $H = J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j = J \sum_{\langle ij \rangle} \big[S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+) \big]$, and keeping terms up to quadratic order in the bosonic operators,

$$S_i^z S_j^z \approx S^2 - S\big(a_i^\dagger a_i + a_j^\dagger a_j\big), \qquad \tfrac{1}{2}\big(S_i^+ S_j^- + S_i^- S_j^+\big) \approx S\big(a_i^\dagger a_j + a_j^\dagger a_i\big),$$

converts the spin Hamiltonian into a bosonic tight-binding Hamiltonian for magnons, whose hopping amplitude is set directly by the exchange coupling J, exactly as advertised: the magnon "hops" between sites with amplitude proportional to JS, in precise analogy with the electronic tight-binding Hamiltonians we have diagonalized all course.

6.15 Magnon dispersions: ferromagnets versus antiferromagnets

Fourier transforming the bosonic Hamiltonian from Section 6.14 on a Bravais lattice with coordination number z and nearest-neighbor structure factor $\gamma(\mathbf{k}) = \frac{1}{z} \sum_{\boldsymbol{\delta}} e^{i\mathbf{k}\cdot\boldsymbol{\delta}}$ gives, for the **ferromagnet** ($J < 0$, writing $|J| = -J$), a diagonal magnon Hamiltonian with dispersion

$$E_{\mathrm{FM}}(\mathbf{k}) = |J|\, S\, z\big(1 - \gamma(\mathbf{k})\big). \tag{6.8}$$

This vanishes at $\mathbf{k} = 0$, as it must (a uniform rotation of every spin costs no energy at all, the Goldstone mode of the broken spin-rotation symmetry), and expanding $\gamma(\mathbf{k})$ for

small $\mathbf{k}$ gives $E_{\mathrm{FM}}(\mathbf{k}) \propto k^2$: a **quadratic** magnon dispersion, exactly the "gapless, quadratic" ferromagnetic magnon the lecture flags.

The **antiferromagnetic** case needs one more step, because the ground state is not simply "all spins up." On a bipartite lattice, apply the Holstein-Primakoff transformation (6.7) separately on the two sublattices, with the roles of S^+ and S^- swapped on the B sublattice (since those spins point the opposite way): $S_i^z = S - a_i^\dagger a_i$ on A, but $S_j^z = -S + b_j^\dagger b_j$ on B. Substituting into $\mathbf{S}_i \cdot \mathbf{S}_j$ for a bond between the two sublattices now produces not just density terms $a^\dagger a$, $b^\dagger b$, but also *anomalous*, particle-number-non-conserving terms $a_i b_j$ and $a_i^\dagger b_j^\dagger$: a signature that the magnon vacuum on top of the Néel state is not simply the bosonic vacuum, and that diagonalizing the Hamiltonian requires a Bogoliubov transformation rather than a plain Fourier transform. Carrying that out gives the antiferromagnetic magnon dispersion

$$E_{\mathrm{AFM}}(\mathbf{k}) = J\,S\,z\,\sqrt{1-\gamma(\mathbf{k})^2}. \tag{6.9}$$

Near $\mathbf{k} = 0$, $1 - \gamma(\mathbf{k})^2 \propto k^2$, so $E_{\mathrm{AFM}}(\mathbf{k}) \propto |k|$: a **linear**, Dirac-cone-like dispersion, gapless at the same point but linear rather than quadratic. This is exactly the "quadratic in ferromagnets, linear in antiferromagnets" rule of thumb from the lecture, now derived rather than merely quoted; it is also, incidentally, one more example (alongside the Dirac cones of session 2 and 3) of a linear dispersion arising from the structure of a symmetry rather than being put in by hand. Figure 6.6 plots (6.8) and (6.9) together, with the zone centre magnified so the parabola and the cone can be told apart.

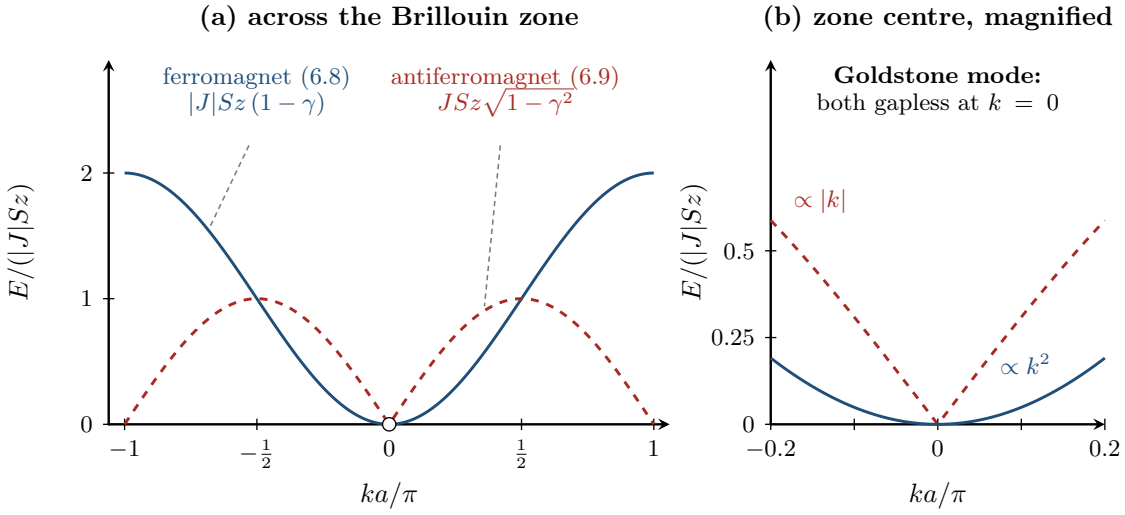


Figure 6.6: Magnon dispersions (6.8) and (6.9) for a chain ($z = 2$, $\gamma(\mathbf{k}) = \cos ka$), both in units of $|J|Sz$. Both vanish at $\mathbf{k} = 0$: a uniform rotation of every spin costs nothing, so the magnon is the Goldstone mode of the broken spin-rotation symmetry and no amount of exchange can gap it. What differs is *how* they vanish. The ferromagnet's $1 - \gamma(\mathbf{k})$ is quadratic in k; the antiferromagnet's $\sqrt{1-\gamma(\mathbf{k})^2}$ is the square root of the same quantity and so is linear, a Dirac-like cone rather than a parabola. The right panel magnifies the zone centre, where the difference is unmistakable. That extra square root is precisely the fingerprint of the Bogoliubov transformation the Néel state forced on us: the anomalous $a_i b_j$ terms are what turn a parabola into a cone.

6.16 Topological magnons

Once magnons have a genuine band structure, a dispersion $E(\mathbf{k})$ over the Brillouin zone, the same question we have already asked twice about *electronic* bands becomes available to ask about *magnon* bands: can that band structure carry a nontrivial topological invariant? This

is precisely the Chern-number machinery from sessions 3 and 4 (Berry curvature $\Omega(\mathbf{k})$ of a Bloch band, integrated over the Brillouin zone to give an integer,

$$C = \frac{1}{2\pi} \int_{\mathrm{BZ}} \Omega(\mathbf{k})\, d^2k, \tag{6.10}$$

applied a third time, now to the bosonic Bogoliubov Hamiltonian of Section 6.15 instead of an electronic tight-binding Hamiltonian. Adding spin-orbit coupling on top of the plain Heisenberg exchange, of the Dzyaloshinskii-Moriya type from Section 6.12, plays exactly the role that the Haldane model's complex second-neighbor hopping played for electrons back in session 4: it introduces complex phases into the magnon hopping amplitudes, which can open a gap in the magnon band structure at finite energy and endow the bands on either side of that gap with a nonzero Chern number (6.10). When such a topological magnon Hamiltonian is put on a ribbon geometry, bulk-boundary correspondence guarantees chiral magnon edge modes living in the gap, propagating in one direction only along each edge, the magnon analogue of the chiral electronic edge states of a Chern insulator. These modes carry spin and energy, not charge, but they are topologically protected in exactly the same sense as their electronic counterparts: the same tool we used for electrons in sessions 3 and 4, now applied to magnon bands.

Notebook check-in: `session6.ipynb`, "(Exercise) Topological magnons in 2D magnets"

Build a magnon Hamiltonian with a Dzyaloshinskii-Moriya-type spin-orbit term on top of Heisenberg exchange, compute its Chern number via (6.10), and check for chiral edge magnon modes on a ribbon, directly reproducing the bulk-boundary correspondence argument above in the magnon setting.

A brief note on scope before we stop: the companion notebook goes further than this session's lecture in a few places, most visibly its exercises on electronic and magnetic symmetry breaking in graphene multilayers, and its closing sections on many-body spin response functions. Those are genuine, worthwhile extensions of today's material, but since neither the transcript nor the slides for this session narrate them, we have not built the corresponding derivations into these notes; treat them as independent notebook exploration rather than as material these notes derive.

6.17 Closing summary

So far in this session we have:

- Rebuilt Hubbard-model magnetism numerically in full depth, including the direct competition between ferromagnetic and antiferromagnetic self-consistent solutions and the Stoner threshold (6.3) that decides whether magnetism appears at all.
- Derived the Heisenberg model rigorously from the strongly correlated Hubbard limit, getting the exact coefficient $J = 4t^2/U$ (6.4) and its lattice generalization (6.5), including how frustration and superexchange modify that simple picture.
- Built the magnon spectrum from first principles via the Holstein-Primakoff transformation (6.6)–(6.7), getting the quadratic ferromagnetic dispersion (6.8) and linear antiferromagnetic dispersion (6.9), and then extended session 3 and 4's topological machinery a third time to show that magnon bands can be topological too.

And that is exactly where the live lecture leaves us: right after topological magnons, with no narrated take-home slide. This is a genuine feature of this year's recording, not a gap in

these notes, so we end here too rather than inventing a summary sentence the lecture never says. Every magnet we have built today, from the mean-field exchange field all the way to the topological magnon edge states, is a state that breaks time-reversal symmetry: some net, staggered, or noncollinear spin arrangement that looks different run backwards in time from how it looks run forwards.

Two threads are left dangling on purpose, both flagged along the way rather than resolved here. The Kitaev interaction of Section 6.12, singled out explicitly as "important for quantum spin liquids," is picked up from two different directions in the sessions ahead: session 7 treats the frustrated, kagome-type Hamiltonians it belongs to as the paradigm case where mean-field theory simply fails, while session 8 returns to the Kitaev model itself as a route to an exactly solvable quantum spin liquid. And the spinon, previewed only by name and by its spin-1/2, charge-neutral quantum numbers back in Section 6.1, gets the same full treatment magnons received today, only in session 8, once we turn from magnets that break time-reversal symmetry to quantum magnets that do not.

Session 7

Quantum Many-Body Entanglement and Tensor Networks

Session 6 ran the Hubbard-model magnetism of session 1 through in full numerical depth, built up the Heisenberg-model strong-coupling limit and Holstein-Primakoff magnons, and applied the topological machinery of sessions 3 and 4 a third time, now to magnon bands. Along the way it dropped a Kitaev- interaction aside flagged as important for quantum spin liquids, materials like herbertsmithite, whose local moments sit on a kagome lattice and refuse to order even at zero temperature, and it stopped right there, on "topological magnons," without circling back for a closing summary. That abrupt stop is not a mid-sentence handoff into today; it is simply a session that ran out of material. But the thread it leaves dangling is exactly where we pick up. Unlike a ferromagnet or an antiferromagnet, a quantum spin liquid does not break time-reversal symmetry: its spins keep fluctuating, locally, no matter how cold we make the system, and the ground state ends up being one enormous entangled object rather than a simple pattern of arrows. That is precisely the kind of Hamiltonian on which every tool this course has built so far quietly breaks down. Mean-field theory, our workhorse since session 1, is a search for a symmetry-broken order parameter: a nonzero magnetization, a nonzero charge imbalance, a nonzero pairing amplitude. A quantum spin liquid has none of these by definition, so self-consistency has nothing to converge on and simply hands back zero, telling us nothing at all about the actual ground state underneath. The same failure shows up in a second family of materials, fractional Chern insulators, where the interactions are again too strong for any effective single-particle description to capture what is going on.

Today we build the tool that attacks this regime head-on rather than presupposing a broken symmetry: tensor networks, and specifically matrix product states (MPS). This is mature technology; it already produces the leading quantitative evidence for spin liquid physics on the triangular and kagome lattices, the very Hamiltonians session 6 teased. Getting there means first noticing something that has been sitting underneath the last several sessions without being said explicitly, that fermionic and spin Hamiltonians are literally the same physics in two languages (the Jordan-Wigner bridge), and then asking a question that turns out not to be "how do we diagonalize a huge matrix" but "how do we even *store* a many-body wave function in the first place." Entanglement, and a quantity called the entanglement entropy, is the right lens for answering that question, and it is where the bulk of today's session lives.

7.1 Two regimes of many-body physics

Let us return to the generic interacting Hamiltonian that has organized the whole course,

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j \;+\; \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j c_k^\dagger c_l. \tag{7.1}$$

The kinetic term is free and exactly solvable; the interaction term is, in general, not. Every session so far has attacked (7.1) the same way: decouple the quartic term by mean field, replace it with an effective bilinear Hamiltonian, and read off whatever symmetry the resulting order parameter breaks. That strategy defines what we can call **weakly correlated matter**: magnets, superconductors, and charge density waves are all, in this sense, single-particle physics in disguise, once the right symmetry-broken background is found.

Today we are explicitly in the opposite regime, **strongly correlated matter**, where interactions are strong enough that no effective single-particle Hamiltonian reproduces the ground state, no matter how we choose the decoupling. Quantum spin liquids and fractional Chern insulators are our two standing examples, and more generally, getting an accurate phase diagram of a genuinely interacting model runs into the same wall. In both cases we cannot shortcut the problem with an ansatz for what symmetry breaks; we need something closer to an exact numerical solution, or at least a controlled approximation that does not assume symmetry breaking from the outset. That is exactly what tensor networks provide.

7.2 Fermions and spins are the same physics: the Jordan-Wigner bridge

Before building the tool, it is worth noticing something that has been sitting underneath the last several sessions without being said explicitly. We have used two apparently different families of many-body Hamiltonian: fermionic Hamiltonians of the form (7.1), with density-density interactions V_{ijkl}, in sessions 1 through 5, and spin Hamiltonians, above all the Heisenberg model $H = \sum_{ij} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j$, in session 6's magnon discussion and again today for quantum spin liquids. These are not two different subjects; they are the same physics written in two different languages, and today is where we make that translation precise.

Look first at a single site. A spinless fermionic site has a two-dimensional local Hilbert space, empty or full; a spin-1/2 site also has a two-dimensional local Hilbert space, down or up. The obvious guess is $c_i^\dagger \leftrightarrow S_i^+$ and $c_i \leftrightarrow S_i^-$, and on a single site this guess is exactly right. It fails, however, the moment we have more than one site, because fermionic many-body wave functions are antisymmetric under particle exchange while spin many-body wave functions are symmetric (bosonic): the naive identification gets the relative signs between sites wrong. Fixing this requires attaching to each site an extra, occupation-dependent phase, a "string" running over every site to its left, so that exchanging two fermions produces exactly the minus sign the spin operators do not have on their own. This is the **Jordan-Wigner transformation**:

$$S_i^+ = c_i^\dagger K_i, \qquad S_i^- = c_i K_i, \qquad S_i^z = n_i - \tfrac{1}{2}, \qquad K_i = \exp\left(i\pi \sum_{j<i} n_j \right), \tag{7.2}$$

with $n_j = c_j^\dagger c_j$ the fermion number operator on site j. The string operator K_i is built entirely from occupation numbers to the left of site i, so it commutes with everything local to sites $j \geq i$ and simply counts, mod 2, how many fermions sit to the left. One can check directly that (7.2), built this way, reproduces the full $SU(2)$ commutation algebra of spin-1/2 operators on every site and between every pair of sites, using only the fermionic anticommutation relations for $c_i, c_i^\dagger$ plus the string. The slogan the lecture uses for this is exactly right: *spin equals fermion times string.* Figure 7.1 shows what that string actually does: it counts

the fermions sitting to the left of site i and returns their parity, so changing a single occupation anywhere in that region flips the sign of S_i^+.

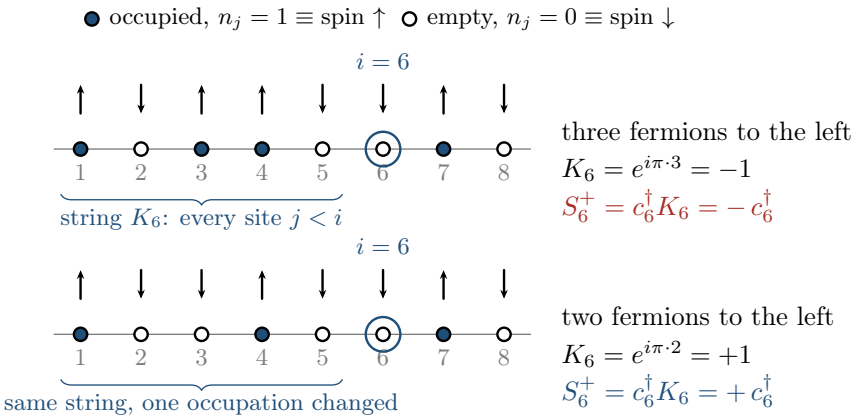


Figure 7.1: Spin equals fermion times string. Site by site the dictionary is trivial: an empty site is a down spin, an occupied site an up spin, $S_i^z = n_i - \frac{1}{2}$. It is the *relative* signs between sites that the naive identification gets wrong, because fermion operators on different sites anticommute while spin operators on different sites commute. The string K_i of (7.2) repairs exactly this: it contributes a factor -1 once for every fermion sitting to the left of site i. The two chains drawn here differ only in the occupations to the left of the marked site $i = 6$, three fermions above and two below, and that parity alone flips the sign of $S_6^+ = c_6^\dagger K_6$. The string is genuinely nonlocal, and what it carries is the fermionic exchange sign that spins do not have on their own.

This is the course's one explicit bridge between the two languages we have used throughout: the fermionic operators of sessions 1 through 5 and the spin operators of session 6 and (as we will see) session 8 are related by (7.2), not just analogous to each other. It also explains a genuinely practical choice we make for the rest of today: since interacting fermionic and interacting spin Hamiltonians are mathematically equivalent, we are free to work entirely in the spin representation from here on, and we will, because spin wave functions are symmetric and that symmetry is exactly what makes the entanglement bookkeeping below clean.

7.3 The real bottleneck: storing the wave function

With the spin language settled, let us ask directly why a generic interacting many-body Hamiltonian is hard. It is tempting to say "because diagonalizing a huge matrix is slow," but that is not quite the heart of the problem; the heart of the problem is more basic than diagonalization, it is simply *storage*.

Take the simplest nontrivial case, the antiferromagnetic Heisenberg dimer, $H = \mathbf{S}_0 \cdot \mathbf{S}_1$ on two spin-1/2 sites. Its ground state is the unique singlet,

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|{\uparrow\downarrow}\rangle - |{\downarrow\uparrow}\rangle \right), \tag{7.3}$$

a state that breaks no symmetry (every single-site spin expectation value vanishes) and is, as we will make precise shortly, maximally entangled. The question of the whole session is how this kind of state generalizes to a macroscopic system with many sites.

Write a general spin-1/2 wave function in the natural product basis, one up/down label s_i per site:

$$|\Psi\rangle = \sum_{s_1,\ldots,s_L} C(s_1,\ldots,s_L)\, |s_1,\ldots,s_L\rangle, \qquad s_i \in \{\uparrow,\downarrow\}. \tag{7.4}$$

For $L = 2$ there are four basis states, up-up, up-down, down-up, down-down, so four coefficients C; the Hamiltonian is a 4×4 matrix. Add one site, $L = 3$, and the count doubles to eight. Add another, $L = 4$, and it doubles again to sixteen. In general the dimension of the Hilbert space, and hence the number of coefficients $C(s_1, \ldots, s_L)$ needed to write down the ground state exactly, is $d = 2^L$: an exponential wall. Even if we never build the full $2^L \times 2^L$ Hamiltonian matrix and only ever generate matrix elements on demand, we still cannot avoid storing the 2^L coefficients of the ground state itself once we have found it. That storage cost, not the cost of diagonalization, is the actual obstruction, and it is the single most important point of the whole session. Tensor networks are one answer to this storage question; the lecture also names, in passing, **neural network quantum states** as a second, machine-learning-imported answer to the exact same question, but does not develop that second tool this session. Session 10 returns to neural network quantum states in depth, from the machine-learning side; today's material is entirely about the first answer, tensor networks, and specifically matrix product states.

Notebook check-in: `session7.ipynb`, "Energy of the Heisenberg model as a function of the system size" (Core)

This exercise has you compute the Heisenberg-chain ground-state energy per site by exact diagonalization as you grow the system, using the `dmrgpy` library (new for this session, alongside `pyqula`). Watch how quickly exact diagonalization stops being feasible; that wall is exactly the 2^L scaling of (7.4), made concrete rather than abstract.

7.4 The matrix product state ansatz

The proposal that fixes this is disarmingly simple. Instead of storing $C(s_1, \ldots, s_L)$ as one enormous array of 2^L numbers, write it as a product of matrices, one matrix per possible value of each local spin:

$$C(s_1, s_2, \ldots, s_L) \;\approx\; M_1^{s_1} M_2^{s_2} \cdots M_L^{s_L}, \tag{7.5}$$

where each $M_i^{s_i}$ is an $M \times M$ matrix (a vector at the two open ends, for a finite open chain), M the **bond dimension**, a free parameter we choose. There are two matrices per site for a spin-1/2 chain, one for $s_i = \uparrow$ and one for $s_i = \downarrow$, so the total number of parameters needed to specify the state is proportional to $L \times M^2$: linear in the system size, in stark contrast to the 2^L of the exact representation (7.4). This is the **matrix product state** (MPS) ansatz. It is, on its face, just a choice, and a fairly arbitrary-looking one; the two questions we spend the rest of Sections 7.5 through 7.8 answering are why it is a *good* choice, and what exactly it is capturing (and failing to capture) about the physical state.

It is worth knowing up front that this is not a toy proposal. Matrix product states currently give the most accurate available results for two of the paradigmatic hard problems in quantum materials: the doped two-dimensional Hubbard model with second-neighbor hopping (implicated in unconventional superconductivity), and the two-dimensional frustrated Heisenberg model on the triangular and kagome lattices, which is where the strongest numerical evidence for genuine quantum spin liquid physics, session 6's herbertsmithite teaser, actually comes from. Both examples are shown as labeled figure panels on slide 15 of the deck ("Solving the 2D Hubbard model at finite doping" and "Solving the 2D Heisenberg model in frustrated lattices"), underscoring that this is squarely a working tool for the strongly correlated regime, not just a formal device.

7.5 Measuring entanglement: the entanglement entropy

To see why (7.5) works, we need a quantitative handle on how entangled a many-body state actually is. Split the system into two complementary pieces, subsystem A and subsystem B. A state is **not entangled** ("a product state") if it factorizes as $|\Psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$; the state $|\uparrow\downarrow\rangle$, and a product of two independently x-polarized single-site states, are both examples. The singlet (7.3) is the opposite extreme: it cannot be written as any such product, and is our working example of an **entangled** state.

To turn "entangled or not" into a number, define the full density matrix $\rho = |\Psi\rangle\langle\Psi|$ and trace out subsystem B to obtain the **reduced density matrix** of A, $\rho_A = \mathrm{Tr}_B\, \rho$. The **entanglement entropy** is then the von Neumann entropy of ρ_A,

$$S(\rho_A) = -\,\mathrm{Tr}(\rho_A \log \rho_A)\,. \tag{7.6}$$

If $|\Psi\rangle$ is a product state, $\rho_A = |\psi_A\rangle\langle\psi_A|$ is a rank-one projector, its only nonzero eigenvalue is 1, and $S = 0$ immediately: no entanglement, zero entropy, exactly as we would want.

Let us check (7.6) explicitly on the singlet (7.3). Writing $\rho = |\Psi\rangle\langle\Psi|$ out in the product basis and tracing over site 1 (our subsystem B) leaves

$$\rho_A = \tfrac{1}{2}\,|\uparrow\rangle\langle\uparrow| + \tfrac{1}{2}\,|\downarrow\rangle\langle\downarrow|,$$

a maximally mixed single-site density matrix, whose two eigenvalues are both $1/2$. Then

$$S = -\left(\tfrac{1}{2}\log\tfrac{1}{2} + \tfrac{1}{2}\log\tfrac{1}{2}\right) = \log 2. \tag{7.7}$$

This is the maximum possible entanglement entropy a single spin-1/2 can carry (a d-dimensional local Hilbert space can carry at most $\log d$), which is exactly the sense in which the singlet is "maximally entangled": every last bit of the local Hilbert space's capacity for entanglement is used.

7.6 A solvable warm-up: entanglement entropy of a dimerized chain

Two exactly solvable models let us see how (7.6)-(7.7) scale with system size, and they are the key to the whole session, so it is worth working through them carefully.

Take a one-dimensional chain of spin-1/2's, but couple them only in disjoint pairs: an antiferromagnetic bond between sites 0 and 1, another between 2 and 3, another between 4 and 5, and so on, with *no* coupling between different pairs,

$$H = \sum_n J\,\mathbf{S}_{2n}\cdot\mathbf{S}_{2n+1}.$$

This Hamiltonian is trivial to solve exactly, because it is just a sum of independent Heisenberg dimers, and its ground state is therefore an exact tensor product of singlets, one per dimer bond:

$$|\Psi\rangle = |\text{singlet}\rangle_{01} \otimes |\text{singlet}\rangle_{23} \otimes |\text{singlet}\rangle_{45} \otimes \cdots\,. \tag{7.8}$$

Physically this is the minimal model of a gapped quantum spin liquid: the ground state breaks no symmetry, and it is separated from the lowest triplet excitation by a large gap (the singlet-triplet splitting on a single dimer), so it is about as far from criticality as a spin model can be.

Now bipartition the chain and ask for the entanglement entropy. Because (7.8) is a tensor product over disjoint bonds, the reduced density matrix ρ_A factorizes bond by bond: any dimer

sitting entirely inside A (or entirely inside B) contributes a pure state to ρ_A and therefore contributes exactly zero to S, by the same argument as Section 7.5's product-state case. The only bonds that contribute are the ones the bipartition boundary actually cuts, and each cut bond contributes precisely the single-dimer result (7.7), $\log 2$, since tracing out one half of a singlet always leaves the same maximally mixed single-site density matrix regardless of what else is going on elsewhere in the chain. So if the boundary cuts n_{cut} bonds, $S = n_{\text{cut}} \log 2$.

Take a small subsystem A containing just two dimers: the boundary cuts exactly two bonds (one on each side), so $S = 2 \log 2$. Take a much larger subsystem A, many dimers wide: the boundary still only touches two bonds, the ones immediately at its two edges, so $S = 2 \log 2$ again, unchanged. **The entanglement entropy of this state does not grow with the size of the subsystem**; it is set entirely by the two points where the one-dimensional boundary sits, however far apart those points are.

7.7 Two dimensions and the area law

Now repeat the same construction one dimension up. Take a two-dimensional lattice and group sites into disjoint clusters of four (tetramers) instead of pairs, coupled internally but not to each other, so the ground state is again an exact tensor product, now of tetramer ground states rather than dimer singlets. The same factorization argument applies: a cluster fully inside A or fully inside B contributes nothing, and only clusters straddling the bipartition boundary contribute to S.

The difference is purely geometric. Compare a small bipartition A, whose boundary cuts eight bonds, against a much larger bipartition B, whose longer boundary cuts many more. Because the entropy again accumulates one contribution per cut bond, S is now set by the *length of the boundary*, and in two dimensions that length genuinely grows as the subsystem grows, unlike the 1D case in Section 7.6 where the boundary was always exactly two points no matter how large the subsystem got. This is the crucial contrast: **in one dimension the entanglement entropy of a gapped ground state saturates**, independent of subsystem size; **in two (or more) dimensions it grows with the size of the boundary**, $S_A \sim L_A \sim \partial A$ and $S_B \sim L_B \sim \partial B$, so $S_B > S_A$ for the larger bipartition. Figure 7.2 puts the two cases side by side, with the cut bonds marked so that the counting can be done by eye in each.

This behavior has a name, the **area law**: for the ground state of a gapped, local Hamiltonian, the entanglement entropy of a subsystem scales with the length (in 1D) or area (in 2D and higher) of the boundary separating it from the rest of the system,

$$S_A \;\sim\; \alpha\, |\partial A|, \tag{7.9}$$

rather than with the *volume* of A itself. This is genuinely restrictive. A uniformly random state drawn from the full 2^L-dimensional Hilbert space would typically have S_A close to its maximum possible value, $|A| \log 2$ for a spin-1/2 subsystem of size $|A|$: a volume law. Gapped ground states are wildly atypical states in exactly this sense; they occupy only the tiny corner of Hilbert space satisfying the area law, and that is precisely the corner both dimer and tetramer models above sit in exactly, and that realistic gapped Hamiltonians sit in approximately.

7.8 Why MPS work in 1D, and where they stop working

The area law is the missing piece that explains why the ansatz (7.5) is a sensible one. Consider an MPS at a given bond, and use it to build the Schmidt decomposition of the state across a bipartition there:

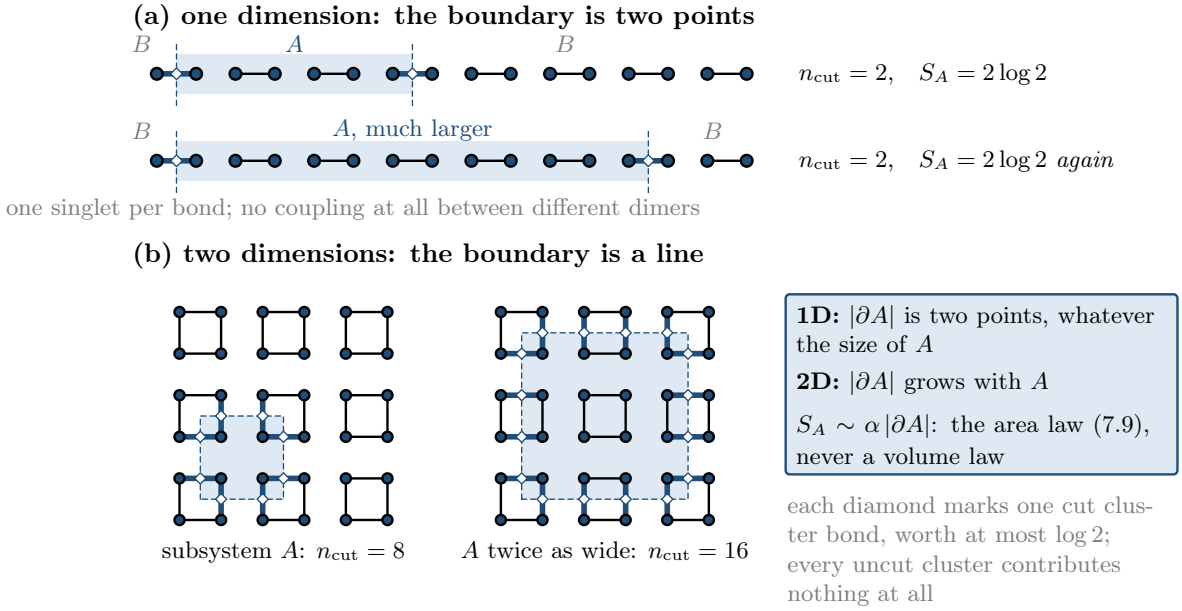


Figure 7.2: Why the entanglement entropy counts boundary, not volume. In both panels the ground state is an exact product of decoupled cluster ground states, so a cluster lying entirely inside A or entirely inside B is in a pure state and contributes nothing; only the clusters the bipartition actually cuts contribute, and the cut bonds are the highlighted ones. (a) The dimerized chain: each cut singlet contributes exactly the single-dimer result $\log 2$ of (7.7), and the boundary of a block is two points no matter how long the block is, so enlarging A from two interior dimers to five leaves $S_A = 2 \log 2$ unchanged. The entropy saturates. (b) The same construction with four-site clusters on a square lattice. Each straddling cluster still contributes only a bounded amount, at most $\log 2$ per cut bond, but now the boundary is a curve whose length grows with the subsystem: doubling the linear size of A doubles the number of cut bonds, eight to sixteen, and S_A grows in proportion. That is the area law (7.9), $S_A \sim \alpha\,|\partial A|$, and the contrast between the two panels is exactly why matrix product states are a one-dimensional tool.

$$|\Psi\rangle = \sum_{\alpha=1}^{M} \lambda_\alpha \, |\alpha\rangle_A \otimes |\alpha\rangle_B,$$

with at most M terms, because the internal (virtual) index running through that bond only takes M values. The reduced density matrix $\rho_A = \sum_\alpha \lambda_\alpha^2 |\alpha\rangle_A\langle\alpha|_A$ therefore has rank at most M, and among all rank-$\leq M$ density matrices the entropy (7.6) is maximized when every eigenvalue is equal, $\lambda_\alpha^2 = 1/M$, giving

$$S \;\leq\; \log M. \tag{7.10}$$

An MPS of bond dimension M, in other words, can represent entanglement entropy at most $\log M$ across any given bond, growing only logarithmically with M. Compare this to (7.9): a 1D gapped ground state has entanglement entropy that *saturates* at a finite, subsystem-size-independent value, so a modest, fixed bond dimension is already enough to capture it essentially exactly. That is the precise, quantitative reason matrix product states are such an excellent ansatz specifically in one dimension: the area law caps exactly the quantity the bond dimension caps. Geometrically, we can picture the whole many-body Hilbert space as a large set, of which MPS parameterize only a tiny corner, precisely the corner of area-law states, reached with a bond dimension that stays modest ($M \sim 10$, 100) as long as we stay in the area-law regime, and that would have to blow up ($M \sim 10^{L/40}$ or worse) to reach a generic, volume-law state. Figure 7.3 draws the comparison directly: the ceiling $\log M$ set by (7.10) against the entropy actually demanded by a gapped, a critical and a generic state as the subsystem grows.

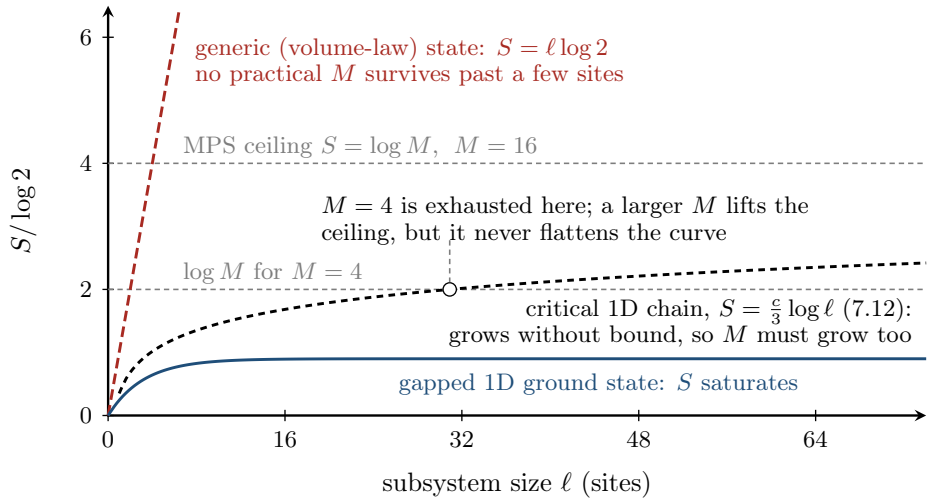


Figure 7.3: The bond dimension is a ceiling on entanglement, and that is the whole story of when matrix product states work. An MPS with bond dimension M admits at most M Schmidt terms across any bond, so by (7.10) it can carry at most $S = \log M$ there: the two horizontal lines. A gapped one-dimensional ground state obeys the area law of Section 7.6, its entropy saturating at a value set by the correlation length and not by the subsystem size, so it slips under a fixed, modest ceiling and stays there for any chain length. A critical chain violates the area law logarithmically, (7.12), so it eventually pierces any fixed ceiling and M must be raised as the system grows, slowly but without bound. A generic state of the 2^L-dimensional Hilbert space obeys a volume law, $S = \ell \log 2$, and exhausts any usable M after a handful of sites, which is why long time evolution, highly excited states, and two-dimensional systems are all out of reach. Drawn in bits, $S/\log 2$, with central charge $c = 1$; the curves are schematic, not fits to a particular model.

The same argument tells us immediately where MPS must fail: whenever the true entanglement entropy exceeds what any practical bond dimension can supply through (7.10).

Concretely, this happens for long-time dynamics (entanglement grows *linearly* with time after a quench, not logarithmically), for genuinely higher-dimensional systems (the area law's boundary length itself grows with system size, per Section 7.7, so M has to grow correspondingly, often exponentially in the subsystem's linear size), for highly excited states (the area law is a ground-state statement, not a generic eigenstate one), and for states far from equilibrium generally. Where the physics stays close to a gapped 1D ground state, though, MPS remains usable well beyond the pure ground-state problem: it extends, at a cost, to time evolution, excited states, and open quantum systems, in 1D routinely and sometimes in 2D.

Notebook check-in: `session7.ipynb`, "Near linear scaling with system size with matrix product states" (Core) and "The role of the bond dimension in tensor networks" (Core)

The first exercise has you confirm directly that the DMRG ground-state cost in `dmrgpy` grows linearly with system size, the $L \times M^2$ scaling of (7.5) rather than the 2^L scaling of (7.4). The second has you scan the bond dimension itself and watch the ground-state energy fluctuation shrink and saturate: that saturation point is where M has become large enough to satisfy the bound (7.10) for the state at hand.

Notebook check-in: `session7.ipynb`, "Entanglement entropy of the ground state" (Core), "Size dependence of the entanglement entropy" (Challenge), and "Bond dimension dependence of the entanglement entropy" (Challenge)

These three exercises let you reproduce Sections 7.5 through 7.8 numerically: compute the bond-resolved entanglement entropy of a Heisenberg chain's MPS ground state, check that it barely grows with system size (the 1D area-law saturation of Section 7.6, in a gapless rather than gapped chain this time), and check that it saturates once the bond dimension is large enough to satisfy (7.10). The quiz for the first of these also asks you to swap in an Ising model and watch the entropy change, a quick way to see the area law respond to a genuinely different (in this case classical, zero-entanglement) ground state.

7.9 Tensor diagrams: notation for networks and operators

Matrix product states come with a convenient graphical notation that makes both the ansatz and the algorithms built on it much easier to reason about. A scalar (rank-0 tensor) is drawn as a bare point; a vector (rank-1) is a point with one dangling leg; a matrix (rank-2) is a point with two legs; a rank-3 tensor is a point with three legs, and so on for any rank. Each leg represents one index of the tensor, and joining two legs together between two tensors means summing over that shared index (contracting it).

An MPS is then simply a horizontal chain of such points, one per site, each with one *physical* leg dangling downward (the index s_i, the actual local spin value) and one or two *virtual* legs connecting it to its neighbors along the chain, carrying the bond-dimension index that gets summed over. A **matrix product operator** (MPO) represents an operator, most importantly the Hamiltonian itself, in the same spirit: each site now carries *two* physical legs instead of one, corresponding to the two indices σ_i, σ_i' of the operator, plus the same virtual legs linking neighboring sites. Applying an MPO to an MPS (computing $H|\Psi\rangle$, say) is then just gluing the two diagrams together along the physical legs and contracting: multiplying every pair of matrices that share an index and summing over it. Done naively, this operation multiplies the bond dimensions of the two objects together, which would defeat the whole point; in practice one performs a singular value decomposition on the result and keeps only the dominant singular values, truncating back down to a manageable bond dimension while discarding the least important part of the Hilbert space being represented. This truncation step is the practical engine behind essentially every MPS algorithm, including the ground-state

search we turn to next. Figure 7.4 collects the notation, the ansatz (7.5) redrawn in it, and this contract-then-truncate step in one place.

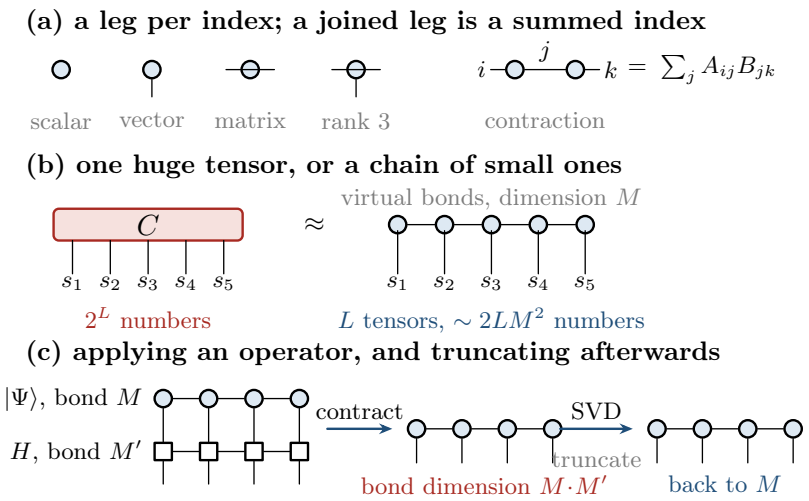


Figure 7.4: The graphical notation, and what the MPS ansatz looks like in it. (a) A tensor is a dot and each of its indices is a leg, so a scalar is a bare point, a vector has one leg, a matrix two, and so on; joining two legs means summing over that shared index. (b) The exact coefficient tensor of (7.4) is a single object with one leg per site, and it hides 2^L independent numbers. The matrix product state (7.5) replaces it by a chain of small tensors, each carrying one physical leg pointing down and virtual legs of dimension M linking it to its neighbours, for $\sim 2LM^2$ numbers in total: the exponential wall is traded for a chain that grows linearly. (c) An operator, the Hamiltonian above all, becomes a matrix product operator with two physical legs per site. Applying it to a state means gluing the diagrams along the shared physical legs and contracting them away, which multiplies the two bond dimensions together; a singular value decomposition keeping only the largest singular values then truncates the result back to a workable M. That truncation is the engine inside every MPS algorithm, DMRG included.

7.10 Finding the ground state: the variational principle and DMRG

Having an ansatz is only useful if we can actually optimize it. Treat the MPS matrices $\{M_i^{s_i}\}$ as variational parameters and minimize the Rayleigh quotient

$$E[\Psi] = \frac{\langle\Psi|H|\Psi\rangle}{\langle\Psi|\Psi\rangle}, \tag{7.11}$$

equivalently, introducing a Lagrange multiplier λ to enforce normalization, minimize $\langle\Psi|H|\Psi\rangle - \lambda\,(\langle\Psi|\Psi\rangle - 1)$ with respect to every matrix in the network. Diagrammatically, differentiating with respect to one particular tensor in the network (say the matrix at site k) is simply "erasing" that tensor from the diagram, since every tensor appears linearly (once in the bra layer, once in the ket layer) in the full contraction. What remains, after erasing site k's tensor from both the numerator and denominator diagrams, is exactly an eigenvalue problem for that single tensor: an effective, reduced Hamiltonian acting on site k's indices, whose ground eigenvector is the locally optimal replacement for $M_k^{s_k}$. Figure 7.5 shows the erasure and the sweep it generates.

This suggests, and is, an iterative algorithm: fix every tensor except site 1's, solve the local eigenvalue problem there, update $M_1^{s_1}$ to the resulting eigenvector; move to site 2, fix everything else, solve the local eigenproblem there, update; continue sweeping through the

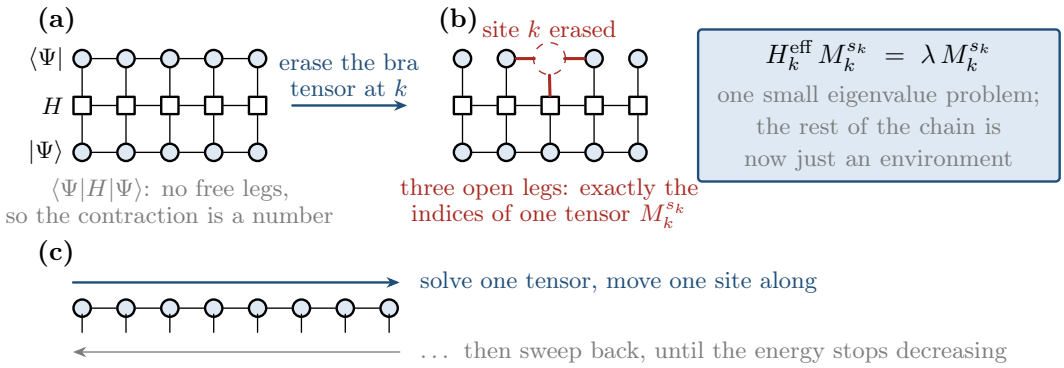


Figure 7.5: How DMRG turns an exponentially large minimization into a sequence of tiny ones. (a) The numerator of the Rayleigh quotient (7.11) is a single closed network: the ket MPS, the Hamiltonian as an MPO, and the bra MPS, all contracted together, leaving no free legs and hence a number. (b) Every tensor enters that network linearly, once in the bra layer and once in the ket layer, so differentiating with respect to the bra tensor at site k simply deletes it. What is left has exactly three open legs, the two virtual bonds and the physical index of site k, which are precisely the indices of the tensor $M_k^{s_k}$ we are solving for: everything to the left and right of k has been reduced to an environment, and the stationarity condition is an ordinary eigenvalue problem of the size of a single tensor. (c) Solve it, move one site along, solve again, and sweep back and forth until the energy stops falling. That is the density matrix renormalization group; at no point is anything of size 2^L ever built.

chain, site by site, until the energy stops decreasing. This sweeping procedure is exactly the **density matrix renormalization group (DMRG)** algorithm. Its efficiency comes from precisely the feature we built (7.11) to exploit: at every step we only ever optimize one matrix at a time, a small, tractable linear-algebra problem, rather than the full exponentially large ground-state search all at once.

Notebook check-in: `session7.ipynb`, "Low energy excitations of a Heisenberg model" (Core)

This exercise asks you to look at the matrix elements between the DMRG ground state and the first excited state of a Heisenberg chain, i.e. what the sweeping procedure of Section 7.10 finds one level up from the state that directly minimizes (7.11).

7.11 Beyond MPS: MERA and PEPS

Matrix product states are the simplest member of a much larger family called **tensor networks**; nothing forces the tensors to sit in a 1D chain, and the very same underlying coefficient tensor C from (7.4) can just as well be redrawn as a 2D grid or an arbitrary graph of connections. Richer connectivity trades optimization simplicity, MPS is the easiest tensor network to optimize, for a larger representable class of states. Two architectures are worth naming specifically.

The **multi-scale entanglement renormalization ansatz (MERA)** organizes its tensors hierarchically across length scales rather than along a single chain, and this extra structure lets it support more entanglement than an MPS built from a comparable number of parameters. That matters because the area law (7.9) is strictly a *gapped*-system statement; exactly at a critical point (a continuous quantum phase transition, typically described by a conformal field theory in 1+1 dimensions), the ground state entanglement entropy no longer saturates but instead grows logarithmically with the size of the subsystem,

$$S(\ell) \;=\; \frac{c}{3}\log \ell \;+\; \text{const.}, \tag{7.12}$$

with c the CFT's central charge. This is a mild but genuine violation of the strict area law (7.9), and it is exactly the kind of mild violation MERA's extra hierarchical structure is built to accommodate, which is why MERA is the tensor network of choice specifically at criticality.

Projected entangled pair states (PEPS) are the direct 2D generalization of the MPS idea: instead of tensors coupled in a line, they sit on the nodes of a genuine 2D lattice, coupled to all of their lattice neighbors. This is a natural and promising architecture for 2D quantum many-body problems, but it comes at a real cost: there is no analogue of DMRG's clean, single-tensor local eigenvalue problem for PEPS, because contracting a general 2D tensor network is itself computationally hard (unlike the 1D contraction that made Section 7.10's local eigenproblem tractable in the first place). PEPS optimization remains an active, harder problem. Figure 7.6 draws both architectures, and makes visible where each one's extra structure comes from and what it costs.

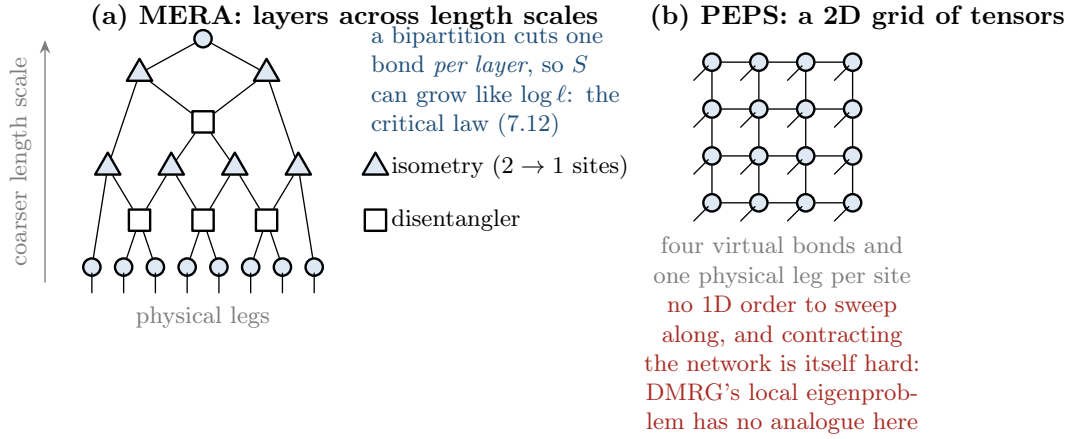


Figure 7.6: The two tensor networks beyond the chain, and what their geometry buys. Left: the multi-scale entanglement renormalization ansatz. Its tensors are stacked in layers, each layer first removing the short-range entanglement between neighbouring blocks (the square disentanglers) and then coarse-graining two sites into one (the triangular isometries), so that the network describes the state at successively longer length scales rather than at successive positions along a chain. A bipartition now cuts one bond in every layer instead of one bond in total, which is precisely how MERA supports the logarithmically growing entropy (7.12) of a critical chain that an MPS of fixed bond dimension cannot reach. Right: projected entangled pair states, the honest 2D generalization of (7.5), with one tensor per lattice site carrying a physical leg and four virtual bonds. The picture also shows the difficulty: there is no way to sweep along such a network one tensor at a time with the rest reduced to a simple environment, and contracting it exactly is itself a hard problem, which is why PEPS has no clean analogue of DMRG.

This is also where this year's lecture ends. Unlike some previous editions of this course, this year's recording stops right after MERA and PEPS, with no continuation into neural network quantum states; the single mention in Section 7.3 is as far as the topic goes today. Keep the tensor-network machinery built across Sections 7.4 through 7.11 in mind regardless: session 10 reuses the same MPS/MERA/PEPS architectures directly, repurposed as machine-learning classifiers, and returns to neural network quantum states properly, from the machine-learning side.

Notebook check-in: `session7.ipynb`, "The entanglement entropy as a signal for phase transitions" (Challenge)

This exercise has you track the entanglement entropy of the transverse-field Ising chain's ground state as you sweep the transverse field through its quantum critical point, using `dmrgpy`'s MPS machinery. Away from criticality you should see area-law saturation as in Section 7.8; near the transition, watch for the entropy failing to saturate the way (7.12) predicts, and for the quiz's own question about which regime is hardest for MPS to capture, which is really asking you to connect back to why MERA exists at all.

7.12 Closing summary

- Strongly correlated matter, quantum spin liquids and fractional Chern insulators among them, is exactly the regime where the mean-field machinery of sessions 1 through 6 has no order parameter to converge on, per (7.1) and Section 7.1.
- The **Jordan-Wigner transformation** (7.2) is an exact mapping between fermionic and spin-1/2 Hamiltonians, "spin equals fermion times string"; it is the bridge between the fermionic language of sessions 1-5 and the spin language of sessions 6 and 8, and it is why we can freely work in the spin representation for the rest of the session.
- The real obstruction to solving many-body Hamiltonians is **storage**: a system of L spins needs 2^L coefficients (7.4) to specify its wave function exactly.
- The **matrix product state** ansatz (7.5) replaces those 2^L coefficients with $O(LM^2)$ parameters, trading exactness for a controllable approximation set by the bond dimension M.
- The **entanglement entropy** (7.6), computed explicitly for the singlet in (7.7), quantifies how entangled a state is; exactly solvable dimer and tetramer models show that gapped ground states obey an **area law** (7.9), entropy set by the boundary length rather than the subsystem volume.
- MPS entanglement is itself bounded, $S \leq \log M$ (7.10), which is exactly why MPS is so effective in 1D (where the area law saturates) and why it needs exponentially large M in higher dimensions or away from the ground state.
- Ground states are found variationally, by minimizing (7.11) one tensor at a time; this sweeping procedure is the **DMRG** algorithm. **MERA** and **PEPS** extend the tensor-network idea to criticality (via the logarithmic entropy correction (7.12)) and to genuine 2D lattices, respectively, at the cost of DMRG's simple local optimization.
- Neural network quantum states were named once, alongside tensor networks, as a second answer to the same storage problem, but were not developed this session; session 10 takes them up properly from the machine-learning side, and also returns directly to this session's MPS/MERA/PEPS machinery as ready-made architectures for machine-learning classifiers.

Session 8

Quantum Magnetism, Spinons and Kondo Physics

Session 6 spent its whole runtime on magnets that break time-reversal symmetry: ferromagnets, antiferromagnets, and, in its final stretch, the Holstein-Primakoff bosonization of these ordered states into magnons, pushed all the way to a topological magnon band structure with its own chiral edge modes. That session ended abruptly, right on "topological magnons," with no closing recap, but it did leave one loose thread along the way: a passing aside about Kitaev-type interactions, flagged at the time as "important for quantum spin liquids" without being followed up. This session is where that thread gets picked up properly, not as a resumed sentence but as its own self-contained lecture, with its own welcome and its own roadmap: minimal models of quantum magnetism, the excitations that quantum spin liquids (QSLs) support (spinons and gauge excitations), and then Kondo systems and Kondo lattice models, where a completely different physical setup turns out to be solvable with exactly the same trick. The thread that ties the whole session together is the one that has been running under this entire course since session 1: take a many-body Hamiltonian that mean-field theory cannot handle, find the right auxiliary-particle reformulation, and mean-field *that* instead.

We set $\hbar = 1$ throughout for the spin operators, matching the convention already implicit in session 6.

8.1 Two classes of magnet, and where each one lives

Every magnetic material we could point to falls, in this lecture's language, into one of two classes. **Classical magnets** break time-reversal symmetry: conventional ferromagnets, antiferromagnets, spin spirals. From the electron's perspective these are describable by mean-field theory, exactly as built in sessions 1 and 6, and this is where most magnetic materials in nature actually sit: iron, copper oxide, or any material with a well-defined Néel temperature.

Quantum magnets are the opposite: they admit no mean-field description in terms of electrons at all, and their ground states carry long-range entanglement. That entanglement is not a side detail, it is what gives rise to excitations that have no counterpart in a symmetry-broken magnet. Two quite different physical realizations of quantum magnetism are the subject of this session:

- **Quantum spin liquids (QSLs)**, the first class we build up in detail. The best-known candidate material is herbertsmithite, a compound whose magnetic ions sit on a kagome lattice of spin-1/2 sites.
- **Heavy-fermion Kondo insulators**, often rare-earth compounds, with samarium hexaboride (SmB_6) as the standard example. We build the theory for this class in the second

half of the session.

8.2 Two emergent excitations: magnons versus spinons

Strong many-body interactions between electrons are what give magnetic materials their emergent, collective excitations in the first place. In a symmetry-broken magnet, that collective excitation is the **magnon**: spin 1, no charge, because the strong electronic repulsion freezes the charge degree of freedom completely and only spin is left to fluctuate. In a quantum magnet, the excitation is qualitatively different: the **spinon**, spin 1/2, also no charge. It is useful to picture a spinon as literally the spin half of an electron. Push that picture one step further and you get the standard fractionalization language for a QSL: the electron splits into a **chargon**, carrying the charge, and a **spinon**, carrying the spin.

There is a simple pictorial argument for why the two excitations differ by exactly a factor of two in spin. Take a ferromagnet: the ground state is all spins up, $|\uparrow\uparrow\uparrow\cdots\rangle$. Flip one spin to build an excited state, and the difference in total spin between ground and excited state is $\Delta S = 1$: a (local) magnon. Now take a quantum spin liquid: the ground state is a collection of singlets everywhere, no single spin pointing anywhere in particular. Build an excited state by leaving one spin unpaired instead of pairing it into a singlet, and the difference in total spin is $\Delta S = 1/2$: pictorially, a spinon. This picture is deliberately oversimplified (spinons are genuinely more subtle than "an unpaired spin"), but it is exactly the right starting intuition for why symmetry-broken magnets and quantum spin liquids host fundamentally different excitations. Figure 8.1 puts the two counting arguments side by side.

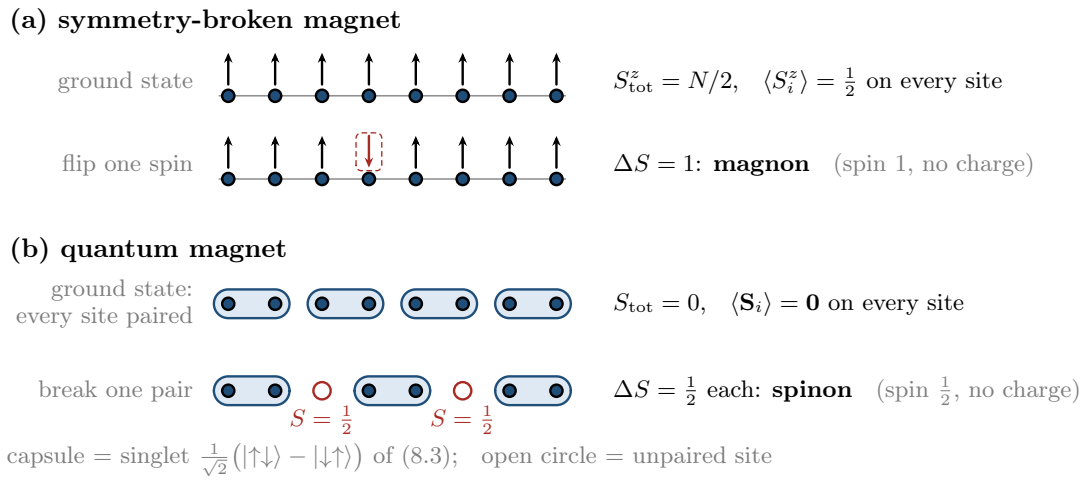


Figure 8.1: Why the two kinds of magnet host excitations differing by exactly a factor of two in spin. (a) In a symmetry-broken magnet every site carries a definite spin (arrows). Flipping one costs $\Delta S = 1$: that excitation is the magnon of session 6. (The single flipped spin drawn here is only the ΔS bookkeeping; the actual magnon is the delocalized precession of figure 6.5, not one overturned site.) (b) In a quantum magnet no site carries a definite spin at all; the sites are paired into singlets (8.3) (capsules), so $\langle \mathbf{S}_i \rangle = 0$ everywhere and $S_{\rm tot} = 0$. Breaking one pair leaves *unpaired* sites (open circles), and each one separately carries $S = 1/2$: a spinon, half a magnon. Spinons are therefore always created in pairs, and whether the pair stays bound or comes apart is the whole content of figure 8.3.

8.3 Two minimal dimers: classical order parameter versus quantum singlet

The cleanest way into quantum magnetism is to look first at the smallest possible magnet, two spins, and compare two different Hamiltonians on it.

The first is the **Ising dimer**,

$$\mathcal{H}_{\text{Ising}} = S_0^z S_1^z, \tag{8.1}$$

coupling only the z-components of two spin-1/2's. This Hamiltonian has two degenerate ground states, related by time reversal:

$$|GS_1\rangle = |\uparrow\downarrow\rangle, \qquad |GS_2\rangle = |\downarrow\uparrow\rangle.$$

Both are eigenstates with the same energy, but each one *individually* breaks time-reversal symmetry (time reversal flips every spin, so it swaps $|GS_1\rangle$ and $|GS_2\rangle$ into one another); only the two-state degenerate pair together respects it. A macroscopic, symmetry-broken antiferromagnet, of exactly the kind built in session 6, is just an infinite-site version of this same story, and the same argument runs equally well for ferromagnets and spin spirals.

Now replace the Ising coupling by the full, rotationally symmetric **quantum Heisenberg dimer**,

$$\mathcal{H}_{\text{Heis}} = \mathbf{S}_0 \cdot \mathbf{S}_1. \tag{8.2}$$

This is what a half-filled, strongly-interacting Hubbard model reduces to in the large-U limit (session 1's derivation). Solving it with $\mathbf{S}_0, \mathbf{S}_1$ as genuine spin-1/2 operators, two spin-1/2's combine into a singlet ($S = 0$) and a triplet ($S = 1$), with $\mathbf{S}_0 \cdot \mathbf{S}_1 = -3/4$ on the singlet and $+1/4$ on the triplet, so the unique ground state is the **singlet**,

$$|GS\rangle = \frac{1}{\sqrt{2}}\Big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\Big), \qquad \langle \mathbf{S}_i \rangle = 0. \tag{8.3}$$

This ground state is qualitatively different from the Ising case in three ways that matter for everything that follows: it is *uniquely* the ground state, with no degeneracy to spontaneously choose between; it is *time-reversal symmetric*, since its net magnetization vanishes on the nose and time reversal maps $|GS\rangle$ to itself; and it is *maximally entangled* (the two terms in (8.3) carry equal weight, the largest entanglement two qubits can hold). Can we have a macroscopic version of this same ground state? That question is exactly where quantum spin liquids come from.

8.4 Frustration: the route to a quantum spin liquid

Before building a macroscopic version of (8.3), we need to know when a Heisenberg model's ground state actually stays time-reversal symmetric like this, rather than spontaneously ordering the way session 1 and session 6's magnets did. The rule of thumb is **frustration**. The paradigm case is a triangular plaquette with antiferromagnetic coupling on all three bonds: any two spins can point oppositely, but the third bond is then forced to sit between two spins that are already antiparallel to each other, and no single spin configuration satisfies all three bonds at once. Geometric frustration of this kind produces many classically degenerate ground states (same energy, different spin configurations), and once quantum fluctuations are switched on, they mix these degenerate configurations into a single quantum superposition that, generically, does not break time-reversal symmetry.

This is why the **triangular** and **kagome** lattices, both built entirely out of triangles, are the standard hunting ground for quantum spin liquid physics: an antiferromagnetic Heisenberg model on either lattice is, in the right coupling regime, a genuine candidate for a QSL ground state. And this is precisely why herbertsmithite, a kagome-lattice compound of spin-1/2 sites, keeps coming up as the best-known real material candidate. Figure 8.2 collects the geometry behind the whole argument: an unfrustrated plaquette, a frustrated triangle with its degenerate configurations, and the kagome lattice built entirely out of them.

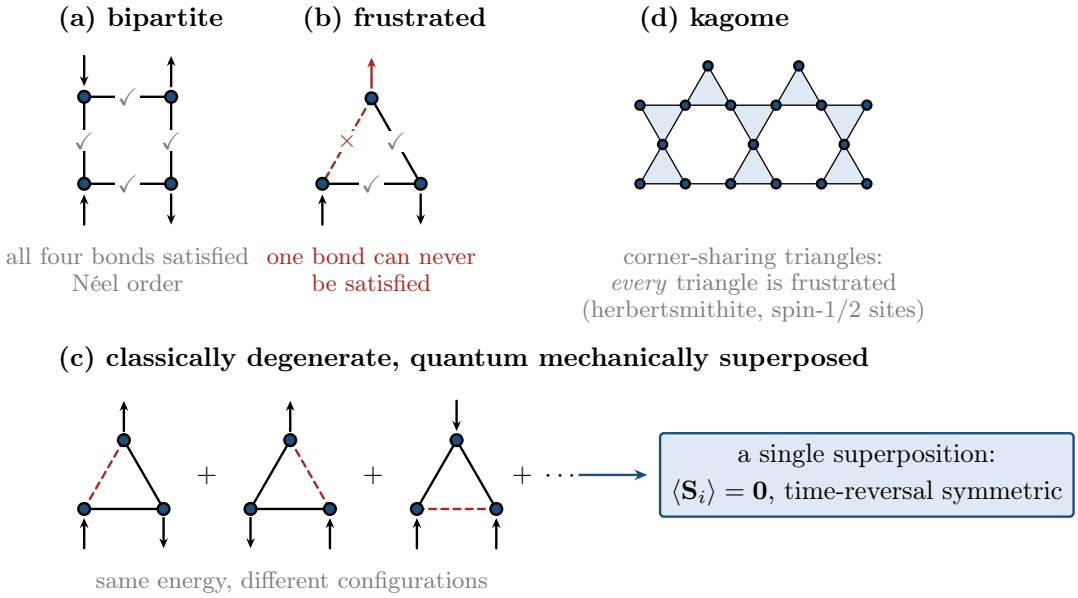


Figure 8.2: Frustration, with the spins drawn as Ising (collinear) variables throughout, which is how the argument in the text is phrased; allowing them to rotate freely instead resolves the triangle into the 120° noncollinear state of figure 6.3, and the degeneracy discussed here is the Ising one. (a) On a bipartite plaquette every antiferromagnetic bond can be satisfied at once (✓), and the ground state is the Néel state: two configurations, related by a global flip, each breaking time-reversal symmetry. (b) On a triangle it cannot be done: once two spins are antiparallel the third bond is stuck (×), whichever way the last spin points. (c) The frustrated bond can sit on any of the three edges, so the classical ground state is degenerate rather than unique; quantum fluctuations mix the degenerate configurations into a single superposition with $\langle \mathbf{S}_i \rangle = \mathbf{0}$, which does *not* break time-reversal symmetry. (d) The kagome lattice of herbertsmithite is built entirely from corner-sharing triangles, so every one of them is frustrated at once and the classical degeneracy becomes macroscopic: this is why triangular and kagome lattices are the hunting ground for quantum spin liquids.

8.5 A first stop short of a spin liquid: the dimerized Heisenberg model

Before tackling a genuinely frustrated Hamiltonian, it is worth building the simplest possible *macroscopic* quantum magnet and checking explicitly that its ground state does what (8.3) promised. Take a **fully dimerized Heisenberg model**: perfectly decoupled dimers, arranged in a chain or on a two-dimensional lattice,

$$H = \sum_n J\, \mathbf{S}_{2n} \cdot \mathbf{S}_{2n+1}, \qquad J > 0. \tag{8.4}$$

Because every dimer is completely decoupled from its neighbors, the many-body ground state can be written down exactly, as a simple product of singlets over the dimer pairs,

$$|\Psi\rangle \sim \bigotimes_n \Big(|\uparrow_{2n}\downarrow_{2n+1}\rangle - |\downarrow_{2n}\uparrow_{2n+1}\rangle\Big), \qquad \langle \mathbf{S}_i \rangle = 0 \text{ on every site.} \tag{8.5}$$

This is, in one sense, a trivial Hamiltonian, effectively a collection of independent zero-dimensional problems. But it already establishes something important: a macroscopic Heisenberg Hamiltonian genuinely *can* have a ground state that does not break time-reversal symmetry, exactly generalizing the two-site singlet (8.3) to an extensive system. Turning on small couplings between the dimers deforms this state smoothly without breaking time-reversal symmetry either.

There is, however, a crucial distinction to draw here, because this state is *not yet* a quantum spin liquid. Its fundamental excitations are still spin-1 excitations: break one singlet into its triplet partner, and you get a **triplon**, a local singlet-to-triplet transition, structurally the same object as a magnon in everything but name. A state built this way, out of a fixed, static pattern of singlets, is called a **valence bond solid (VBS)**. Getting to a genuine quantum spin liquid, with spin-1/2 excitations rather than spin-1 triplons, requires going a step further than a static dimer covering: it requires the singlets themselves to fluctuate and resonate across the whole lattice, which is exactly the frustrated case we return to next. Figure 8.3 draws that contrast explicitly.

8.6 The QSL ground state: the resonating valence bond wavefunction

For a genuinely frustrated Heisenberg Hamiltonian $\mathcal{H} = \sum_{ij} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j$, the expected ground state is no longer a single, fixed dimer covering like (8.5). Instead it is a **linear combination of singlets over every possible pairing of sites**, each pairing carrying its own amplitude, a genuinely complicated many-body wavefunction. One could try to attack this directly (write it as a tensor network, as session 7 does for the general strongly correlated case), but there is a more physically transparent route: a variational ansatz built on an analogy to superconductivity.

The idea is that singlets in a QSL "feel" a lot like Cooper pairs in an s-wave superconductor: both are pairwise singlet correlations. A BCS-type wavefunction is easy to write at the mean-field level in the right basis (Nambu space, exactly as in session 5), so we start from a BCS-paired state built out of the same auxiliary fermions $f_{i\sigma}$ that we will justify properly in Section 8.8,

$$|\Psi_{\mathrm{BCS}}\rangle = \prod_{\mathbf{k}} \Big(u_{\mathbf{k}} + v_{\mathbf{k}}\, f^\dagger_{\mathbf{k}\uparrow} f^\dagger_{-\mathbf{k}\downarrow}\Big)|0\rangle, \tag{8.6}$$

with $u_{\mathbf{k}}, v_{\mathbf{k}}$ variational parameters exactly as in session 5's BCS state. Left on its own, (8.6) is a superposition of states with every possible fermion occupation per site (empty, singly, doubly occupied), which is not a physical spin state at all: a QSL enforces strong local interactions, so single occupation per site is required, with no charge fluctuations allowed between sites. We enforce this by applying a **Gutzwiller projector** onto the singly-occupied subspace,

$$|\Psi_{\mathrm{RVB}}\rangle = P_G\, |\Psi_{\mathrm{BCS}}\rangle, \qquad P_G = \prod_i n_i(2-n_i), \qquad n_i = \sum_\sigma f^\dagger_{i\sigma} f_{i\sigma}, \tag{8.7}$$

which evaluates to 1 on states with exactly one fermion per site and to 0 otherwise. What survives is a genuine many-body spin wavefunction: a superposition over ways of pairing up sites into singlets, in the spirit of the single-dimer singlet (8.3), now spread across an entire frustrated lattice. This construction is named exactly for what it describes: the **resonating**

valence bond (RVB) variational wavefunction, and it turns out to be a genuinely good description of quantum spin liquid physics. Figure 8.3 shows what separates this resonating state, and the excitations it supports, from the frozen dimer covering of Section 8.5.

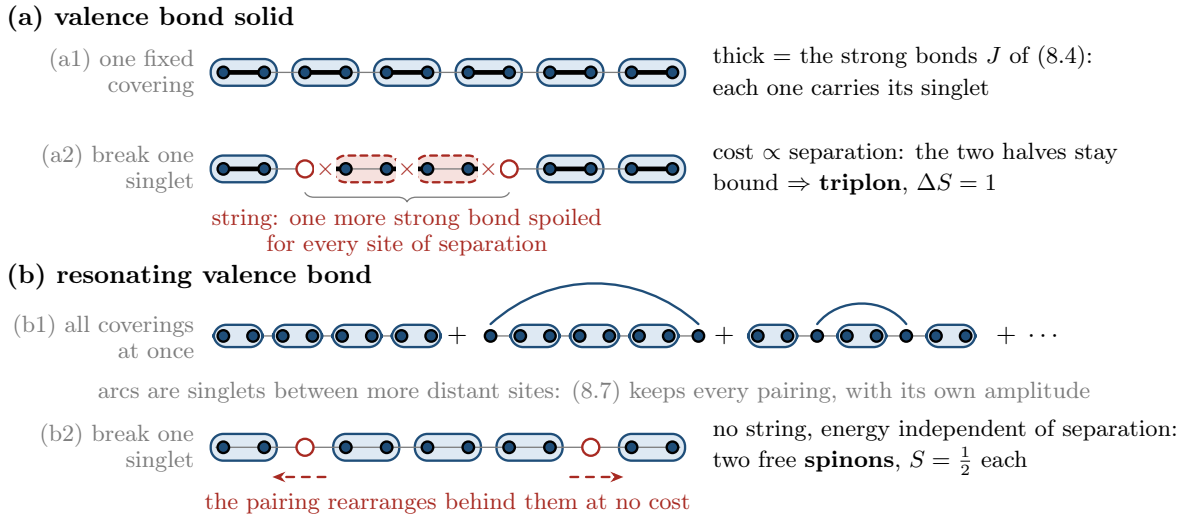


Figure 8.3: The difference between a valence bond solid and a quantum spin liquid is not whether the ground state is built from singlets – both are – but whether the singlet pattern is frozen. (a1) The fully dimerized model (8.4) has one fixed covering, (8.5). (a2) Separating the two unpaired spins of figure 8.1b leaves every strong bond in between without its singlet, so the energy cost grows with the separation: the pair is confined and only the bound object propagates, the spin-1 triplon. (b1) The RVB state (8.7) is a superposition over *all* pairings, near and far. (b2) Now the background re-pairs behind the excitations at no cost, the confining string never forms, and the two halves separate as independent spin-1/2 spinons. Frustration is what makes the superposition in (b1), rather than any single covering, the ground state.

8.7 Why the magnon (Holstein-Primakoff) approach fails for QSL excitations

Writing down the exact QSL ground state is hard, so the natural next move, exactly as in session 6, is to ask about excitations *on top of* the ground state rather than the ground state itself. Our only tool for that so far is the **Holstein-Primakoff transformation**, which represents

$$S_i^z = S - a_i^\dagger a_i, \qquad S_i^+ \approx \sqrt{2S}\, a_i, \qquad S_i^- \approx \sqrt{2S}\, a_i^\dagger, \tag{8.8}$$

linearized under the assumption that the bosonic magnon population is small, $\langle a_i^\dagger a_i \rangle \ll S$: an excellent approximation near a fully (or nearly) polarized magnet, exactly the regime session 6 used it in.

A QSL sits in the opposite limit. Its defining property, $\langle \mathbf{S}_i \rangle = 0$ on every site, forces, through (8.8),

$$\langle a_i^\dagger a_i \rangle = S, \tag{8.9}$$

precisely the regime where the small-boson-density condition behind the linearized Holstein-Primakoff mapping fails outright. There is no ordered reference state for a QSL to be "close to" and expand around, so the whole magnon machinery that worked for session 6's ordered magnets simply is not the right tool here. We need a genuinely different approximation for the excitations of a quantum spin liquid.

Notebook check-in: `session8.ipynb`, "Magnetization of a large Heisenberg model" (Core)

Diagonalize a moderately large quantum Heisenberg chain and confirm numerically that the local magnetization stays zero on every site, even in the presence of a small external field, as long as that field is weaker than the finite-size gap to the neighboring total-spin sector: the numerical face of the defining property behind (8.4) and (8.9).

8.8 The parton construction: Abrikosov fermions and the single-occupancy constraint

The construction that makes QSL excitations tractable represents each spin not as a bosonic fluctuation around an ordered reference state, but as an auxiliary **fermion** field $f_{i\sigma}$ ($\sigma = \uparrow, \downarrow$) living at every site: a particle carrying spin 1/2 but no electric charge, exactly the spinon of Section 8.2, now written down formally. These are called **Abrikosov fermions**, and the operator identity that defines the parton transformation is

$$\mathbf{S}_i = \frac{1}{2} \sum_{\alpha\beta} f_{i\alpha}^\dagger \, \boldsymbol{\sigma}_{\alpha\beta} \, f_{i\beta}, \tag{8.10}$$

with $\boldsymbol{\sigma}$ the vector of Pauli matrices. One can check directly, from the fermionic anticommutation relations, that (8.10) reproduces the correct $SU(2)$ spin algebra $[S_i^a, S_i^b] = i\varepsilon^{abc} S_i^c$ on the full four-state Fock space of $f_{i\uparrow}, f_{i\downarrow}$ (empty, singly occupied up, singly occupied down, doubly occupied), with no extra input needed. The catch is that this transformation *artificially enlarges* the local Hilbert space: a physical spin-1/2 has only two states, while $f_{i\uparrow}, f_{i\downarrow}$ give four. We remove the two unphysical extra states (empty and doubly occupied, both of which (8.10) sends to $\mathbf{S}_i = 0$ anyway) by imposing, by hand, the **single-occupancy constraint**

$$\sum_\sigma f_{i\sigma}^\dagger f_{i\sigma} = 1 \qquad \text{at every site } i. \tag{8.11}$$

This is exactly the fermionic analogue of the Gutzwiller projection (8.7) applied earlier to the many-body wavefunction: projecting the many-body state onto the singly-occupied subspace and constraining the fermion filling to one per site are the same statement, seen from the wavefunction side versus the operator side. Figure 8.4a shows exactly which two states the constraint has to throw away, and why they were never spin states to begin with.

With (8.11) enforced, (8.10) is an *exact* rewriting of a spin-1/2 operator in fermionic language, and we can substitute it directly into the Heisenberg Hamiltonian $\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \, \mathbf{S}_i \cdot \mathbf{S}_j$. Using the Pauli-matrix Fierz identity $\sum_a \sigma^a_{\alpha\beta}\sigma^a_{\gamma\delta} = 2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}$ and the constraint (8.11) on both sites gives the clean identity

$$\mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{4} - \frac{1}{2} \chi_{ij}^\dagger \chi_{ij}, \qquad \chi_{ij} \equiv \sum_\sigma f_{i\sigma}^\dagger f_{j\sigma}, \tag{8.12}$$

with χ_{ij} the spinon "bond" (hopping) operator between sites i and j. Substituting (8.12) turns the original, bilinear-in-spin Heisenberg Hamiltonian into a Hamiltonian that is *quartic* in the auxiliary fermions,

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\frac{1}{4} - \frac{1}{2} \chi_{ij}^\dagger \chi_{ij} \right) = \text{const} - \frac{1}{2} \sum_{\langle ij \rangle} J_{ij} \, \chi_{ij}^\dagger \chi_{ij}. \tag{8.13}$$

This is exactly the same kind of difficulty session 1 and session 6 faced with the electronic Hubbard interaction: a quartic term with no direct diagonalization. We treat it the same way, by **mean-field decoupling**,

$$\chi_{ij}^{\dagger}\chi_{ij} \;\approx\; \langle \chi_{ij}^{\dagger} \rangle \, \chi_{ij} + \chi_{ij}^{\dagger} \, \langle \chi_{ij} \rangle - |\langle \chi_{ij} \rangle|^2. \tag{8.14}$$

8.9 The mean-field spinon Hamiltonian and its dispersion classes

Absorbing the couplings and expectation value of (8.14) into a single mean-field parameter χ_{ij} per bond, and enforcing the constraint (8.11) on average with a Lagrange multiplier if needed, the mean-field decoupling of Section 8.8 collapses the original spin problem into a genuine, diagonalizable tight-binding-like Hamiltonian for the auxiliary fermions,

$$\mathcal{H}_{\text{MF}} = \sum_{ij,s} \chi_{ij} \, f_{i,s}^{\dagger} f_{j,s}. \tag{8.15}$$

It is worth pausing on what makes this mean-field step different from every other mean-field decoupling in this course so far: it does not need to break time-reversal symmetry. Choosing the same χ_{ij} for spin up and spin down keeps $\langle \mathbf{S}_i \rangle = 0$ exactly, so the resulting ground state is time-reversal symmetric by construction, just as (8.4) and (8.9) require. Equation (8.15) is nothing but a tight-binding model for the auxiliary fermions, with hoppings χ_{ij} fixed self-consistently by the lattice geometry and the exchange couplings J_{ij}. Its excitations are the **spinons**: emergent, spin-1/2, charge-neutral fermionic quasiparticles, produced by exactly the same "quartic interaction, mean-field decouple, diagonalize a bilinear Hamiltonian" recipe used for electrons throughout this course, just transplanted into the spin sector. Figure 8.4b is the resulting picture: spinons hopping on the lattice with amplitude χ_{ij}, at half filling because of (8.11).

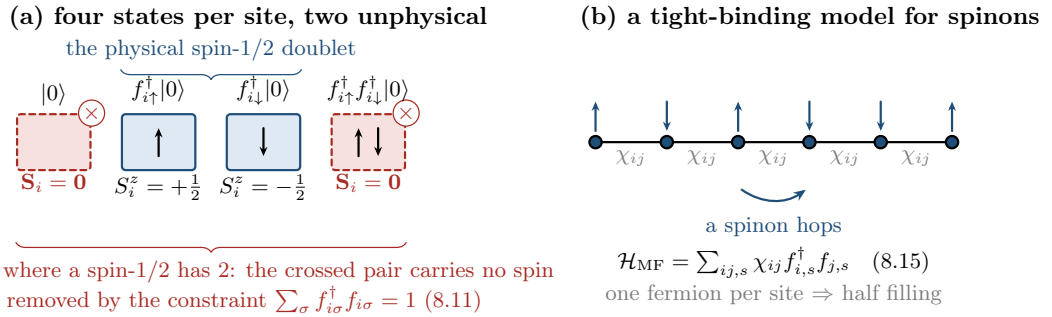


Figure 8.4: What the Abrikosov-fermion rewriting (8.10) costs, and what it buys. (a) A site's four fermionic states. The two singly-occupied ones reproduce the physical spin-1/2 doublet exactly; the empty and doubly-occupied ones are extra, unphysical states that (8.10) maps to $\mathbf{S}_i = \mathbf{0}$ and that the constraint (8.11) – the operator-side twin of the Gutzwiller projector (8.7) – removes by hand. (b) What is left after the mean-field decoupling (8.14): an ordinary tight-binding problem (8.15) for charge-neutral, spin-1/2 fermions hopping with the self-consistent bond amplitude χ_{ij}. One fermion per site is the reason the resulting spinon model must be studied at half filling, and choosing the same χ_{ij} for both spin species keeps $\langle \mathbf{S}_i \rangle = \mathbf{0}$, so unlike the magnetic mean fields of sessions 1 and 6 this one breaks no time-reversal symmetry.

Diagonalizing (8.15) on a given lattice produces a genuine spinon band structure, and depending on the resulting mean-field parameters, three qualitatively different outcomes are possible:

- **Gapless spinons**, with excitations arbitrarily close to zero energy.
- **Gapped spinons**, with a finite energy cost to create a spinon.

- **Dirac spinons**, with a linear energy-momentum relation at isolated points in the Brillouin zone.

Notebook check-in: `session8.ipynb`, "Spinon Fermi surface in gapless quantum spin liquids" (Core)

Build the spinon tight-binding Hamiltonian (8.15) on the triangular lattice and watch its Fermi surface evolve as the ratio of hopping amplitudes t_2/t_1 changes. The quiz asks you to redo it on the square and kagome lattices, and to justify why the effective spinon model must sit at half filling: that is exactly the single-occupancy constraint (8.11) at the mean-field level.

8.10 Emergent gauge symmetry and vison excitations

There is one more essential piece to the QSL story, and it is a direct consequence of the fact that we introduced fermions to represent a spin. Consider a local, site-dependent phase rotation of the auxiliary fermions,

$$f_{i\sigma} \;\to\; e^{i\theta_i} f_{i\sigma}, \qquad f^\dagger_{i\sigma} \to e^{-i\theta_i} f^\dagger_{i\sigma}. \tag{8.16}$$

Both the physical spin operator (8.10) and the single-occupancy constraint (8.11) are completely unaffected by (8.16): the phase $e^{i\theta_i}$ cancels between $f^\dagger_{i\alpha}$ and $f_{i\beta}$ in every term, so the Hamiltonian and the physical spin operators are exactly invariant. But the mean-field bond variable χ_{ij} is *not*: it picks up a relative phase, $\chi_{ij} \to e^{i(\theta_i-\theta_j)}\chi_{ij}$. This is precisely the structure of a lattice gauge theory: χ_{ij} behaves like a link variable, and (8.16) is a local ("gauge") transformation under which the physical spin theory is invariant while the mean-field saddle point of Section 8.9 is not. Any two choices of χ_{ij} related by (8.16) describe exactly the same physical quantum spin liquid; the mean-field decoupling only ever fixes a gauge-equivalence class, not a unique set of hoppings. This redundancy is the origin of an **emergent gauge theory** for quantum spin liquids.

Once this gauge structure is recognized, χ_{ij} becomes a genuinely dynamical field on the lattice links, capable of carrying its own topological defects, in direct analogy to how the superconducting order parameter's gauge degree of freedom can host vortices. Write the mean-field Hamiltonian as $\mathcal{H} = \sum_{ij,s} |\chi_{ij}| e^{i\theta_{ij}} f^\dagger_{i,s} f_{j,s}$, and define the flux around a plaquette as the product of phases picked up going around its boundary, $\Phi = \prod e^{i\theta_{ij}}$. A concrete example: on a square lattice, imagine hoppings around every elementary loop pick up phase $\Phi = +1$ (a ground state), except one loop, which picks up $\Phi = -1$. That single flipped plaquette is an excited state, a topological defect in the emergent gauge field, called a **vison**: the gauge sector's own excitation, generally coupled to the matter sector's spinon excitations built in Section 8.9. A quantum spin liquid, in this language, is characterized not just by its spinon dispersion but by an emergent gauge theory and the vison excitations that theory supports, a genuinely new organizing principle with no counterpart anywhere in the symmetry-broken magnets of session 6. There is no notebook exercise built around this construction; it stands as a purely conceptual, formal piece of the QSL picture. Figure 8.5 draws both halves of it: why the flux, and not the link variable, is the physical quantity, and what a single flipped flux looks like on the lattice.

8.11 From a single Kondo impurity to the Kondo lattice

Having built the machinery for quantum spin liquids, the lecture turns to a second, physically quite different setting that turns out to be solvable with exactly the same trick: the **Kondo lattice model**, which arises whenever a metal and a correlated, local-moment system coexist.

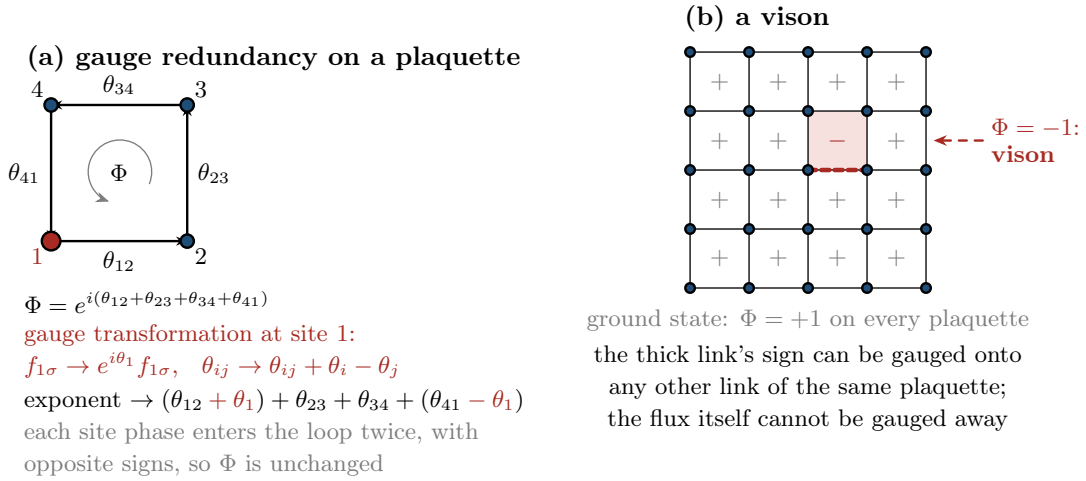


Figure 8.5: The emergent gauge theory of the parton construction, and its excitation. (a) The mean-field bond variable lives on the links, $\chi_{ij} = |\chi_{ij}|e^{i\theta_{ij}}$, and the local phase rotation (8.16) shifts every link phase by $\theta_i - \theta_j$. Going once around a plaquette, each site phase is picked up twice with opposite signs, so the flux $\Phi = \prod e^{i\theta_{ij}}$ is untouched: the link variables are pure gauge, the flux is physical. This is exactly why two mean-field solutions related by (8.16) describe the same spin liquid. (b) The flux is therefore what labels the state. With $\Phi = +1$ on every plaquette as the ground state, flipping a single plaquette to $\Phi = -1$ costs energy and cannot be undone by any gauge transformation: that localized defect of the gauge field is a **vison**. Which link carries the minus sign is a gauge choice and can be moved around at will; the shaded plaquette cannot.

Start with the simplest possible case: a single spin-1/2 magnetic impurity sitting inside a non-interacting two-dimensional electron gas. The total Hamiltonian is a free, dispersive kinetic term for the conduction electrons, plus a Heisenberg-like coupling between the impurity spin and the local conduction-electron spin density at the impurity site,

$$H = -t \sum_{\langle ij\rangle,\sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.} \right) + H_K, \qquad H_K = \sum_{\alpha\beta} \left(c^\dagger_{0\beta}\, \boldsymbol{\sigma}_{\beta\alpha}\, c_{0\alpha} \right) \cdot \mathbf{S}, \tag{8.17}$$

the **Kondo coupling** H_K. Solving this exactly is a genuinely hard many-body problem, but the physical picture of its ground state is simple to state even without the full solution: the impurity spin forms a singlet, structurally the same object as (8.3), with some effective conduction electron in its vicinity,

$$|GS\rangle \sim \frac{1}{\sqrt{2}} \Big(|\Uparrow\downarrow\rangle - |\Downarrow\uparrow\rangle \Big),$$

while the rest of the electron gas behaves as an effectively non-interacting Fermi sea. This turns out to be a genuinely good approximation to the many-body ground state, and the resulting state has total spin zero and does not break time-reversal symmetry: despite hosting a real local moment, this is already, in the sense of Section 8.4, a quantum magnet.

Now generalize to a whole **lattice** of magnetic impurities, one per site, each coupled to the same conduction-electron bath,

$$H = -t \sum_{\langle ij\rangle,\sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.} \right) + J \sum_{j,\alpha\beta} \left(c^\dagger_{j\beta}\, \boldsymbol{\sigma}_{\beta\alpha}\, c_{j\alpha} \right) \cdot \mathbf{S}_j. \tag{8.18}$$

Every impurity again wants to form a singlet with a nearby conduction electron, but with many impurities competing for the same electron sea, the screening electrons end up shared, fractionally, between different impurities' singlets. The result is a state of massive, lattice-wide

entanglement between all the magnetic sites and the electron gas, while the total spin remains exactly zero.

Notebook check-in: `session8.ipynb`, "Magnetization of a large Heisenberg model with a magnetic impurity" (Challenge) and "Magnetization of a dimerized Heisenberg model with a magnetic impurity" (Challenge)

These exercises show, numerically and in real space, how a single magnetic impurity's local field decays with distance into an otherwise gapless or dimerized quantum magnet: the gapless chain shows algebraic, staggered decay, tracing the correlation profile of a critical spin liquid; the dimerized (valence-bond-solid) chain of Section 8.5 instead shows exponential decay over a finite correlation length, because its ground state is gapped. This is the same screening-cloud physics behind (8.17), viewed from the real-space, single-chain side rather than the itinerant-electron side.

8.12 Solving the Kondo lattice: the same parton trick, now with a hybridization

The lecture makes the parallel to Section 8.8 explicit: this problem is solved the same way we solved the QSL. Replace every impurity spin by an Abrikosov fermion, $\mathbf{S}_j = \frac{1}{2} f^\dagger_{j\alpha} \boldsymbol{\sigma}_{\alpha\beta} f_{j\beta}$, subject to the same single-occupancy constraint (8.11), written here in the equivalent traceless form

$$S_{\alpha\beta}(j) = f^\dagger_{j\alpha} f_{j\beta} - \frac{n_f(j)}{N}\delta_{\alpha\beta}. \tag{8.19}$$

Substituting into (8.18) turns the Kondo coupling into a genuine quartic interaction, now coupling the auxiliary f fermions not to another site's f fermion (as in (8.13)) but to the *real* conduction electrons $c_{j\sigma}$ at the same site,

$$H \sim \sum_{\mathbf{k}\alpha} \varepsilon_{\mathbf{k}}\, c^\dagger_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} \;-\; J \sum_{j,\alpha\beta} \Big(c^\dagger_{j\beta} f_{j\beta}\Big)\Big(f^\dagger_{j\alpha} c_{j\alpha}\Big). \tag{8.20}$$

Exactly as in (8.14), we mean-field decouple this quartic term, taking the expectation value $\langle c^\dagger_j f_j \rangle$, and Fourier transform to momentum space. This produces a genuinely bilinear Hamiltonian mixing real electrons and auxiliary fermions,

$$H \sim \sum_{\mathbf{k}\alpha} \varepsilon_{\mathbf{k}}\, c^\dagger_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} \;-\; \gamma_K \sum_{\mathbf{k},\alpha\beta} f^\dagger_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} + \text{h.c.}, \tag{8.21}$$

with $\varepsilon_{\mathbf{k}}$ the **conduction band dispersion** and γ_K the **Kondo hybridization**, the single self-consistently determined mean-field parameter carrying all of the information about the strength of the original Kondo coupling J.

8.13 Electronic structure of the Kondo lattice: the Kondo insulator

Equation (8.21) is, per spin, an ordinary two-band Bloch Hamiltonian in the basis $(c_{\mathbf{k}\alpha}, f_{\mathbf{k}\alpha})$. When $\gamma_K = 0$, there is no mixing at all: the spectrum is simply the ordinary, dispersive conduction band $\varepsilon_{\mathbf{k}}$, together with a perfectly **flat**, zero-energy band of f-fermion levels (flat because these auxiliary fermions carry no hopping of their own once the Kondo coupling is switched off). Switching the Kondo coupling back on makes γ_K finite, and the flat and dispersive bands undergo an **anti-crossing**: wherever the flat spinon band would have crossed the conduction band, a **gap** opens instead, and the bigger the Kondo coupling, the bigger this

gap becomes. This is confirmed directly by comparing the spectra at increasing J_K: at $J_K = 0$ the bands simply cross; already by $J_K = 0.25$ a visible gap has opened at the crossing; by $J_K = 0.5$ the gap is substantially larger. Figure 8.6 shows the two ends of that comparison, and where the Fermi level sits in each.

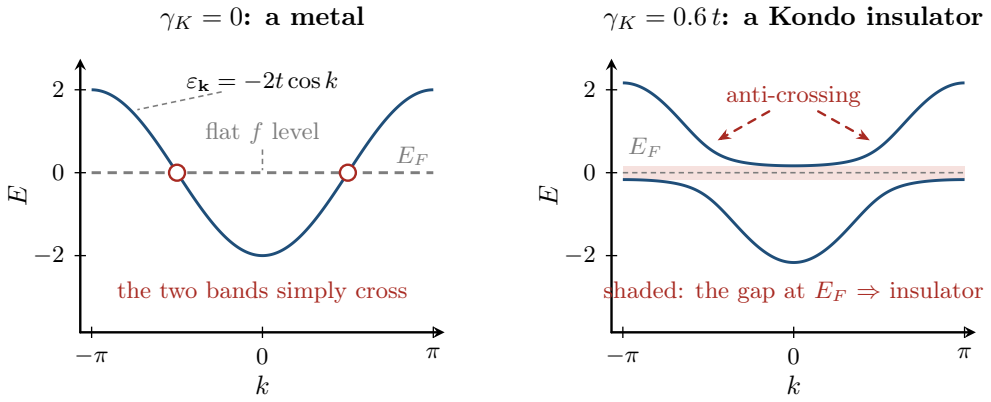


Figure 8.6: Why a lattice of local moments insulates. Left: with the Kondo coupling switched off, (8.21) is a dispersive conduction band $\varepsilon_{\mathbf{k}}$ together with a perfectly flat band of f levels – flat because the auxiliary fermions have no hopping of their own – and the two simply cross at the Fermi level, which is a metal. Right: a finite hybridization γ_K forbids the crossing. The bands repel, the states near the crossing become mixtures of conduction electron and local moment, and a gap opens exactly where the crossing was. With one f fermion per site from the constraint (8.11) and a half-filled conduction band, the lower hybridized band is exactly full: the Fermi level lands inside the gap and the metal has become a **Kondo insulator**. The gap grows with γ_K, and so with the Kondo coupling J_K; the nearly flat band edges it leaves behind are what makes the surviving quasiparticles heavy.

Physically, the ground state of this hybridized system is, by construction, non-magnetic: total spin zero, time-reversal symmetric, exactly as it was for the single impurity (8.17) and for the quantum spin liquid of Section 8.9. The consequence for the electronic structure, though, is new: a lattice of Kondo impurities converts what would otherwise be an ordinary metal into an **insulator**, because every electron sitting at the Fermi surface ends up forming a bound state with a magnetic impurity. This is the **Kondo insulator**, also called a heavy-fermion insulator, and SmB_6, introduced already in Section 8.1, is the standard real-material example.

It is worth noting the resemblance to a rather different construction from session 5: superconductors also open a gap in their quasiparticle spectrum through hybridization, there between electrons and holes rather than between electrons and auxiliary fermions. The two settings are physically very different, but the *mechanism* by which the single-particle spectrum develops a gap is closely analogous, and so is the way the effect turns on: the temperature scale at which the Kondo-insulator gap becomes visible depends on the Kondo coupling in a strongly non-linear way, the same qualitative non-linear dependence familiar from the superconducting gap's dependence on the pairing interaction.

Notebook check-in: `session8.ipynb`, "Quantum magnetism in the Kondo lattice" (Core) and "Spinon dispersion in the Kondo lattice" (Challenge)

The first exercise builds (8.21) directly on a van der Waals-type lattice: identify the flat (Kondo-site) band, the anti-crossing with the conduction band, and check what happens to the gap as you move the chemical potential. The second adds a direct exchange coupling between the Kondo sites themselves, giving the otherwise-flat spinon band its own dispersion, the lattice version of the Doniach picture: the Kondo coupling J_K favors a local screened singlet while the direct exchange J favors magnetic correlations between the local moments, and their competition controls which regime, screened or magnetic, is stable.

8.14 Closing summary

The lecture's own closing recap, reconstructed here with the equations it was pointing back to:

- Magnetism, in every form met across sessions 1, 6 and today, arises from strong repulsive interactions between electrons.
- Frustrated magnetic models (triangular and kagome lattices, Section 8.4) can display **quantum spin liquid** behavior: a ground state with zero local spin expectation value on every site (8.4), maximally entangled, generalizing the Heisenberg dimer singlet (8.3) through the valence-bond-solid state (8.5) to a genuinely resonating, frustrated ground state (8.6)-(8.7); conventional mean-field theory cannot capture it by construction.
- Because a QSL's zero magnetization is the opposite regime from the one Holstein-Primakoff needs (8.9), its fundamental excitations are not magnons but **spinons**, with $S = 1/2$: emergent, charge-neutral fermions obtained from the Abrikosov/parton construction, (8.10)-(8.15), the electron mean-field recipe of sessions 1 and 6 transplanted into the spin sector. Spinon bands can be gapless, gapped, or Dirac-like (Section 8.9).
- The parton construction carries an emergent gauge redundancy (8.16): the mean-field bond variable is a link field whose topological defects are gauge excitations called **visons**, generally coupled to the spinons (Section 8.10).
- Quantum entanglement mixing spin and fermion degrees of freedom also arises in **Kondo systems**: the same parton trick, applied to a lattice of local moments embedded in a metal, (8.17)-(8.21), turns the Kondo exchange coupling into a mean-field **hybridization** γ_K that gaps the conduction band at the anti-crossing and converts a metal into a **Kondo insulator** (heavy-fermion insulator), with the gap growing monotonically with the Kondo coupling strength (Section 8.13).

Session 9 picks this construction straight back up: the Abrikosov-fermion/spinon machinery built here gets reused directly, now applied to a single magnetic impurity sitting inside an otherwise ordinary metal, with an explicit callback to a Friedel-oscillation result derived earlier in that same session.

Session 9

Excitations and Defects in Quantum Materials

Session 8 left us with a small zoo of freshly built ground states: quantum spin liquids described by mean-field spinons, and Kondo lattices whose local moments, screened by conduction electrons through the very same Abrikosov-fermion trick, turn a metal into a heavy-fermion insulator. Every session up to this point has followed the same basic move: take a many-body Hamiltonian, find a controlled or approximate reformulation, and read off the ground state that comes out. Today we do something different with all of that machinery. Rather than building one more ground state, we are going to *probe* the ground states we already have. The idea is simple to state: take a quantum material, metal, superconductor, topological insulator, quantum magnet, whichever one you like, disturb it a little, and watch how it responds. It turns out that the response itself, computed entirely from the undisturbed ground state, tells us an enormous amount about what that ground state actually is. We will run this one idea, linear response, as a probe over essentially every system class the course has built so far: ordinary metals, session 5's superconductors, sessions 3 and 4's topological matter, and session 8's quantum magnets, and in each case we will see the response function acting as a diagnostic, reading off Fermi surfaces, superconducting pairing symmetries, topological invariants, and spin gaps, all from the same underlying formalism. Along the way we will also see the Stoner criterion for itinerant magnetism fall directly out of the same response-function language, reconnecting to session 1 and session 6's magnetism, but now viewed as an *instability* of the paramagnetic response rather than as a mean-field ground-state ansatz we imposed by hand. Today's session is, in that sense, the course's diagnostic and synthesis session: less a new building block than a new lens trained on everything we have already built.

9.1 Framing: response functions as a probe of the ground state

Let us set up the general experiment we have in mind, because the same setup is going to recur in every example today. We take a quantum material with some ground state we understand (or think we understand), and we make a small perturbation at position x and time t. Then we measure some observable at a different position x' and a different (typically later) time t'. The relation between what we did at (x, t) and what we observe at (x', t') is encoded in a **response function**, and once we know that function completely, we know how the material reacts to a perturbation applied anywhere, at any time.

The power of this framing is that it is completely observable-agnostic. The same formalism covers:

- a perturbation in the local electrostatic potential, measured as a change in the local

charge density, giving the **dielectric function**;

- a local magnetic moment or field, measured as a change in the local magnetization elsewhere, giving the **magnetic susceptibility** (equivalently, the spin-fluctuation response);
- a local temperature change, measured as a change in temperature elsewhere, giving the **thermal conductance**.

There are also two quite different ways of actually running this experiment in the lab, and it is worth having both in mind, because we will use each type of example today. One route is a genuine physical perturbation: substitute one atom of a metal for a different element, and you have created a real, localized impurity. As we will see in detail in Section 9.6, the resulting **Friedel oscillations** in the electronic density directly reveal the material's Fermi surface. The other route is that the measurement itself is the perturbation: fire a neutron, whose own magnetic moment couples to the local moments in a magnetic material, and the scattering cross-section you record is a direct readout of the material's magnetic fluctuations (inelastic neutron scattering). These look like very different experiments, but the underlying formalism, and the kind of information you extract, is the same in both cases.

Finally, a simplification that will make almost everything below tractable: whenever the unperturbed Hamiltonian has translational symmetry in space and is time-independent, the response function can only depend on $x' - x$ and $t' - t$, not on x, t, x', t' separately. That is precisely the situation in which Fourier transforming to reciprocal space and frequency, (q, ω), is the natural language, and it is the language we will use for essentially the rest of this session.

9.2 A classical warm-up: the response of a classical magnet

Before bringing in any quantum many-body machinery, it is worth seeing the whole logic of "response predicts the ground state" play out in a setting we already know how to solve exactly: a classical Ising magnet at finite temperature, treated in mean-field theory (recall Section 1.12 for the mean-field recipe itself). The question we want to ask is genuinely useful: sitting at a temperature well above any ordering transition, is there a measurement we can make *right there*, in the disordered phase, that tells us what kind of order the system is heading toward as we cool it down?

The relevant response is the **magnetic susceptibility**, $\chi = \partial M / \partial H$, the linear-response coefficient relating an applied field H to the magnetization M it induces. Solving the mean-field Ising model at high temperature gives three distinct behaviors, depending on the sign and strength of the exchange coupling:

$$\chi(T) \;\propto\; \begin{cases} \dfrac{1}{T} & \text{paramagnet (no exchange coupling), diverges only at } T = 0, \\ \dfrac{1}{T - T_c} & \text{ferromagnet, diverges at a positive } T_c, \\ \dfrac{1}{T + T_c} & \text{antiferromagnet, apparent divergence at a negative temperature.} \end{cases} \tag{9.1}$$

The physical picture behind (9.1) is worth holding on to, because we are going to meet it again, in a fully quantum guise, in Section 9.5. In the paramagnetic regime, thermal fluctuations dominate and the susceptibility stays finite at any $T > 0$: nothing is trying to order. In the ferromagnetic case, fluctuations *reinforce* the applied field, so as $T \to T_c^+$ the system polarizes more and more easily, and χ genuinely diverges right at the transition. In the antiferromagnetic case, fluctuations *fight* the applied field (favoring staggered rather than uniform

order), which shows up as the opposite sign in the denominator: the susceptibility looks like it is heading toward a divergence at a temperature below absolute zero, which of course it never reaches, but the sign alone already tells you the ordering tendency is antiferromagnetic rather than ferromagnetic. Experimentally, this is a genuinely practical diagnostic: plot $\chi^{-1}(T)$, and a straight line extrapolating to a positive intercept signals an incipient ferromagnet, a negative intercept an incipient antiferromagnet, all measured entirely from the disordered high-temperature phase. This is the conceptual template for the entire session: **the response of the disordered/uniform state, computed without ever solving for the ordered state itself, already contains the information about which ordered state is coming.**

9.3 Linear response and perturbation theory, done properly

Let us now build the quantum version of this idea carefully, since everything that follows rests on it. Split the full Hamiltonian into an unperturbed piece H_0 (a metal, a superconductor, a quantum magnet, whatever system we are studying) and a perturbation V, with a bookkeeping coupling constant λ that we will eventually set to one:

$$H(\lambda) = H_0 + \lambda V, \qquad \lambda = 0 \text{ (pristine)}, \quad \lambda = 1 \text{ (full perturbation)}. \tag{9.2}$$

Every observable of the system, energies, densities, magnetizations, wavefunctions themselves, becomes a function of λ once we fix $H(\lambda)$. If that function is smooth enough to Taylor-expand around $\lambda = 0$ (we will come back to exactly when this assumption fails), then for any observable A,

$$A(\lambda) = A^{(0)} + \lambda\, \chi_A + O(\lambda^2), \qquad \chi_A \equiv \left.\frac{\partial A}{\partial \lambda}\right|_{\lambda=0},$$

and by definition χ_A, the first-order coefficient, *is* the response function we have been talking about all session. To actually compute χ_A for a generic observable, we first need the perturbed wavefunctions. Expanding an unperturbed eigenstate $|n\rangle$ of H_0 (energy $E_n^{(0)}$) in the full basis $\{|m\rangle\}$ of H_0, standard Rayleigh-Schrödinger perturbation theory gives, to first order in λ and away from any degeneracy,

$$|n(\lambda)\rangle = |n\rangle + \lambda \sum_{m \neq n} \frac{\langle m|V|n\rangle}{E_n^{(0)} - E_m^{(0)}}\, |m\rangle + O(\lambda^2). \tag{9.3}$$

The structure of (9.3) is worth keeping firmly in view, because it is the engine behind every derivation in the rest of this session: a perturbation mixes a state $|n\rangle$ with every other state $|m\rangle$, weighted by the matrix element of V between them, and suppressed by the inverse energy gap between the two states. States close in energy mix strongly; states far apart barely mix at all. Once we have (9.3), any observable, in particular any density or magnetization, can be written down to first order in λ by inserting the perturbed states.

When does this break down? The whole construction in (9.3) presupposes that $A(\lambda)$ is an analytic function of λ near $\lambda = 0$, and that assumption is not automatic. Three paradigmatic quantum states are known to be intrinsically non-perturbative in exactly this sense, worth flagging explicitly since they mark the boundary of everything derived below:

- **An impurity in a metal, the orthogonality catastrophe.** In the thermodynamic limit, the full many-body wavefunction of a metal with an impurity switched on becomes exactly orthogonal to the wavefunction with the impurity switched off, no matter how weak the impurity potential is.

- **A quantum spin in a metal, the Kondo effect.** The emergent Kondo temperature is a non-perturbative function of the Kondo coupling (it involves an essential singularity, not a power series, in the coupling constant).
- **Superconductivity in the BCS limit.** The superconducting gap is likewise non-perturbative in the attractive coupling constant.

For the rest of this session we restrict ourselves to the regime where perturbation theory *does* hold, which is where (9.2) and (9.3) give a controlled, systematic way of computing responses.

9.4 The Fermi sea and the Lindhard function

To make (9.3) concrete, let us specialize to the case that will occupy most of today: a non-interacting electronic Hamiltonian in diagonal form, $H_0 = \sum_k \varepsilon(k)\, c_k^\dagger c_k$. Its many-body ground state is the Fermi sea: fill every single-particle state with $\varepsilon(k)$ below the Fermi energy, leave every state above it empty. Correspondingly, $\langle c_k^\dagger c_k \rangle = n_F(\varepsilon(k))$, equal to 1 for occupied states and 0 for unoccupied ones (at $T = 0$; more generally the Fermi-Dirac function), and the set of k-points sitting exactly at the Fermi energy defines the **Fermi surface**. This surface is going to be the single most important geometric object in everything that follows, because a small perturbation can only rearrange states close to it: moving an electron from deep in the Fermi sea to deep above it costs far too much energy to matter at first order.

Now apply (9.3) with a specific, physically motivated perturbation: a local shift of the on-site energy, i.e. a local change in the chemical potential at one point in the material. The observable we track is the change in local electronic density, $n(x) - n_0(x)$. Using translational symmetry, Fourier transform to q-space and insert the perturbed wavefunctions (9.3) into the density operator; after some algebra (each term comes from a matrix element $\langle k+q|V|k\rangle$ divided by the energy denominator $\varepsilon(k+q) - \varepsilon(k)$, exactly the structure of (9.3)) one obtains the **static Lindhard function**,

$$\chi_0(q) = \frac{1}{N} \sum_k \frac{n_F(\varepsilon(k)) - n_F(\varepsilon(k+q))}{\varepsilon(k+q) - \varepsilon(k)}. \tag{9.4}$$

It is worth checking (9.4) makes sense before using it. As $q \to 0$, expanding both numerator and denominator to first order in q gives $\chi_0(q \to 0) \to -\sum_k n_F'(\varepsilon(k))/N$, which at $T = 0$ is exactly $D(E_F)$, the density of states per site at the Fermi level: a positive, finite number, as it should be. Equation (9.4) is non-zero only when k is occupied and $k+q$ is unoccupied (or vice versa); if both are occupied or both empty, the numerator vanishes. So the response is built entirely out of states straddling the Fermi surface, exactly as the general argument above anticipated.

Instabilities and nesting. Equation (9.4) has a striking feature in one dimension. If there exists a wavevector $q^\star$ connecting an occupied state at k to an empty state at $k + q^\star$ of the *same* energy, the denominator in (9.4) vanishes right at that term, and $\chi_0(q^\star)$ diverges. For a generic 1D band the Fermi surface is just the two points $\pm k_F$, and the single nesting vector connecting them is $q^\star = 2k_F$ (figure 9.1a,b); summing (9.4) over the whole band gives a logarithmic divergence, $\chi_0(2k_F) \sim \ln(W/T)$ (with W a bandwidth cutoff), which is exactly the Peierls instability that drives one-dimensional metals to dimerize into a charge density wave. Physically: a response function that diverges means an arbitrarily weak perturbation at that wavevector produces an arbitrarily large rearrangement of the ground state, i.e. the ground state we started from is unstable to that particular fluctuation. This is the central strategic idea we will keep reusing: to predict a new phase of matter, look for the wavevector at which the response function of the old phase diverges.

Everything above generalizes straightforwardly to finite frequency: repeating the derivation with a time-dependent perturbation $\propto e^{-i\omega t}$ gives the (retarded) **dynamical Lindhard function**,

$$\chi_0(q,\omega) = \frac{1}{N}\sum_k \frac{n_F(\varepsilon(k)) - n_F(\varepsilon(k+q))}{\varepsilon(k+q) - \varepsilon(k) - \hbar\omega - i0^+}, \tag{9.5}$$

which reduces to (9.4) at $\omega = 0$. For a one-dimensional chain, plotting $\operatorname{Im}\chi_0(q,\omega)$ over the whole (q,ω) plane produces a characteristic surface of divergences that traces out the particle-hole continuum of the band structure; this ω-q plot is slide-only material, the spoken lecture describes its shape in words ("a whole collection of ω versus q values where there are divergences") but never draws it, so the figure itself is worth opening the slides for.

The same logic extends cleanly to two dimensions. For a square lattice with dispersion $\varepsilon(k) = -2t(\cos k_x + \cos k_y)$, low filling gives a small, roughly circular Fermi surface, and $\chi_0(q,0)$ is correspondingly large for q inside a circle set by the Fermi surface's diameter and small outside it: the response function is, quite literally, a picture of the Fermi surface. This is the physical basis of **quasiparticle interference (QPI)**: real STM experiments put impurities into a 2D metal, image the resulting modulation of the local density of states, and invert the observed wavelength to read off the Fermi surface. At half filling, however, the square lattice's particular dispersion gives $\varepsilon(k+Q) = -\varepsilon(k)$ for $Q = (\pi,\pi)$: a *perfect* nesting condition connecting enormous portions of the Fermi surface (which is itself a square of straight lines at half filling, not a circle) all at the same wavevector simultaneously. Equation (9.4) then develops a massive divergence exactly at $q = (\pi,\pi)$, signaling a strong instability there. Figure 9.1 collects the whole argument in one place: the single 1D nesting vector, the logarithmic divergence it produces in $\chi_0(q)$, and the straight-edged 2D Fermi surface that $Q = (\pi,\pi)$ nests all at once. The half-filling Fermi surface shape and the explicit nesting vector are shown on the slides more concretely than described in speech; it is worth looking at that figure directly to see the straight parallel edges that make the (π,π) nesting so effective.

Notebook check-in: `session9.ipynb`, "The charge-charge response function" (Challenge)

Plot $\operatorname{Im}\chi_0(q,\omega)$ for a tunable band structure and chemical potential μ, exactly as in (9.5). Watch the divergence track the Fermi surface as you change μ, and see whether you can predict, from the band structure alone, which q-vector the instability will sit at, before running the calculation.

9.5 Interactions as a perturbation: RPA, plasmons, and the Stoner instability

So far the perturbation has been an external impurity. Nothing stops us from treating the electron-electron interaction *itself* as the perturbation, using χ_0, the non-interacting response we just built, as input, and asking for the response χ of the full interacting system. We need two things: a rule connecting χ to χ_0, and a criterion for reading instabilities off of χ.

From χ_0 to χ: the random phase approximation. Consider a contact (Hubbard-like) interaction of strength U coupling to the density. The physical picture is a diagrammatic one: the response to a probe is the bare response χ_0, plus the bare response followed by one interaction event and another bare response, plus two interaction events, and so on, an infinite ladder of terms. This is most efficiently captured not by summing the series term by term but by writing down the self-consistent equation the full response must satisfy,

$$\chi(q,\omega) = \chi_0(q,\omega) + \chi_0(q,\omega)\,U\,\chi(q,\omega),$$

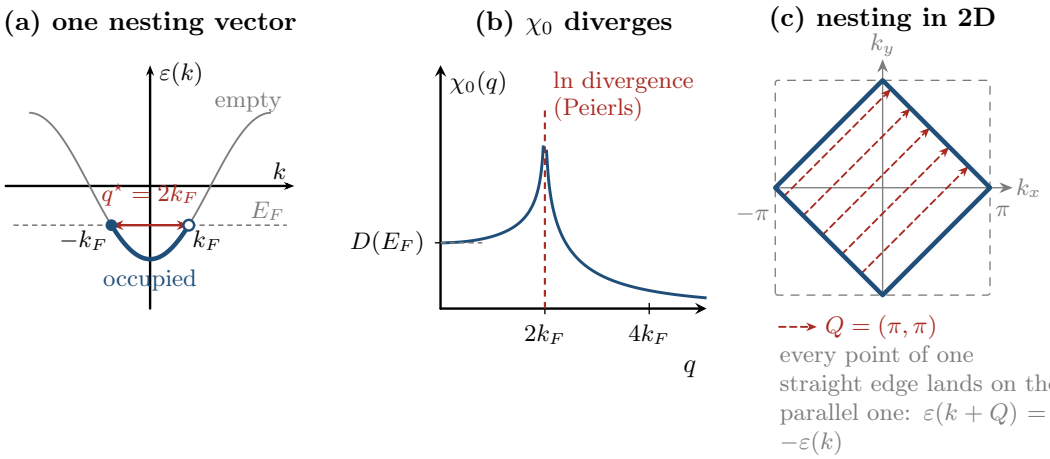


Figure 9.1: Nesting is what makes a response function diverge. (a) A one-dimensional band $\varepsilon(k) = -2t\cos k$ at partial filling: the occupied states (thick) end at $\pm k_F$, and the single wavevector $q^\star = 2k_F$ connects one Fermi point to the other, so the energy denominator of (9.4) vanishes for that term. (b) The resulting static Lindhard function (the closed form quoted in Section 9.6): finite and equal to $D(E_F)$ as $q \to 0$, exactly as the $q \to 0$ check of (9.4) requires, and logarithmically divergent at $q^\star = 2k_F$ — the Peierls instability. (c) In two dimensions the criterion becomes geometric. At half filling the square lattice's Fermi surface is the straight-edged diamond $|k_x| + |k_y| = \pi$, and because $\varepsilon(k+Q) = -\varepsilon(k)$ for $Q = (\pi, \pi)$, that one vector maps *every* point of one edge onto the parallel opposite edge simultaneously. This is why the divergence at (π, π) is so much stronger than the one-dimensional one: whole segments of Fermi surface nest, not isolated points. At low filling the same Fermi surface is a small circle, which no single q can nest this way.

which solves algebraically to give the **random phase approximation (RPA)**,

$$\chi(q,\omega) = \frac{\chi_0(q,\omega)}{1 - U\,\chi_0(q,\omega)}. \tag{9.6}$$

Equation (9.6) is a curious hybrid: derived order by order in U, it is perturbative, but the act of resumming the whole geometric series makes it non-perturbative, capturing physics (a genuine pole, i.e. a genuine phase transition) that no finite order in U alone ever could.

Two distinct sources of divergence. The interacting response (9.6) can diverge for two quite different reasons, and it is worth telling them apart because they correspond to two different emergent quasiparticles.

From χ_0 itself, exactly as in Section 9.4: an electron-hole excitation, a single-particle process, moving one electron from an occupied to an empty state.

From the RPA denominator, $1 - U\chi_0 = 0$, which is a genuinely new, collective phenomenon coming from the interaction. For a long-range Coulomb interaction, $U \to v(q)$, with a strong q-dependence (diverging as $q \to 0$), the pole of (9.6) as a function of ω at fixed q defines a new quasiparticle, the **plasmon**: a collective charge oscillation of the whole electron gas. Its dispersion depends on the screening and dimensionality of the system, but in the standard RPA (Bohm-Pines) treatment of the three-dimensional electron gas, at long wavelength it takes the form

$$\omega_{\mathrm{pl}}(q)^2 = \omega_p^2 + \tfrac{3}{5} v_F^2\, q^2 + O(q^4), \tag{9.7}$$

with ω_p the classical plasma frequency; the plasmon survives even at $q = 0$, unlike the particle-hole continuum, because it is a genuinely collective mode of the whole charge density rather than a single-particle excitation. Figure 9.2 puts both kinds of divergence on the same (q, ω) plane, and shows explicitly the small-q window in which the pole exists and no single-particle excitation does.

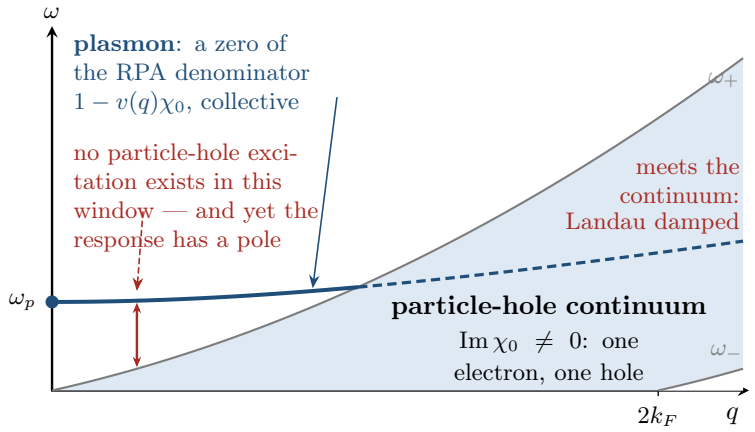


Figure 9.2: The two divergences of the RPA response (9.6), drawn in the (q, ω) plane for the three-dimensional electron gas. The shaded region is where $\mathrm{Im}\,\chi_0(q,\omega) \neq 0$: the *particle-hole continuum*, every (q,ω) at which a single electron can be lifted from an occupied state k to an empty state $k+q$. Its upper boundary is $\hbar\omega_+ = \hbar v_F q + \hbar^2 q^2/2m$, and beyond $2k_F$ a lower boundary $\hbar\omega_- = \hbar^2 q^2/2m - \hbar v_F q$ opens up; both follow from the energy denominator of (9.5), and for $q < 2k_F$ the continuum reaches all the way down to $\omega = 0$ — the same statement as χ_0 being built entirely from states straddling the Fermi surface. The plasmon (9.7) is a different object: it is a zero of the RPA *denominator* $1 - v(q)\chi_0$, so it exists at ω_p even as $q \to 0$, in a window where no single-particle excitation exists at all. That window, marked here, is the whole content of the phrase "collective mode, not a particle-hole pair". Once the branch meets the continuum it can decay into particle-hole pairs and is Landau damped (dashed).

The Stoner criterion, as a response-function instability. Now apply the same RPA logic to the *spin* channel of a Hubbard-type interaction, so χ_0 in (9.6) becomes the non-interacting spin susceptibility. The interacting spin response diverges precisely when

$$U\,\chi_0(q,0) = 1. \tag{9.8}$$

At $q = 0$, using the $q \to 0$ limit of (9.4), $\chi_0(0,0) = D(E_F)$, so (9.8) reduces to $U\,D(E_F) = 1$: this is exactly the **Stoner criterion** for itinerant ferromagnetism, an interacting metal becomes magnetic once the product of the density of states at the Fermi level and the interaction strength crosses one. But (9.8) is more general than the $q = 0$ special case: for a Fermi surface with strong nesting, such as the half-filled square lattice of Section 9.4, $\chi_0(q,0)$ is *maximized* not at $q = 0$ but at the nesting vector $Q = (\pi, \pi)$, where it is already logarithmically enhanced. The instability threshold in (9.8) is then reached first at $Q = (\pi,\pi)$ rather than at $q = 0$: the system develops a *staggered*, antiferromagnetic instability rather than a uniform ferromagnetic one, at a smaller value of U than the naive Stoner estimate would suggest.

It is worth pausing on this, because it is exactly the reconnection the course has been building toward. Sessions 1 and 6 derived magnetism by writing down the Hubbard model, applying mean-field theory directly to the *ordered* state, and self-consistently solving for a non-zero magnetization m (Section 1.13's Equation 1.7). Today we have derived the *same physics*, the same threshold $U\,D(E_F) = 1$, the same preference for antiferromagnetism at half filling, without ever writing down an ordered mean-field ansatz: we asked instead when the *paramagnetic*, disordered response function develops a divergence, exactly the same logic as Section 9.2's classical susceptibility. Mean-field theory and the response-function view are two routes to the same instability, one computed from the broken-symmetry side, one computed entirely from the symmetric side by watching it become unstable.

The same machinery, applied to different response channels and different signs of interaction, classifies a whole family of instabilities that is worth laying out explicitly (this three-way

classification is presented as a clean summary table on the slides, more systematically than the spoken walkthrough, which mainly worked the magnetic case through in detail):

Instability	Response channel	Interaction sign
Magnetism	spin response	repulsive
Dimerization / charge density wave	charge-phonon response	repulsive
Superconductivity	electron-electron scattering response	attractive

Notebook check-in: `session9.ipynb`, "The charge-charge response function" (Challenge)

The same notebook section used above for nesting extends, in its second part, to the interacting case: add a Hubbard interaction U and an external staggered (antiferromagnetic) field to a dimerized chain, and watch what happens to the response as U increases. This is the notebook-scale version of equation (9.8), the Stoner instability appearing as U crosses the threshold set by χ_0.

9.6 Impurities as diagnostics I: Friedel oscillations in a metal

We now turn to the most physical realization of a perturbation: an actual local impurity. Four case studies follow, one for each broad class of ground state we have built over the course: metals, superconductors, topological matter, and quantum magnets. In each case the diagnostic logic is the same: perturb locally, and read the nature of the ground state off of how the perturbation's effect decays in space.

Start with a metal. An impurity is a local on-site energy shift, and adding one to a translationally symmetric Hamiltonian necessarily breaks that symmetry, but note that perturbation theory itself never required translational symmetry in the first place: equations (9.2) and (9.3) hold for any H_0 and any V. Applying them directly to the local density of states gives a response proportional to matrix elements between Fermi-surface states, and for a generic (parabolic-like) band the resulting oscillation is set entirely by the Fermi wavevector.

To see this rigorously, note that the static Lindhard function of a one-dimensional free-electron gas has the closed form

$$\chi_0(q) = \frac{m}{\pi\hbar^2 q} \ln\left|\frac{2k_F + q}{2k_F - q}\right|,$$

which is smooth everywhere except for a logarithmic non-analyticity exactly at $q = 2k_F$, the very same nesting singularity identified in Section 9.4. A standard fact of Fourier analysis is that a non-analyticity of a function $\chi_0(q)$ at a wavevector $q^\star$ produces an algebraically decaying oscillatory tail in its real-space Fourier transform, with period fixed by $q^\star$ and an envelope set by the *type* of the singularity. Applying this to the impurity-induced density change $\delta n(x) = \int \frac{dq}{2\pi}\, e^{iqx}\, \chi_0(q)\, v_0$ (Born approximation, impurity strength v_0) gives the **Friedel oscillation**,

$$\delta n(x) \xrightarrow{|x|\to\infty} A(v_0)\, \frac{\cos\big(2k_F|x| + \varphi_0\big)}{|x|^{\,d}} \qquad (d = 1, 2, 3), \tag{9.9}$$

with a universal oscillation wavevector fixed purely by the Fermi surface ($2k_F$ in 1D, or the Fermi-surface diameter more generally), an algebraic envelope whose power depends on the dimensionality d, and a non-universal amplitude and phase set by the impurity strength and microscopic details. This is a genuinely elegant closure of the loop opened in Section 9.4: the very same $q = 2k_F$ singularity in $\chi_0(q)$ that we identified there as an instability channel is, here, what produces a long-ranged, power-law-decaying real-space oscillation, rather than an actual instability, because a single local impurity is far too weak a perturbation to reorganize

the whole Fermi sea; the divergence in $\chi_0(q)$ just makes its Fourier partner decay slowly rather than exponentially.

Concretely, for a one-dimensional tight-binding chain with an impurity at site 0, the local density of states oscillates around the impurity with wavelength π/k_F (half the Fermi wavelength $\lambda_F = 2\pi/k_F$, since the modulation is set by $2k_F$ rather than by k_F), and raising the chemical potential shortens k_F^{-1} and speeds up the oscillation correspondingly; the same phenomenon appears, with the same physical content, in a two-dimensional square lattice, and this real-space LDOS modulation around impurities is exactly what an STM measures in practice (this is quasiparticle interference again, seen this time in real rather than reciprocal space). The explicit LDOS-versus-position plots at two different fillings, for both the 1D and the 2D case, are genuinely clearer on the slides than in words: the "oscillations get faster as k_F grows" comparison is a visual one, worth looking at directly rather than only reading the formula. As a complementary check, the slides (and notebook) also show the *insulating* case: an impurity placed in a gapped 1D chain produces an LDOS that converges to its uniform value very quickly rather than oscillating with a slow power-law envelope, a preview of the gapped-versus-gapless diagnostic that we will make fully general in Section 9.9 and draw in figure 9.5.

Notebook check-in: `session9.ipynb`, "Impurity in a 1d metal" (Core), "Impurities in a 2d metal" (Core), and "Impurity in an electronic insulator" (Core)

Place a single impurity in a 1D chain, then a 2D square lattice, and watch the LDOS oscillate as in (9.9). Change the chemical potential and confirm that the oscillation wavelength tracks π/k_F exactly as predicted; this is the cleanest place in the whole session to see a response formula and a numerical experiment agree digit for digit. The third section repeats the exercise with the chemical potential inside a gap instead of a partially filled band: confirm that the LDOS now converges to its bulk value within a site or two, with no slowly-decaying oscillatory tail, exactly the gapped-versus-gapless contrast that Section 9.9 makes fully general.

9.7 Impurities as diagnostics II: magnetic impurities in a superconductor

Superconductors behave completely differently under a local perturbation, because a superconductor is, for single-particle excitations, a bulk insulator: there is a finite gap Δ, so there is no Fermi surface to produce Friedel-type oscillations. The interesting physics instead concerns the distinction between magnetic and non-magnetic impurities.

Recall that an s-wave Cooper pair binds one spin-up and one spin-down electron. A **non-magnetic** impurity, an ordinary local potential, does not distinguish spin up from spin down: it does not break time-reversal symmetry, and so it cannot lower the energy cost of breaking any particular Cooper pair. Concretely, the density of states of a conventional s-wave superconductor with and without a non-magnetic impurity is essentially unchanged, gap and all: the slides show the two spectra directly overlapping.

A **magnetic** impurity is entirely different: it couples with strength J to the local spin, distinguishes up from down, and therefore *does* break time-reversal symmetry. This lowers the energy cost of breaking one specific Cooper pair and produces a genuine bound state inside the gap. In the classical-spin (Born) approximation this is the **Yu-Shiba-Rusinov** problem (the transcript does not name it, but the setup matches it exactly): with $\alpha \equiv \pi\nu_0 JS$ (where ν_0 is the normal-state density of states at the Fermi level and S the impurity spin), the in-gap bound state energy is

$$E_b(J) = \Delta\,\frac{1-\alpha^2}{1+\alpha^2}. \tag{9.10}$$

Equation (9.10) reproduces exactly the qualitative curve the slides show as a function of J: at $J = 0$ ($\alpha = 0$), $E_b = \Delta$, sitting right at the gap edge, i.e. no genuine bound state yet. As J increases, α grows, and E_b moves down through the gap; at $\alpha = 1$ (i.e. $\pi\nu_0 JS = 1$) the state crosses zero energy exactly, and for $\alpha > 1$ the numerator of (9.10) turns negative, so the bound state re-emerges on the opposite side of zero. This crossing is a genuine quantum phase transition: below it the many-body ground state has total spin $S_{\text{tot}} = 0$ (every Cooper pair intact, the impurity spin screened into the condensate), above it $S_{\text{tot}} = 1/2$ (one Cooper pair broken, leaving a single dangling spin). A superconductor with a single, tunable magnetic impurity is therefore a minimal, exactly solvable model of a quantum phase transition, driven entirely by a local perturbation. Both halves of the story are drawn in figure 9.3: the in-gap states that a magnetic impurity adds to an otherwise untouched BCS spectrum, and the trajectory (9.10) traces across the gap as α grows.

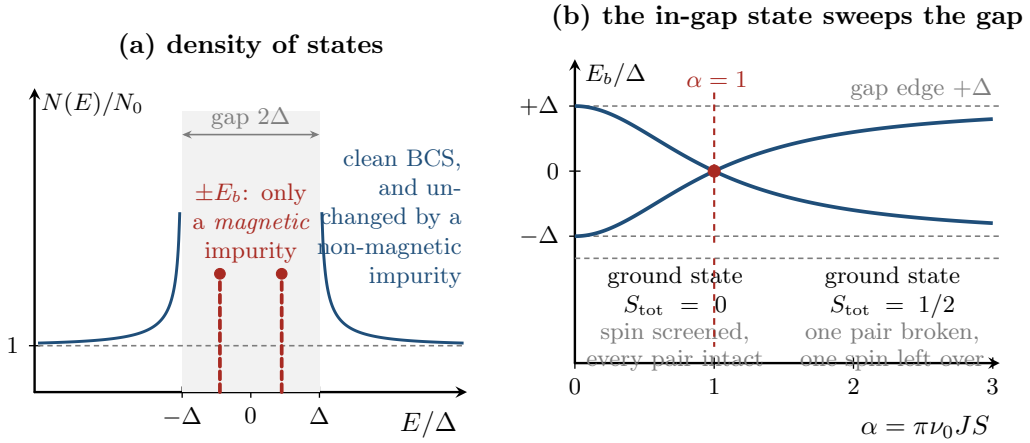


Figure 9.3: A magnetic impurity in an s-wave superconductor. (a) The single-particle density of states. The BCS continuum, gapped by Δ with coherence peaks at the gap edges, is left essentially untouched by a *non-magnetic* impurity: it does not distinguish up from down, so it cannot make breaking any particular Cooper pair cheaper. A *magnetic* impurity does distinguish them, breaks time-reversal symmetry, and pulls a pair of bound states into the gap at $\pm E_b$. (b) Where those states sit, from (9.10), as a function of $\alpha = \pi\nu_0 JS$. At $\alpha = 0$ they sit exactly at the gap edges, which is to say there is no bound state yet; as the exchange grows they sweep across the gap and cross at $\alpha = 1$. The crossing is not a level repulsion that failed to open — it is a genuine quantum phase transition of the many-body ground state, from $S_{\text{tot}} = 0$ (every Cooper pair intact, the impurity spin screened into the condensate) to $S_{\text{tot}} = 1/2$ (one pair broken, one dangling spin left behind). A single tunable impurity is thus an exactly solvable quantum phase transition.

Notebook check-in: `session9.ipynb`, "Magnetic impurities in a conventional superconductor" (Core) and "Single magnetic impurity in a superconductor" (Challenge)

Build a 1D superconducting chain with a magnetic impurity of tunable exchange J and track the in-gap state's energy exactly as in (9.10). Confirm the sign change at the crossing, and check that a non-magnetic impurity (set $J = 0$, vary only the local potential) leaves the gap edges essentially untouched, as the qualitative argument above predicts.

9.8 Impurities as diagnostics III: vacancies in topological matter

The strongest possible local impurity is a **vacancy**: remove a site entirely, equivalently send its on-site potential to infinity. This is where the bulk-boundary correspondence built in sessions

3 and 4 (Chern number, Zak phase, and the guarantee of gapless modes at any boundary of a topologically non-trivial phase) pays off directly. Topological boundary modes are known to survive around a large hole cut into a topological material; shrink that hole all the way down to a single missing site, and the argument does not stop working. A vacancy is, in effect, a boundary of zero size, and bulk-boundary correspondence still applies: whenever the bulk carries a non-trivial topological invariant, a vacancy is guaranteed to host an in-gap bound state, as long as the vacancy itself preserves the symmetry protecting that invariant.

This gives a genuinely useful diagnostic, because a fully gapped bulk spectrum on its own cannot distinguish a topological phase from a trivial one. Take a **chiral, spin-triplet topological superconductor** (the class studied via Majorana modes and Chern-number classification in session 5): its bulk spectrum, with no impurities, is fully gapped and looks, on its own, indistinguishable from a trivial s-wave superconductor. But a vacancy is a non-magnetic perturbation (it does not distinguish spin, so it preserves time-reversal symmetry, even in the extreme limit where the local potential diverges), and once it is inserted, an in-gap mode appears. The same logic runs for a **Chern insulator**: a vacancy in a system with finite Chern number produces an in-gap mode, while the same vacancy in a topologically trivial insulator produces nothing of the sort. In both cases, the observable signature is identical and simple: *if introducing vacancies produces in-gap modes, the underlying phase is topological; if it does not, the phase is trivial*, regardless of what the clean bulk spectrum alone would have told you. Figure 9.4 draws both the geometric argument (a hole shrunk down to a single site) and the three spectra that make the diagnostic usable.

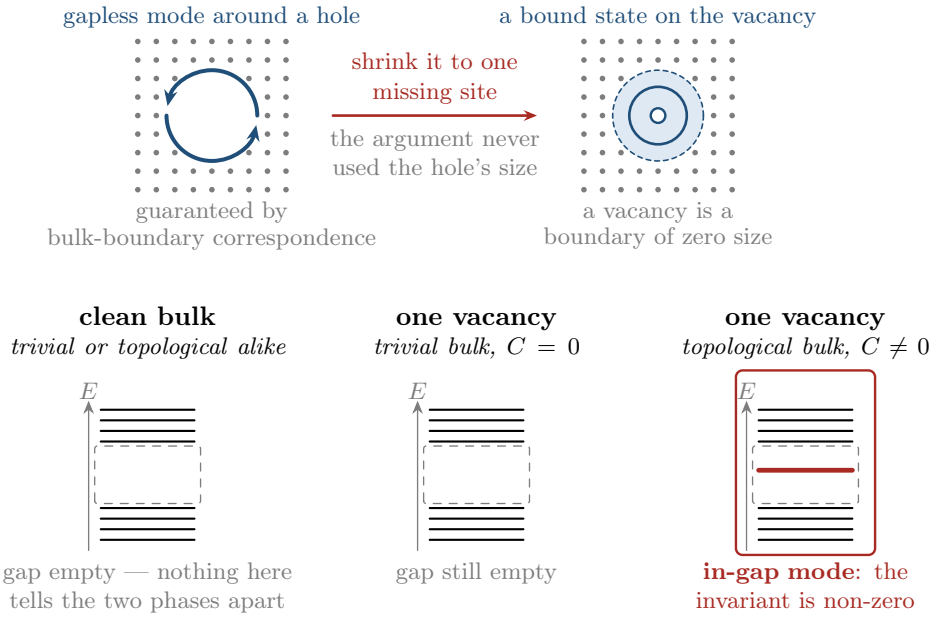


Figure 9.4: Why a single missing atom detects topology. Top: cut a large hole into a topologically non-trivial insulator or superconductor and bulk-boundary correspondence guarantees gapless modes running around its edge. Nothing in that argument refers to the hole's size, so shrinking it down does not switch it off — a vacancy is a boundary of zero size, and it binds an in-gap mode. Bottom: this is exactly the diagnostic that a clean spectrum cannot give. A trivial and a topological insulator with the same gap have bulk spectra that look identical, so nothing there distinguishes them (left). Add a vacancy and only the topological one develops a state inside the gap (right). The rule is as simple as it looks: vacancies produce in-gap modes if and only if the bulk invariant is non-zero, provided the vacancy itself respects the symmetry protecting that invariant.

Notebook check-in: `session9.ipynb`, "Non-magnetic impurities in a topological superconductor" (Challenge), "DOS of non-magnetic impurities in a topological superconductor" (Challenge), "Single vacancy in a topological superconductor" (Challenge), "Vacancies in a topological insulator" (Challenge), and "Single vacancy in a topological insulator" (Challenge)

Build a chiral p-wave superconductor and a Chern (Haldane-model) insulator, each with a single vacancy, and confirm the in-gap mode appears in both. The notebook's own suggested check is a good one: swap the pairing symmetry to conventional s-wave, or the topological gap to a trivial sublattice-imbalance gap, and watch the in-gap mode disappear, precisely because the bulk invariant that guaranteed it via bulk-boundary correspondence is now zero.

9.9 Impurities as diagnostics IV: screening in quantum magnets

The last case study returns us to session 8's construction directly. Take a frustrated Heisenberg model realizing a quantum-spin-liquid-like ground state: zero net magnetization on every site, and, crucially, a ground state that does not break time-reversal symmetry. Add one local magnetic perturbation and ask how the surrounding spins screen it.

For $S = 1/2$, we do not need to redo the construction from scratch, session 8 already built exactly the tool we need: represent each spin by a pair of auxiliary (Abrikosov) fermions, $S_i = \frac{1}{2}\sum_{\alpha\beta} f^\dagger_{i\alpha}\boldsymbol{\sigma}_{\alpha\beta} f_{i\beta}$, subject to the single-occupancy constraint $\sum_\sigma f^\dagger_{i\sigma} f_{i\sigma} = 1$ at every site, and mean-field decouple the resulting quartic Hamiltonian. As derived in session 8, this turns the Heisenberg model plus a local magnetic perturbation into a bilinear, tight-binding-like Hamiltonian for the spinons, with an effective on-site term at the impurity site that mixes with the spinon band. We are not going to re-derive that mapping here; see session 8's notes (its Abrikosov-fermion construction) for the full mean-field decoupling.

What matters for us today is what this mapping buys us: it turns the "screen a magnetic moment embedded in a quantum spin liquid" problem into *exactly* the same mathematical problem as Section 9.6's "impurity in a one-dimensional electron gas," just with electrons relabeled as spinons. The induced spin correlation $\langle S_0 \cdot S_n \rangle$ as a function of distance n from the impurity therefore inherits the same structure as (9.9): an oscillation set by the antiferromagnetic exchange, decaying with distance according to whether the underlying spinon band is gapped or gapless.

This is where the comparison between $S = 1/2$ and $S = 1$ Heisenberg chains becomes sharp. Both realize a spin-liquid-like ground state with zero on-site magnetization in the absence of an impurity, and both develop an oscillating induced moment once the impurity is switched on, but the decay is qualitatively different:

$$\langle S_0 \cdot S_n \rangle \;\sim\; \begin{cases} \dfrac{(-1)^n}{n} & S = 1/2 \quad \text{(power-law, gapless)}, \\ (-1)^n\, e^{-n/\xi} & S = 1 \quad \text{(exponential, gapped)}. \end{cases} \tag{9.11}$$

The reason is exactly the reasoning behind (9.9) once more: the spin-1/2 Heisenberg chain maps, via the Abrikosov-fermion construction, onto a gapless spinon tight-binding chain, with a genuine Fermi surface (two points, as in Section 9.4's 1D metal), so the impurity response has to decay algebraically, precisely as a metal's Friedel oscillations do. The spin-1 chain, in contrast, has an excitation gap (the Haldane gap, not named explicitly in the transcript but exactly what opens for integer-spin antiferromagnetic chains), which sets a finite correlation length ξ and forces the impurity response to decay exponentially, exactly as charge screening does inside a band insulator.

This gives us a completely general diagnostic, one that ties together every case study in this session: *power-law decay of an impurity response signals a gapless ground state; expo-*

nential decay signals a gapped one, and the exponential's prefactor is literally the correlation length. Section 9.6's metal (power law), the insulating chain noted at the end of that section (exponential), and the two quantum-magnet cases here are all instances of the same underlying statement, drawn side by side in figure 9.5.

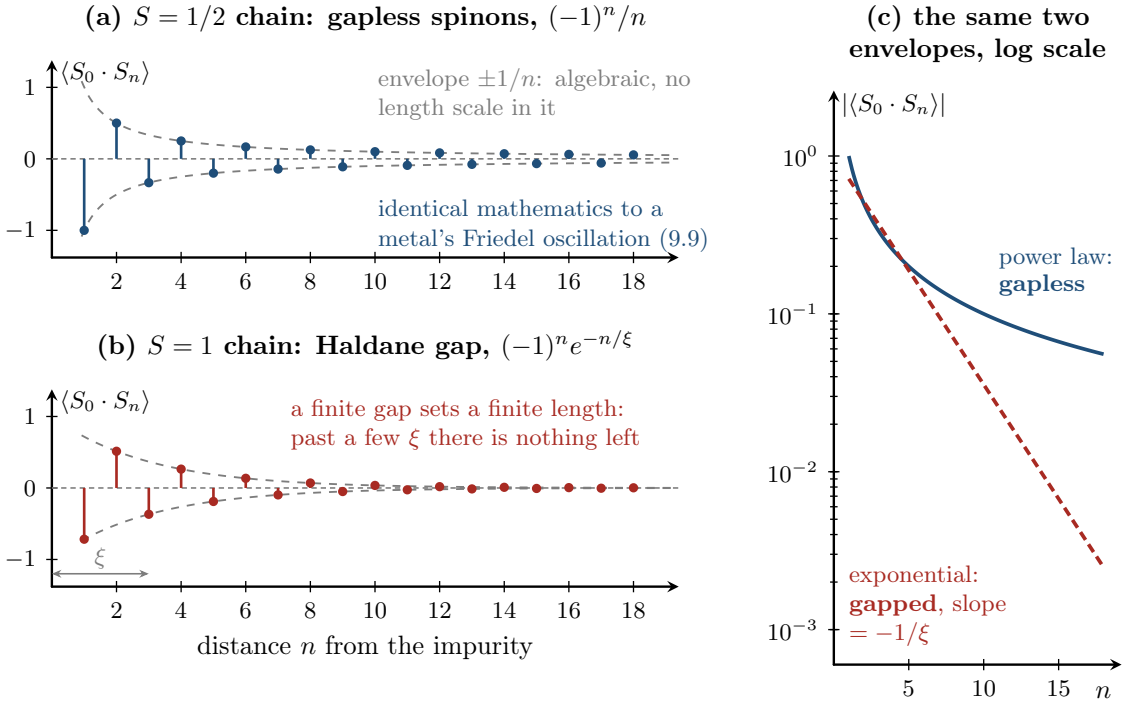


Figure 9.5: The one diagnostic that ties the whole session together, drawn for the two Heisenberg chains of (9.11). Both look the same close to the impurity: an induced moment that alternates in sign from site to site, because the exchange is antiferromagnetic. What distinguishes them is only how the envelope falls off. (a) The $S = 1/2$ chain maps, through session 8's Abrikosov-fermion construction, onto a *gapless* spinon band with a genuine (two-point) Fermi surface, so its envelope decays as the power law $1/n$ — the same mathematics, and the same reason, as the Friedel oscillations (9.9) of a metal. (b) The $S = 1$ chain has the Haldane gap, which fixes a finite correlation length ξ and forces an exponential envelope $e^{-n/\xi}$, exactly as charge screening decays inside a band insulator. (c) On a logarithmic vertical scale the two become trivially distinguishable: an exponential is a straight line of slope $-1/\xi$, a power law is not. Read the statement in both directions — a measured decay law is a measurement of whether the ground state is gapped.

This particular sub-topic, screening in quantum magnets, is slides-and-lecture material with no direct counterpart in `session9.ipynb`: none of the notebook's sections build a spinon/Abrikosov-fermion chain, so there is no hands-on check-in that reproduces (9.11) itself.

Notebook check-in: `session9.ipynb`, "The impact of a strong impurity" (Core)

Not a spin model, but the closest available exercise in spirit: it compares how a dimerized (more strongly gapped versus more weakly gapped) chain responds to a strong local impurity. Run it with (9.11) in mind and confirm the same principle in the fermionic language, the more strongly gapped chain screens the impurity over a shorter length scale.

9.10 Disorder and Anderson localization

Everything so far has concerned a *single* impurity. What happens once we scatter many impurities throughout the whole material? A single impurity barely dents the ground state;

many impurities, especially placed randomly, can change it completely, because they break translational symmetry throughout the entire system rather than just locally. It is worth laying out the full spectrum of possibilities, since there are genuinely three distinct regimes of translational symmetry to consider, not just "symmetric" and "not":

1. **Periodic.** Full translational symmetry; every eigenstate is extended, with weight $\sim 1/N$ on each of N sites.

2. **Quasiperiodic.** Not periodic (no repeating unit cell), but not random either: the potential follows a fixed, deterministic, non-repeating rule. Eigenstates here can be extended, critical (power-law localized, not even normalizable in the strict sense), or fully localized, depending on the potential's strength.

3. **Strongly (randomly) disordered.** Eigenstates are localized, each with a finite localization length.

Quasiperiodic case. The standard model here (an instance of the Aubry-André-Harper family, though not named as such in the transcript) is a 1D tight-binding chain with an on-site modulation at an irrational wavevector,

$$H = -t \sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.} \right) + V_0 \sum_n \cos(2\pi q\, n + \phi)\, c_n^\dagger c_n, \qquad q \notin \mathbb{Q}.$$

Because q is irrational, the potential never exactly repeats, so the chain has no translational symmetry at all, yet it is not random either: the value of the potential at every site is completely determined once q and ϕ are fixed. As V_0 is tuned up from zero, the eigenstates cross over from extended to marginal, power-law critical behavior right around the model's self-dual point ($V_0 = 2t$ in the plain Aubry-André model), and on to exponentially localized for V_0 well above it.

The diagnostic tool for telling these regimes apart is the **inverse participation ratio (IPR)**,

$$\text{IPR}(\psi) = \sum_i |\psi_i|^4, \qquad \sum_i |\psi_i|^2 = 1. \tag{9.12}$$

An extended state with weight $\sim 1/N$ per site gives IPR $\sim N \cdot (1/N)^2 = 1/N$, vanishing as the system grows. A localized state, with $O(1)$ weight on $O(1)$ sites, gives IPR $\sim O(1)$, independent of system size. A critical state sits in between, with IPR following a non-trivial power law in N, a multifractal signature that is neither of the two clean limits. Figure 9.6 shows all three eigenstates side by side, together with the N-scaling of (9.12) that separates them quantitatively.

Fully disordered case. Now let the on-site potential be genuinely random, of strength W, $H = -t \sum_n (c_n^\dagger c_{n+1} + \text{h.c.}) + \sum_n w_n\, c_n^\dagger c_n$ with w_n drawn uniformly from $[-W, W]$. At $W = 0$ this is an ordinary clean metal with the band structure of the periodic hopping alone; as $W \to \infty$ every eigenstate collapses onto a single site. In dimension $d \geq 3$, there is a genuine **Anderson localization transition** at a finite critical disorder W_c, a true mobility edge separating extended, metallic states from localized ones, exactly the kind of "critical point" described on the slides. Below the transition (or at any energy on the extended side of a mobility edge) the system conducts; above it, states exist at every energy, so there is *no spectral gap* in the density of states, but those states cannot carry current, so there *is* a transport gap: the **Anderson insulator** is an insulator with no gap in its spectrum at all, a genuinely unusual state of matter compared to every band or Mott insulator we have met so far in this course.

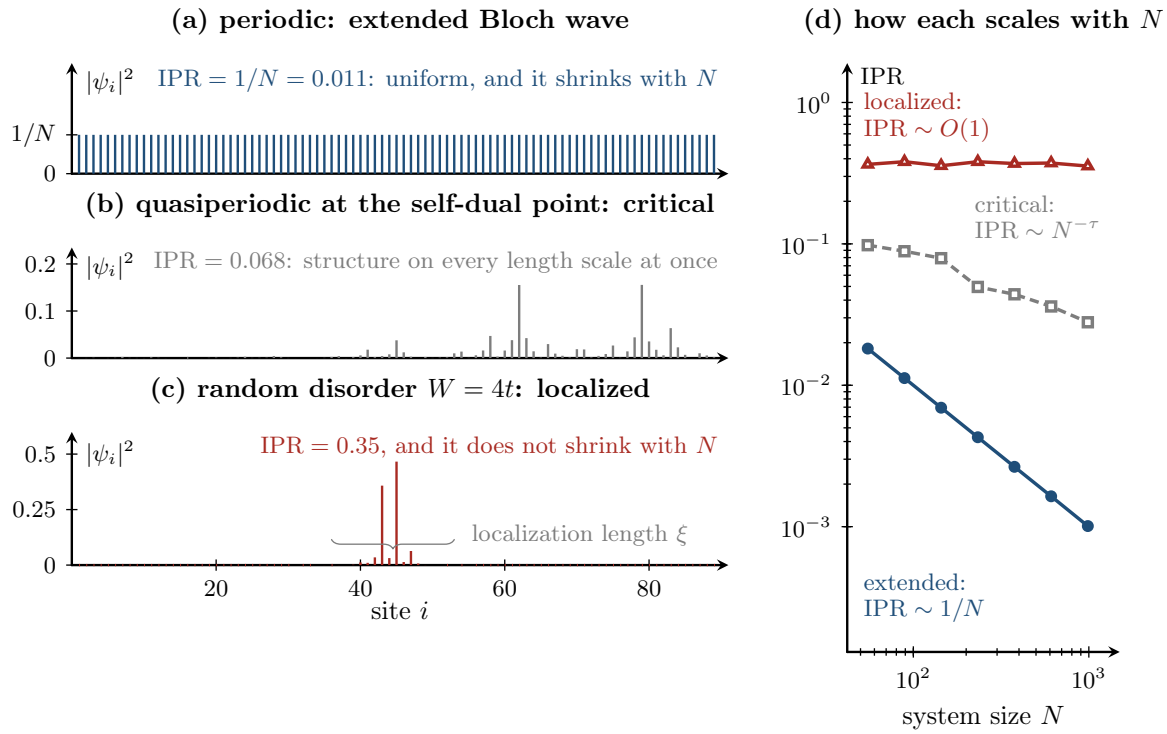


Figure 9.6: The three regimes of Section 9.10, shown as what they actually do to an eigenstate of a chain of $N = 89$ sites. (a) With a periodic potential every eigenstate is a Bloch wave: weight exactly $1/N$ on every site, and that height shrinks as the chain grows. (b) At the self-dual point of the quasiperiodic model the state is neither: it has structure on every length scale at once, large spikes with smaller copies of the same pattern between them — critical, or multifractal. (c) With genuinely random disorder the state collapses onto a handful of sites and is strictly zero, to numerical precision, over most of the chain. *Note the three different vertical scales*; that is precisely why one needs a number rather than an eye, which is what the inverse participation ratio (9.12) provides. (d) It separates the three cleanly by how it scales with system size: IPR $\sim 1/N$ for extended states, IPR $\sim O(1)$ for localized ones, and an anomalous power $N^{-\tau}$ with $0 < \tau < 1$ for the critical state in between — three straight lines of three different slopes on a log-log plot.

The special role of one dimension. In $d = 1$ (and, by the scaling theory of localization, marginally in $d = 2$ as well), there is no true metallic phase surviving at any non-zero disorder in the thermodynamic limit: an arbitrarily small $W > 0$ eventually localizes every state. What survives is a finite localization length $\xi(W)$, which for weak disorder near the band center of the 1D model scales as $\xi(W) \propto (t/W)^2$, diverging as $W \to 0$. A finite sample shorter than $\xi(W)$ still looks metallic in practice; only once the sample length exceeds $\xi(W)$ does the localized, insulating character actually show up in transport. This is the precise sense in which "weak disorder in 1D" and "genuinely metallic" are never quite the same statement, even though they can look identical at short length scales.

Notebook check-in: `session9.ipynb`, "The impact of disorder in different systems" (Core), "Extended and localized states in quasicrystals" (Challenge), and "Localization due to random disorder" (Core)

Sweep the quasiperiodic modulation strength and separately the random disorder strength W, and compute (9.12) for the resulting eigenstates in each case. Watch IPR cross over from $1/N$-like to $O(1)$ behavior as you increase the potential strength, and try to locate, by eye, where the extended-to-localized crossover happens in each of the two models; compare that to the critical/mobility-edge language above.

9.11 Closing summary

- A perturbation applied to a quantum material and measured elsewhere defines a **response function**; under translational and time-translation symmetry it is most naturally expressed in (q, ω)-space (Section 9.1).
- A classical warm-up already carries the whole session's logic: the sign of the divergence in the high-temperature magnetic susceptibility (9.1) predicts, from the disordered phase alone, whether the low-temperature order will be ferromagnetic or antiferromagnetic.
- Linear response theory (9.2)-(9.3) computes the leading effect of any perturbation from the unperturbed ground state alone, but breaks down for the orthogonality catastrophe, the Kondo effect, and the BCS gap, none of which are perturbatively connected to the unperturbed state.
- For a Fermi sea, the static and dynamical Lindhard functions (9.4)-(9.5) are built entirely from matrix elements between occupied and empty states near the Fermi surface, and diverge whenever a nesting vector connects large portions of that surface, signaling an instability (Section 9.4).
- Summing the interaction to all orders via the random phase approximation (9.6) reveals two distinct sources of divergence: electron-hole excitations from χ_0 itself, and collective plasmon excitations (9.7) from the RPA denominator.
- The Stoner criterion (9.8), $U\chi_0(q, 0) = 1$, recovers session 1 and session 6's magnetism as an *instability of the paramagnetic response function*, including the preference for antiferromagnetism at a nested (half-filled) Fermi surface, rather than as a mean-field ordered ansatz imposed by hand.
- A local impurity in a metal produces Friedel oscillations (9.9), a power-law-decaying, $2k_F$-periodic density modulation that directly images the Fermi surface, the same physics an STM measures via quasiparticle interference.

- A magnetic impurity in a superconductor creates a Yu-Shiba-Rusinov in-gap bound state (9.10), whose energy crosses zero at a genuine $S_{\text{tot}} = 0 \to 1/2$ quantum phase transition; a non-magnetic impurity leaves the spectrum essentially untouched.
- A vacancy in a topological superconductor or Chern insulator produces an in-gap mode purely by bulk-boundary correspondence, a diagnostic for topology that a clean, fully gapped bulk spectrum alone cannot provide (Section 9.8).
- Screening of a magnetic impurity in a quantum spin liquid reuses session 8's Abrikosov-fermion mean-field construction directly, and inherits the metal-impurity mathematics of Section 9.6: the induced spin correlation (9.11) decays as a power law for the gapless spin-1/2 chain and exponentially for the gapped spin-1 chain, giving a fully general gapped-versus-gapless diagnostic from impurity response alone.
- Many impurities at once can break translational symmetry throughout the whole system, giving rise to Anderson localization: a genuine mobility edge in $d \geq 3$, diagnosed by the inverse participation ratio (9.12), and a spectral-gap-free but transport-gapped Anderson insulator; in one dimension, arbitrarily weak disorder localizes every state, with a diverging localization length as the disorder strength goes to zero (Section 9.10).

Every session so far has gone in the same direction: start from a Hamiltonian, and compute what it predicts, whether an observable, a response function, or a ground state. Session 10 turns that arrow around. Having spent this whole course computing observables from Hamiltonians, next time we ask the inverse question: given only observed data, can we infer the Hamiltonian that produced it, and machine learning turns out to be exactly the right tool for making that question precise.

Session 10

Machine Learning Quantum Materials

Session 9 closed by turning linear response into a diagnostic tool and running it across essentially every system class the course had built: metals, superconductors, topological matter, quantum magnets. That session was a synthesis session in a narrow sense, one formalism applied everywhere. Today's session is a synthesis in a much broader sense: we are going to step back from condensed matter for a moment and ask what happens when we let machine learning talk to everything we have built over the last nine sessions, in both directions. Sometimes machine-learning architectures will turn out to be exactly the right tool for a hard many-body problem; sometimes, rather wonderfully, the reverse will be true, and a many-body technique we already know will turn out to be exactly the right architecture for a machine-learning problem. This is also, quite deliberately, the last session of the course, and before we start we should be upfront about one practical difference from every session before it: **there is no companion Jupyter notebook for this topic**. The `jupyter-notebooks/` folder stops at `session9.ipynb`, so today's notes carry no "try it in the notebook" callouts anywhere, not because the material is any less hands-on in spirit, but because that particular piece of infrastructure was not built for this session. Treat today's derivations as the thing to sit with directly on the page.

The session closes the course's arc in two deliberate ways, and it is worth naming both before we start. First, we are going to build neural network quantum states from the ground up. This is the first time the course meets the idea in full: a neural network standing in for the exponentially many coefficients of a many-body wavefunction, trained by turning the variational principle itself into a loss function, occupying its own place in the hierarchy of variational many-body states alongside matrix product states and projected entangled pair states, and reused a second time for reconstructing wavefunctions from a handful of measurements. Second, and this is the genuine full-circle moment, we are going to meet Hamiltonian learning, which is the literal inverse of the very first move this course ever made. Session 1 opened by saying: give us a Hamiltonian, and we integrate the Schrödinger equation forward to get observables. Today's closing topic starts from observables and asks for the Hamiltonian. Ten sessions after we first wrote that forward arrow down, we get to reverse it.

10.1 Two directions of traffic between machine learning and many-body physics

Let us set the map for today, because the session genuinely runs in two directions and it helps to know which way we are facing at each point. In one direction, we use machine-learning architectures, mostly neural networks, to attack many-body problems: representing wavefunctions too large to store, reconstructing states from a handful of measurements, classifying phases of matter, learning Hamiltonians from experimental observables. In the other direction,

we use many-body architectures, tensor networks in particular, as machine-learning tools in their own right, applied to ordinary tasks like image classification. And running underneath both directions is a third theme, arguably the most practically important one: combining machine learning with the condensed-matter machinery we already have (density functional theory, phase classification, Hamiltonian extraction) to attack problems that are hard for either approach alone. The order we will follow, matching how the lecture actually unfolded, is: neural networks and how they learn, neural network quantum states and tomography, tensor networks repurposed as machine learning, machine learning inside density functional theory, phase classification, and finally Hamiltonian learning.

10.2 Neural networks as function approximators

Machine learning is a large field, and it is worth being precise about which corner of it we are using today. Three broad task types are worth naming. **Supervised learning** maps labeled inputs to outputs, image classification being the standard example, and it is the one we will build on for the rest of the session. **Generative learning** models a probability distribution directly, so that one can *sample* new instances of it; a generative adversarial network trained on photographs of cats, for instance, can produce photorealistic images of cats that never existed. **Reinforcement learning** trains an agent to choose actions that maximize a cumulative reward, given the state of its environment. We will not develop generative or reinforcement learning further today (though both have real uses elsewhere in quantum materials); the GAN cat-face example is worth keeping in mind purely as a reminder that supervised learning is not the whole of the field, even though it is the whole of today's toolkit.

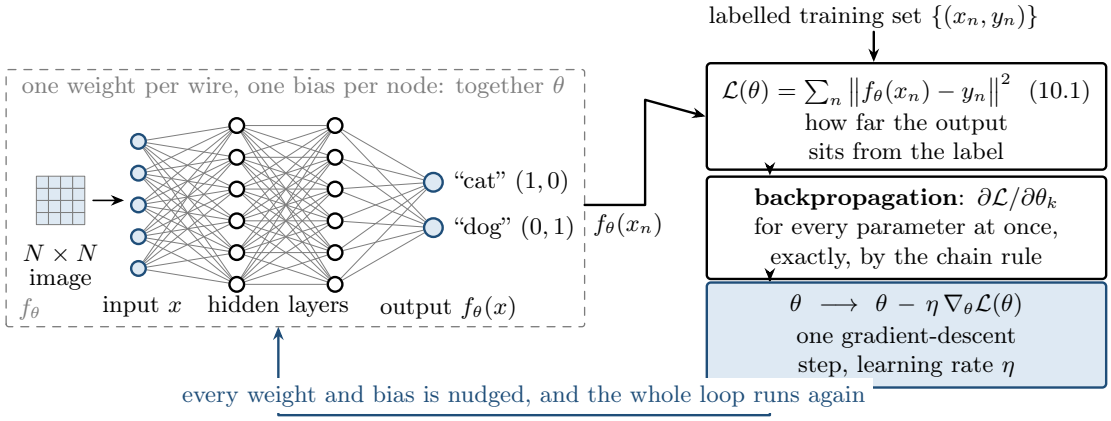
A deep neural network is best thought of as a very flexible, parametric function. It takes an input (an array of numbers), passes it through a sequence of layers, and produces an output (another array of numbers). Within a layer, each node's value is a nonlinear function of the values of all nodes in the previous layer, combined with a set of free parameters: the *weights* (one per connection) and *biases* (one per node). The architecture, the number and width of the layers, sets how complex a function the network *can* represent; the specific values of the weights and biases, collectively θ, set which function it *does* represent at any given moment. Training a network means searching over θ for the values that make the network implement the function we actually want. Concretely, we define a **loss function** $\mathcal{L}(\theta)$ that measures how far the network's output is from the desired one, and we minimize it, typically using backpropagation (an efficient, exact way of computing $\partial\mathcal{L}/\partial\theta$ via the chain rule) combined with gradient descent:

$$\mathcal{L}(\theta) = \sum_n \left\| f_\theta(x_n) - y_n \right\|^2, \qquad \theta \;\longrightarrow\; \theta - \eta\, \nabla_\theta \mathcal{L}(\theta), \tag{10.1}$$

where $\{(x_n, y_n)\}$ is a set of labeled training examples and η is the learning rate. Every use of a neural network in this session, however exotic the physics attached to it, is a variation on (10.1): identify the right input, the right output, and the right loss, and the training procedure is always the same gradient-descent machinery.

The concrete example to keep in mind, because we will reuse its exact architecture twice more today, is cat-versus-dog image classification. An image is an $N \times N$ matrix of pixel values; we flatten it into the network's input layer, and the output layer has just two nodes, with the target $(1, 0)$ for "cat" and $(0, 1)$ for "dog." We train on a labeled set of images, minimizing (10.1) with f_θ the network's output and y_n the one-hot label, and once trained, the network classifies new, unlabeled images. This is a genuinely reusable template: nothing about it depends on the input being a photograph, only on it being an array of numbers with a categorical label attached. We will meet the very same input-matrix-to-one-hot-output structure again

in Section 10.6, with a spin configuration standing in for a photograph. Figure 10.1 draws the template once and, underneath it, the four later constructions of this session that reuse it unchanged apart from the choice of input, output and loss.



The same loop, four more times in this session — only the three columns change

construction	input	output	loss, minimized over θ
NNQS ground state §10.3	a spin configuration s	the coefficient $C_\theta(s)$	the variational energy $E[\theta]$, (10.3)
state tomography §10.3	a spin configuration s	modulus and phase of $C_\theta(s)$	$-\sum_s n(s) \ln p_\theta(s)$
phase classification §10.6	a spin snapshot	a one-hot phase label	(10.1), entirely unchanged
Hamiltonian learning §10.7	an observable $\mathcal{O}$	the parameters λ	$\sum_n \|\mathcal{N}_\theta(\mathcal{O}_n) - \lambda_n\|^2$, (10.12)

Figure 10.1: The one construction the rest of this session keeps reusing. **Left:** a deep network is nothing but a parametric function f_θ. The architecture — how many layers, how wide — fixes what functions it *can* represent; the weights and biases θ fix which one it *does* represent right now. **Right:** training is the loop of (10.1): evaluate the output, measure it against the label with a loss, get $\partial\mathcal{L}/\partial\theta$ exactly by backpropagation, take a gradient step, repeat. **Bottom:** every later construction in this session is that same loop with a different choice of input, output and loss — nothing else changes. Read each row against the section it names: the physics sits entirely in the three columns, never in the machinery.

10.3 Neural network quantum states

We meet neural network quantum states (NNQS) here for the first time in the course, and we get to treat them immediately with the full machine-learning machinery of Section 10.2. The one-sentence idea is that a neural network can stand in for the exponentially many coefficients of a many-body wavefunction, occupying its own place in the hierarchy of variational many-body states this course has been building since session 7's matrix product states, and today's job is to make that idea precise: what exactly the network computes, what loss function trains it, and where it sits relative to the ansätze we already know.

Start from the difficulty we have met before, most recently in session 7's discussion of entanglement growth: a wavefunction for L spins,

$$|\psi\rangle = \sum_{s_1,\ldots,s_L} C(s_1,\ldots,s_L)\,|s_1,\ldots,s_L\rangle, \tag{10.2}$$

is specified by 2^L coefficients, and storing them explicitly is hopeless once L is much bigger

than 30 or 40. The neural network quantum state (NNQS) idea is to stop storing $C(s_1, \ldots, s_L)$ and instead *generate* it: build a neural network C_θ that takes a spin configuration $(s_1, \ldots, s_L)$ as input (formally, an L-pixel "image") and returns the coefficient, or, for a genuinely complex wavefunction, its real and imaginary parts (equivalently, its modulus and phase) as two output numbers. This is exactly the matrix-product-state move from Section 10.4 below, a compact generator standing in for exponentially many numbers, but built from a neural network's parameters rather than a chain of matrices.

The question is then how to find the right θ for the ground state, and here is where Section 10.2's machinery reappears essentially unchanged. The variational principle tells us that

$$E[\theta] \;=\; \frac{\langle\psi_\theta|H|\psi_\theta\rangle}{\langle\psi_\theta|\psi_\theta\rangle} \;=\; \frac{\sum_{s,s'} C_\theta^*(s)\, H_{ss'}\, C_\theta(s')}{\sum_s |C_\theta(s)|^2} \;\geq\; E_0, \tag{10.3}$$

with equality only when $|\psi_\theta\rangle$ coincides with the true ground state (or, in practice, the closest state the ansatz can reach). Equation (10.3) *is* the loss function of (10.1): the "observable output" the network is being trained to minimize is nothing but the system's energy. To make (10.3) computable without summing over all 2^L configurations, one samples s from the Born distribution $p(s) = |C_\theta(s)|^2 / \sum_{s'} |C_\theta(s')|^2$ and evaluates it as an average over the sampled local energy,

$$E_{\mathrm{loc}}(s) = \sum_{s'} H_{ss'}\, \frac{C_\theta(s')}{C_\theta(s)}, \qquad E[\theta] = \big\langle E_{\mathrm{loc}}(s) \big\rangle_{s\sim p}, \tag{10.4}$$

which only ever needs H evaluated between configurations connected by the (typically sparse) Hamiltonian, not the full $2^L \times 2^L$ matrix. Training then proceeds exactly as in (10.1): compute the energy and its gradient with respect to the network parameters,

$$\frac{\partial E}{\partial \theta_k} = 2\,\mathrm{Re}\Big[\big\langle E_{\mathrm{loc}}(s)\, O_k^*(s)\big\rangle - \big\langle E_{\mathrm{loc}}(s)\big\rangle\big\langle O_k^*(s)\big\rangle\Big], \qquad O_k(s) \equiv \frac{\partial \ln C_\theta(s)}{\partial \theta_k}, \tag{10.5}$$

and update $\theta_k \to \theta_k - \eta\, \partial E/\partial\theta_k$ repeatedly, with the same learning rate η as in any other neural-network training, until the gradient vanishes and the energy has converged to its variational minimum.

It is worth being precise about where NNQS sits relative to the other variational ansätze this course has used. The slides make this into an explicit nested picture: draw the full many-body Hilbert space as one region, and inside it a smaller region for every state an NNQS of a given architecture can reach; projected entangled pair states sit inside that NNQS region, matrix product states (session 7's construction) sit inside the PEPS region in turn, and the parameterizations of gapped one-dimensional systems sit innermost of all (a matrix product state is, after all, just a particular, highly structured choice of "network"). NNQS is therefore the most expressive of the variational families this course has used, precisely because it imposes the least structure on how the coefficients depend on the configuration. Equation (10.3) is the loss function and (10.4)-(10.5) the training procedure that make this picture operational rather than just structural. Figure 10.2 collects all three pieces: what the network actually computes, how the sampled energy estimate is assembled, and that nested picture drawn out.

10.3.1 Quantum state tomography

A second, genuinely distinct payoff of the same NNQS ansatz is **quantum state tomography**. Suppose we have a real experimental system in some unknown state $|\psi\rangle$, and the only access we have to it is through projective measurement: measure the state of every spin, and the system collapses to one particular configuration s with probability $|C(s)|^2$. Repeating the

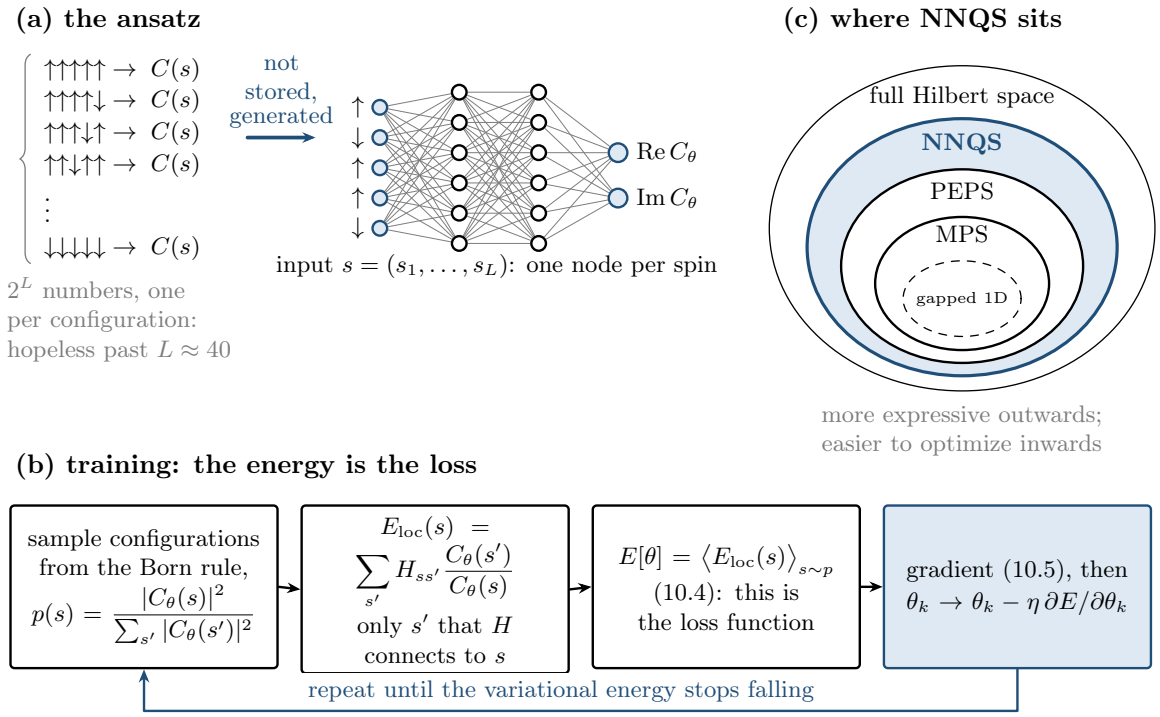


Figure 10.2: Neural network quantum states. **(a)** What the ansatz actually is: the exponentially long table of coefficients in (10.2) is not compressed, it is simply never written down. The configuration s is fed in as an L-pixel "image" and the network returns the one coefficient $C_\theta(s)$ that was asked for — the same silhouette as figure 10.1, with only the input and output relabelled. **(b)** Training: the variational energy (10.3) is the loss, and it is estimated by sampling configurations from the Born distribution and averaging the local energy (10.4). This is what makes the scheme work at all: $E_{\mathrm{loc}}(s)$ only ever needs the handful of matrix elements $H_{ss'}$ that a sparse Hamiltonian actually connects to s, never the full $2^L \times 2^L$ matrix. **(c)** The nesting the slides draw: a matrix product state is a highly structured special case of a PEPS, a PEPS is in turn the kind of state a large enough NNQS can absorb, and NNQS sits outermost because it imposes the least structure of all on how C depends on s. Read this as the lecture's organizing picture rather than as a theorem — the outer inclusion is a statement about sufficiently expressive networks, not about every architecture — and note that expressiveness is not free: the ordering runs the other way for how easy each family is to optimize.

measurement many times generates a set of observed configurations, distributed according to $|C(s)|^2$, but we never observe $C(s)$ itself, only its modulus squared, and doing this for all 2^L configurations separately is exactly the exponential problem we started with.

The fix is structurally identical to the ground-state search above: parameterize the unknown amplitude and phase with a network, for instance a restricted Boltzmann machine, or two separate networks, one for $|C_\theta(s)|$ and one for its phase, and fit θ by **maximum likelihood** against the observed measurement outcomes, rather than by minimizing an energy. Concretely, one maximizes $\sum_s n(s) \ln p_\theta(s)$ over θ, where $n(s)$ counts how often configuration s was observed and $p_\theta(s) = |C_\theta(s)|^2 / \sum_{s'} |C_\theta(s')|^2$ is the network's predicted probability. This reconstructs a full approximation to the unknown wavefunction using far fewer than 2^L measurements, which is precisely what makes it useful as an experimental tool: tomography with a genuinely exponential measurement budget is not an experiment anyone can run.

This is slides-and-transcript material with no notebook counterpart, in keeping with this session's overall absence of a companion notebook; there is no `sessionN.ipynb` cell to point you to here, so treat (10.3)-(10.5) and the tomography likelihood above as the complete, self-contained derivation.

10.4 The many-body wavefunction as a big-data problem: tensor networks for machine learning

We now run the previous section's logic in reverse. Instead of using a machine-learning architecture to solve a many-body problem, we treat the many-body wavefunction itself as an instance of a big-data problem, and repurpose a many-body compression technique, tensor networks, as a machine-learning architecture.

Recall the matrix product state (MPS) ansatz, which will already be familiar from session 7: rather than storing all 2^L coefficients of (10.2) independently, we assume they factorize into a product of per-site matrices,

$$C(s_1, \ldots, s_L) = \mathrm{Tr}\left[A^{s_1} A^{s_2} \cdots A^{s_L}\right], \tag{10.6}$$

where for each physical value of s_i (spin up or down, say) there is a distinct $D \times D$ matrix A^{s_i}, and D, the *bond dimension*, is a free parameter we choose. The number of free parameters in (10.6) scales as $L \times d \times D^2$ (d the local Hilbert space dimension), linear in system size rather than exponential, exactly the same compression idea as the NNQS of Section 10.3, just parameterized by matrix elements instead of network weights. The same compression trick generalizes beyond the strict one-dimensional MPS geometry: a *tree tensor network* rearranges the same matrices into a tree rather than a chain, the *multiscale renormalization ansatz* (MERA) adds extra layers of coarse-graining on top of a tree structure, and *projected entangled pair states* (PEPS) generalize the whole construction to a genuinely two-dimensional grid. For ground-state search these architectures are not interchangeable (an MPS is substantially easier to optimize than a PEPS, which is exactly why one-dimensional problems favor MPS and two-dimensional problems often still resist it); but if all we want is a flexible, efficiently-parameterized function of a high-dimensional input, as in machine learning, the distinction between architectures matters much less, and any of them will do.

This is a direct, literal reuse of session 7's own architectures, not just an analogy: the same matrix product states, tree tensor networks, MERA, and PEPS built there to compress ground-state wavefunctions are picked up here unchanged and repurposed as function approximators. This reframing has a name, "quantum-many-body-inspired machine learning," and a clean worked example: image classification with tensor networks. Map each pixel of an N-pixel image to a two-dimensional vector (white $\to$ spin up, black $\to$ some other reference direction, say), and take the tensor product of all N pixel-vectors to get a single vector in a 2^N-dimensional

space, exactly the same exponentially large space that a many-body wavefunction with N spins would live in. The classification task then becomes the search for a *linear* operator on that 2^N-dimensional space that maps each image to its one-hot label, and a tensor network (a tree-tensor-network shape works well here) is exactly the efficient, structured parameterization of such a linear operator that (10.6) and its generalizations already gave us. Applying the tensor network to the tensor-product image, then a final small tensor to project down to two output numbers, reproduces precisely the cat-versus-dog structure of Section 10.2, now with "purse versus shirt" as the worked example and the pixel tensor-product standing in for the flattened input matrix. Figure 10.3 draws the four geometries as tensor diagrams and then the classifier built out of one of them.

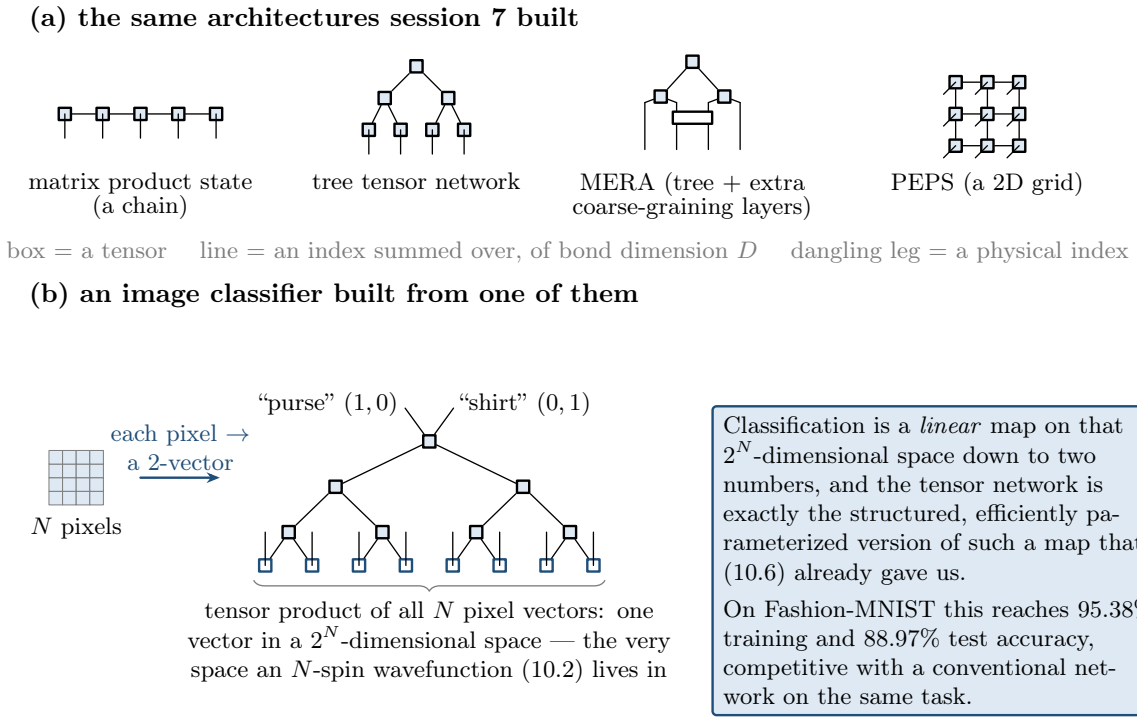


Figure 10.3: Tensor networks used the other way round. **(a)** The four geometries the text names, drawn as tensor diagrams: every box is a tensor, every line an index summed over, and the dangling legs are the physical indices. The bond dimension D carried by the internal lines is what (10.6) calls a free parameter. For ground-state search these are emphatically not interchangeable — an MPS optimizes easily, a PEPS does not — but as flexible parameterized functions of a high-dimensional input, which is all a classifier needs, any of them will serve. **(b)** Image classification with a tree tensor network. Each pixel becomes a two-component vector, and their tensor product is a single vector in a 2^N-dimensional space — exactly the space an N-spin wavefunction of (10.2) lives in. Classification is then a *linear* map on that space down to two numbers, and the tensor network is precisely the efficient, structured parameterization of such a map that (10.6) already handed us. Note what has and has not changed from figure 10.1: the input is still an array with a categorical label, the output is still one-hot, only the function in between is now built from matrices instead of layers of nodes.

This is not a toy result. Applied to the Fashion-MNIST dataset, a standard benchmark of clothing images, this tensor-network classifier reaches **95.38% accuracy on the training set and 88.97% on the test set**, competitive with conventional neural network architectures on the same task. (These numbers appear only on the slides, not narrated in the recording beyond "very high accuracy," and are exactly the kind of concrete figure worth pulling into the written notes rather than leaving implicit.) The broader point to take away is symmetric with Section 10.3's: a variational architecture built to compress many-body wavefunctions is, mathematically, just another kind of neural network, and it is directly usable as one.

10.5 Machine learning in density functional theory

We turn now to a different problem, and a much older piece of condensed-matter machinery: density functional theory (DFT). The motivation is familiar from Section 10.3, but posed for real materials rather than lattice models. Given a material's Hamiltonian,

$$H = \sum_i \left(-\frac{\hbar^2 \nabla_i^2}{2m} \right) + \sum_i v_{\text{ext}}(\mathbf{r}_i) + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|},$$

with v_{ext} the potential of the nuclei, the many-body ground state $\psi_0(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ is, of course, completely intractable to obtain directly for a realistic number of electrons. DFT sidesteps this by asking a smaller question: not for the full wavefunction, but for the much lower-dimensional ground-state electronic density $n(\mathbf{r}) = N \int |\psi_0|^2 \, d^3 r_2 \cdots d^3 r_N$.

The **Hohenberg-Kohn theorem** is what makes this legitimate rather than merely convenient: for a given external potential $v_{\text{ext}}(\mathbf{r})$, the ground-state density $n(\mathbf{r})$ determines $v_{\text{ext}}(\mathbf{r})$ uniquely (up to an additive constant), and hence determines the full Hamiltonian and the full many-body wavefunction. The arrow $\psi_0 \to n$ is trivial, an integral; the arrow $n \to \psi_0$ is the nontrivial, but *guaranteed to exist*, content of the theorem. Consequently the ground-state energy is a functional of the density alone,

$$E[n] = F[n] + \int d^3 r \, v_{\text{ext}}(\mathbf{r}) \, n(\mathbf{r}), \tag{10.7}$$

with $F[n] = T[n] + V_{ee}[n]$ a *universal* functional (the same for every system, independent of v_{ext}) whose exact mathematical form nobody knows.

The **Kohn-Sham construction** makes (10.7) usable in practice. We introduce a fictitious system of non-interacting electrons chosen to reproduce the *same* density $n(\mathbf{r})$ as the real, interacting one, governed by an effective single-particle Hamiltonian

$$H_{\text{KS}} = -\frac{\hbar^2 \nabla^2}{2m} + v_{\text{eff}}(\mathbf{r}), \qquad v_{\text{eff}}(\mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \underbrace{e^2 \int d^3 r' \, \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}}_{\text{Hartree (mean-field) term}} + \underbrace{\frac{\delta E_{\text{xc}}[n]}{\delta n(\mathbf{r})}}_{v_{\text{xc}}(\mathbf{r})}. \tag{10.8}$$

The first two pieces of v_{eff} are exactly a mean-field decoupling, structurally the same move as session 1's mean-field theory, applied here to the Coulomb interaction rather than a lattice Hubbard term. The last piece, the exchange-correlation potential v_{xc}, is defined to absorb everything the mean-field approximation misses, and it is precisely as unknown as $F[n]$ was. In practice one approximates $E_{\text{xc}}[n]$ (LDA, GGA, meta-GGA, and further refinements) and solves (10.8) self-consistently: guess a density, build H_{KS}, diagonalize it for the Kohn-Sham orbitals, recompute the density from those orbitals, and repeat until convergence.

Two distinct places to bring machine learning into this loop were discussed, and it is worth keeping them separate, because they solve different problems:

- **Learning the functional itself.** Replace or augment the LDA/GGA/meta-GGA approximation to $E_{\text{xc}}[n]$ (or its functional derivative v_{xc}) with a neural-network parameterization, trained to match higher-accuracy reference calculations. This can genuinely improve accuracy, but it does *not* remove the self-consistency loop: one still has to diagonalize H_{KS} at every iteration, which remains the expensive step for large systems regardless of how good v_{xc} is.
- **Learning the whole map.** A bolder use of the Hohenberg-Kohn theorem is to skip the self-consistency loop altogether: train a neural network to map atomic positions

(equivalently, $v_{\text{ext}}(\mathbf{r})$) directly to the ground-state density $n(\mathbf{r})$, with no diagonalization at inference time at all. This is the idea behind a "machine learning potential," already an established technique in quantum chemistry and condensed matter: train the network on conventional Kohn-Sham calculations for small, tractable systems, then apply it to much larger systems where a direct Kohn-Sham calculation would be too costly to run at all.

Figure 10.4 places both of these on the self-consistency loop itself, which is the quickest way to see why only the second of them changes what is computationally reachable.

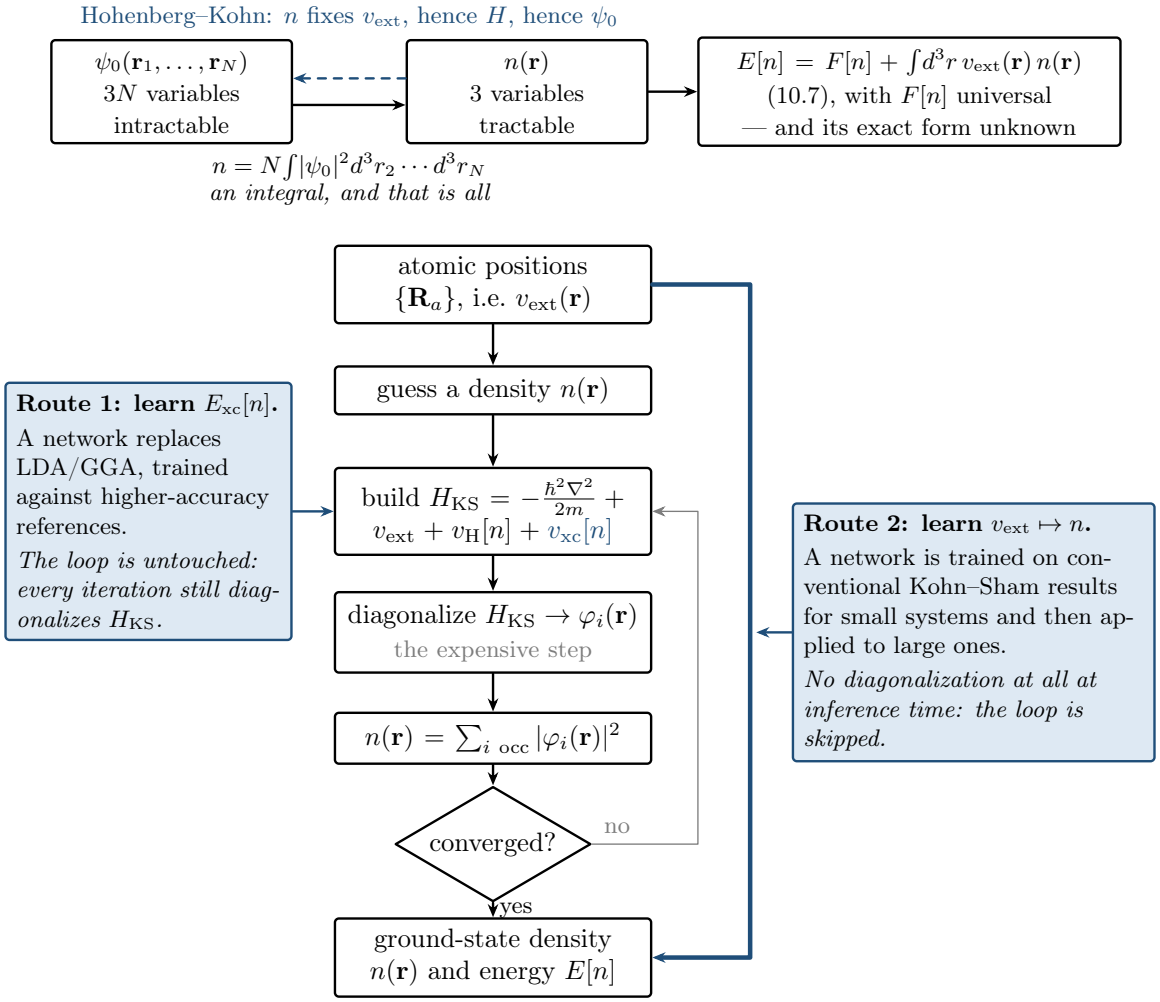


Figure 10.4: Where machine learning enters density functional theory, and why the two entry points are not the same idea. **Top:** the Hohenberg–Kohn reduction is deliberately lopsided. Going from the many-body ground state to the density is a $3N \to 3$ integral anyone can do; coming back is guaranteed to be possible but comes with no formula, and that missing formula is the whole opening. **Bottom:** the Kohn–Sham loop (10.8) turns the problem into a self-consistent single-particle one. **Route 1** replaces the unknown exchange–correlation piece by a trained network: more accurate, but the diagonalization stays inside the loop and it is the diagonalization, not v_{xc}, that limits the system size. **Route 2** learns the Hohenberg–Kohn map $v_{\text{ext}} \mapsto n$ outright, trained on conventional Kohn–Sham results for small systems, and then does no diagonalization at all — the whole loop is bypassed. Only route 2 changes what is computationally possible; route 1 changes only how good the answer is.

This too is slides-and-transcript material with no notebook counterpart; DFT calculations were never part of this course's numerical toolkit (`pyqula` is a tight-binding and mean-field library, not a DFT code), so there is nothing to point you to beyond the derivation above.

10.6 Phase classification: reusing the local-order-parameter-versus-topological-order split

We come now to a section that is, in a real sense, a review of the whole course's classification scheme, seen through a machine-learning lens. Throughout this course we have used two quite different tools to tell phases apart. Sessions 1, 5, and 6 used a **local order parameter**: a quantity built from a simple, local expectation value (charge imbalance for the CDW, magnetization for the magnet, the pairing amplitude for the superconductor) that is zero in one phase and nonzero in the other. Sessions 3 and 4 instead used a **topological invariant** (the Chern number, the Zak phase) precisely because the phases in question are *not* distinguished by any local quantity at all; the distinction only becomes visible non-locally. Today's phase-classification example runs exactly this contrast as an "easy" and a "hard" machine-learning worked example, and the lecture itself frames it that way.

The easy case: the Ising model. Take classical Ising spins $\sigma_i = \pm 1$ on a square lattice with the standard nearest-neighbor Hamiltonian and a local order parameter,

$$H_{\text{Ising}} = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \qquad m = \frac{1}{N} \sum_i \sigma_i, \tag{10.9}$$

which orders ferromagnetically ($|m| \to 1$) below a critical temperature T_c and disorders ($m \to 0$) above it. Given only a single snapshot of the spin configuration, at some unknown temperature, the phase is easy to read off even by eye: sum the spins and look at whether the result is close to $\pm N$ or close to 0. It is nonetheless a clean, controlled test case for machine learning: train the very same cat-versus-dog architecture from Section 10.2 (input, the $N \times N$ spin snapshot; output, a one-hot ordered/disordered label) on labeled snapshots below and above T_c, and the classifier sharpens into a genuinely sharp transition as the system size grows (the concrete sweep shown on the slides runs $L = 10$ up to $L = 60$), reproducing what we already know analytically.

The hard case: the Ising gauge theory. Now replace the pairwise coupling with a **plaquette** interaction, the product of the four spins around each elementary square of the lattice,

$$H_{\text{IGT}} = -J \sum_p \prod_{i \in \partial p} \sigma_i, \tag{10.10}$$

with the sum running over plaquettes p and the product over the four spins sitting on the four bonds that bound that plaquette; note that the spins here live on the *links* of the lattice, not on its sites. At zero temperature the ground state satisfies $\prod_{i \in \partial p} \sigma_i = +1$ on *every* plaquette, but this constraint is compatible with a macroscopic number of distinct spin configurations (flipping the four spins on the bonds that meet at any single vertex, for instance, leaves every plaquette product unchanged, since each of the four plaquettes touching that vertex contains exactly two of those four bonds and every other plaquette contains none — this local gauge transformation is what gives the model its name). At infinite temperature, by contrast, plaquettes with product $+1$ and -1 occur with equal probability. Crucially, both the ordered, constraint-satisfying configurations and the disordered ones can look, individually, like a featureless 50/50 mix of up and down spins: the naive order parameter (10.9) that worked perfectly for the Ising model is *useless* here, exactly because this is a topological rather than a local-order transition, the same structural obstruction that forced sessions 3 and 4 away from local order parameters and toward Chern numbers and Zak phases. What (10.10) reproduces at the level of a toy classical model is the essential feature of those earlier examples: the low-temperature and high-temperature phases are **topologically distinct**, but nothing local tells them apart.

A convolutional neural network, trained on labeled low-T and high-T configurations with the same supervised architecture as before, succeeds where the naive summary statistic fails; it is, in effect, learning a nonlocal function of the spin configuration that plays the role a topological invariant would play analytically. One further physical subtlety is worth stating precisely, since it is easy to blur: the two phases here are separated by a **size-dependent crossover**, not a true finite-temperature phase transition, so there is no sharp T_c to converge to as the system grows, unlike the genuine Ising transition of (10.9). What makes the machine-learning result genuinely useful is that the trained network, having only ever seen one system size during training (the slides show it trained at a single size and evaluated across $L = 4$ up to $L = 28$), correctly captures this crossover across *other* sizes and temperatures it was never trained on. Figure 10.5 sets the easy and the hard case side by side in real space, which is where the difference between them is actually visible.

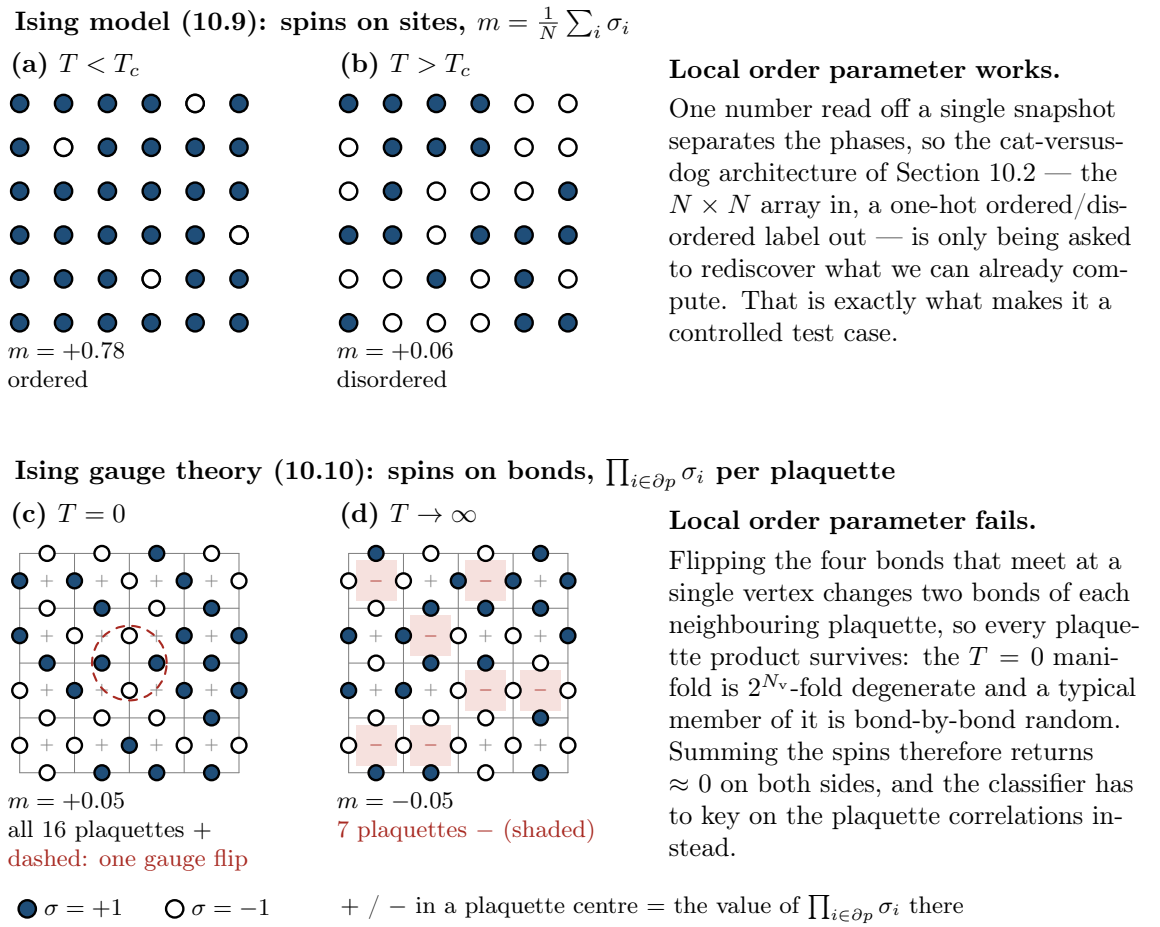


Figure 10.5: Why one phase-classification problem is easy and the other is hard. **(a,b)** For the Ising model (10.9) the magnetization of a single snapshot already separates the phases, and a network trained on such snapshots is only rediscovering the local-order-parameter logic of sessions 1, 5 and 6. **(c,d)** For the Ising gauge theory (10.10) the spins live on the *bonds*, and the $T = 0$ condition $\prod_{i\in\partial p}\sigma_i = +1$ leaves an exponentially large degenerate manifold: a typical ground state is bond-by-bond random, so the magnetization is ≈ 0 in both columns and no single-bond quantity tells them apart. What differs is the four-spin plaquette structure, marked in the plaquette centres, and finding a function of the raw configuration sensitive to it is what the convolutional network has to do for itself. This is the same obstruction that forced sessions 3 and 4 from local order parameters to Chern numbers and Zak phases.

10.7 Hamiltonian learning: the inverse of session 1's opening move

We come, finally, to the piece of today's material that closes the loop most explicitly. Go back to session 1, Section 1.4: every quantum mechanical system is governed by the Schrödinger equation, and given a Hamiltonian (and an initial state) we can integrate forward and obtain, in principle, every observable the system will ever show us: density, magnetization, response functions, and so on. That forward arrow, Hamiltonian $\rightarrow$ observable, is what essentially this entire course has been built to compute, one physical setting at a time.

But it is not the arrow an experimentalist actually gets to use. A real measurement never hands you the Hamiltonian; it hands you an observable directly, a density, a conductivity, an ARPES spectrum, a neutron-scattering structure factor, and the Hamiltonian is exactly the thing you do *not* have and would like to infer. **Hamiltonian learning** poses this explicitly as the inverse problem: the slides write the Hamiltonian itself as $H(\lambda_i, w_i)$, splitting its dependence into the physical parameters of interest λ (magnetic or electric fields, hopping amplitudes, many-body interaction strengths) and a set of "hidden," or nuisance, variables w one does *not* want to fit (smaller marginal parameters, disorder realizations, artifacts of whichever computational method generated the training data, plain randomness). Writing $\mu \equiv w$ for that hidden-variable vector, the forward direction is the map

$$\mathcal{O} = \mathcal{O}\big[H(\lambda, \mu)\big], \tag{10.11}$$

and Hamiltonian learning asks for the inverse map, $\mathcal{O} \mapsto \lambda$, marginalizing over μ as best we can rather than trying to fit it. This is a genuinely difficult inverse problem in general (many different λ can be compatible with the same measured $\mathcal{O}$, and real experimental data to train on is scarce), and the practical strategy is, once again, to lean on theory to make up for the shortage of experiment. Simulate the observable $\mathcal{O}$ from many different, *known* Hamiltonians $H(\lambda_n)$ spanning the parameter range of physical interest (the same forward calculations, density, magnetization, spectroscopy, that earlier sessions taught us to compute directly), and train a neural network $\mathcal{N}_\theta$ on this synthetic data to invert (10.11), minimizing

$$\mathcal{L}(\theta) = \sum_n \big\| \mathcal{N}_\theta\big(\mathcal{O}[H(\lambda_n)]\big) - \lambda_n \big\|^2 . \tag{10.12}$$

Once trained on simulated pairs where the answer λ_n is known by construction, the same network is applied to a genuine experimental observable $\mathcal{O}_{\text{exp}}$ (a two-dimensional spectroscopy of a quantum magnet is the concrete example the lecture used), and $\hat{\lambda} = \mathcal{N}_\theta(\mathcal{O}_{\text{exp}})$ gives an estimate of the physical Hamiltonian parameters, for instance the first-, second-, and third-neighbor exchange couplings of that magnet, extracted directly from a measurement with no closed-form inversion formula available at all. Figure 10.6 lays out both stages, the synthetic training set and the deployment on real data, against the forward arrow they are jointly inverting.

This is the literal mirror image of session 1's opening framing, and it is worth sitting with that symmetry for a moment. We spent nine sessions building increasingly sophisticated ways to go from a Hamiltonian to an observable: mean-field decoupling, Bloch's theorem, topological invariants, Landau levels, Nambu spinors, magnon spectra, tensor-network ground states, parton constructions, and response functions. Hamiltonian learning does not abandon any of that; it uses exactly that forward machinery, run many times over on synthetic data, as the training set for learning to run the process in reverse. The course's method and the course's final method are, in that sense, the same method, pointed in opposite directions.

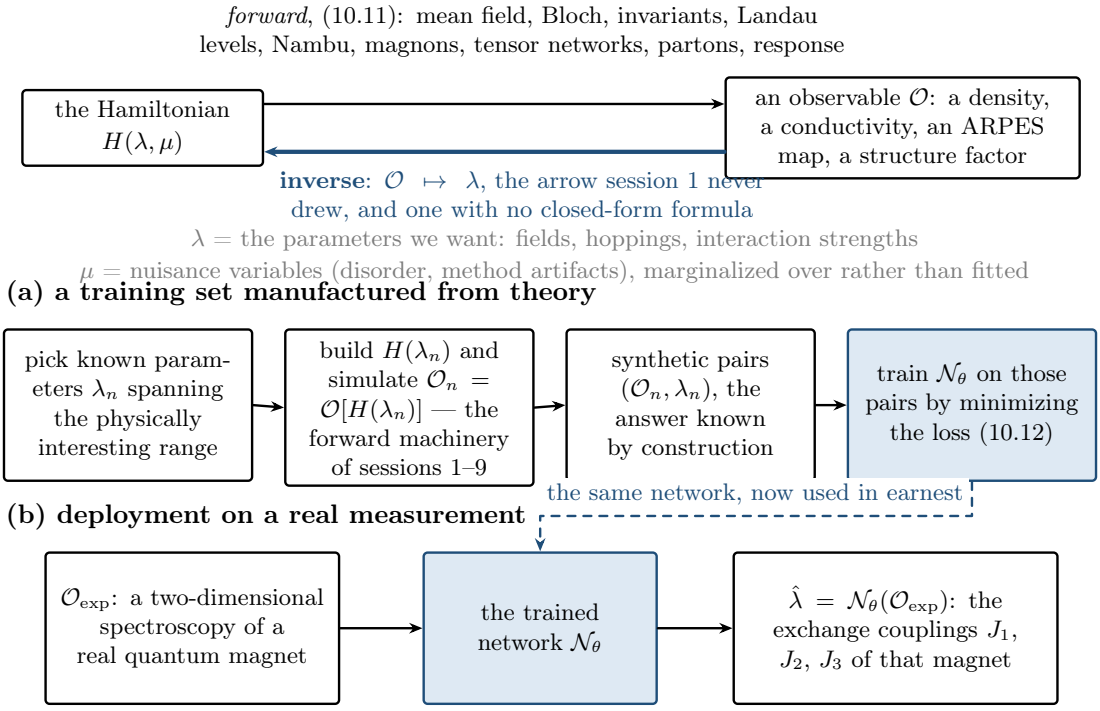


Figure 10.6: Hamiltonian learning, and why it closes the course's loop rather than stepping outside it. **Top:** the forward arrow $H \mapsto \mathcal{O}$ of (10.11) is what sessions 1 to 9 spent their entire time building; the arrow an experiment actually needs runs the other way, and no closed-form inversion of it exists. **(a)** The way round the shortage of experimental training data is to manufacture it: simulate $\mathcal{O}$ from many *known* Hamiltonians, so that every training pair arrives with its answer λ_n attached by construction, and fit the network with (10.12). **(b)** Only then is the trained network pointed at a real measurement. Notice what the training set is made of: the forward machinery of this whole course, run many times over. The course's final method and its first method are the same method, pointed in opposite directions.

10.8 Closing summary

- A deep neural network is a parametric function trained by minimizing a loss function via gradient descent, equation (10.1); every other construction in this session is a specific choice of input, output, and loss built on top of that one template.
- **Neural network quantum states** replace explicit storage of a wavefunction's 2^L coefficients with a network $C_\theta(s)$, equation (10.2), trained by treating the variational energy itself as the loss function, equations (10.3)-(10.5); NNQS sits above matrix product states and PEPS in the variational hierarchy, imposing the least structure of any ansatz this course has used. The same ansatz, fitted by maximum likelihood instead of energy minimization, solves quantum state tomography with far fewer than 2^L measurements.
- **Tensor networks**, built from the matrix product state ansatz (10.6) and its tree/MERA/PEPS generalizations, can be run in reverse as machine-learning architectures; applied to image classification (Fashion-MNIST), this reaches 95.38% training and 88.97% test accuracy, competitive with conventional neural networks.
- **Density functional theory** rests on the Hohenberg-Kohn functional (10.7) and the Kohn-Sham self-consistency loop (10.8); machine learning enters either by learning the unknown exchange-correlation piece of (10.8) (self-consistency loop unchanged) or by learning the entire potential-to-density map directly (self-consistency loop removed entirely).
- **Phase classification** reuses this course's local-order-parameter-versus-topological-order split one more time: the Ising model (10.9) is the easy, locally-detectable case (sessions 1/5/6's logic), while the Ising gauge theory (10.10) is the hard, topologically-distinct-phases-with-no- local-order-parameter case (sessions 3/4's logic), and a trained network succeeds where the naive order parameter fails.
- **Hamiltonian learning** poses the inference of a Hamiltonian from measured observables, equation (10.11), as the literal inverse of session 1's opening move; training a network on synthetic forward data, equation (10.12), and applying it to real experimental observables lets us close that loop in practice, not just in principle.

We opened this course by asking how a handful of identical atoms, arranged into a lattice, could produce a phenomenology as rich as the one we spent ten sessions touring: charge density waves, magnets, superconductors, topological insulators, fractional excitations, spin liquids, and more, none of it visible in a single atom on its own. Answering that question took us through second quantization and mean-field theory, through Bloch's theorem and topological invariants, through Nambu spinors and magnon spectra, through tensor networks and parton constructions, and through linear response as a diagnostic across all of it. Today's session closed that arc twice over: once by showing that a neural network can stand in for the many-body wavefunction itself, and once by showing that the entire forward machinery of the course, Hamiltonian in, observable out, can be run backward to infer the Hamiltonians that real materials actually have. That is a fitting place to leave things: not with a final answer, but with a general-purpose way of asking the question again, for whichever quantum material comes next.